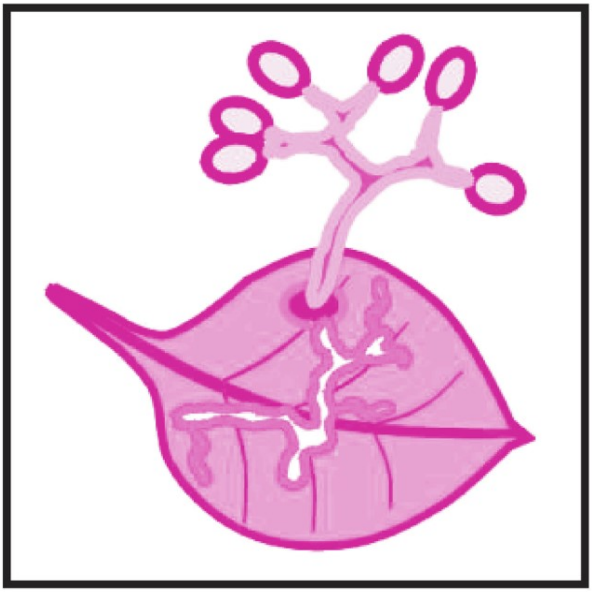
Developments in Plant Pathology

Advances in Downy Mildew Research

Volume 2

Edited by
Peter Spencer-Phillips and
Michael Jeger



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ADVANCES IN DOWNY MILDEW RESEARCH – VOLUME 2

Developments in Plant Pathology

VOLUME 16

The titles published in this series are listed at the end of this volume.

Advances in Downy Mildew Research – Volume 2

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PREFACE

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This second volume of Advances in Downy Mildew Research has arisen from novel data presented at the Downy Mildews Workshop at the 8th International Congress of Plant Pathology (ICPP) held in Christchurch, New Zealand in February 2003, together with more general overview contributions.

Emphasis in this volume is on the biology of compatible interactions, forecasting and epidemiology, host specialisation, genetic variability amongst pathogen populations, novel methods for detection and systematics, and induced resistance. Two chapters focus on the related oomycete *Albugo candida*, which shares many pathogenic characteristics with the downy mildews. Much can be learnt by comparing different pathosystems, and it is important to maintain this breadth when a small number of model systems, such as downy mildew on *Arabidopsis thaliana*, are commanding the most research attention. Contributions on specific downy mildews focus on *Bremia lactucae*, *Peronospora destructor*, *Peronospora sparsa*, *Peronospora viciae*, *Plasmopara halstedii*, *Plasmopara viticola, Pseudoperonospora cubensis* and *Sclerospora graminicola*. Review chapters on compatibility, forecasting and systematics consider a broader range of downy mildew fungi, and compare them with other oomycete and biotrophic pathogens.

As with the previous volume, the aim has been to address a variety of aspects rather than encompass all downy mildew research. It is envisaged that other topics and pathogens will be included in future volumes linked to the five year ICPP cycle. Areas based on molecular technology are undergoing the greatest change due to a very substantial investment of resources. Thus the genetics of virulence, avirulence and host resistance, phylogeny and nomenclature, rapid in-field diagnostics, cell signalling and molecular mechanisms of compatibility are where many significant advances are occurring. These advances have the potential to underpin improved control measures, which are more durable, sustainable and safer for both consumers and the environment. They also promise to help provide the increased crop yields required to feed hungry people in the developing nations. With some 17% of the world fungicide market directed towards controlling downy mildew diseases (Gisi, 2002), let alone other oomycetes, significant progress is essential.

It is important that advances in knowledge and technology are integrated and applied in disease control. For the latter to be successful, there is a continued need for parallel research and practical expertise at the basic biology, disease control and epidemiological levels. We look forward to learning what has been achieved in the next volume of this series and at the 2008 ICPP in Torino, Italy.

Gisi U. (2002) Chemical control of downy mildews. In *Advances in Downy Mildew Research*, P.T.N. Spencer-Phillips, U. Gisi, A. Lebeda, eds. Kluwer Academic Publishers, Dordrecht, The Netherlands, pp. 119-159.

THE COMPATIBLE INTERACTION IN DOWNY MILDEW INFECTIONS

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1. INTRODUCTION

Downy mildews are oomycetes belonging to the orders Peronosporales and Sclerosporales in the class Peronosporomycetes, within the kingdom Straminipila, according to Dick (2002a, 2002b). The taxonomy of this group of organisms, however, is under continuous revision. For example, *Peronospora parasitica* causing downy mildew of brassicas is now known correctly as *Hyaloperonospora parasitica* (Constantinescu and Fatehi, 2002), and will be referred to as such throughout this review. Further revisions will follow shortly: Choi and co-workers (2003) recently suggested that *H. parasitica* should be divided into several distinct species, based on sequence analysis of the ITS region of rDNA.

Although these oomycetes do not map phylogenetically to the kingdom Fungi, they are considered to be fungi in the broadest sense because they share a life-style based on a hyphal and mycelial mode of growth, an absorptive heterotrophic nutrition and reproduction via spores (Dick, 2002a; Diéguez-Uribeondo *et al*., 2004). The downy mildew fungi are also examples of endophytic biotrophs forming an intercellular mycelium, typically with intracellular haustoria, and gaining nutrients from living host cells. They are obligate to this biotrophic mode of nutrition and none have been grown in axenic culture, although hyphae of *Peronospora viciae* isolated from pea leaves are able to accumulate carbon from glucose *in vitro* (El-Gariani and Spencer-Phillips, this volume). They share these characteristics with the Albuginaceae (monogeneric, ie all *Albugo* spp.), another family of obligate biotrophic pathogens within the Peronosporales

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that cause white rust or white blister diseases (see Meekes *et al.* and Gilijamse *et al.*, this volume). In some respects, they also share much with the biotrophic phase of hemi-biotrophic *Phytophthora* species such as *Phytophthora infestans* (Hohl, 1991). Indeed it appears that the largest downy mildew genus, *Peronospora*, is more closely related to *Phytophthora* than it is to both the other downy mildew genera and the Albuginaceae (Cooke *et al.*, 2002; Voglmayr, 2003).

The fact that downy mildew fungi form intercellular hyphae and intracellular haustoria has meant that inevitably they are compared with the haustorial forming epiphytic and endophytic powdery mildew (ascomycete) and the endophytic rust (basidiomycete) fungi. It should be remembered, however, that downy mildews do not always form haustoria (Fraymouth, 1956), and their intercellular hyphae also provide an interface for nutrient uptake and general inter-communication, as in other non-haustorial biotrophs (Spencer-Phillips, 1997). Indeed Dick (2002a) has pointed out that the development of haustoria may have evolved separately even within the oomycetes, so that haustoria may have a different physiology and function in each of the genera *Albugo*, *Peronospora* and *Sclerospora*.

Whilst it is tempting to assume that molecular mechanisms of biotrophy will have evolved in parallel with morphological features (*viz.* haustoria), there are many examples in the history of biology where this approach has been misleading. In fungal systematics, similar morphological features used previously for taxonomic and nomenclature purposes do not fit with contemporary phylogenic concepts based on molecular data. In relation to physiological function, the debate concerning the apoplastic versus symplastic pathways for phloem loading provides another example that is potentially relevant to strategies for nutrient acquisition by biotrophic fungi (Spencer-Phillips, 1997). These alternative pathways were championed by opposing research teams, with each group working with their favoured model plant species. It was realised subsequently that different plants might use different mechanisms, as exemplified by this statement from Bourquin *et al.* (1990): “either one of these two concepts... is not valid, or phloem loading occurs via the symplastic pathway in some species and via the apoplastic pathway in other species”. The latter, with a plethora of intermediates, is now accepted widely (Lalonde *et al.*, 2003).

Thus it is important to retain an open mind about mechanisms of compatibility in downy mildews, and to ensure that they are studied in parallel with other oomycetes, as well as the rusts and powdery mildews. An additional reason for a special focus on downy mildew diseases is that they cause significant crop losses (Clark and Spencer-Phillips, 2000), and account for approximately 17% of the world fungicide market (Gisi, 2002).

It is encouraging, therefore, that one of the main model species for plant molecular biology, *Arabidopsis thaliana*, is susceptible to both *Albugo candida* and *Hyaloperonospora parasitica* infections. This has resulted in downy mildew diseases leading the field in some aspects of research into host-pathogen interactions (Mauch-Mani, 2002; Slusarenko and Schlaich, 2003). There remains a need, however, to work with a range of downy mildew diseases and hosts that are important crops, because there is considerable diversity within this group of pathogens as mentioned above.

The present review focuses on aspects of the compatible interaction between downy mildew pathogens and their hosts, and expands on parts of an earlier overview of downy mildew biology by Clark and Spencer-Phillips (2000). A recent review of resistant interactions and the mechanisms of resistance is provided by Mauch-Mani (2002), whilst evolutionary aspects together with a critical analysis of the key features of the downy mildew pathogens are considered by Dick (2002a). Detailed descriptions of symptoms, the formation, survival and germination of asexual spores and oospores, and various other aspects of the biology of individual downy mildew pathogens are given in chapters in Spencer (1981), the IMI descriptions published by CAB International Mycological Institute (Kew, Surrey, UK), and the Compendium of Plant Diseases series published by the American Phytopathological Society (eg see Biddle, 2001).

2. COMPATIBILITY: SYMPTOMS AND DETECTION OF INFECTION

Infected plant tissue can appear to be perfectly healthy under some growth conditions, with host cells infected by haustoria remaining alive and appearing (superficially, at least) to be functioning normally. In the laboratory, pea leaves with localised secondary lesions of *Peronospora viciae* can remain green even during sporulation, and may only senesce at the same rate as uninfected leaves (Clark, 1989). In some compatible interactions between *Arabidopsis thaliana* and *H. parasitica*, no symptoms are visible before sporulation (Parker *et al.*, 1996). Other downy mildews (and sporulating field infections of pea) usually give rise to varying amounts of chlorotic and necrotic patches of shoot tissue, especially in the later stages of infection, although it has been suggested that necrosis is the result of secondary infecting microorganisms. For example, *Alternaria alternata* has been found in downy mildew lesions of cucurbits (Bains and Singh, 1996).

It is important to note that the terms 'compatible' and 'incompatible' are not absolute for all interactions. For a particular plant species and downy mildew pathogen, different host accessions and pathogen isolates can show a range of interactions which vary through a spectrum from completely incompatible to completely compatible. The *A. thaliana*/*H. parasitica* pathosystem provides a good example (eg Holub *et al.*, 1994). Highly incompatible interactions result in extensive or localised but rapid host cell damage (a hypersensitive reaction) and no sporulation, whilst a highly compatible interaction would show no necrosis and support profuse sporulation of the pathogen. An intermediate interaction would display varying amounts of host cell damage and some sporulation. Tissue specificity has also been described, with different levels of compatibility seen in leaves versus cotyledons in the *H. parasitica* infections of *Arabidopsis* (Aarts *et al.*, 1998). The term 'compatible' is therefore used here to refer to the most compatible interactions between particular host and downy mildew species.

Lack of macroscopically visible symptoms, at least in early stages of infection and in some pathosystems until asexual sporulation commences (eg see Gilles and Kennedy, Xu and Pettitt, this volume), has important implications for assessment in the field. This is because the disease severity keys used to determine the degree of resistance of cultivars to downy mildews measure either symptoms or sporulation (Dixon, 1981; Thakur *et al.*, 1998). Field infections by *P. viciae* are usually symptomless before sporulation (Biddle, 2001) and, theoretically at least, infections could result in oospore production but no asexual sporulation. Certainly oospore densities can be similar in cultivars from different resistance categories (van der Gaag and Frinking, 1996). An investigation of *P. viciae* infections of cv Early Onward, classed as resistant (Dixon, 1981), showed limited growth (Figure 1) within plant tissues with infrequent sporulation (Clark and Spencer-Phillips, 1994) and it is clear that the extent of pathogen growth within host tissues is not reflected by conventional keys. It should be noted that occasional colonies in the resistant cv Early Onward grew to the same extent as in the susceptible cv Krupp Pelushka, and presumably it was these colonies that gave rise to sporulating infections. Most did not reach this stage of development. A factor associated with the reduced growth in the resistant cv was a significant difference in frequency of haustoria. At 2 days post inoculation, hyphal length and haustorial frequency in the susceptible cv was 2.4 and 4.8 fold greater respectively than in the resistant cv, and by 4 days this had increased to 10.2 fold for hyphal length and 10.6 fold for haustoria, as indicated in Figure 2. At six days post-inoculation, hyphae had accumulated 4.2 times more ethanol-insoluble label from $^{14}CO_2$ supplied to infected leaves in the susceptible than in the resistant cv (Clark and Spencer-Phillips, 1993).

Alternative methods for detecting infection from bulked samples include using antisera to downy mildews and ELISA, as has been applied to pea seed (Corbiere *et al.*, 1995), *Sclerospora graminicola* on pearl millet (Shailasree *et al.*, 2001) and *Plasmopara halstedii* on sunflower (Liese *et al.*, 1982). Note that few crops harbour more than one downy mildew species (although maize is an exception; Sharma *et al.*, 1993) which reduces problems relating to species-specificity. Infections of the same host by two oomycete pathogens simultaneously (Achar, 1993), however, does pose a potential problem.

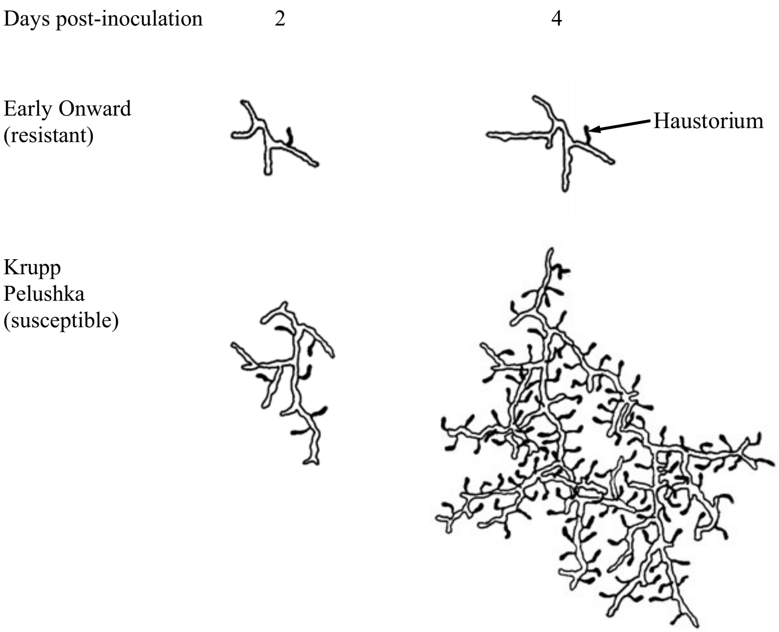

Figure 1. Diagrammatic representation of *Peronospora viciae* colonies at 2 and 4 days post-inoculation of *Pisum sativum* leaves of the resistant cv Early Onward and the susceptible cv Krupp Pelushka. Traced from images of colonies but adjusted to mean hyphal length and haustorial frequency data of Clark and Spencer-Phillips (1994).

Refinement of these techniques is underway in a number of laboratories. With *Peronospora viciae*, for example, proteomics is being employed to select key marker proteins expressed at early stages of infection (Amey *et al.*, 2003). The genes encoding selected proteins will be cloned and expressed so that highly specific monoclonal antibodies can be produced. These can then be deployed in either lateral flow immunoassay devices (Danks and Barker, 2000) or biosensors (eg Pemberton and Hart, 2001) if greater sensitivity is required. If these aims are realised, then growers and advisors will have a cheap, reliable diagnostic tool for use in the field to detect infections before symptoms appear. This would inform the targeted deployment of fungicides, with considerable benefits to the consumer and the environment.

Other detection methods, such as fatty acid profiling for infection of sunflower (Spring and Haas, this volume), also are being developed for seed samples. Interestingly, both a polymerase chain reaction (PCR) diagnostic test (Roeckel-Drevet *et al.*, 1999) and monoclonal antibodies for ELISA (Bouterige *et al.*, 2000) have been developed to detect *P. halstedii*. The monoclonal antibodies detect all spore structures, but not hyphae, from all seven races of *P. halstedii*. The PCR test detects infections prior to sporulation in all races tested, but species specificity might be a problem as cross-reactivity was found following Southern hybridisation (of the PCR gel) for numerous non-target species. Development of reliable tests for this fungus in seed shipments are vital to prevent metalaxyl-resistant isolates of virulent races from spreading from France and the USA to Eastern Europe, where sunflower is the major oil-seed crop. In addition to a lack of resistant cultivars, beyond metalaxyl there are no preventative measures and there are no cures for crops infected by such isolates (Roeckel-Drevet *et al.*, 1999; Bouterige *et al.*, 2000). Molecular tools used in downy mildew systematics (see Spring, this volume) also have potential for application in disease detection systems. Recent examples include use of an internal transcribed spacer amplicon as a molecular marker for identification of *H. parasitica* (Casimiro *et al.*, 2004).

Oospores, conidia and sporangia can lead to either localised lesions or systemic infections and all can result in premature death of a proportion of infected plants. In pea, systemic infections are found more commonly after oospore infection and occur when the whole of the growing shoot becomes infected via the apical meristem as it emerges from a germinating seed. A young plant with an apical meristem infection continues to grow for a time in tandem with the downy mildew mycelium, which consequently infects all tissues resulting from the meristem. Profuse sporulation follows, with either stunted growth or distortion and early death of the plant occurring (Mence and Pegg, 1971). Systemically infected pearl millet plants show signs of hormonal imbalance with leafy growths produced in place of floral parts, to give the so-called "crazy top" or "green ear" syndrome (Jeger *et al.*, 1998).

Most plant species become more resistant with age. Sorghum, for example, becomes completely resistant to systemic infection a few weeks after seed germination (Bock *et al.*, 1998). Additionally, the systemic and local states of infection can interchange. Systemic infection can occur when hyphae grow from a leaf lesion through the stem to the shoot apical meristem (Taylor *et al.*, 1990). Alternatively, plants may outgrow a systemic infection which is then localized to the lower regions of the plant (Mence and Pegg, 1971). This is particularly pronounced in the "recovery resistance" of millet, sorghum and maize (Singh and Talukdar, 1996). Here uninfected parts of a plant can compensate for early infection by continued growth to produce perfectly healthy leaves, tillers and seed sets. This can occur even after systemic infection, and is interesting because the plant allows the fungus to complete its asexual lifecycle (apparently no oospores are produced; Jeger *et al.*, 1998). This presumably means that the selection pressure for a change in virulence is reduced or eliminated completely. Pearl millet cultivars selected for this trait are being used for the production of commercial hybrids in India, and show recoveries of between 90 and 98% in areas heavily infested with downy mildew, with 70-98% incidence.

The mechanism of recovery resistance is unknown, but it might be related to systemic acquired resistance, and perhaps to the phenomenon whereby seedlings grown from seeds of systemically infected pea plants never become systemically infected themselves (Stegmark, 1994; Lebeda and Schwinn, 1994). It should be noted, in contrast, that infections by *Albugo candida* can predispose horse-radish leaves to infection by *H. parasitica* (Saharan *et al.*, 1997).

Another type of symptom is exemplified by sorghum infected by *Peronosclerospora sorghi*. Here, whitish stripes can appear on the leaves where rows of oospores have been produced between the vascular strands. These later become necrotic and shred, releasing oospores into the air (Jeger *et al.*, 1998). Alternatively oospores can remain in necrotic plant tissue to infect via the soil in a future season.

Infections by *P. viciae* can result in a visible whitish mycelium on the internal surfaces of pea pods (Biddle, 2001), but this is not thought to lead to infection of plants grown from seed from these pods. Seed transmission of downy mildew diseases in other hosts is considered in the next section.

3. THE INFECTION CYCLE

3.1 Oospores

Oospores are large (25-50 μm diameter) thick-walled sexual spores which form within plant tissue. In *P. viciae*, oospores form within the leaves, stems and seed coats, and along the internal pod walls of pea plants (Falloon and Sutherland, 1996). They are the survival propagules and initiate the primary infections. Oospores result from the fusion of oogonia and antheridia from two different mating types in heterothallic species (eg *S. graminicola*), or from the same mycelium in homothallic species (eg *P. sorghi*). Some species (eg *H. parasitica*) have both heterothallic and homothallic isolates (van der Gaag and Frinking, 1996). After meiosis only one nucleus remains within the oospore.

Within dried plant material and in soil, oospores can survive many years; oospores of *Peronospora destructor* in onion debris are known to germinate well after 25 years of outdoor storage (Stegmark, 1994). Oospores germinate to form a germ tube that either infects directly using a germ tube or, in some species such as *Plasmopara viticola*, by producing a secondary sporangium and subsequently zoospores (Vercesi *et al.*, 1999). Oospores germinate either sporadically when environmental conditions are favourable or in response to root exudates (eg in *Peronosclerospora sorghi*; Gowda and Bhat, 1986) and infect germinating seedlings and sometimes older plants (van der Gaag and Frinking, 1997).

Mycelia (and perhaps oospores) within seed coats can form another source of inoculum. Mycelium within cabbage seeds has been shown to infect developing seedlings and is thought to be a major source of inoculum in South Africa (Achar, 1995). Mycelium of *Peronosclerospora sorghi* has been detected in the scutellum of maize embryos, and this may be a source of inoculum in this crop in Nigeria (Adenle and Cardwell, 2000). In contrast, although affecting the quality and germinability of the seeds, mycelium within pea seed coats has never been found to infect seedlings growing from these seeds (Stegmark, 1994).

3.2 Conidia and sporangia

Asexual sporulation results in the production of spores 15-30 μm in diameter. They may contain a number of nuclei derived both by mitosis within the developing spore and by subsequent migration from the conidiophore or sporangiophore (Tommerup, 1981). The spores are termed sporangia, if they produce zoospores internally which are then released and encyst to germinate via a germ tube (eg *S. graminicola*), or conidia if they germinate directly via a germ tube (eg *P. viciae*). A single conidium can be sufficient to initiate an infection (eg of Kohlrabi by *H. parasitica*; Saharan *et al.*, 1997).

Conidia or sporangia are produced on the ends of stalked conidiophores or sporangiophores, which usually protrude through the open stomata of leaves or stems (Figure 2). *P. halstedii* and *Plasmopara lactucae-radicis* are the only two species reported to sporulate on roots (Stanghellini *et al.*, 1990). Davison (1968) described five stages in the production of conidiophores of *H. parasitica*: formation of primordia, emergence of unbranched conidiophores; branch formation; development of conidia; formation of a septum that delimits mature conidia. The spores are generally released by a twisting motion of the sporangiophore or conidiophore during changes in humidity. They are ephemeral with a life span of only a few hours or days, depending on the humidity once released, and will not survive lack of liquid water after germination. It is not surprising, therefore, that the downy mildew mycelium, which is protected within the host plant, is programmed not to sporulate unless humid conditions prevail. In some species, for example millet downy mildew (*S. graminicola*), leaf wetness is an absolute requirement for sporulation. Each species also has its own temperature and light regime requirements for asexual sporulation, which may differ from those required for oospore production (eg in *P. viciae;* van der Gaag and Frinking, 1996). A full consideration of the environmental conditions favouring sporulation in downy mildew diseases is beyond the scope of this review.

One interesting, unexplained and often unpublished characteristic of asexual sporulation in compatible interactions is that successive transfers of a downy mildew isolate on the same host cultivar may lead to a progressive reduction in asexual sporulation (eg for *Peronospora hyoscyami* see: Johnson, 1988). In our experience with *P. viciae*, conidial isolates need to be maintained on a panel of host cultivars, and spores harvested to produce a mixture that is then used to inoculate the next batch of plants in order to ensure the isolate is not lost due to lack of sporulation (Ashton, Clark and Spencer-Phillips, unpublished). This suggests that an isolate can become adapted to a particular cv in a compatible interaction: it becomes more biotrophic and less damaging by reducing its demand on the host for the significant quantities of nutrients required to support sporulation. Microscopy of inoculated tissues shows that the pathogen still has an extensive intercellular mycelium despite reduced spore production.

The asexual spores are transferred from host to host to produce secondary infections by rain splash and can be dispersed hundreds of kilometres by wind (Bock *et al.*, 1997). Epidemics therefore tend to occur with cool or moderate temperatures and a humid environment, typically in the spring and autumn in temperate countries, or during the rainy season in the tropics. Asexual propagation can be extremely rapid, the cycle sometimes taking as little as three days (72 hours) after infection in *H. parasitica* on *Arabidopsis* (Bittner-Eddy *et* al., 2003), and four and a half days (108 hours) in *P. viciae* on pea (Ott, 2004).

In related oomycete pathogens, host factors which emanate from stomata attract and control development of the zoospores (eg: *Aphanomyces cochlioides*, Takayama *et al.*, 1998; *Phytophthora sojae*, Connolly *et al.*, 1999). With downy mildews, however, such chemical or physical topography factors have yet to be determined (Kiefer *et al.*, 2002). Evidence for host influence on germination of conidia is provided by data showing differential germination rates of *Peronospora hyoscyami* on different *Nicotiana* species, which might be explained by availablility of cations (Johnson, 1988).

A downy mildew conidium or encysted zoospore germinates to give rise to a germ tube which then forms an appressorium over the intended point of entry into a healthy plant leaf or stem. Germination of spores can occur *in vitro*, providing a useful model system to study these early stages of the infection process (eg the role of actin and tubulin in establishing polarity during germination of encysted zoopores of *Plasmopara viticola*; Riemann *et al.*, 2002). Development *in vitro* proceeds through to the appressorium stage in some species such as *P. viciae* (Clark and Spencer-Phillips, 1994), where the proportion of germ tubes producing appressoria was only slightly less (95%) on glass than on a susceptible host (99%). Such data have been interpreted to indicate that host topographical signals are not essential for the differentiation of appressoria (Carzaniga *et al.*, 2001).

The point of entry into host tissues could be, depending on the downy mildew species, either a stoma, the upper epidermal cell wall or the anticlinal wall between two epidermal cells (Figure 2). In *P. manshurica* on soybean, virtually all penetration was via anticlinal cell walls (Riggle, 1977), whist in *P. destructor* on onion penetration was reported to be entirely via stomata (Kofoet and Zinkernagel, 1991). Approximately 20% of infections by *P. viciae* on pea leaves were via stomata, in experiments that compared two cultivars with a two-fold difference in the number of stomata per unit area and in the total circumference of aperture available for penetration (Clark and Spencer-Phillips, 1994). This suggested that

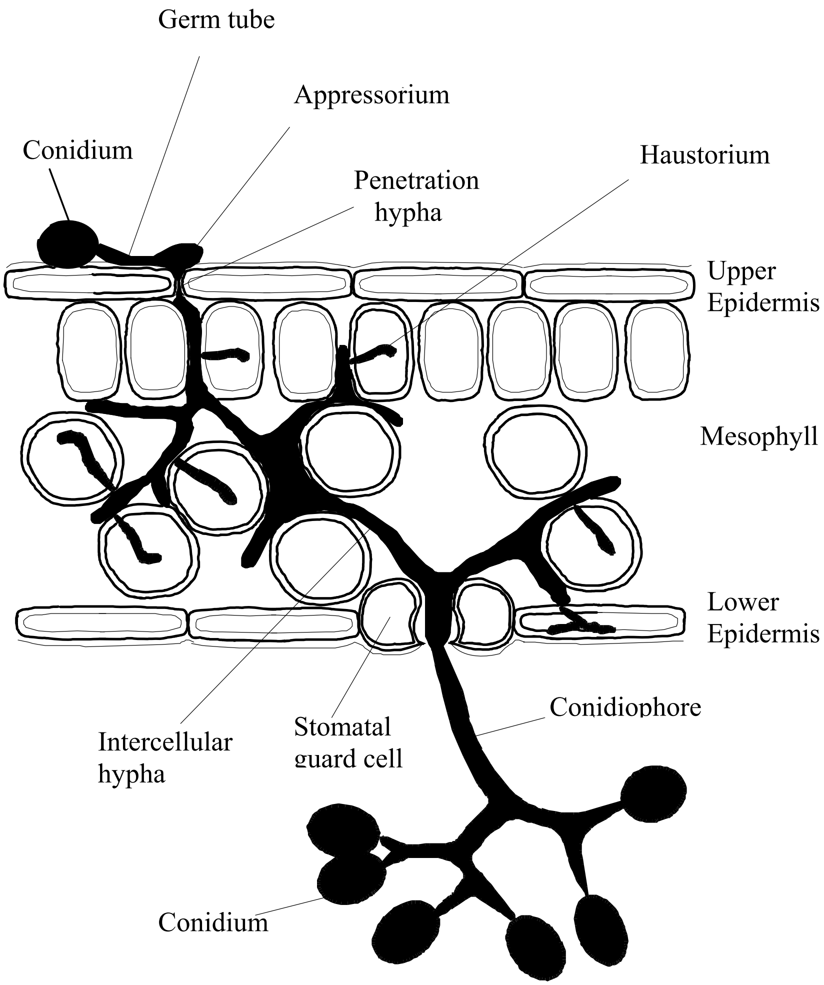

Figure 2. Diagrammatic representation of the asexual infection cycle of *Peronospora viciae* in a pea leaf. The conidium germinates to form a germ tube; a penetration hypha develops from under an appressorium and penetrates either via epidermal cell walls or through a stoma. Intercellular hyphae can vary in width considerably, sometimes filling intercellular spaces. Haustoria are intracellular structures formed within host cells. Asexual conidia form mostly on the lower surface of the leaf on conidiophores which emerge through stomata. Reproduced from Clark and Spencer-Phillips (2000), with permission from Academic Press.

penetration via stomata was a fixed response by a proportion of conidia, and not a random or host-regulated response.

Germ tubes of *H. parasitica* conidia can produce multiple appressoria on both artificial and host surfaces, which is thought to reflect a series of unsuccessful attempts at penetration (Carzaniga *et al*., 2001). These authors also characterised extracellular matrices produced by germ tubes and appressoria of *H. parasitica*, which had previously been described as a mucilaginous sheath by Saharan *et al*. (1997). The adhesive nature of the carbohydrate and proteinaceous matrices suggested that they contributed to attachment of germlings to the surface of the host.

3.3 The endophytic mycelium

A penetration hypha grows from the underside of the appressorium into the plant tissue, passing through the stoma or epidermal cell walls to give rise to an intercellular hypha (Figure 2). In some species, such as *Bremia lactucae*, secondary spore-like structures termed infection vesicles are produced, and a hypha grows out from these (Tommerup, 1981).

The endophytic hyphae grow by branching that is often described as dichotomous, and form a coenocytic mycelium with apparently indeterminate growth. A rigorous topological analysis of 48 hour old colonies of *P. viciae*, however, has shown that their hyphae have a branching pattern that is intermediate between herringbone and random, and certainly not dichotomous (Ott *et al*., 2003). Figure 3 shows the different appearance of a topological structure, such as a fungal colony, with herringbone, random and dichotomous topologies. It has been suggested that the initial herringbone growth observed reflects a strategy for overcoming host resistance, achieving rapid colonisation of infected tissue in order to out-compete other colonies of the same or different fungi, and therefore to maximise the potential for nutrient acquisition (Ott *et al*., 2003). It would also increase the chance of finding a compatible mating type in heterothallic isolates.

Topological analysis of older *P. viciae* colonies (Ott, 2004), however, indicates that whilst colonies at 60 hours post-inoculation have a topology that is not statistically different to 48 hour old colonies, a significant shift towards a more random and dichotomous topology occurs at 108 hours (4.5 days post-inoculation). This shift coincides with formation of the first conidiophores. A dichotomous topology would increase the surface area in contact with plant cells for nutrient uptake at this time of peak demand on the host. It would also increase the chance of contact with stomata, hence optimising the potential for production of conidiophore primordia and the dispersal of spores. The possibility that the pattern of branching reflects the phloem loading mechanism operating in the infected tissue now merits investigation (Spencer-Phillips, 1997; Ott *et al*., 2003). There may also be different branching patterns in different parts of the same host tissue, perhaps tending towards herringbone in regions of low nutrient supply, but dichotomous in areas where the nutrient supply is greater (such as adjacent to veins).

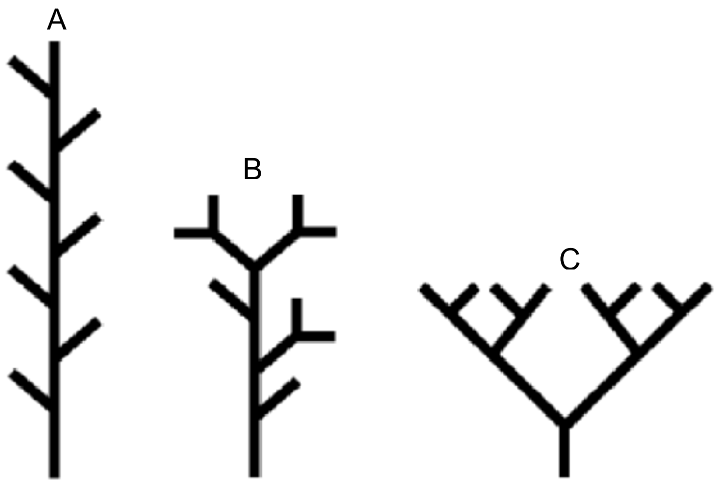

Figure 3. Schematic representation of a topological structure such as a fungal hyphal system of magnitude 8 (ie with 8 hyphal apices), representing (A) herringbone, (B) random (intermediate) and (C) dichotomous topologies. Reproduced from Ott *et al*. (2003), with permission from Cambridge University Press.

Although hyphal width can be fairly constant, it can also vary by about five fold with the tubular wall expanded to fill the intercellular space. This has been recorded by Mangin (1895) and Fraymouth (1956) for several species of downy mildews, and particularly species that did not appear to form haustoria readily. It has also been noted in *Peronospora rubi* infections of Tummelberry to occur mainly in petioles and veins, again where haustoria are less frequent (Williamson *et al.*, 1995). These authors described broad, fan-shaped and multi-branched "fasciated hyphae", associated with veins in leaves. The formation of hyphae with different characteristics within the same host has been described as "hyphal dimorphism" by Zimmer and co-workers (1990) in *Peronospora ducometi* infections of buckwheat (*Fagopyrum esculentum*). They recorded narrow (5-12 µm, with mean of 8.6 µm) hyphae that formed haustoria in stems, leaves, flowers and seeds, and also broad (14-44 µm, mean 27 µm) hyphae where haustoria were absent in petioles and flowers. Downy mildew hyphae are typically broader (10-12 µm) in species that infect dicotyledonous hosts than those infecting monocots (3-5 µm; Dick, 2002a).

At intervals along the hyphae, specialised intracellular structures called haustoria typically are formed within adjacent plant cells. These are finger-like, branched or globose (depending on species) projections of determinate growth and are formed as follows. A penetration peg grows through the plant cell wall and invaginates the plant plasma membrane, which enlarges to encompass the growing haustorium (Figure 4). The haustorium therefore remains extra-cytoplasmic. The invaginated region of host membrane is called the extrahaustorial membrane, and the space between this and the haustorial wall is called the extrahaustorial matrix. The haustorial cytoplasm is therefore surrounded by the haustorial plasma membrane, haustorial wall, extrahaustorial matrix and extrahaustorial membrane. These form a complex interface presumably for nutrient and molecular exchange between host and pathogen, and perhaps especially for molecules that are not able to traverse the apoplast between intercellular hyphae and plant cells. The specialised structure of haustoria and their interface with infected plant cells has been the subject of many reviews (eg see Perfect and Green, 2001) and will not be considered here.

Suffice to say that haustoria of downy mildew pathogens have not had the same attention as those of the rusts and powdery mildews. The downy mildews certainly merit a detailed investigation similar to the pioneering work undertaken with rust fungi by Mendgen's group, that has advanced significantly knowledge of haustorial function at the molecular level (Voegele and Mendgen, 2003).

All downy mildews set up a symbiotic relationship with the plant tissue, with host cells remaining alive at least for a short time and often through the sporulation phase of the infection cycle. Hyphae become attenuated and contents become sparse and autolysed when conidiophores and sporangiophores mature or oogonia form. Hyphae can be sealed by a plug of callose-like material in older infections (Fraymouth, 1956), and can become septate after treatment with some fungicides (El-Gariani and Spencer-Phillips, this volume).

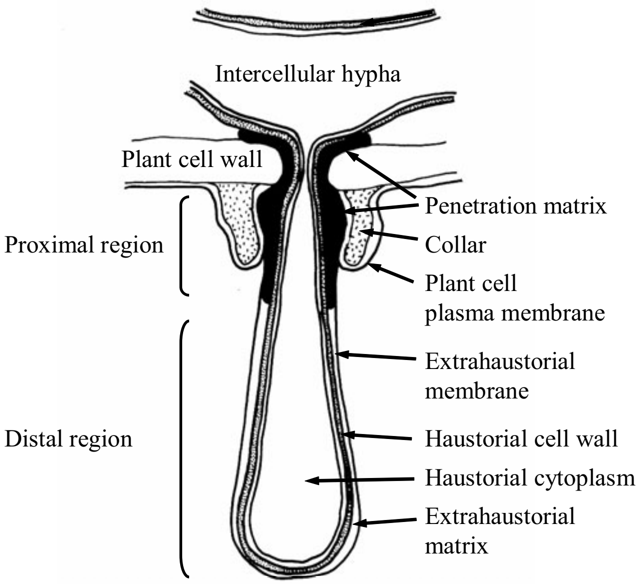

Figure 4. Diagram of a downy mildew haustorium and its interface with an infected plant cell, based on data from Hickey and Coffey (1978), Beakes *et al*. (1982) and Ashton, Beale and Spencer-Phillips (unpublished) for *Peronospora viciae* infections of pea. The distal and proximal regions of the haustorium are indicated.

4. HOST-PATHOGEN INTERACTIONS

All types of plant resistance occur against downy mildews. Plant resistance can be classified according to whether it is monogenic, polygenic, race-specific (including gene-for-gene or vertical resistance) or race non-specific, non-host or general (also known as field, horizontal, partial or quantitative) resistance, and all of these have been found to operate against downy mildew pathogens (Mauch-Mani, 2002). The most intensively studied is race-specific resistance and at least 23 dominant plant resistance R genes against downy mildew (*Dm* genes) have been found in the lettuce population, each with a corresponding downy mildew avirulence gene (Meyers *et al.*, 1998). Not surprisingly, therefore, a large proportion of downy mildew research has been directed towards elucidating the complex interactions that these resistant interactions represent.

Unfortunately, however, this has resulted to some extent in a lack of knowledge about the compatible interaction, the basic biology of which is still incompletely understood. In part this is due to the fact that the study of downy mildews is particularly difficult because they cannot be grown in axenic culture, and problems with germination, infection and storage. There are also many homothallic isolates which do not interbreed, and this has severely hampered genetic studies. Recently, however, two homothallic isolates of *H. parasitica* grown on the same *A. thaliana* plant were found to out-cross, which will initiate rapid progress in the study of this interaction (Gunn *et al.*, 2002). Hyphal anastomosis, which would allow mixing of nuclei from different sporangial populations, has been described in related oomycete *Phytophthora* species (Stephenson *et al.*, 1974). It is postulated to have occurred in *P. rubi* infections (Williamson *et al.*, 1995) but is thought to be uncommon in oomycetes (Tommerup, 1981).

Downy mildew mycelia and spores are diploid, and a treatment to produce a completely homozygous oospore has not yet been developed. This means that somatic recombination, shown to occur in oomycetes, will cause genetic drift in cultured isolates and may account for race changes observed in *Bremia* species (Crute and Norwood, 1980; Crute and Dixon, 1981), and the progressive reduction in sporulation referred to above in section 3.2. This has resulted in sharp contrast between the genetic control of host plants and that of the downy mildew used to infect them. Even in sophisticated research into disease resistance the downy mildew isolates can be grown only using simple and imprecise methods, such as from an inoculation from a ‘known’ population of oospores and then single sporangial isolates selected by growth on particular host accessions (eg McDowell *et al.*, 1998). It is quite likely that repetition of this process would lead to isolates that are not identical genetically.

Despite the difficulties, genetic analyses of downy mildew isolates have begun. DNA fingerprinting using microsatellites can now distinguish cultivar-specific pathotypes of *S. graminicola* (Sastry *et al.*, 1995) and the first oomycete, *Phytophthora infestans*, was transformed reliably by Judelson *et al.* in 1991. Several genes from *B. lactucae* and *Phytophthora megasperma* have been sequenced and the promoters transferred to other oomycetes (Judelson *et al.*, 1992). The genome size and complexity of *B. lactucae* has been elucidated and, surprisingly, 65% of the nuclear DNA was found to consist of repetitive sequences (Francis *et al.*, 1990). A genetic map of this species has now been published, using data from molecular markers and avirulence genes (Sicard *et al.*, 2003).

An unusual discovery is the presence of ssRNA mycoviruses in *S. macrospora* and *P. halstedii* (Mayhew *et al.*, 1992).

4.1 Physiology and biochemistry

In the compatible interaction, host cell damage is minimal, with no evidence of toxin production and only very localised production of cell wall-degrading enzymes during the penetration of plant cells walls (eg by *B. lactucae*; van Pelt-Heerschap and Smit-Bakker, 1993). As the infection progresses, changes in translocation patterns, hormonal levels and host-cell permeability typify a biotrophic interaction. Sucrose and other nutrients are diverted from source leaves to the new infection sink created by a localised lesion (Ayres *et al*., 1996). Other changes include increases in the activity of several enzymes including invertases, α-glucosidase, ribonuclease, β–1,3-glucanase and chitinase isozymes (Cachinero *et al.*, 1996) and peroxidase (Reuveni, 1998). Enzyme activities are not always stimulated, with amylase and invertase activites both reduced within *H. parasitica* infected *Raphanus sativus* tissues compared to healthy controls (Achar, 1994).

Infection by biotrophic pathogens can result in changes to all the principal metabolic processes of host tissues. Photosynthesis, respiration, and translocation can be affected throughout the whole infected plant and not just at sites of local infection (Kosuge, 1978). Some production of phytoalexins (eg the coumarin scopoletin in sunflower, Spring *et al.*, 1991) and flavanoids (eg in grape, Dai *et al.*, 1995) is also stimulated in compatible as well as resistant interactions. The respiration of cabbage cotyledons infected by *H. parasitica* rose sharply soon after infection, and at sporulation was almost double that of uninoculated control plants (Thornton and Cooke, 1974). Additionally, areas of increased synthesis or rapid accumulation of photoassimilates surrounded necrotic zones and were compared to the green islands of rust infections. A sink was created and nutrients from adjacent uninfected tissue were transferred to infection sites and used to support sporulation by *P. cubensis* (Perl *et al.*, 1972).

The mechanisms by which fungi within leaves affect photoassimilation remain uncertain and are possibly associated with altered water and solute relations within the leaf. It has been argued that the changes are not likely to be beneficial to the fungus (Farrar and Lewis, 1987). Bhatia and Thakur (1994), however, have shown an increase in the total sugar and protein content following infection of a highly susceptible cultivar of pearl millet with *S. graminicola*.

Invertases are key enzymes likely to be involved in sugar accumulation. Extracellular invertases are normally present in apoplastic solutions and have a key role in phloem unloading (Roitsch *et al*., 2003). Up-regulation of these invertases is a common response to infection (Hall and Williams, 2000). This leads to changes in levels of soluble sugars that are available for pathogen nutrition, and also appears to be involved in the induction of host responses such as increased callose synthesis, peroxidase activity and salicylic acid content (see Chou *et al*., 2000). Increased apoplastic invertase would increase the concentration of glucose available for pathogen nutrition, and reduce the amount of sucrose removed from infected leaves by phloem loading. This may be particularly significant in downy mildew infections of hosts where sucrose is the main translocate, and is exported from leaves by an apoplastic pathway (Spencer-Phillips, 1997). A resulting effect would be the "sugar feeding" that has been reported for some downy mildew diseases (Bhatia and Thakur, 1994). The recent discovery that *P. viciae* hyphae can utilise glucose but not sucrose as a carbon source *in vitro* (El-Gariani and Spencer-Phillips, this volume) supports this notion, although it should be stressed that the role of invertases has not been elucidated fully in downy mildew infections. A progressive increase in hexose content of the apoplast as infection proceeds and invertase activity increases may also result in hyphal topology switching from herringbone-like to a more dichotomous form (Figure 3), as the nutrient supply becomes more favourable (Ott *et al*., 2003).

A significant factor relating to the role of invertases in biotrophic infections seems to be whether hexose concentrations increase within infected tissues. In one of the few downy mildews where sugar concentrations and invertase activity have been studied, Brem *et al*. (1986) showed that host hexoses did not accumulate, but the activity of soluble invertase was stimulated in *Plasmopara viticola* infections of vine leaves. In contrast, a comprehensive study of the related oomycete, *Albugo candida*, showed that the sucrose, glucose and fructose accumulated significantly within infected *Arabidopsis* tissues compared to healthy controls (Chou *et al*., 2000). Concentrations were enhanced most in later stages of infection, which presumably aided carbon supply for pathogen sporulation. This was paralleled by a reduction in starch content within the infected region but an increase within uninfected regions of the same leaves, compared to uninfected controls. Chou *et al*. (2000) also showed that wall-bound and soluble invertases had a greater activity within infected tissues than in healthy controls. Whilst soluble invertase activity declined as both healthy and infected leaves aged, the wall bound invertase activity increased 40-fold within infected tissue alone and was shown to be of host origin. This provides a mechanism whereby source tissues are converted to sinks. Of particular interest was a new isoform of the soluble invertase that appeared in later stages of infection: it was suggested that this was of pathogen origin. Elucidating the location of this fungal enzyme, either within the hyphal cytoplasm or secreted into the host, will provide key information on mechanisms of biotrophy operating in this pathosystem.

Host cell membrane permeability can be increased, for example in *H. parasitica* infections of *Brassica* species (Kluczowski and Lucas, 1982), but this may be associated with later, more necrotrophic phases of the interaction, perhaps as the result of non-biotrophic secondary invading microorganisms. Severe damage to the host coincides with sporulation in many species of downy mildew, not least because this is when the pathogen's demand on the host for nutrients is maximal. In onion, up to 10% of stomata can be blocked with conidiophores bearing conidia, which can amount to 5% of the dry weight of a leaf being produced each night (Lucas and Sherriff, 1988).

As mentioned above, systemic infections can lead to changes in plant morphology but the causal details of many of these processes are unknown. It is interesting that an auxin-metabolising enzyme, only found in sunflower infected by downy mildew, caused growth retardation which could be reversed by application of exogenous gibberellin, but not auxin (Benz and Spring, 1995). The unusual preference of *H. parasitica* for radish galls caused by *A. candida* (resulting in dual infection) causes hyperauxinity in hypertrophic tissues (Achar, 1993).

Two biochemical markers for resistance seem to be the level of peroxidase activity in cultivars of muskmelon (*Cucumis melo*), lettuce or millet (Sreedhara *et al.*, 1995), and that of lipoxygenase in millet (Nagarathna *et al.*, 1992). Pre-infection levels are higher in resistant cultivars, but peroxidase activity does increase after infection in both resistant and susceptible cultivars. There is increasing evidence that the level of peroxidase is related to age-related resistance in a number of host-pathogen interactions including *P. viticola* on grape vines. Here, older leaves are found to have a partial resistance with reduced lesion areas and sporulation, and a higher level of peroxidase activity than younger leaves. Significantly, a reverse age effect was found with the related oomycete pathogen *P. infestans* on potato. In the latter, peroxidase activity is higher in the younger leaves which are more resistant to late blight than in mature, more susceptible leaves (Reuveni, 1998).

One action of peroxidases is in the production of isodityrosine bonds linking hydroxyproline-rich glycoproteins. These are normally present in the cell walls of plants and, when insoluble, form one of the major polymer systems (Cooper *et al.*, 1987). Amounts of hydroxyproline-rich glycoproteins were shown to increase after infection in 18 out of 20 compatible or incompatible host-pathogen interactions (Mazau and Esquerre-Tugaye, 1986), as a response to tissue damage (Chrispeels *et al.*, 1974) and in response to ethene (Ridge and Osborne, 1970). They are also believed to be important in systemic acquired resistance in pea downy mildew (Taylor, 1986). Hydroxyproline-rich glycoproteins probably provide the template for the polymerization of lignin-like phenols and other polymers. Treament with sodium chlorite (which breaks isodityrosine bonds) and pectic enzymes effectively macerated tissue of a susceptible pea cv infected by *P. viciae*, which was resistant to maceration with pectic enzymes alone (Clark and Spencer-Phillips, 1990).

It seems that phytoalexin production and hydroxyproline-rich glycoprotein and lignin bonding are all part of the general response to infection mediated by peroxidases, and are ineffective against a fungal race virulent to that particular cultivar. Of course, it is possible that it is the rate at which these are released that contributes towards resistance, as postulated for *P. viticola* infections (Dai *et al.*, 1995). In *A. thaliana*, four phytoalexin genes are required for resistance to *H. parasitica* (Glazebrook *et al.*, 1997). It has been shown that while accumulation of the phytoalexin camalexin varies greatly between *A. thaliana* accessions, it can be induced by many biotic and abiotic factors and there is no significant correlation between resistance and camalexin accumulation (Mert-Turk *et al.*, 2003).

Another aspect of the general response to infection seems to be systemic acquired resistance, whereby plant tissues remote from the site of infection become more resistant to the same, or to other, pathogens. Systemic acquired resistance occurs after infection by either virulent or avirulent pathogens. In biotrophs, it involves signalling, salicylic acid elicitation (rather than the jasmonate pathways used against necrotrophs; McDowell *et al.*, 2000) and a network of complex downstream interactions including 3 classes of proteins: SAR, NIR and SIR, of which NIR are proline-rich (Rairdan *et al.*, 2001). Is it possible that systemic acquired resistance in the compatible interaction might benefit the downy mildew pathogen by inhibiting the growth of other parasites?

One reaction of the plant to the formation of haustoria (which must, in any case, include a large number of complex biochemical responses) is the formation of a callose collar or ring round the entry point of the haustorium (Figure 4), which remains as a callose papilla if haustorial penetration fails (Sharada *et al.*, 1995). Perhaps not surprisingly, electron microscopic studies have shown that changes occur in plastid size and structure, the plasma membrane, cell wall, mitochondria, ribosomes and polyribosomal density following haustorial penetration both within infected and in adjacent uninfected cells. These sub-cellular effects reflect changes in respiration, photosynthesis, protein synthesis and other cellular processes (Thakur and Murty, 1992).

At the light microscope level, infected cells can appear to have a faint brown colour and autofluorescence when unstained, and can be stained with Trypan Blue (Clark and Spencer-Phillips, 1994). Enzyme-catalysed browning reactions are a common feature of the response of plant cells to invasion by pathogens (Kuc, 1972). Browning results from oxidative polymerization of low molecular weight phenolic compounds to yield tannin-like substances containing quinoid groups, capable of precipitating proteins and cross-linking to other polymers (Swain, 1977). Autofluorescence is probably due to deposition of phenolics in the cell walls (Stumpf and Heath, 1985). Substantial numbers of these cells have been seen in a compatible interaction between *P. viciae* and the pea cv Krupp Pelushka, in which the cells were only lightly stained by Trypan Blue (although colonies within this interaction had less necrotic cells per length of hyphae than colonies within the resistant cv Early Onward; Clark and Spencer-Phillips, 1994). It is not known whether these cells were undergoing necrosis as part of a hypersensitive response, or whether they were undergoing a mild reaction which resulted in recovery and maintenance of the compatible interaction.

It should be noted here that the walls of oomycetes contain cellulose (Aronson *et al.*, 1967) and seem to be related to plant cell walls. Therefore materials destined for plant cell walls in uninfected plants might reasonably be expected to find suitable binding sites within the hyphal walls of downy mildew fungi. Indeed there is evidence that hyphal walls are lignified in *H. parasitica* infections (see Slusarenko and Schlaich, 2003). Presumably downy mildew hyphae have developed ways to cope with this environment. Oomycete hyphae store the sugar trehalose (not polyols such as mannitol, which are stored by other fungi; Pfyffer *et al.*, 1990), to the extent that it has been found to be a suitable marker for *P. viticola* infections of *Vitis vinifera* (Brem *et al.*, 1986).

4.2 Nutrient transfer from host to pathogen

Research into the transfer of organic nutrients to powdery mildew and rust fungi is more advanced than for downy mildew infections. The mechanisms proposed for rusts and powdery mildews have been reviewed extensively (eg: Manners and Gay, 1982; Spencer-Phillips, 1997; Voegele and Mendgen, 2003), and will only be considered in outline here to set the scene for what is known about downy mildews.

ATPase driven, proton co-transport of organic nutrients is thought to be the most significant mechanism for nutrient transfer from host to pathogen, based on current evidence. Membrane bound ATPases are proton pumps that transport protons across a membrane, creating a chemiosmotic potential which can be used to drive the symport or antiport of other molecules via membrane-bound carrier proteins. Proton-translocating ATPases in plasma membranes and tonoplasts of higher plants and fungi are directly involved in many physiological processes such as pH regulation, stomatal opening and phloem loading. In *Blumeria graminis* (cereal powdery mildew) infections, ATPase activity on epidermal cell wall-lining plasma membranes caused proton extrusion from host cells that is coupled to proton extrusion from haustoria (Gay *et al.*, 1987). This energised the rapid influx of organic nutrients. The same mechanism is assumed to operate in all powdery mildews. In this model, coupling relies strictly on the membrane domain structure and neckbands found in *B. graminis* and *Erysiphe pisi* (pea powdery mildew) haustoria (Spencer-Phillips and Gay; 1981; Woods and Gay, 1983).

The necks of these haustoria are characterised by regions with encircling bands of material with unusual staining properties (see Perfect and Green, 2001). These are the neckbands, and they delimit two structurally and functionally distinct domains of both the host and pathogen plasma membranes. High ATPase activity was found on the haustorial plasma membranes but the plasma membranes of appressoria and epiphytic hyphae lacked activity, with a sharp transition at the neckband region (Spencer-Phillips and Gay, 1981). The plasma membrane of infected host cells also showed a sharp transition at the neckband: the wall-lining domain had normal high levels of ATPase activity whereas the invaginated domain, the extrahaustorial membrane, showed no ATPase activity.

Whilst haustoria of rusts and downy mildews share most structural components with those of powdery mildews as a result of convergent evolution, significant differences justify caution when predicting primary function in these unrelated phyla. The fact that rusts and downy mildews are endophytic, and therefore have access to mesophyll cells (rather than epidermal cells) and to the apoplast (rather than surface exudates) may or may not account for structural differences observed. Nevertheless, variations on the above model for *Blumeria/Erysiphe* haustoria have been proposed for these infections.

Host membrane domains and neckbands have been found in several rust infections. ATPase activity at the haustorial plasma membrane was not detected by enzyme cytochemistry (see Perfect and Green, 2001), but has been detected at this location in dikaryotic haustoria of *Uromyces fabae* by other methods (Voegele and Mendgen, 2003). Woods and Gay (1983) found that ATPases could not be detected at haustorial plasma membranes or extrahaustorial membranes in infections by the oomycete *Albugo candida*. By staining with chromic and phosphotungstic acids, it was shown that glycoproteins (such as ATPases) were only present on the wall-lining domain of the host plasma membrane, but no neckband could be distinguished in the lettuce downy mildew pathogen *B. lactucae* (Woods *et al.*, 1988). Indistinct transitions from one membrane domain to another in the rust *Puccinia poarum* and *B. lactucae* have lead to suggestions that membrane fluidity was restricted by some means other than a neckband (Woods and Gay, 1987). It was suggested that lateral diffusion of the constituents of the two domains was prevented by reduced fluidity of the extrahaustorial membrane due to tight adhesion to an osmiophilic layer (equivalent to an extrahaustorial matrix) over the entire surface of the haustorium. Neckbands have been visualized, however, using histochemical staining techniques in certain rust infections where previously no such structures were thought to exist (Woods and Gay, 1983), and care should be taken in the interpretation of these results. Possibly other neckband-like structures do not contain the electron-dense material such as silicon and ferric pyrophosphate found (by electron-probe X-ray analysis) in the neckbands of the rust *Puccinia coronata* f.sp. *avenae* (Chong and Harder, 1980).

In *P. viciae*, the extrahaustorial membrane differed from the wall-lining plasma membrane in its staining reaction with phosphotungstic-chromic acid (Hickey and Coffey, 1978). Preliminary investigations (Beale *et al.*, 1990) have shown that two domains of fungal ATPase activity can be distinguished in most haustoria with a transition (and hence a neckband-like structure) part way along the digit-like haustorium. ATPase activity was absent from the distal portion (see Figure 4) of the haustorial plasma membrane. The host plasma membrane also had two domains with no activity at the distal region of the extrahaustorial membrane. It is possible therefore that components similar to powdery mildew haustoria are also present in *P. viciae* infections. It should be noted that some haustoria had ATPase activity present along the whole length of their plasma membrane and the extrahaustorial membrane, which suggests the absence of a neckband-like structure. Possibly these haustoria may have been juvenile or in a similar abnormal category to those described by Woods (1985).

In other interactions, haustoria can be found in various states of development, either because they are juvenile or because in some cells haustoria never establish an effective relationship. For example, the neckbands of rust haustoria are not impermeable in young infections

(Heath, 1976). Again, the plasma membrane of some cells infected by *B. lactucae* haustoria become completely 'abnormal' (ie similar to the extrahaustorial membrane in other cells; Woods, 1985), and these cases may be the result of failure to establish a normal, compatible relationship with the host cell.

Further structural studies are needed to investigate the possibility of neckbands in downy mildew haustoria, and physiological data is required to show whether or not the extrahaustorial matrices are sealed from the apoplast in *Peronospora* infections. There is some evidence, reported in Ayres *et* al. (1996), that solutes cannot diffuse readily from the host apoplast to the extrahaustorial matrix in *P.viciae* infections. This restriction of movement might be a role for the penetration matrix (Figure 4). The identity and location of the membrane-bound carrier proteins, whether they are sucrose/proton or glucose/proton symporters, needs to be established. Further useful information on the identity of transported sugars and the transport pathway would be provided if pathogen invertases were localised. The ability of *P.viciae* hyphae to accumulate label from glucose but not sucrose *in vitro* (El-Gariani and Spencer-Phillips, this volume) suggests that pathogen invertases are most likely to be extracellular.

Two possibilities for the function of downy mildew haustoria present themselves. Either (a) their primary function is the uptake of organic nutrients, or (b) they have some other, as yet unknown, function and organic nutrients are taken up primarily from the apoplast by the hyphae. It is also possible that the relative function of hyphae and haustoria varies, depending on the stage of infection and the host tissue infected.

In case (a), haustoria with no domain structure and therefore with opposing ATPases on the haustorial and extrahaustorial membranes, could still function if the fungal ATPase activity was much greater than the host's. With a domain structure in which there was no ATPase activity on haustorial or extrahaustorial membranes, the haustoria would rely on diffusion to extract photoassimilates from the mesophyll cell and, where necessary, rely upon the host's ATPase and transporter proteins to extract nutrients from the apoplast (Gay and Woods, 1987). Possibly the mechanism by which host ATPases on the extrahaustorial membrane are switched off, also switches off those on the adjacent haustorial membrane.

There is evidence for case (b), however, in that carbon from sucrose can be transferred to intercellular hyphae when the route via infected host cells and haustoria is blocked by the sucrose transport inhibiter PCMBS (Clark and Spencer-Phillips, 1993). Experiments showing that *P. viciae* hyphae separated from living plant cells can also accumulate label from glucose (El-Gariani and Spencer-Phillips, this volume) adds to this evidence. Enzyme cytochemistry showing intercellular hyphae with high ATPase activity (Beale *et al.*, 1990) indicates how the uptake of organic nutrients might be energised across the hyphal plasma membrane.

The jury is therefore still out on the primary function of downy mildew haustoria (Spencer-Phillips, 1997) and other roles can be envisaged. For example, the mechanisms by which downy mildews influence the physiology of the plant, by re-routing nutrients from other leaves, is not known and may well require an interface across which large signalling molecules are able to pass directly into the plant cells in order to affect DNA transcription and other metabolic processes.

With dikaryotic hyphae of the rust fungus *U. fabae*, a hexose transporter expressed specifically at the haustorial plasma membrane transports both glucose and fructose (Voegele *et al.*, 2001). Amino acid transporters are also present at the haustorial membrane, and recent evidence now indicates that amino acids may be accumulated by intercellular hyphae too (Voegele and Mendgen, 2003). Haustorial and hyphal routes for nutrient accumulation may also operate in downy mildews, as suggested by Takahashi *et al.* (1977) in an electron microscopy autoradiography study of ^{14}C uptake by *Pseudoperonospora cubensis* hyphae in cucumber leaves. Until we have more data on haustorial function, a question of fundamental biology, the processes involved in these biotrophic, compatible interactions will remain unresolved.

5. FUTURE DEVELOPMENTS

It is widely recognised that mechanisms of biotrophy, and hence compatibility, in downy mildew infections merit increased attention, with recent authors (eg Lebeda and Schwinn, 1994; Spencer-Phillips, 1997; Slusarenko and Schlaich, 2003) essentially re-iterating the point made by Ingram (1981).

Advances are being made, and the future is promising in this era of functional genomics. The goal of understanding the function of every gene and gene product involved in a particular pathosystem should provide the holistic picture that has eluded researchers in the past. For oomycetes, van West *et al.* (2003) provide a useful review of the techniques available. Molecular techniques such as suppressive subtractive hybridization have been used to identify a number of genes induced during the compatible interaction between *Phytophthora infestans* and potato leaves (Beyer *et al.*, 2001), and in the *H. parasitica*/*Arabidopsis* interaction (Bittner-Eddy *et al.*, 2003). On the basis that it is proteins that deliver biological function encoded by genes, proteomics could prove to be a key tool in elucidating aspects of host-pathogen interaction. These techniques are already yielding information about the spectrum of proteins that are either up- or down-regulated in compatible downy mildew infections (Amey *et al.*, 2003).

A truly holistic picture of compatibility also requires techniques to locate gene products and metabolites at specific sites of cellular interaction, through to understanding the growth and reproductive strategy of the whole mycelium. Advanced techniques in microscopy, such as quantitative bioluminescence and single photon imaging to resolve sugar concentrations at single cell level (Borisjuk *et al.*, 2002), are particularly powerful tools. We can expect further exciting developments in downy mildew biology in the near future.

6. ACKNOWLEDGEMENTS

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FORECASTING DOWNY MILDEW DISEASES

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1. INTRODUCTION

Within the oomycetes, the Peronosporales consist of many fungal pathogens capable of causing substantial losses to important crops, such as, for example, *Plasmopara viticola* on grapes and *Bremia lactucae* on lettuce (Spencer, 1981). Growers are faced with many problems in controlling downy mildew diseases: (i) many varieties are not completely resistant to downy mildew infections or major gene resistances bred into cultivars can become ineffective within a few seasons by adaptation of downy mildew populations; (ii) downy mildew pathogens often have a short period of latency following infection leading to rapid disease increases over time, (iii) populations of downy mildew pathogens can acquire resistance to fungicides; (iv) governments and supermarket chains have imposed limits on the number of fungicide applications during crop production and fungicide residues at the time of harvesting; and (v) the number of registered fungicides has been substantially reduced. Disease forecasting systems could assist growers to achieve effective control of downy mildew within these constraints. Forecasting models could predict when crops are at risk of significant downy mildew development so that they can be treated to avoid or minimise such risks.

The main aim of downy mildew forecasting models is to provide effective and sustainable disease control and to reduce fungicide applications by timing them better and omitting sprays when weather conditions are unfavourable for downy mildew. Considerable successes in reducing fungicide applications while maintaining effective downy mildew control have been achieved by spraying according to weather-based models. One example of success is the Californian model for lettuce downy mildew which was able to reduce fungicide sprays by 67% while maintaining a good level of downy mildew control by applying fungicides only when infections were predicted on days of prolonged morning leaf wetness

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(Scherm *et al.*, 1995). An additional benefit for growers is that the cost of downy mildew control can be reduced.

A reduction in fungicide usage puts downy mildew populations under less selection pressure for resistant individuals. Fungicides, which are prone to downy mildew populations acquiring resistance, could remain effective for longer if fungicides are applied less frequently according to a downy mildew forecast. There are numerous examples of resistance to metalaxyl. In lettuce downy mildew, metalaxyl resistance has been found in the UK (Crute, 1987; Crute *et al.*, 1987), Australia (Wicks *et al.*, 1994) and California (Schettini *et al.*, 1991). Especially, new products such as Strobilurins, which are prone to downy mildew populations acquiring resistance, need to be limited in their use.

It has also been suggested that the application of less fungicides in forecast-directed downy mildew control programs could reduce fungicide residues in the harvested product. However, in grape production in Italy it was demonstrated that there were no differences in fungicide residues between forecast-based treated grapevines and vines that were treated with fungicides every 20 days (Gozzini *et al.*, 1995). Fungicides residues were most likely not reduced, because they are more affected by the time between the last fungicide spray and harvest, than by the total number of fungicide sprays that were applied.

In the current climate where growers are continuously scrutinised by government and supermarket chains for safe and environmentally friendly production of crops, forecasting models are favoured by growers to provide a quality assurance for applying fungicides only when required. Quality assurance of crop produce is very important for growers to inform supermarkets on how they are meeting their quality standards. Especially in the UK, most growers are under direct contract with supermarket chains to deliver crop produce and need to provide detailed information on how their crops were produced.

This review will focus on forecasting of downy mildew on grapevines (*Plasmopara viticola*) and lettuce (*Bremia lactucae*), two economically important world-wide occurring downy mildew pathogens (Pearson and Goheen, 1988; van Bruggen and Scherm, 1997), which differ in their epidemiology, and for which substantial information on forecasting has been published. Oospores are the most important source of primary inoculum of *P. viticola* at the start of the season, whereas the oospores of *B. lactucae* are considered insignificant in contributing to lettuce downy mildew epidemics. *P. viticola* sporangia germinate by releasing zoospores which encyst and then infect a plant, whereas *B. lactucae* sporangia germinate directly with a germ tube. The pathogens also differ in their mechanism of dispersal in that *P. viticola* sporangia are mainly released during rain showers and dispersed within microdroplets by wind, whereas *B. lactucae* sporangia are released by the onset of light at sunrise, the increases in temperature and decreases in RH, and then dispersed by wind. Lettuce crops are short-lived and need to be replanted continuously, whereas grapevines often remain in the same plot for many years. Several forecasting models, which have been developed for these downy mildew pathogens, are reviewed in this chapter and results on evaluating the application of these models in field crops is reviewed. Which elements of the models appear to be most successful in predicting disease development? How well can these models reduce fungicide sprays while maintaining effective disease control? This chapter will try to answer these questions and give advice on further development to improve forecasting models for downy mildew diseases in crops.

2. EPIDEMIOLOGY

2.1 Grapevine downy mildew

American grape, *Vitis lambrusca*, is less susceptible than *Vitis vinifera* (Gadoury *et al*., 1997). This affects the relationships between downy mildew development and weather variables. A longer leaf wetness duration is required to obtain the same level of infection on *V. lambrusca* and a longer duration of high RH is required for sporulation on *V. lambrusca* (Lalancette *et al*., 1988a; Lalancette *et al*., 1988b).

2.1.1 Sources of inoculum and primary infection

In vineyards in Europe it was found that *P. viticola* hibernates during the winter, when the grapevines bear no leaves, as oospores (Baldacci, 1947; Gehmann, 1987; Sung, 1990; Burruano *et al*., 1994; Vavassori, 1994; Vercesi, 1997). These oospores were frequently found to be produced within heavily infected leaves during late summer, and are thought to be the main source of inoculum infecting the new grapevine leaves in spring and early summer. Observations of oospore maturation and germination in Italy suggested great variation between locations within the same season and variation between seasons (Vercesi, 1997). This suggests that variation in weather patterns, but possibly also variation in soil type might affect oospore maturation and germination. Several studies have investigated the effects of weather on oospore maturation and germination, but the author is not aware of any studies investigating the effect of soil type.

Early studies found that oospores mature when the minimum daily temperature is above 10°C, and will germinate when heavy rainfall of more than 10 mm rainfall per day occurs and the young vine shoots are at least 10 cm in length (Baldacci, 1947). This was also called the ‘three ten’ rule. This information has been used by growers to time their first fungicide spray in the new season. However, although it is an easy rule that can give an indication of when the primary infection is likely to occur, it was found to be inaccurate in field trials in Italy (Vercesi, 1997). Predictions of oospore germination according to the ‘three ten’ rule were more than 15 days too early. Later studies have identified more complex effects of temperature and rainfall on oospore maturation and germination.

The maturation of oospores takes several months from their formation within infected leaves in August/September (Sung, 1990) to their germination in early spring until mid June (Vavassori, 1994). It was found that oospores mature only when daily average temperatures are at least 8°C (Gehmann, 1987). Oospores were ready to germinate when the temperature sum of such days from the 1st of January had reached the value of 160-170°C. In a different study, the amount of rainfall between the time of oospore formation in September and February was found to be associated with the date of oospore maturity (Sung, 1990). However, the effects of rainfall were studied in only three seasons and no exact relationship between the amount of rainfall during this period and the date of oospore maturation was found. Later studies by Hill (2000) found that oospore maturation is affected by temperature and vapour pressure deficit, but the actual relationships are not described. Thus, oospore maturation is affected by temperature and the availability of water.

Mature oospores germinate by forming a sporangiospore, which then needs to be dispersed and release its zoospores for primary infection to occur. Germination is initiated by heavy rainfall events of at least 8-10 mm of rain in a day (Gehmann, 1987). Sporangiospores are then formed at least three days after such rainfall events, but in May and June the germination may take longer and at least 6-10 days were required before sporangiospores were formed (Burruano *et al.*, 1994). This suggests that the higher temperatures in May/June may delay germination. In northern Italy, rainfall was found to induce oospore germination until the middle of June (Vavassori, 1994). Vercesi (1997) observed in vineyard studies that at least two rain events were required before primary infections were observed on grapevine leaves. This suggested that after formation of sporangiospores by germination of oospores a second rainfall event is required to disperse the sporangia and release the zoospores to cause the actual primary infections.

2.1.2 Conditions for infection

Sporangia of *P. viticola* germinate by releasing zoospores, which encyst and infect readily when they come in contact with host tissues. Germination and infection only occur when leaves are wet, and the duration of wetness required for infection is affected by temperature. Blaeser and Weltzien (1979) found a linear relationship between the minimum duration of leaf wetness required for sporangia to germinate and temperature. They found that a sum of hourly temperatures when leaves are wet of 50°C·h was required for infection of *V. vinifera*. Similar results were obtained for infection of *V. lambrusca* where a minimum of 2-2.5 h of wetness was required for infection at 15, 20 and 25°C and a minimum of 3.5-5.5 h at 10°C; but levels of disease were lower than reported for *V. vinifera* (Lalancette *et al.*, 1988a). Little disease was observed at 5 and 30°C.

The latent period of *P. viticola* was found to vary between 3-4 and 15-23 days and was affected by temperature and air humidity (Mueller and Sleumer, 1934; Goidanich *et al.*, 1957; Dai *et al.*, 1995). The shortest latent periods of 3-4 days were observed at 17 to 24°C, and latent periods increased when temperature increased to 29°C and decreased to 11°C. Latent periods were generally 2 to 4 days shorter at 100% RH than at 80-90% RH.

2.1.3 Sporulation

Sporulation of *P. viticola* occurs only during darkness at night (Mueller and Sleumer, 1934; Brook, 1979), and is clearly inhibited by light (Arens, 1929; Yarwood, 1937). In particular, blue-green and near UV-light was found to inhibit sporulation, whereas red or far-red light did not (Brook, 1979; Rumbolz *et al.*, 2002). Although light inhibited sporangia formation it did not inhibit the development of sporangiophores. In studies where humidity was controlled by salt solutions, it was found that at least 98% RH is required during the period of darkness for sporangia to be produced on detached leaves (Blaeser and Weltzien, 1978). In a similar study by Leu and Wu (1982) it was found that sporangia were produced when RH was at least 93%. They also observed that sporangiophores are produced at humidities as low as 33% RH, even though sporangia were not produced below 93% RH. In vineyards, sporulation was also observed when the measured RH was 75-100% (Arens, 1929; Mueller and Sleumer, 1934). Temperature was found to affect the quantity of sporangia produced. Most sporangia were produced between 17 and 27°C and only few sporangia were produced at 15 and 29°C (Blaeser and Weltzien, 1978). On *V. lambrusca*, large numbers of sporangia were also produced at 15°C, but sporangia were not produced at 10°C (Lalancette *et al.*, 1988b). However, in Swiss vineyards, sporulation was also observed following nights when the temperature was, on average, 10°C (Siegfried *et al.*, 2001). Also, in northern Italy, sporulation was observed at temperatures below 15°C (Vercesi, 1997). Thus, sporulation of *P. viticola* is favoured by relatively warm and humid nights, but recent observations in vineyards suggest sporulation can also occur at temperatures as low as 10°C.

Sporangia are produced after 6 to 10 h (Arens, 1929; Mueller and Sleumer, 1934; Rumbolz *et al.*, 2002) at high humidity during the dark period at night. For *V. lambrusca* it was found that the rate of sporulation and thus the time to first appearance of sporangia during periods of high humidity was affected by temperature (Lalancette *et al.*, 1988b). A similar relationship between temperature and rate of sporulation could also exist for *V. vinifera*, but has to our knowledge not been investigated. Even though the sporangia produced on *V. lambrusca* appeared morphologically fully developed after 7.5 h at high humidity they had not fully matured. After 7.5 h of high humidity almost none of the sporangia germinated and germination percentage increased with increasing duration of high humidity to 50-70% at 15-25°C after 12 h of high humidity, which suggested further maturation after their production.

Sporangiophores are produced at low humidity (Leu and Wu, 1982) and are not inhibited by light (Rumbolz *et al.*, 2002), whereas sporangia are formed only at high humidity in the dark. Sporangiophores could therefore already be produced even when conditions are not favourable for sporangia production. Sporangia could be produced more rapidly on such readily developed sporangiophores when conditions become favourable for sporangia production. Thus, the minimum duration of high humidity required for sporangia production within a vineyard may be shorter than the minimum duration reported here from published research and would be valuable information for a forecasting model. However, this aspect appears not to have been investigated.

2.1.4 Release, dispersal and survival of sporangia

Measurements in vineyards of spore deposition on microscope slides coated with vaseline have shown that sporangia were only released during rainfall events (Blaeser and Weltzien, 1978). In later studies with a spore trap, which was specially designed to collect air and rain droplets, it was confirmed that rainfall events were required to release the sporangia of *P. viticola* (Kast, 1994). Kast found a correlation of numbers of trapped sporangia with rainfall, but he found no correlation with windspeed, temperature or humidity. However, we cannot exclude that sporangia are also released by wind, because Blaeser and Weltzien (1978) found that sporangia were released when wind speed was greater than 9 m s^{-1} in wind tunnel studies. Furthermore, it has been shown that the trapping efficiency of spore traps could be significantly reduced by high wind speeds (Wakeham *et al.*, 2003), thus air-borne sporangia may not have been detected by Kast's spore trap during periods of high wind speeds.

Sporangia of *P. viticola* were found to survive longer at low temperatures and high humidities in studies in which sporangia were exposed to humidity-controlled air above concentrated salt solutions (Blaeser and Weltzien, 1978). At 30°C, sporangia survived for only 6 h and this increased to more than 10 days at 10°C at 100% RH or increased to 3 days at 10°C at 30% RH. Sporangia that were removed from the leaves and kept at 30% RH, retained their germination capacity for 6 h. Later studies found that sporangia can survive for longer even at very low humidities. Sporangia kept at 22°C at 30% RH were still able to cause infections after nine days (Kast and Stark-Urnau, 1999). The effects of solar radiation on survival of *P. viticola* sporangia appear not to have been studied, even though for other downy mildew pathogens, such as *B. lactucae* (Wu *et al.*, 2000), solar radiation was found to rapidly reduce viability of sporangia.

2.2 Lettuce downy mildew

2.2.1 Sources of inoculum and primary infection

Lettuce crops are grown through most of the year in lower latitudes, such as in California. Under these circumstances, overlap between crops secures a 'green bridge' by which the pathogen can spread from a mature infected crop to a newly planted crop, in which it will cause primary infections. Studies on the spatial distribution of lettuce downy mildew in the Salinas Valley in California found a relatively short range of influence and a clustered spatial distribution (Wu *et al.*, 2001c). From these results it was concluded that the availability of inoculum is less important than environmental factors on lettuce downy mildew epidemics, and it suggests that inoculum sources are widely available either from nearby lettuce crops or from wild lettuce, such as *L. serriola* (van Bruggen and Scherm, 1997). At higher latitudes, such as in Canada or in northern Europe, lettuce crops are grown from early spring to late summer/autumn, but are not grown outdoors during winter. In these regions *B. lactucae* is thought to survive the winter as mycelium in infected crop debris (Kushalappa, 2001). Contaminated transplants, which are raised in glasshouses during the winter, could also contribute to primary infections in spring. Infection from contaminated seed may be very rare. In a study in California it was reported that 6 out of 23000 seeds were contaminated with *B. lactucae* (0.3‰), but the risk of infection was not determined (van Bruggen and Scherm, 1997).

Oospores are frequently produced in heavily infected lettuce leaves, and both mating types were found to be present in many parts of the world (van Bruggen and Scherm, 1997). Oospores were found to be present in infected leaves collected from five out of six sites in the UK in 1972 and 1973 (Fletcher, 1976). Debris, which contained oospores, was mixed with soil in the same study. In the first year two out of 6538 and in the second year one out of 9216 seedlings were infected (probability of infection was 0.1-0.3‰), but it cannot be excluded that these infections may have been caused by contaminated seed and not by oospores. The role of oospores in causing downy mildew epidemics is, therefore, thought to be negligible, but oospores are thought to be important in generating genetic variation by recombination in *B. lactucae* populations.

2.2.2 Conditions for infection

Germination of sporangia of *B. lactucae* occurs when leaves are wet and is favoured by cooler temperatures. Observations of germination of sporangia on leaves in a controlled-environment experiment showed that germination was optimal over a broad range of temperatures from 5-20°C (Scherm and van Bruggen, 1993). Less than 20% of sporangia germinated at 25°C, and sporangia did not germinate at 30°C. Earlier work by Powlesland (1954) found that germination was optimal at around 10°C, but that germ tube elongation was greatest at 15°C. This finding, that germination is favoured by cooler temperatures than those favourable for germ tube elongation, was confirmed by Verhoeff (1960). He found that percentage germination of sporangia in aqueous suspensions was highest at 0 to 14°C, where germination occurred within 1-2 h and most sporangia had germinated after 4 h, and that germ tube growth was greatest at 7 to 20°C.

In the controlled-environment experiment by Scherm and van Bruggen (1993), infection was not observed at 25 and 30°C, and this reflected their findings that little or no germination occurred at these temperatures. They found that *B. lactucae* can achieve high levels of infection after 4 h of leaf wetness at 15°C. This finding corresponded well with their field observations that extended morning leaf wetness (from 6:00 am) of on average 4.2 h following the production of new sporangia was associated with infection (Scherm and van Bruggen, 1994b). Morning leaf wetness was on average 1.9 h on days when infection did not occur. The earlier work by Powlesland (1954) also supported these findings. He found infection to occur at 2 to 20°C, but not at 25°C. Furthermore, he also found that a minimum leaf wetness duration of 4 h was required for infection to occur at 15 to 21°C. In field experiments in California it was found that not only can high temperatures during spore germination and penetration inhibit infection, but also during the first 4 h after penetration (Wu *et al.*, 2002). Disease incidence was greatly reduced if temperatures exceeded 20°C during leaf penetration, and disease incidence was also reduced if temperatures exceeded 25°C during the first 4 h post penetration.

Studies by Verhoeff (1960) of the latent period of *B. lactucae* in temperature-controlled cabinets found that the latent period of *B. lactucae* is affected by temperature with latent period being shortest at 17 to 25°C. Latent period increased when temperature decreased below this optimum. A latent period of up to 34 days was observed at 6°C. In California, latent period was studied under fluctuating temperatures occurring in field plots (Scherm and van Bruggen, 1994a). Under these fluctuating temperature conditions, temperature had little or no effect on latent period, and latent periods observed at low temperatures were much shorter than the expected latent periods according to Verhoeff's data for constant temperatures. This suggests that daily fluctuations in temperature actually stimulate the rate of growth of *B. lactucae*. Thus, great care should be taken in using a latent period model, based on data from constant temperature studies, to predict when new sporangia are produced after infection in field crops.

In field trials in Canada it was found that latent period, as the time to first release of sporangia, was related to temperature during periods of high humidity at night (Carisse and Philion, 2002). Sporangia were first collected by spore traps in field plots in two seasons when 200 degree-night hours of RH > 95% had accumulated. A good sigmoid relationship was found between degree-night hours of RH > 95% and relative numbers of sporangia in the air. This suggests that periods of high humidity at night greatly affect the duration of the latent period. This could also explain why Scherm and van Bruggen (1994a) had found little or no effect of temperature in their trials. Temperatures during the night and very early morning, when humidity is generally high in crops, are less variable from day to day, and would thus result in little effect of temperature on latent period in lettuce field crops.

Light also affects infection of lettuce by *B. lactucae*. Raffray and Sequeira (1971) found that when lettuce plants were exposed to continuous light for seven days after inoculation with *B. lactucae* sporangia in water-saturated air and were then given a dark treatment of 8 h, sporulation was not observed, whereas when they were given an initial dark period of 24 h and then kept under continuous light for the next six days they did sporulate during the following dark period. Close microscopic examination of the plants that were inoculated and exposed to continuous light without an initial dark period revealed no mycelium of *B. lactucae* within the leaf tissues. These results suggest that light can have an inhibitory effect on infection during the first 24 h after inoculation. Recent infection studies by the author with *B. lactucae* have found a leaf disease incidence of 10% or less on seedlings, which were inoculated with dry sporangia and then wetted for 8 h in light, whereas leaf disease incidence was 70% on inoculated seedlings, which were wetted for 8 h in darkness (T. Gilles, unpublished results). These findings confirm an inhibitory effect of light during the early infection stages.

2.2.3 Sporulation

Sporulation was found to occur during periods of darkness, but was inhibited by light (Yarwood, 1937). More detailed examination of the inhibition by light found that small quantities of sporangia were produced at low light intensities of 142 µmol $m^{-2}s^{-1}$, but that sporulation was completely inhibited at a light intensity of 356 µmol $m^{-2}s^{-1}$ (Raffray and Sequeira, 1971). In particular, green light of around 526 nm was found to inhibit sporulation. Blue-violet light had no inhibitory effect on sporulation.

High humidity and temperatures ranging from 4 to 20°C are required for sporulation of *B. lactucae*. In field trials in Canada sporulation occurred when the night humidity was greater than 95% RH (Carisse and Philion, 2002). Under controlled humidity conditions above salt-solutions, sporulation was found to occur even at 90% RH (Powlesland, 1954). During such periods of high humidity and darkness the first sporangia were produced in 2 h and a maximum was reached in 6 h in laboratory studies (Raffray and Sequeira, 1971). Temperatures of 1 to 2°C and 25°C were found to inhibit sporulation in controlled-temperature cabinets, whereas sporulation did occur at 4 to 20°C (Powlesland, 1954).

2.2.4 Release, dispersal and survival of sporangia

The release of *B. lactucae* is triggered by the initiation of solar light and reduction in humidity during early morning hours and continues during the hours of daylight. In a controlled-environment experiment it was shown that light initiation and reduction in RH can independently trigger release of sporangia (Su *et al.*, 2000). Thus, the increase in light in the morning by sunrise could trigger release of sporangia even when humidity is high and the crop is wet, and thereby allowing infections to occur directly after spore release, such as data for field plots in California suggested (Scherm and van Bruggen, 1995). This was supported by further evidence from other field trials in California in which it was found that a radiation threshold of 41 μmol m^{-2} s^{-1} for release of sporangia followed by morning leaf wetness of 3 h or longer was associated with days on which infection occurred (Wu *et al.*, 2002). However, most sporangia were found to be released between 9:00 and 16:00 local time in Canada, California and the UK (Fletcher, 1976; Scherm and van Bruggen, 1995; Carisse and Philion, 2002), when humidity is low and the crop is dry. These sporangia need to survive until the evening/night when high humidity and crop wetness occur again to cause infection.

Under Californian summer conditions, the majority of the sporangia that are released during the day will not survive until the evening when conditions are again favourable for infection, because they are killed by high solar radiation levels. In particular UV-B reduced the viability of sporangia significantly (Wu *et al.*, 2000). It was found that most of *B. lactucae* sporangia were killed when the total UV-dose during a day increased to *ca.* 0.5 MJ m^{-2}. In comparison, on most clear summer days in coastal California total UV-dose was *ca.* 0.95 MJ m^{-2}, suggesting that survival of sporangia is highly unlikely. At higher latitudes and under cloud cover the level of UV-radiation is reduced and sporangia are more likely to survive. In Canada, for example, up to 80% of sporangia were still viable after 9 h of exposure to radiation on overcast days (Bhaskara-Reddy *et al.*, 1996). Even on clear days some of the sporangia were still able to survive under Canadian conditions. For example, 21% of sporangia were viable after exposure to 30 MJ m^{-2} total solar radiation over 9 h.

Earlier studies found that high temperatures could also reduce viability of sporangia over time. In laboratory studies by Verhoeff (1960) sporangia survived more than 100 days at 2°C, but survival was reduced to 20 days at 21°C. The level of humidity had little effect on survival of sporangia. A later laboratory trial also suggested that survival of sporangia was reduced by high temperature, but not affected by humidity (Wu *et al.*, 2000). Sporangia survived for more than 12 h at 23°C, but survived for not more than 2 to 5 h at 31°C.

3. FORECASTING MODELS

3.1 Grape downy mildew forecasters

A wide range of forecasting systems has been proposed for grapevine downy mildew (Table 1). Some of these systems have been evaluated in field trials to determine the extent to which fungicide applications have been reduced and disease control secured.

Table 1. Characterisation and evaluation of forecasting systems for grapevine downy mildew (*Plasmopara viticola*).

Model	References	Basis of model	Input	Evaluation	Fungicide reduction	Disease control
EPI (l'Etat Potentiel d'Infection)	Strizyk, 1983a,b Maurin, 1983 Ronzon, 1987 Molot, 1986 Molot *et al.*, 1987 Rocque, 1983 Trevoux, 1985	The accumulation of conditions favourable to oospore maturation and secondary infection are compared to averages, minima and maxima of historical data for many seasons (preferably 20-30 years) for a certain vineyard or region. When accumulated values for favourable conditions for disease development increase above or below the average for a certain location, than disease risk is regarded as high or low, respectively.	Temp. RH Rainfall	historical data: Portugal 1970-99 Gomes and Amaro, 2001 Field trials: Italy 1989-95 in 2 vineyards Vercesi, 1997	30% reduction reduction (not quantified)	NA good, except in 1993
PRO (Plasmopara Risk Oppenheim)	Hill, 1990	Detailed epidemiological model, which predicts downy mildew risk based on the occurrence of favourable weather conditions for oospore maturation and primary infection (Baldacci, 1947), secondary infection and latent period (Mueller and Sleumer, 1934), sporulation (Blaeser and Weltzien, 1978) and dispersal of sporangia (Hill, 1990).	Temp. RH Rainfall Wetness	Field trials: Italy 1989-95 in 2 vineyards Vercesi, 1997	reduction (not quantified)	good, except in 1995
Milvit	Magnien *et al.*, 1991	Quantitative simulation model of numbers of sporangia based on infection/survival of sporangia (Ravaz, 1914), incubation period (Mueller and Sleumer, 1934), sporulation (Ravaz, 1914) and dispersal of sporangia.	Temp. RH Wetness	Field trials: France 1988-93 Muckensturm, 1995	25% reduction	good, not different from standard practices
Envirocaster	Madden *et al.*, 2000	Epidemiological model predicting downy mildew risk in relation to spore load (resulting from predicted sporulation and survival of sporangia) and infection.	Temp. RH Wetness	Field trials: Ohio, USA 1989-1995	51% reduction	Good, not different from standard practices

Table 1 (continued). Characterisation and evaluation of forecasting systems for grapevine downy mildew (*Plasmopara viticola*).

Model	**References**	**Basis of model**	**Input**	**Evaluation**	**Fungicide reduction**	**Disease control**
Plasmo - crop growth model	Orlandini *et al.*, 1993 Rosa *et al.*, 1993	Quantitative simulation model of diseased leaf area based on sporulation (Lalancette *et al.*, 1988b), survival of sporangia (Blaeser and Wetlzien, 1978), infection (Lalancette *et al.*, 1988a) and incubation (Rosa *et al.*, 1995)	Temp. RH Wetness	Field trials: Italy 1990-93 Orlandini *et al.*, 1993 Rosa *et al.*, 1995	40% reduction	good control
Plasmo + crop growth model	Orlandini and Rosa, 1997 Rosa and Orlandini, 1997	In a later version, the simulation of secondary infection now interacts with a leaf area growth model (temperature relationship) (Sall, 1980) to simulate disease intensity as infected tissue per total available tissue	Observed disease severity and leaf area	Not tested	NA	NA
Vinemild	Blaise and Gessler, 1990 Blaise *et al.*, 1996	Complex quantitative simulation model, which combines information on the occurrence of favourable conditions for disease development (sporulation, sporangial dispersal, infection) and host growth into a progeny/parent ratio model (Jeger, 1986) to predict disease severity, which is then fed into a dry matter assimilation model (Wermelinger *et al.*, 1991; Blaise *et al.*, 1999)	Temp. RH Rainfall Observed disease severity	Historical data: Switzerland 1988-90 Blaise *et al.*, 1996 Evaluation of yield prediction 1988-96 Dietrich *et al.*, 1998	NA NA	NA NA
Lufft HP-100	Siegfried *et al.*, 2001	Epidemiological model, which predicts downy mildew risk based on the occurrence of favourable conditions for oospore maturation (Gehmann, 1987), primary infection (Mueller and Sleumer, 1934) and secondary infection (Mueller and Sleumer, 1934; Blaeser and Weltzien, 1979)	Temp. Rainfall Wetness	Field trials in 4 locations in Switzerland in 1995-2000 Viret *et al.*, 2001	28% reduction	Good
SIMPO	Hill, 2000	Prediction of oospore maturation and germination (Hill, 1998)	Temp. RH Rainfall	Field trials: 6 sites in Italy and Germany	(not yet published)	NA

3.1.1 EPI Potential State of Infection (l'Etat Potentiel d'Infection) model

The EPI model uses a database of monthly and daily meteorological data over several seasons, preferably 20 to 30 seasons. EPI compares the accumulation of weather conditions, which are favourable to disease development, in the current season to the average, minimum and maximum of such accumulated values for a historical database. EPI gives estimates of the risk of a severe epidemic by accumulating favourable conditions for oospore maturation, the 'potential' phase, and by accumulating conditions favourable for secondary infections, the 'kinetic' phase. The potential phase, which is based on the effects of rainfall on oospore maturation during winter and spring (Maurin, 1983; Molot, 1986; Ronzon, 1987), is calculated monthly from October to March. The kinetic phase, which is based on the conditions required for secondary infections (Rocque, 1983; Trevoux, 1985; Molot *et al.*, 1987), is calculated daily from April to August. The EPI model is based on the concept that by studying the conditions required for disease development over several years, the averages will get closer to the average conditions required for disease development when more years of historical data are used (Strizyk, 1983a; Strizyk, 1983b), which enables predictions of risk to become more accurate. The 'potential' phase model, which predicts risk in relation to oospore maturation, is used to delay the onset of fungicide spray programs (Raynal, 2001). Fungicide treatments are also avoided when disease development predicted by the 'kinetic' phase model is below the average. EPI was tested on historical meteorological and disease development data from 1970 to 1999 for a vineyard in the Bairrada wine region in Portugal (Gomes and Amaro, 2001). It was found that on average a reduction of two to three sprays per season could have been achieved by spraying according to the EPI model. However, it is not known whether the level of disease control would have been sufficient, because the EPI model was run on historical data only.

In Italy, the EPI model was modified according to findings that oospores can mature and germinate until June (Vercesi, 1997). In the earlier version of EPI maturation and germination of oospores was considered to occur only until March. Therefore, the program was amended to predict the effects of rain on oospore maturation with the model by Ronzon (1987) until the end of June. The modified model was named EPI Plasmopara and was tested in vineyards in northern Italy from 1989 until 1995. It was found that EPI Plasmopara was accurate in predicting the level of risk during the early epidemic stages, but was inaccurate in predicting the risk of epidemics that had a low initial risk and became a severe epidemic by secondary infection cycles. In 1994, EPI predicted reduced disease development during the early epidemic phase and less fungicide sprays were applied without any significant reduction in grape production. However, in 1993 EPI also predicted a reduced disease development during the early epidemic phase, but failed to predict significant increases in disease severity by secondary infection cycles later in the season, resulting in significant losses by disease. This suggests that predictions of risk by the 'kinetic' phase model are not very reliable and could lead to under-prediction of disease risk and fungicides not being applied when required potentially leading to severe epidemics.

3.1.2 PRO – Plasmopara Risk Oppenheim

The Plasmopara Risk Oppenheim model (Hill, 1990) predicts primary infection by the 'three ten rule' (Baldacci, 1947; see 2.1.1). This information is then used to apply the first fungicide spray of the season. Disease development by secondary infections is then simulated using the relationship between the length of the incubation period and temperature (Mueller and Sleumer, 1934), the conditions required for sporulation (Blaeser and Weltzien, 1978) and the conditions for dispersal of sporangia (Hill, 1990). The literature is unclear about when applications of fungicide sprays are advised in relation to predictions of pathogen growth, sporulation and dispersal of sporangia.

Field trials in northern Italy from 1989 until 1995, in which the accuracy of the model was tested, have shown that predictions of the date of oospore maturation and occurrence of primary infections were inaccurate (Vercesi, 1997). Primary infections in the vineyards were frequently more than 15 days later than the dates predicted by the 'three ten rule' in PRO. In one season, predictions by the sporulation model failed, because sporulation occurred at temperatures below the fixed threshold temperature of 15°C in the model. The PRO model always resulted in a reduction of fungicide sprays in comparison with the traditional calendar-based method, but disease control was not always effective. In three different vine-growing areas within the same season, PRO gave poor control of downy mildew, which developed rapidly early in the season. However, in seasons with late downy mildew development, PRO was very effective in controlling downy mildew on grapevines.

3.1.3 Milvit

The Milvit model, which was developed by the French Plant Protection Service, predicts disease development by simulating different stages in the *P. viticola* infection cycle (Magnien *et al.*, 1991). The model predicts infection and survival of sporangia, incubation period, sporulation and dispersal of sporangia in different compartments using temperature and relative humidity data as input. Disease progress is simulated by linking the contribution of each compartment to each other in terms of numbers of sporangia. The risk of downy mildew is expressed as the prediction of the number of sporangia in the simulation. This simulation is started after the maturation and primary infections are observed or predicted by the potential phase part of the EPI model (Maurin, 1983; Molot, 1986; Ronzon, 1987).

From the literature it is not clear when fungicide applications are advised in relation to risk predictions by Milvit. However, trials using the model in vineyards in France from 1988 until 1993 were reported to reduce fungicide sprays by, on average, two sprays from the eight sprays used by traditional methods without significant differences in downy mildew control (Muckensturm, 1995).

3.1.4 Plasmo

Plasmo is a simulation model, in which an epidemiological model has been combined with a plant growth model (Figure 1) (Orlandini and Rosa, 1997; Rosa and Orlandini, 1997). The increase in infected leaf area is simulated with models describing the effects of temperature, relative humidity and leaf wetness on sporulation (Lalancette *et al.*, 1988b), survival of conidia (Blaeser and Weltzien, 1978), infection (Lalancette *et al*, 1988a) and period of incubation (Rosa *et al.*, 1995). A model that describes grapevine leaf growth in relation to temperature (Sall, 1980) is then combined with the epidemiological model to simulate the severity of disease, as infected leaf tissue per total amount of leaf tissue. There are several parameters in Plasmo that need to be calibrated when it is used for the first time in a certain vineyard. These calibrated parameters then take account of the specific conditions that affect epidemic progress in that particular vineyard. The value of these calibrated parameters is relatively stable between seasons according to information from trials by Rosa *et al.* (1995). To start the simulation the grower needs to input a measurement of disease intensity and the amount of leaf area per vine. The criteria for application of fungicides in relation to simulation outputs are not discussed in any of the publications on Plasmo.

The latest version of Plasmo (Orlandini and Rosa, 1997; Rosa and Orlandini, 1997) has not been tested in field trials for its ability to reduce fungicides and provide good control of downy mildew. An earlier version of Plasmo (Orlandini *et al.*, 1993; Rosa *et al.*, 1993; Rosa *et al.*, 1995), which did not include the leaf area growth model, was tested in field trials between 1990 and 1993 in several vineyards in Tuscany in Italy. The earlier version of Plasmo was capable of reducing fungicide applications by 39% without increasing disease incidence when compared to plots, which were treated according to traditional calendar-based methods.

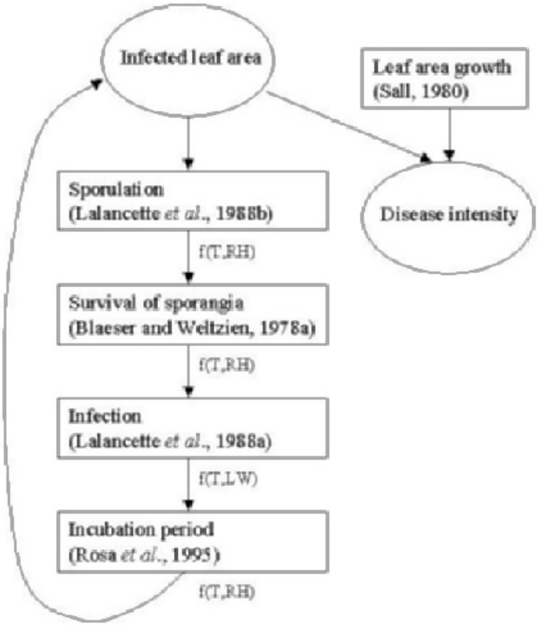

Figure 1. Schematic diagram of PLASMO (Rosa and Orlandini, 1997), a forecasting system for grapevine downy mildew (*Plasmopara viticola*) in which an epidemic model interacts with a crop growth model to predict disease intensity.

3.1.5 Vinemild

Vinemild is a simulation model, in which a mechanistic model of the asexual life cycle and a host growth model both feed into a progeny/parent ratio model to give an output of the proportion of diseased leaf area (Blaise and Gessler, 1990, 1992; Blaise *et al.*, 1996). The rate of increase in diseased leaf area is considered to be related to the proportion of uninfected leaf area, the amount of infected leaf area at the time of infection and the actual rate of epidemic progress depending on favourable weather conditions and the susceptibility of available host tissues. Photosynthesis is then considered to be reduced by a factor proportional to the amount of diseased leaf area. This information is then fed into a dry matter assimilation and allocation model (Wermelinger *et al.*, 1991), so that the effects on yield can be simulated. The philosophy behind this simulation model is that fungicides should be applied to avoid yield losses, but not to avoid downy mildew entirely (Blaise *et al.*, 1996).

The input for Vinemild are crop meteorological data, such as temperature, relative humidity and rainfall, and at least one observation of of the proportion of diseased leaf area in a vineyard at the onset of an epidemic. However, before Vinemild can be used in a particular vineyard or region, some of the model parameters need to be calibrated. According to Blaise *et al.* (1996) only one season of assessments of disease development and collection of meteorological data is sufficient to calibrate parameters for a particular vineyard or region. They suggest that these calibrated parameters do not vary greatly between seasons. However, they have not presented any data on the variability of calibrated parameters between seasons, even though simulations were run over seven seasons from 1988 to 1996 (Blaise and Gessler, 1990; Blaise *et al.*, 1996; Dietrich *et al.*, 1998). Over all seasons, Vinemild was accurate in simulating the early and explosive phases of downy mildew epidemics in vineyards, but always overpredicted disease severity and total leaf area in the final phase. This suggests that there are other factors affecting late epidemic and crop development, which are currently unknown.

The accuracy of the yield model in Vinemild was also tested in these field trials (Dietrich *et al.*, 1998) and the results from these trials suggested that simulated fruit dry mass reflected measured sugar contents reasonably well, but did not accurately predict the level of fruit mass. Later studies found that the earlier model in Vinemild, which described the increase in fruit mass as an exponential curve with day-degrees, gave a poor fit to measurements of increase in fruit mass over time in three trials in 1995, 1996 and 1997 (Blaise *et al.*, 1999). A double exponential curve gave a better fit to data for the increase in fruit dry matter with day-degrees over these three seasons. Thus, two phases of rapid exponential growth in fruit mass are separated by a phase of slow growth. This is not uncommon in fruit development. The growth of peach fruits follows a similar pattern (Genard and Bruchou, 1993). Results of trials investigating the accuracy of the new yield model have not yet been presented. Field trials conducted by Jermini *et al.* (2001) suggested that predictions of fruit yield are also complicated by mobilisation of sugars from starch reserves in the roots of grapevines. In downy mildew infected grapevines potential losses in fruit yield are compensated by a strong mobilisation of assimilates from the roots.

So far, the Vinemild model has been used as a research tool to investigate the interaction between downy mildew epidemics and crop growth and to understand its effect on fruit yield. This has resulted in guidelines on the economic injury level, above which vineyards need to be treated with fungicides to avoid yield losses (Jermini *et al.*, 2001). At the beginning of berry ripening a disease severity level of up to 5% will not affect yields, and this economic injury level increases to 10% severity at the end of the first ripening phase. In a 1998 field trial, downy mildew was kept below the economic injury by only three fungicide sprays. In comparison, seven sprays were applied to control downy mildew in the standard spray schedule.

The original aim of the model was to devise a system which identifies when fungicide sprays need to be applied to avoid yield losses. The finding that yield losses due to downy mildew infections are strongly compensated by mobilisation of assimilates puts a question mark on whether spraying to avoid yield losses is appropriate, because it is likely that if grapevines mobilise assimilates from roots too frequently it will weaken the plants. Possibly, the aim of Vinemild needs to be changed to predicting when to apply fungicides to avoid a level of disease severity that will mobilise assimilates.

3.1.6 Lufft HP-100

The Lufft HP-100 is a small weather station, in which an epidemiological model is integrated (Siegfried *et al.*, 2001). The grower obtains outputs of the model on the screen of the weather station in the field. The epidemiological model predicts the time it takes for oospores to mature in relation to temperature according to Gehmann (1987), and predicts when the actual primary infections occur in relation to heavy rainfall based on the findings by Mueller and Sleumer (1934). Sporulation is predicted to occur when the conditions for sporulation, which were found by Blaeser and Weltzien (1979) are met. Thus, the model determines when environmental conditions required for disease development have been met. The weather station measures the environmental input parameters for the model, such as temperature, relative humidity, rainfall and leaf wetness duration.

The Lufft HP-100 system for grapevine downy mildew was tested in four vineyards in Switzerland over 6 seasons from 1995 to 2000 (Viret *et al.*, 2001). In downy mildew control programs, in which fungicides were applied according to predictions of downy mildew risk by the Lufft HP-100 system, the number of treatments were reduced by, on average, 28%. Disease control was good and in some years even better than the standard calendar-based spray method. However, the criteria for application of fungicides were not explained in the published material.

3.1.7 SIMPO

The model SIMPO (Hill, 2000) predicts only one element of the grapevine downy mildew epidemic, namely the oospore maturation. Infections by oospores are considered to be very important, and are thought to lead to heavy infections just before flowering in certain seasons. Therefore, understanding when infections by oospores occur will enable growers to make the right decisions on when to apply fungicides to protect the crop. Oospore maturation and germination are predicted on the basis of measurements of temperature, relative humidity and precipitation. A field evaluation study is in progress, but results have not yet been published.

3.1.8 Envirocaster

The prediction of risk of downy mildew infection in vineyards by Envirocaster is based on non-linear models fitted to data from controlled-environment experiments on infection (Lalancette *et al.*, 1987; 1988a) and sporulation conditions of *P. viticola* on *V. lambrusca* (Lalancette *et al.*, 1988b), the American grapevine (Madden *et al.*, 2000). Survival of sporangia is also predicted by the Blaeser and Weltzien (1978, 1979) model, which predicts sporangial survival in relation to temperature and relative humidity. At each infection event the total spore load, which is calculated from predicted sporulation and survival of sporangia, over a period of twelve days is considered. Measurements of temperature, wetness duration and relative humidity are input into these models and the output is expressed as percentages between 0 and 100%. Fungicides are applied when the predicted risk of sporulation and infection is greater than 25%, and the time between the previous application is more than 14 days.

Envirocaster was tested over seven seasons in the vineyards of Ohio State University (Madden *et al.*, 2000). The forecasting system was successful in reducing fungicide sprays by one to six sprays (a median of three sprays) per season in comparison with four to ten applications by standard practices. Good control of downy mildew was achieved and the level of disease incidence was not different from that achieved by standard practices. Interestingly, in these trials the effectiveness of models that used only the infection or the sporulation model was tested. Both simplified models reduced spray applications in comparison to standard practices and achieved good control of downy mildew, but the number of spray applications was greater than those applied according to the complete Envirocaster model. Thus, including more models describing relationships between *P. viticola* development processes and the environment could potentially reduce fungicide sprays even further. Oospore maturation is one such important process, which is currently not described by a model in Envirocaster. If delays in maturation of oospores and primary infections could be predicted by Envirocaster then the number of fungicide applications could be reduced even further in some seasons.

3.1.9 New-leaf appearance model

The new-leaf appearance model is an alternative crop-based concept for a model to rationalise fungicide sprays in relation to the duration of the protective activity of fungicides sprayed onto grapevine leaves and the rate of development of new leaves on grapevines. Field trials had shown that several fungicides still control downy mildew for a considerable period of time after their application on leaves (Bleyer *et al.*, 2001). For example, the fungicide metiram was active for 21 days after its application on leaves in a crop (Huber *et al.*, 2000). However, most fungicides are not systemic and did not show any activity on newly developed leaves after the application. The aim of this model is to apply fungicides only when a certain number of new leaves are predicted to have developed by a growth model (Schultz, 1992) and weather conditions are favourable to disease development as determined by an epidemiological model (Bleyer and Huber, 1995). In a trial in 2000, it was found that fungicides could be reduced significantly without affecting disease control by spraying after four new leaves are predicted to have developed following the last fungicide treatment. Further information on trials over several seasons are required to conclude how effective this model system is in reducing fungicide inputs and securing downy mildew control.

3.2 Lettuce downy mildew forecasters

3.2.1 Californian infection models

The Californian lettuce downy mildew forecasting models are based on findings that the occurrence of infection in lettuce field crops is associated with prolonged morning leaf wetness. Scherm and van Bruggen (1994b) found a significant difference in duration of morning leaf wetness from 06:00 between infection and non-infection days. On days of infection, morning leaf wetness duration averaged 4.2 h, whereas on days, when infection did not occur, morning leaf wetness persisted for 1.9 h, on average. In another study, it was found that sporangia were released in small numbers early in the morning and caused infection when morning leaf wetness duration persisted for at least 3 h (Scherm and van Bruggen, 1995).

Based on these findings a model was developed, which predicted infection events when morning leaf wetness persisted for 4 h from 06:00 (Scherm *et al.*, 1995). This model advises the application of a fungicide spray within one day after the prediction of an infection event. Fungicides were not applied if new infection events were predicted within ten days after the previous spray. This model was evaluated in seven trials during summer and autumn of 1993 and 1994 on three locations in coastal California. The application of fungicides was reduced by 67% by applying according to the forecast in comparison with calendar-based spray applications (three applications per crop). Disease intensity was not different from that obtained for crops that were treated according to the calendar-based schedule. The Californian infection model seemed accurate even under Canadian conditions. In one trial in one season, the number of protectant sprays was reduced from 6 to 4 by spraying according to the Californian morning leaf wetness model (LWD from 3:00 until 10:00 am) without any increase in disease severity when compared to the standard fungicide schedule (Philion *et al.*, 1998). However, considerable care should be taken when using forecasting systems that were developed in a completely different geographical location. Morning leaf wetness duration is a good criterium to predict infection in California, where most sporangia are killed during the day (Wu *et al.*, 2002). However, in Canada, a considerable proportion of sporangia is likely to survive daytime conditions, and infections in the evening when the crop becomes wet are likely. Thus, probably different criteria need to be used or added to the Californian to effectively apply it under Canadian conditions.

This original system was later compared to two modified systems (Wu *et al.*, 2001b) in 1996 to 1998. In one modified forecasting system, survival of sporangia during daytime was predicted as well as prediction of infection according to the original system. Spores were considered to survive when daytime temperatures were above 30°C for less than 3 h and when solar radiation levels of greater than 0.3 kWm^{-2} occurred for less than 3 h (Wu *et al.*, 2000). In a second modified system, the occurrence of sporulation events was also predicted as well as prediction of survival of sporangia and infection by the original system. Sporulation was predicted to occur when both relative humidity within the canopy was greater than 80% RH and wind-speed was less than 0.5 $m\ s^{-1}$ for less than 4 h (Su *et al.*, 1998). In four trials between 1996 and 1998, survival of sporangia was never predicted and observations in the field plots confirmed this. Thus, predictions of infection by the first modified system were identical to the original system. Conditions during the night were only rarely found to inhibit sporulation, i.e. sporulation was not observed on only 24 out of 153 nights (16%) and only 6 of these non-sporulation nights were correctly predicted. Conditions always favoured sporulation on nights, which were followed by prolonged morning leaf wetness. Predictions of infection by the second modified system were also identical to the original system. Thus, under Californian conditions it seems that *B. lactucae* sporangia rarely survive the high daytime temperatures and solar radiation levels and that mornings with prolonged leaf wetness are rarely preceded by nights with unfavourable conditions for sporulation. However, under different climatic conditions these factors may play important roles in downy mildew epidemics.

Later work has improved the original infection model. To reduce the number of false negatives the minimum morning leaf wetness required for infection to be predicted was reduced to 3 h based on results from receiver operating curve analysis (Wu *et al.*, 2002). It was found that high temperatures during the 3-h wetness period when penetration occurs and during a 4-h post-penetration period can inhibit the establishment of infection. Upper temperature thresholds of 20 and 22°C for the 3-h penetration and 4-h post-penetration period, respectively, were included in the model. Furthermore, the 3-h morning leaf wetness period was changed to start after sunrise, which was defined as the hour when solar radiation increased above 8 W m^{-2}, instead of at 06:00. In a comparative study between this modified system and the original system it was found that the modified system gave fewer false negative and false positive predictions of infection. For historical data from 14 field studies between 1993 and 1998 the modified model predicted equal or fewer infection events when disease incidence was low, but predicted equal or more infection events when disease incidence was high. However, the accuracy of this improved model in reducing fungicide inputs and maintaining a good level of downy mildew control has not been tested in actual lettuce field trials.

3.2.2 Bremcast

Bremcast is a Canadian forecasting model for lettuce downy mildew and predicts downy mildew development when crop meteorological conditions are favourable for sporulation, survival of sporangia and infection (Kushalappa, 2001). Sporulation is predicted in relation to wetness duration, humidity and temperature during the night (Powlesland, 1954; Tchervenivanova, 1995). Infection was predicted by its interaction with temperature and leaf wetness duration (Scherm and van Bruggen, 1993). The model predicts the survival of sporangia during the day based on previous findings on its relationship with solar radiation, which was fitted to experimental data from a study in Canada (Bhaskara Reddy *et al.*, 1996). Even though results from that study suggested that a considerable number of sporangia can survive the solar radiation dose in Canada, infection is only considered to occur when leaf wetness occurs in the morning. However, it is likely that the sporangia that have survived the daytime conditions will cause infection whenever the crop is wet at night.

The BREMCAST model was evaluated with historical weather and downy mildew disease incidence data for 181 days collected from 13 commercial lettuce fields in California during 1991 and 1992, and for 10 days in a small field plot in Quebec. According to Kushalappa (2001), predictions for disease intensity by BREMCAST were correct in 84% of all predictions. However, the criteria for when a prediction is considered to be correct are unclear from Kushalappa's publication. Field trials will need to be conducted to assess whether timing spray applications according to BREMCAST predictions can reduce fungicide applications.

3.3 Quality of input weather data

The input of weather data is the main driver of forecasting models. Thus, the quality and accuracy of weather data will have great impact on the accuracy of model outputs. Forecast weather data have frequently been used in forecasting models, which predict the risk of downy mildew epidemics based on the occurrence of favourable weather conditions, because it will enable this type of models to give 'true' predictions of infection, whereas when measured weather data are input, disease development is predicted just after the event has occurred. Many of the fungicides for controlling downy mildew are contact fungicides and are most effective when they are applied just before infection events. Furthermore, large farm operations often require some time to schedule a spray in their program of work. Thus, the use of forecast weather data in forecasting models would overcome these problems.

However, forecast weather data are less accurate than measured weather data and therefore introduce another error into model predictions. In the Californian infection model, which predicts infection based on days with prolonged morning leaf wetness, forecasts of leaf wetness were used as input. Even though in seven trials in 1993 and 1994 90% of days were forecast correctly as days with or without prolonged leaf wetness, there were great errors in predicting the time of onset and end of leaf wetness (Scherm *et al.*, 1995). The error of forecasts of morning leaf wetness duration was greater than 2 h for 61.7% of predictions. This resulted in crucial infection events not being predicted by the forecasting model and thus fungicides not being applied at times when they were required. The inaccuracy of using forecast leaf wetness data was confirmed by four trials from 1996 to 1998, where the termination of morning leaf wetness was predicted correctly for only 24 out of 124 days in these three years (Wu *et al.*, 2001b). In a more detailed study, in which leaf wetness duration was predicted by two different weather forecast systems at two locations in California over two seasons, it was found that predictions of leaf wetness by both systems were most inaccurate near the 10:00 threshold time of wetness duration for a prediction of infection (Wu *et al.*, 2001a).

Leaf wetness within a crop is a parameter, which is highly variable within a crop. The measurement of leaf wetness by one or two sensors within a crop might not reflect very well the incidence of wetness within an entire crop canopy. A model has been developed, which simulates the incidence of water droplets throughout a grapevine canopy by simulating dew formation, evaporation and the life time of droplets on leaf surfaces by taking into account the energy balances within a crop, air temperature, wind-speed and humidity (Hoppmann and Wittich, 1997). A good relationship was found between measured wetness duration and simulated wetness duration. Preliminary model outputs suggest that simulated wetness duration may be successfully used as an input parameter in grapevine downy mildew forecasting systems.

Measurements of wetness and humidity are greatly affected by location of the sensors, whether they are located within or outside crops. The humidity within crops is generally higher and is also affected by the density and size of the crop and the amount and size of foliage. Similarly, measured leaf wetness duration is longer within crops, and is also affected by these crop parameters. Thus, preferably forecasts need to be based upon measurements of wetness and humidity within crops to reflect the actual crop microclimate, in which the downy mildew pathogens operate. Forecasting models should also clarify the height and location within the crop at which these weather parameters were measured, because the accuracy of forecast outputs can be greatly affected by this. For example, for the Californian lettuce downy mildew infection model, leaf wetness was measured between lettuce plants at a height of 10 cm above the soil (Scherm *et al.*, 1995). However, in most publications of forecasting models these essential details are not described. Also, measurements of weather, which are made by national weather stations at a distance from the crops concerned, are frequently used as input data for forecasting models, and could increase the error of prediction. One possible method of reducing the error of such weather data is by using extrapolation of data between distant weather stations surrounding the crop (Russo *et al.*, 1989).

Also, sensors themselves can be inaccurate. The commonly used humidity sensors in weather stations are highly inaccurate at measuring high humidities, where they can have an error of ± 2 to 4%. New humidity sensors are being developed that monitor the condensation and evaporation from a temperature-controlled glass mirror (Sorli *et al.*, 2002). These sensors are more accurate at high humidities. Over the range 80-95% RH, the accuracy of these sensors was ± 1%. However, these sensors are very expensive and therefore not used in crop weather stations.

4. DISCUSSION

All forecasting models for grapevine and lettuce downy mildews, which were reviewed in this chapter, were successful in reducing fungicide sprays, and field trials suggested that with most of these forecasters a good level of disease control was obtained by applying fungicides accordingly. The level of disease control was in many cases not different from that obtained by traditional calendar-based schedules. All successful forecasters were based on knowledge on the effects of weather factors on disease development processes. It is difficult to identify which processes were the most important variables to predict to achieve effective disease control with fewer fungicides. However, all successful forecasters predicted infection often in combination with predictions of available spore load (sporulation and/or sporangial survival). Although, geographic region may affect which disease development processes are important. For example, under Californian conditions sporangia of *B. lactucae* only rarely survive daytime solar radiation levels (Wu *et al.*, 2000), whereas in Canada sporangia could survive these daytime conditions more frequently (Bhaskara-Reddy *et al.*, 1996), thus survival of sporangia is an important element in the Canadian forecaster, Bremcast (Kushalappa, 2001), but not in the Californian infection model. Furthermore, all successful forecasters used weather data measured within the crops concerned as input.

Field studies over seven seasons by Madden *et al.* (2000) suggested that less risk periods are identified resulting in less fungicide spray applications, when models predicting different disease development processes are combined. They found that when they applied fungicide sprays only when the risk of infection was high or when spore load was predicted to be high, fungicide sprays were reduced by 31% and 46%, respectively, in comparison with a standard 14-day schedule. A greater reduction in fungicide usage of 51% was obtained, when they applied fungicides to when both the risk of infection and a considerable spore load were moderate to high. However, care should be taken in combining models that predict different disease development processes. The more models are combined, the greater the probability of errors in prediction. If one or more of the combined models frequently give false negative predictions then this will result in underestimation of the risk of disease development. Even though substantial savings in fungicides are made by using such a model, there is a risk that in some seasons significant disease progress is not predicted. This also highlights that forecasters should not only be tested for their ability to reduce fungicides and maintain disease control, but also the accuracy of the underlying models predicting processes such as infection and sporulation need to be tested for their accuracy in field trials. The use of receiver operating curve analysis (Wu *et al.*, 2000) is highly recommended to avoid false negative predictions to occur too frequently. The PRO forecaster, which failed to predict disease development early in the season of 1995 in a field trial in northern Italy, is an example of a forecaster with combined models, in which one of the models is inaccurate, in this case the oospore maturation and primary infection model (Baldacci, 1947; Vercesi, 1997). Of the forecasters reviewed in this chapter, the proposed testing of the accuracy of the models in a forecaster has to my knowledge only been done in great detail with the Californian infection model for *B. lactucae* (Scherm and van Bruggen, 1994b; Wu *et al.*, 2002)

Not all forecasting models were always successful in achieving a good level of disease control, which was at least equal to the level obtained for traditional fungicide schedules. Models could fail to predict disease development when forecast weather data is used as input data. The Californian infection model with input of forecast morning wetness frequently failed to predict important infection events, which led to fungicides not being applied at times when they were required to control lettuce downy mildew (Scherm *et al.*, 1995; Wu *et al.*, 2001a; Wu *et al.*, 2001b). The forecasts of wetness frequently often failed to predict the onset and end of wetness periods, which is an important parameter in the Californian model.

The accuracy of input data is very important for obtaining accurate predictions of disease development, such as the failure to predict infection of *B. lactucae* with forecast wetness data has demonstrated. Wetness and humidity are both important parameters throughout the development of downy mildew pathogens, affecting infection, incubation period, sporulation and release of sporangia. However, it is difficult to measure these parameters accurately. Wetness is highly variable within crop canopies. Thus, measurements by one or two leaf wetness sensors may not accurately reflect the incidence of wetness in a crop. A model simulating the population of water droplets in a grapevine canopy has been devised to predict the incidence of wetness (Hoppmann and Wittich, 1997), but this model has not been widely tested for its accuracy in disease forecasters. The type of humidity sensors used in weather stations are highly inaccurate (± 2 to 4%). It appears that little work has been done to investigate how the inaccuracy of crop environmental sensors affects the accuracy of model outputs. Another aspect is the location of sensors in a crop, which could significantly affect the measurement.

Publications of forecasters frequently fail to provide sufficient detail to enable proper replication of trials by independent research and for implementation of models by advisors and growers. More detailed information is required on where sensors were in relation to the crop when disease forecasts were developed or when they were evaluated in trials, because location of sensors could affect model outputs. Furthermore, the criteria for application of fungicides are often not described or described with insufficient detail. Examples of where such criteria were described in sufficient detail can be found in publications by Madden *et al.* (2000) and Wu *et al.* (2001b).

Downy mildew models that have been developed in a certain region with a certain genotype of the host, are frequently used in forecasters for downy mildew on different host genotypes in different geographical regions. For example, Plasmo (Orlandini *et al.*, 1993; Rosa *et al.*, 1993) uses the Lalancette *et al.* (1988a,b) models for sporulation and infection of *P. viticola* on *V. lambrusca*, a less susceptible American grapevine. These models were developed in studies with isolates of *P. viticola* from Ohio, to predict disease development on *V. vinifera*, European grapevine, which is generally more susceptible, in Italy. Such transfer of models to predict downy mildew in an entirely different environment could increase the inaccuracy of model predictions. There may be significant differences in, for example, threshold temperatures for infection and sporulation, minimum wetness duration required for infection, etc., because of pathogen adaptation to different climatic conditions. On *V. lambrusca*, sporangia were not produced at 10°C (Lalancette *et al.*, 1988b), but sporulation in Swiss vineyards was also observed following nights when the temperature was, on average, 10°C (Siegfried *et al.*, 2001). Thus, great care should be taken in using models that were developed elsewhere in a different cropping environment, in forecasters. In Plasmo (Rosa and Orlandini, 1997) and Vinemild (Blaise and Gessler, 1990) some of the parameters can be calibrated for use in a different vineyard and/or region. These calibrated parameters account for some of the regional and vineyard variability. However, the parameters in these models, which determine the weather thresholds and relationships with weather factors, are fixed. Thus, for example, regional changes in threshold temperature for sporulation cannot be adjusted for. Ideally, models that are describing relationships between disease development processes and weather factors, need to be evaluated for their accuracy of prediction in each different cropping environment and region, and if necessary adjusted.

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CUCURBIT DOWNY MILDEW: A UNIQUE PATHOSYSTEM FOR DISEASE FORECASTING

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1. INTRODUCTION

Downy mildew of cucurbits is caused by *Pseudoperonospora cubensis* (Berk. & M.A. Curtis) Rostovtsev. This disease has a worldwide distribution and probably occurs wherever cucurbits are grown (except unirrigated, very dry climates) and is especially prevalent in areas with a warm, humid climate. For example, in the United States (US) the disease is prevalent in the southeast occurring each year on all commercially grown cucurbits. However, its occurrence in the desert areas of the southwest is very rare.

P. cubensis is an obligate parasite which can grow and multiply in nature only on or in living tissue. The fungus only causes disease on members of the Cucurbitaceae (Palti and Cohen, 1980) which are frost-sensitive plants. Consequently, the pathogen cannot survive in areas where frosts occur that are sufficient to kill cucurbits. For convenience, we refer to the 30th latitude in the southeastern US, as a point below which cucurbit downy mildew can survive winters. Sporangia of *P. cubensis* survive between 22 hours and 16 days depending on temperature, relative humidity, and ultraviolet radiation (Cohen, 1981). This is ample time to survive long-distance dispersal via wind currents which can travel great distances in one to two days. Our hypothesis is that the disease is introduced annually into northern latitudes from below the 30th latitude where it can survive winters. Moreover, we hypothesize that transport of the pathogen can occur over long distances via atmospheric wind currents. Thus, forecasts of the movement of the disease should be possible by tracking long-distance spore movement of *P. cubensis* from known sources of inoculum to other potential sites of infection. This contrasts sharply with most disease forecasting systems which assume the host and inoculum are present and focus efforts on determining the environmental conditions necessary for

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growth and development of the pathogen. This paper describes a forecasting system for cucurbit downy mildew that has been in use since 1998. The benefits and limitations of the system are discussed.

2. MATERIALS AND METHODS

2.1 Disease reports

The forecasting system is dependent on timely and well-documented reports of new downy mildew occurrences. A network of approximately 40 “state representatives” (mostly university-based plant pathologists and horticulturists) from the US and Mexico was established. Each representative was asked to report any occurrence of the disease in their area via an electronic report form on the world wide web (http://www.ces.ncsu.edu/depts/pp/cucurbit/). The disease report form prompts the representative for important information regarding the outbreak, including the location of the source, estimated date of first occurrence, cucurbit host, disease severity, size of outbreak and other source characteristics. The state representative also reports when the source has been eliminated, usually at the end of the cropping cycle. The forecaster uses the most recently reported and important continuing source sites to initiate each new set of forecast trajectories. Multiple source sites that occur close to one another (eg, within the same county) are reported as a single, central location. Forecasts are issued twice weekly and more frequently if urgent situations arise. The system is based on the tobacco blue mold forecasting system developed by C.E. Main (Davis and Main, 1984; Main *et al.*, 2001). Tobacco blue mold (caused by *Peronospora tabacina*) and cucurbit downy mildew share several important biological characters (eg, overwintering of the pathogen) which make the forecasting system applicable to both diseases (Davis and Main, 1986 and 1989).

2.2 Forecast model

Spore transport in the atmosphere is calculated using the HY-SPLIT trajectory model from the National Oceanic and Atmospheric Association’s (NOAA) Air Resources Laboratory (ARL) in Silver Spring, Maryland, USA. For the necessary meteorological input, HY-SPLIT commonly uses outputs of either the ETA model or the AVN (Aviation) model, members of the family of Numerical Weather Prediction (NWP) models used to forecast short-term weather conditions in and around the

United States. These meteorological data are generated initially by the National Center for Environment Prediction (NCEP), a branch of NOAA, then modified by ARL to conform to HY-SPLIT's input requirements. The data of primary interest are the forecast wind fields in the atmospheric boundary layer. In nearly all cases, HY-SPLIT trajectories are provided by ARL via an automated electronic mail system. The trajectory is a plot of the future atmospheric pathway of a "parcel" of air likely to contain spores; in other words, the prediction of the spatial and temporal positions of a spore cloud center for the future two days following release from a source site (the model has recently expanded the duration to 54 hours).

2.3 Forecast

Each forecast includes a map showing a trajectory of atmospheric wind currents starting at the point of the known disease source and its predicted movement over the next 54 hours (Figure 1). Beginning in 2003, the forecast model expanded the duration from 48 to 54 hours. The risk of a disease outbreak is estimated based on the following considerations: temperature, rainfall, cloud cover, survivability during transport (UV radiation and desiccation effects associated with cloud cover), and deposition from actual and potential rainfall. General weather conditions in the eastern US that may influence the movement of spores or provide opportunities for subsequent infection are discussed in a "regional weather" section. An "outlook" section combines these elements describing the likelihood of inoculum spread and disease risk, 54 hours into the future. Each forecast is archived on the internet and at NC State University.

2.4 Map description

The source location is represented on the map (Figure 1) by a star. The trajectory, which represents the centerline of a cloud of airborne spores, is given by a line starting from the source. The triangles on the forecast pathway give the spore cloud position at six-hour intervals. The large triangles are the positions at 00Z or 12Z. [Chronological time is given in Universal Time (UTC) or Zulu time (Z). Both refer to the reference time formerly known as Greenwich Mean Time.] Header labels detail the time and date of the trajectory start, and the time and date of the meteorological forecast file used to compute the trajectory. The latitude and longitude of the disease site used for the source is given along the vertical left edge of the map.

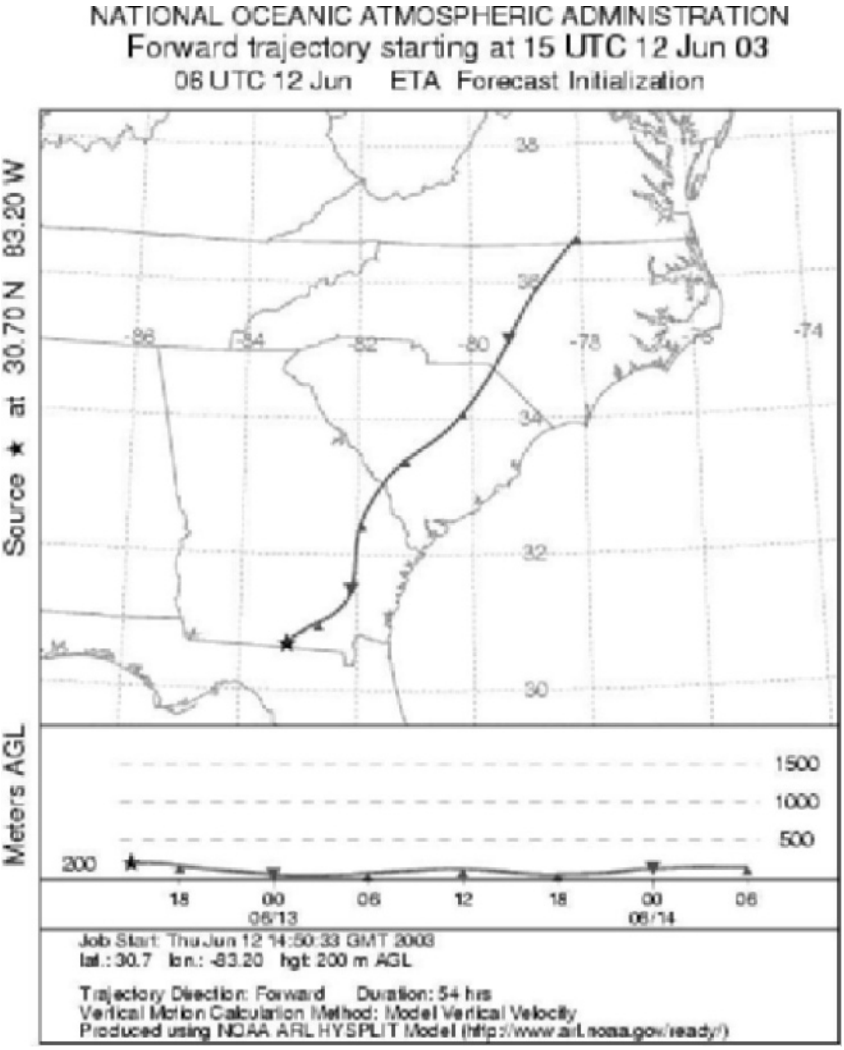

Figure 1. A sample forecast showing air movement from inoculum source to 54 hours into the future. The triangles on the forecast pathway give the spore cloud position at six-hour intervals. The trajectory starts near Lake Park, Georgia (source of spores) and terminates at the Virginia/North Carolina border, a distance of approximately 1860 km. See section 2.4 for more details.

The small rectangular graph underneath the map shows the vertical motion of the spore cloud center. (The vertical motion is determined by using the vertical velocity parameter of the meteorological model.) The solid line represents the pathway. The triangles on the vertical pathway correspond to the triangles on the horizontal trajectory shown in the map above it. The triangle timeposts are labeled underneath the vertical motion graph. The height of the vertical trajectory is given in meters above ground level (meters AGL). Height labels are located to the side of the vertical motion graph.

3. RESULTS AND DISCUSSION

A total of 1121 forecasts were made between 14 May 1998 and 22 October 2002, with an average of 224 forecasts per year (Table 1). Each year forecasts began when disease reports were received from southern Florida or Mexico (i.e., February or March) and terminated after several hard frosts occurred in the southeastern US (i.e., October or November). Reports of the disease were common from southern states along the eastern seaboard (North Carolina and south) and southern Texas. Reports were much less common from northern states. No reports were received from the western US. Thus, from reports of disease occurrence alone (notwithstanding the limitations of this type of data), we can begin to describe the spatial and temporal distribution of cucurbit downy mildew in the US.

From these 1121 forecasts, 19% high-risk situations were predicted, 32% moderate-risk situations and 49% low-risk situations. Forecast trajectories combined with experience suggest that long-range (>160 km) transport scenarios are few. Most forecasts involve localized (<30 km) and/or short-range (<160 km) disease development potential. Because of the widespread distribution of cucurbit crops in the eastern US (Figure 2), the disease is likely to be spreading locally and over long distances at the same time.

Many factors are involved in predicting the future movement of fungal spores and the associated weather conditions. The HY-SPLIT model represents state-of-the-art atmospheric trajectory analysis, but does not reflect the accuracy of the meteorology data (i.e., winds). Good HY-SPLIT trajectories may deviate from the true path by about 15% of the transport distance; many trajectories are off between 30 to 35%. Complex weather situations can produce errors of 50% or more. Research with HY-SPLIT and similar models using historical data of actual epidemic spread provides

Table 1. Chronological listing of five years of forecasts, their disease source locations and durations.

Year	City, State	County	First forecast	Last forecast	No. forecasts
1998	Leesburg, FL	Lake	14 May 98	04 Aug 98	20
	Clinton, NC	Sampson	06 Aug 98	29 Oct 98	20
	Kinston, NC	Lenoir	10 Sep 98	29 Oct 98	15
	Pollocksville, NC	Jones	10 Sep 98	29 Oct 98	15
	Edneyville, NC	Henderson	10 Sep 98	22 Oct 98	13
	New Haven, CT	New Haven	06 Oct 98	29 Oct 98	7
				Sum	90
1999	Immokalee, FL	Collier	02 Mar 99	19 Nov 99	40
	Tifton, GA	Tift	24 Jun 99	19 Nov 99	28
	College Station, TX	Brazos	08 Jul 99	31 Aug 99	12
	Clinton, NC	Sampson	12 Jul 99	01 Nov 99	32
	Jacksonville, NC	Onslow	31 Aug 99	01 Nov 99	22
	Seaboard,clayton, & Stem, NC	N. Hampton, Johnston, Granville	31 Aug 99	01 Nov 99	22
	Salisbury, MD	Wicomico	31 Aug 99	01 Nov 99	22
	Crossville, TN & Blairsville, GA	Cumberland, Union	31 Aug 99	01 Nov 99	22
	Richmond, KY	Madison	31 Aug 99	01 Nov 99	22
	Vincennes, IN	Knox	31 Aug 99	08 Sep 99	3
	Mequon, WI	Ozaukee	31 Aug 99	01 Nov 99	19
	Marlboro, NJ	Monmouth	10 Sep 99	01 Nov 99	18
	Charleston, SC	Charleston	13 Sep 99	19 Nov 99	25
				Sum	287

Table 1, continued. Chronological listing of five years of forecasts, their disease source locations and durations.

2000	Ciudad Mante, Mexico	--	14 Feb 00	13 Jun 00	56
	Colima, Mexico	--	14 Feb 00	13 Jun 00	56
	Immokalee, FL	Collier	04 Apr 00	31 Oct 00	27
	Webster, FL	Sumter	04 May 00	31 Oct 00	47
	Tifton, GA	Tift	15 Jun 00	31 Oct 00	36
	Houston, TX	Harris	15 Jun 00	03 Aug 00	13
	Hendersonville, NC	Henderson	01 Aug 00	05 Oct 00	20
	Mason, OK	Payne	03 Aug 00	05 Oct 00	19
	Newton Grove, NC	Sampson	15 Aug 00	24 Aug 00	4
	Mount Olive, NC	Wayne	29 Aug 00	31 Oct 00	17
	Salisbury, MD	Wicomico	31 Aug 00	31 Oct 00	16
	Blairsville, GA	Walker	05 Sep 00	05 Oct 00	10
	Bridgeton, NJ	Cumberland	21 Sep 00	31 Oct 00	10
	Lake Geneva, WI	Walworth	28 Sep 00	05 Oct 00	3
				Sum	334
2001	Immokalee, FL	Collier	15 Mar 01	16 Jun 01	28
	El Campo, TX	Wharton	14 Jun 01	10 Jun 01	6
	Charleston, SC	Charleston	10 Jun 01	19 Jun 01	2
	Knoxville, TN	Knox	19 Jun 01	23 Oct 01	21
	Stanton, KY	Powell	09 Aug 01	23 Oct 01	20
	Sylvester, GA	Worth	09 Aug 01	25 Oct 01	21
	Goldsboro, NC	Wayne	09 Aug 01	25 Oct 01	21
	Clarksville, TN	Montgomery	16 Aug 01	25 Oct 01	20
	Hartwell, GA	Hart and Union	23 Aug 01	23 Oct 01	17
	South Deerfield, MA	Franklin	23 Aug 01	23 Oct 01	17
	Piggett, AR	Clay	06 Sep 01	25 Oct 01	14
	Vincennes, IN	Knox	11 Sep 01	23 Oct 01	12
	Painter, VA	Accomack	11 Sep 01	23 Oct 01	12
				Sum	211

Table 1, continued. Chronological listing of five years of forecasts, their disease source locations and durations.

2002	Immokalee, FL	Collier	05 Mar 02	27 Jun 02	34
	Charleston, SC	Charleston	23 Apr 02	22 Oct 02	34
	Groveland, FL	Lake	16 May 02	22 Oct 02	35
	Camilla, GA	Mitchell	23 May 02	22 Oct 02	34
	Goldsboro, NC	Edgecombe, Wayne	13 Aug 02	22 Oct 02	19
	Painter, VA	Accomack	22 Aug 02	22 Oct 02	17
	Carney's Point, NJ	Salem	03 Sep 02	22 Oct 02	13
	Vincennes, IN	Knox	05 Sep 02	22 Oct 02	12
	Homestead, FL	Miami-Dade	22 Oct 02	22 Oct 02	1
				Sum	199
				Total	1121
				Annual Average	224

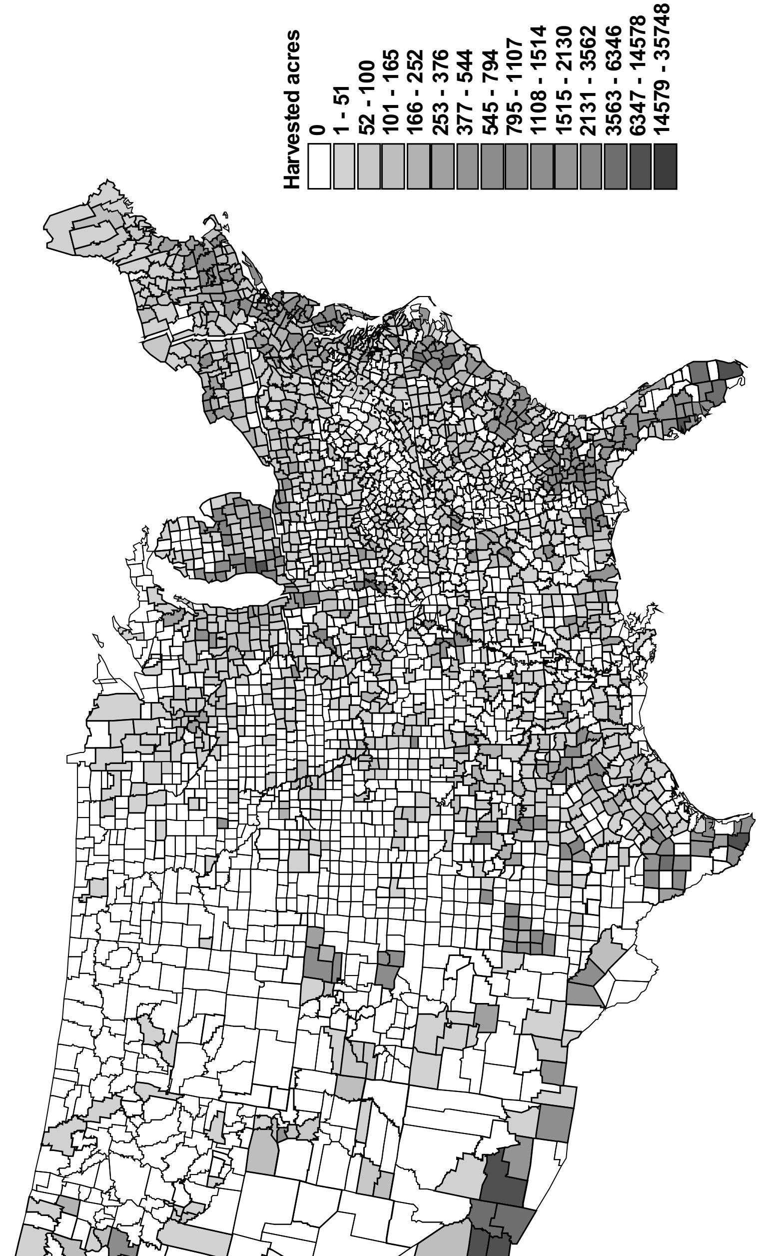

Figure 2. US harvested cucurbit (watermelon, cucumber, cantaloupe, squash, pumpkin and honeydew) acreage by county, 1997. Data source: USDA census of agriculture, 1997. Map created in ArcView by G.J. Holmes and J.A. Thurman, North Carolina State University, Raleigh, USA.

good qualitative guidelines for evaluating the accuracy and usefulness of the forecasts. Each downy mildew forecast includes a measure of the anticipated confidence in its trajectory pathway. It is important that users of this information also continue to pay close attention to local weather conditions associated with the forecasts.

Another important factor in forecasting is the experience of the forecaster. Although the trajectories are generated using mathematical models, the forecaster must intepret them in relation to weather conditions along their path in order to provide risk assessments. This process is inherently subjective and requires that the forecaster skillfully incorporates knowledge of weather, inoculum source and disease biology into the risk assessments. It is our goal to minimize subjectivity where possible, but it cannot be eliminated.

An important limiting factor in the forecasting system is field scouting and reporting. This type of effort is not sustainable in the long term as it requires dependence on individuals for whom this is a low priority. Thus, frequent reminders are necessary to keep awareness high among state representatives. Once state representatives realize that submitting disease reports is extremely simple and requires very little time, they are more inclined to provide reports in the future. This requires several years to achieve.

The benefits of the forecasting system include 54-hour advanced geographic warning of potential disease outbreaks and sensitivity of the fungal population to fungicides (assuming the report from the source has included this). The system requires minimal input once the models are running and the website has been established. The website contains a great deal of information related to downy mildew (eg, epidemiology, management guidelines, diagnostic guide, list of state representatives and other important links), thus serving an important educational role.

3.1 System validation and future direction

How is this type of forecasting system validated? If trajectories predict inoculum movement to an area where susceptible hosts are present/prevalent and favorable environmental conditions exist, the probability of a disease outbreak should be high. If an outbreak occurs, did the system work? If it does not, did it fail? Are repeated introductions of the pathogen necessary so that inoculum can reach a critical threshold necessary for disease to occur? If conditions for disease development are less than optimum, what period of time is necessary before concluding that disease will not develop as a result of a particular inoculum introduction event(s)? We believe that answers to these questions will come from associating actual events with predicted events over multiple years using a space-time statistical model, a direction we intend to take in the future.

The current system should be intensified for at least two seasons to investigate the optimum frequency of forecasting. The assumption is that more reports of disease outbreaks are needed in order to describe and analyze when and where the disease is occurring in the US.

Host specificity exists within *P. cubensis* (Thomas *et al.*, 1987; Lebeda and Gadasová, 2002) and this biological character could be used as a marker to track its movement. A set of host differentials could be planted in strategic locations along the most common route of disease spread (i.e., the south eastern seaboard states - Florida, Georgia, South Carolina, North Carolina and Virginia). These host differential plots must be monitored closely in order to know the precise timing of disease development and the pathovar(s) responsible. Spore trapping and site-specific weather measurements at the source would help assess the concentration and duration of inoculum production. Alternatively, a study of the population genetics of *P. cubensis* could reveal DNA profiles that could be used to track the movement of populations.

New tools in Geographic Information Systems (GIS) can be used to define spatial and temporal distribution of favorable environmental conditions for both disease and host development using historical weather data. Host distribution maps at county-level resolution (Figure 2) could be combined with historical weather data to define the temporal and spatial distribution of the host. This information would add a significant amount information that could be used in the risk analysis of each forecast as well as the validation of the forecasting system.

In the 1940s, C.J. Nusbaum attempted a similar epidemiological experiment. Researchers in 11 Atlantic coastal states reported occurrences of cucurbit downy mildew from Florida to Massachusetts (Nusbaum, 1944 and 1948). Based on the circumstantial evidence this provided, Nusbaum concluded that wind-borne spores were responsible for primary infection.

The difference between what Nusbaum and his colleagues were doing in the 1940s and what we are doing 50 years later is adding prognosis, precision, speed, and availability to the forecasts. The added precision is the result of two improvements: 1) use of meteorological models to actually track spore movement; and 2) a large network of collaborators who report disease outbreaks. Speed of production and availability of forecasts is accomplished through the Internet. However, adequate validation of the system is still needed.

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EVALUATION OF MILIONCAST, A FORECASTER FOR ONION DOWNY MILDEW, WITH HISTORICAL DATA

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1. INTRODUCTION

Downy mildew, which is caused by *Peronospora destructor* (Cook, 1932), is a very common disease of bulb and salad onions. Yield losses of up to 75% in bulb onions have been recorded (Develash and Sugha, 1997). Fungicides to control downy mildew are often applied prophylactically, as frequently as every 10 days. However, such frequent use is very costly to growers, increases environmental risks and pollution, and increases the selection pressure for fungicide resistant isolates within *P. destructor* populations. Furthermore, the number of active compounds that are allowed has been reduced as a result of changes in fungicide regulations. Thus, forecasters are needed to use fungicides more efficiently and effectively by timing the applications better to when actual disease development occurs.

DOWNCAST (Jesperson and Sutton, 1987) was the first forecaster for onion downy mildew, and is based on findings of the conditions required for sporulation and infection in Canada by Hildebrand and Sutton (1982; 1984a; 1984b; 1984c). DOWNCAST was very effective in reducing fungicide applications by 40% without increasing disease severity in comparison with standard practices in field trials in New Zealand (Wright *et al.*, 2002; Chynoweth *et al.*, this volume). Later, ONIMIL (Battilani *et al.*, 1996) and de Visser's DOWNCAST (de Visser, 1998) were developed, which were variations of the original DOWNCAST. However, the sporulation model within these DOWNCAST-based models was found inaccurate under Dutch and UK climatic conditions (de Visser, 1998; Gilles *et al.*, 2004). Therefore, a new forecaster was developed, which was based on sporulation and infection studies with UK isolates of *P. destructor*, and this was named MILIONCAST, an acronym for 'MILdew on onION foreCAST' (Anonymous, 2002; Gilles *et al.*, 2004). In this study, the

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accuracy of MILIONCAST is evaluated with historical data for onion downy mildew epidemics in the UK. A previously developed concept, that observed peaks in increase in disease development within a disease progress curve are caused by infection events followed by a latent period (Gilles *et al.*, 2001), is used in this study to derive a measurement of the accuracy of MILIONCAST from historical disease progress data.

2. MATERIALS AND METHODS

2.1 Historical data

From 1996 to 2001, downy mildew severity was observed in small observation plots of onion 'Armstrong' at Wellesbourne in the UK. The observation plots were 10 m × 10 m in size with five beds of four rows per bed at 35/30/35 cm row spacing. Onion seed was drilled at 34 seeds per m row. The observation plots were exposed to *P. destructor* inoculum to ensure infection by placing infected plants at a distance from the plots. Fifty plants were marked in each plot and percentages of leaf area infected on these plants were assessed at intervals ranging from 3 to 14 days. Weather data were measured at 30 min intervals by automated weather stations with temperature, relative humidity and wetness sensors within each plot. The sensors were located between beds at a 15 cm height above the ground.

2.2 MILIONCAST

MILIONCAST consists of three models: (i) a model to predict sporulation, (ii) a model to predict infection and (iii) a model to predict latent periods. The sporulation model was developed in previous work by Gilles *et al.* (2004) and predicts sporulation quantitatively in relation to temperature and relative humidity during the dark period at night. Sporulation was predicted with 95% confidence when values of greater than 4.15 were calculated for sporulation by the model. In this study, sporulation was considered to have occurred when a value greater than 4.15 was predicted. Eighty-one percent of predictions of sporulation by this model were correct in a trial over 120 days at Wellesbourne, UK.

The infection model is based on data from an infection study under controlled environmental conditions, in which infection of onion leaves by *P. destructor* sporangia was studied in relation to temperature and wetness duration (Anonymous, 2002). A quadratic relationship was found between minimum leaf wetness duration (w_{min}; in h) and temperature (T; in °C) ($P <$ 0.001):

$$w_{\min} = 20.3 - 1.8T + 0.068T^2$$

The minimum wetness period relates to the rate of infection at each temperature. Infection (i) is predicted with input of the 30 min temperature and wetness data when:

$$\sum_{t=0}^{t=i} \frac{1}{2w_{\min}} = 1$$

This model has not been independently tested for its accuracy to predict downy mildew infections in onion crops under outdoor conditions.

The latent period model is based on previous experiments, where *P. destructor* infected onions were kept under constant temperature conditions (Anonymous, 2002). Latent periods were 9 to 11 days between 15 to 23°C and increased with temperature decreasing below 15°C. The inverse latent period (1/L; in day^{-1}) was described well by a gamma distribution function of temperature (T) ($P < 0.001$) (Figure 1):

$$\frac{1}{L} = 3.5 \frac{4.9^{-5.7} T^{4.7} e^{\frac{-T}{4.9}}}{\Gamma(5.7)}$$

The end of the latent period (l), and thus the onset of the infectious period, is predicted with input 30-min temperature data when:

$$\sum_{t=0}^{t=l} \tfrac{1}{48} \frac{1}{L} = 1$$

Predicted and observed latent periods were compared for onion plants infected by *P. destructor*, which were put outdoors on 21 different dates between April and September of 1999. Predicted latent periods on onion 'Armstrong' plants varied, on average, ± 1.9 days when compared to observed latent periods, and were also affected by when good conditions for sporulation occurred after plants became infectious.

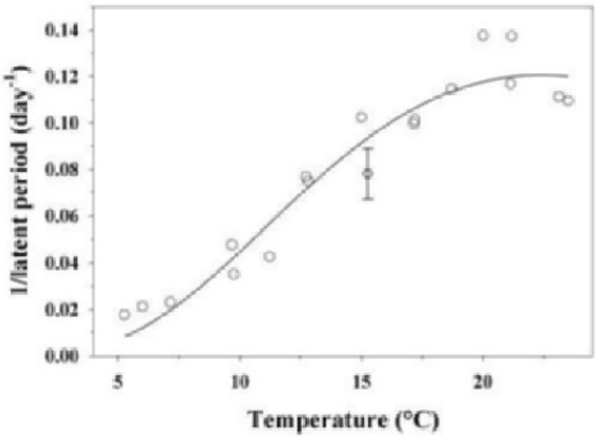

Figure 1. Inverse latent period in relation to temperature. Fit of a model (line) to observed data (○). The error bar on one of the data points is the standard error of observations.

2.3 Evaluation of MILIONCAST accuracy on historical data

Temperature, relative humidity and wetness data, which were recorded at 30 min intervals, were input into MILIONCAST. When sporulation was predicted and followed within 24 h by a prediction of infection, then infection was assumed to occur, and was called a sporulation/infection event (s/i event). Latent periods were calculated when an s/i event was predicted. Latent period calculations were started from the onset of wetness that led to the predicted infection ($t = 0$).

A random model was also used to predict disease development. The random predictions were used to investigate if MILIONCAST was more accurate in identifying when disease development occurs than by random prediction. Random predictions were made for each day after a calculated latent period from the beginning of the weather data. This is the earliest possible day for which MILIONCAST could have made a prediction. Random numbers between 0 and 1 were generated for each day within the computer program Microsoft® Excel. Disease increase was predicted when the random number was below 0.106, which is the average frequency at which s/i events were predicted by MILIONCAST over the six seasons. Random predictions were made 50 times for each season.

Peaks in disease increase were identified from the data for observed disease severity. A peak in increase in disease severity was assumed to have occurred when the increase in disease severity over a time interval between two subsequent assessments was greater than those recorded during the previous and subsequent time intervals. The concept is that such peaks of increase in disease severity are the result of infection events, which resulted in an increase in visible downy mildew symptoms after a period of latency (Gilles *et al.*, 2001).

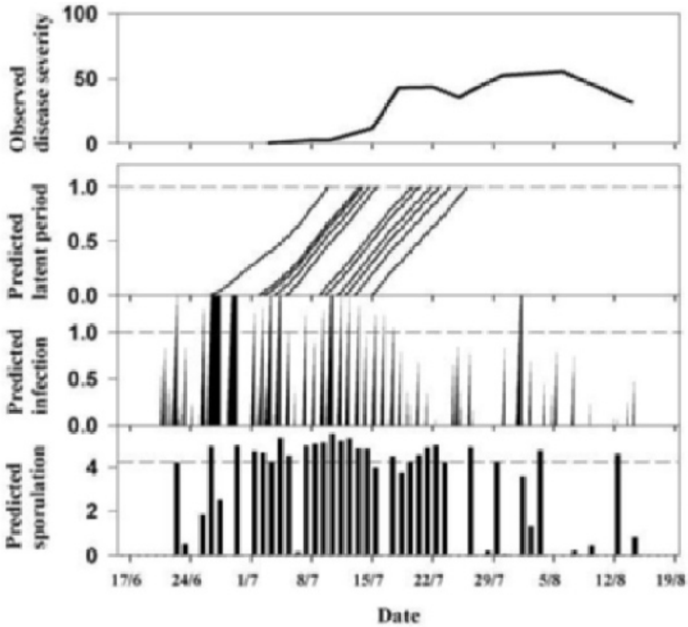

Figure 2. The MILIONCAST forecast output of predicted sporulation and infection events, which are followed by predicted latent periods, in comparison with observed disease severity data for the 1997 season at Wellesbourne.

With this information the accuracy of prediction by MILIONCAST and the random model were tested according to two criteria. Criterion 1: the time of the prediction of s/i events followed by a predicted latent period was within an observed peak in increase in disease severity. Criterion 2: an observed peak in increase in disease severity was predicted by at least one s/i event followed by a latent period. The accuracy of prediction of multiple s/i events on consecutive days (≥ 2 consecutive s/i events) was also evaluated, because such multiple s/i events appeared to be associated with disease increase in high risk seasons (Chynoweth *et al.*, this volume). Each multiple event was regarded as one event to conduct the evaluation for multiple events. The frequencies, at which the two criteria were fulfilled, were calculated for each season.

The frequencies at which criteria 1 and 2 were met by 50 random predictions for each season were compared to the average frequency at which these criteria were met by MILIONCAST over six seasons. A Student's t-test was done to test whether the frequency of success for either criterion 1 or 2 obtained by MILIONCAST was higher than those obtained by random prediction.

3. RESULTS

Predictions of s/i events followed by predictions of the length of the latent period frequently corresponded with observed disease increases in all six seasons. For example, in 1997 conditions were favourable for sporulation and infection in early to mid July on a succession of days (Figure 2). These predictions of s/i events followed by predicted latent periods corresponded with observed disease increases in mid to late July.

On average, 60.5% of predicted s/i events were associated with observed peaks in disease increase (Table 1). This was significantly higher than that achieved by random prediction (56%; $P < 0.001$). However, MILIONCAST was not better than a random model in predicting the periods of risk ($P = 0.21$), and predicted 66.7% of the periods when peaks in disease increase occurred (the random model predicted 65.4% of these periods). A higher percentage of predictions were associated with peaks in disease increase, when predicted multiple s/i events were considered. On average, 66.7% of these were associated with peaks in disease increase, but they predicted only 27.8% of these periods. The prediction of single s/i events was especially important in 1998, 1999 and 2001, where multiple s/i events were not predicted and only single s/i events were associated with peaks in disease increase.

4. DISCUSSION

On average, 60.5% of predictions by MILIONCAST were associated with observed peaks in increase in disease severity, and this was significantly higher than by random prediction (56%). However, although MILIONCAST was more accurate than the random model, the accuracy of prediction appeared not to be very high. Downy mildew symptoms are often difficult to recognise. Frequently, onion leaves, in particular of young onion plants, which appeared healthy, were found to bear sporangiophores under

Table 1. Frequency of successful predictions by MILIONCAST according to criterion 1 and criterion 2 .

Model	**Season**	**Criterion 1**[1]	**Criterion 2**[2]
MILIONCAST	1996	11/16[3]	4/5
MILIONCAST (≥ 2 events)		3/3	3/5
MILIONCAST	1997	7/11	3/3
MILIONCAST (≥ 2 events)		1/2	1/3
MILIONCAST	1998	1/4	1/3
MILIONCAST (≥ 2 events)		0/0	0/3
MILIONCAST	1999	2/2	2/4
MILIONCAST (≥ 2 events)		0/0	0/4
MILIONCAST	2000	4/9	1/2
MILIONCAST (≥ 2 events)		2/4	1/2
MILIONCAST	2001	1/1	1/1
MILIONCAST (≥ 2 events)		0/0	0/1
Average frequency over 6 seasons			
MILIONCAST		60.5%**[4]	66.7%
MILIONCAST (≥ 2 events)		66.7%**[4]	27.8%**[5]
Random model		56.0%	65.4%
s.e. of random prediction		1.40%	1.62%

[1] Criterion 1: a prediction of an s/i event corresponds with an observed peak of increase in disease severity following a predicted latent period.
[2] Criterion 2: a time period when a peak of increase in disease severity was observed is correctly predicted.
[3] Frequency at which a criterion is met, written as: 'number of correct predictions' / 'total number of predictions'.
[4] Frequency is greater than that obtained by random prediction ($P < 0.001$).
[5] Frequency is less than that obtained by random prediction ($P < 0.001$).

close examination without any apparent symptoms such as lesions or clearly visible sporulation. Furthermore, disease assessments were made at rather large intervals (3 to 14 days). These factors could have resulted in a very low accuracy of identifying when peaks of increase in disease severity had occurred and thus an insufficient level of discrimination by the test used in this study. The accuracy of MILIONCAST may thus be higher than that found in this study.

The accuracy of prediction by MILIONCAST could also have been reduced by the compounded error of the three models, which are combined within this forecaster. MILIONCAST predictions are based on the combined prediction by three models: a sporulation model, an infection model and a latent period model. Tests of the sporulation model on outdoor onion plants suggested that the model predicted sporulation events with 81% accuracy (Gilles *et al.*, 2004). The error of the latent period model was, on average, ±1.9 days on infected onion 'Armstrong' plants exposed to field conditions. This error in prediction of latent period may even be greater for other onion cultivars, because they are mainly outcrossing populations of plants. The combined error of these models is likely to be greater than the error of each individual model. For example, the error in prediction of latent period could have led to correctly predicted s/i events not being registered as associated with a peak in increase in disease severity. The increase in forecast error by combining different models could also apply to other onion downy mildew forecasters, such as DOWNCAST (Jesperson and Sutton, 1987), de Visser's DOWNCAST (de Visser, 1998), ONIMIL (Battilani *et al.*, 1996) and ZWIPERO (Friedrich *et al.*, 2003). Thus, although predictions by MILIONCAST described the observed disease progress better than by random prediction over six seasons, the accuracy of the model was found to be relatively low. This could have been caused by either an insufficient level of discrimination by the test used in this study or a large error of prediction, because of the combined inaccuracies of three different models within the forecaster.

Even though field trials in New Zealand, in which the application of DOWNCAST was tested, have shown that onion downy mildew was controlled well by applying fungicides after the prediction of at least three successive s/i events (Wright *et al.*, 2002), this study has shown that, although the probability of a multiple s/i event to be associated with increase in disease severity is higher, single s/i events were clearly associated with disease increase in certain seasons. For example, in 1998, 1999 and 2001, significant increases in disease severity were associated with single s/i events, which were predicted by MILIONCAST. Thus, if MILIONCAST is used in a fungicide spray program, then sprays may need to be considered even after prediction of a single s/i event. It is important to note that s/i events do not occur uniformly over time. Frequently, the weather conditions can be very similar over a series of days resulting in multiple s/i events. Clearly, future field trials are required to determine the best criteria for application of fungicides and to assess whether downy mildew can be controlled effectively by timing fungicides according to MILIONCAST predictions. Also, field trials are required to compare the accuracy of MILIONCAST to other onion downy mildew forecasters.

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USE OF DISEASE FORECASTING MODELS FOR CONTROL OF ONION DOWNY MILDEW IN NEW ZEALAND

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1. INTRODUCTION

Onion downy mildew caused by (*Peronospora destructor* (Berk.) Casp. Ex Berk) can be responsible for large production losses in some years in onion-growing areas in the North Island of New Zealand. Losses arise when damage to foliage results in small size and poor storage quality of onion bulbs (Chupp and Sherf, 1960). The severity of disease is dependent on weather conditions and locality, with severity increasing in wet seasons (Fullerton *et al.,* 1986; Wright, 1992).

Management of onion downy mildew in New Zealand involves applications of protective fungicides at 7-10 day intervals during the growing season, with additional applications of curative (systemic) fungicides when disease risk is perceived to be high (Wright, 1992). Protective fungicides, such as mancozeb, kill sporangia of *P. destructor* on contact (Gunn, 1991), but are only effective if applied before infection occurs (Viranyi, 1981). Curative fungicides, such as metalaxyl, penetrate plant surfaces, killing mycelium of the pathogen within plant tissues (Gunn, 1991). In an effort to avoid development of resistance, New Zealand onion growers are restricted to three applications of curative fungicide per season (voluntary agrochemical guidelines).

The benefits of using disease-risk forecasting models for timing fungicide applications include potential reductions in production costs and in other adverse effects of agrochemical use, such as environmental concerns and negative public perceptions. Forecasting models can be used

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to time curative fungicides after infection conditions have been monitored. They are less suitable for timing protective fungicides, because these need to be applied prior to infection and predictions of site-specific weather variables are seldom sufficiently accurate.

2. ONION DOWNY MILDEW MODELS

The relationship between weather and onion downy mildew development is well documented (Mukerji, 1975; Howard *et al.,* 1994; Schwartz, 1995). Several models for predicting sporulation and infection by this disease are available, mostly based on criteria originally determined by Hildebrand and Sutton (1982, 1984a,b,c). The environmental conditions required for sporulation and infection from the original DOWNCAST model of Jesperson and Sutton (1987) are outlined in Table 1.

Table 1. Environmental conditions required for sporulation and infection of onions by *P. destructor* (from Jesperson and Sutton, 1987).

	Environmental conditions
Sporulation	Temperature during previous day/night < 24°C; and no rain between 0200-0600 hrs; and relative humidity >95% (uninterrupted) between 0200-0600 hrs
Infection	Wetness during 0700-1000 hrs following sporulation; or wetness during 1900-2100 hrs following sporulation

Individual models involve minor variations on these criteria, differing according to local climatic conditions, different instrumentation, or just different interpretations by different researchers.

A comparison of the accuracy of sporulation predictions of various models (Tijs Gilles, pers. comm. 2003) showed that while most models are reasonably accurate (70-80%, when predictions of no sporulation are included), they differ markedly in the number of sporulation events correctly predicted. Out of 50 nights when sporulation was observed on onion plants, the number of correctly predicted sporulation nights ranged from 6 to 37. This highlights the need for improved models, and the care needed when interpreting model outputs. What was common to all models was that the number of predicted events exceeded the number that can be accommodated within resistance management guidelines for curative fungicides.

3. USE OF DOWNCAST IN NEW ZEALAND

Using a modified DOWNCAST model (Whiteman and Beresford, 1998), weather data for several seasons from the Pukekohe onion growing area of New Zealand were compared for numbers and timing of sporulation-infection events. Requirements for sporulation and infection in this study were based on modified DOWNCAST criteria (Table 2).

Table 2. Onion downy mildew sporulation and infection criteria used in New Zealand (Whiteman and Beresford, 1998).

	Environmental conditions
Sporulation	Temperature during previous day/night < 24°C; and <0.2mm rain between 0100-0500 hrs; and relative humidity >95% (uninterrupted) between 0100-0500 hrs
Infection	leaf wetness during 0500-0800 hrs following sporulation; or leaf wetness for three hours between 1900-2400 hrs the evening following sporulation; or leaf wetness for three hours between 1900-2400 hrs the second evening following sporulation

In order to develop a practical response (number of curative fungicide applications) to the sporulation-infection events predicted by the model, the patterns of occurrence of events were assessed over several seasons (Table 3).

Table 3. Frequency of sporulation-infection events over 5-day periods at Pukekohe, New Zealand

Number of events per running 5-day period	Season						
	95-96	96-97	97-98	98-99	00-01	01-02	02-03
1	47	34	46	37	59	38	51
2	43	34	35	4	17	42	29
3	18	24	6	0	7	25	12
4	9	16	4	0	2	11	5
5	1	3	1	0	1	6	1
Number of events each season	47	45	20	9	28	57	36

In seasons with the greatest total number of events, and thus the greatest disease risk, events tended to occur on consecutive days. In seasons with high disease risk (total events as indicated by the model), that is 1995-96, 1996-97, and 2001-02 seasons, approximately half of the sporulation events occurred in groups of 2 or 3 in 5-day periods (Table 3). By grouping sporulation-infection events (e.g. three events in a 5-day period), numbers of spray applications that met the resistance management guidelines could be achieved.

4. MODEL TESTING

In the 2000-01 and 2001-02 seasons, field trials were conducted to investigate the effects of different fungicide timing strategies for controlling downy mildew (Wright *et al.*, 2002). Plots sprayed with mancozeb either weekly (MANC 7) or fortnightly (MANC 14), or weekly from disease onset (M7-ONSET), were supplemented with metalaxyl (Ridomil® Gold MZ WG) applied every 28 days (RID-28), after an infection-sporulation "alert" (three events in 5 days, RID-ALERT), or after alerts following disease onset in the crop (RID-ALERT/ONSET), giving a total of 12 treatments, replicated four times. Three metalaxyl applications were made per treatment. Three months after harvest, onion bulbs from each plot were graded and weighed.

The first season of the trial (2000-01) was a low-risk year for downy mildew (Table 3), and consequently no differences in mean bulb weight were found among the treatment plots (the exception being the untreated control plots having lower bulb weights and higher leaf damage) (Wright *et al.*, 2002). The second season was more conducive to downy mildew, with statistically significant differences on bulb yield being found for mancozeb and metalaxyl strategies (Figure 1).

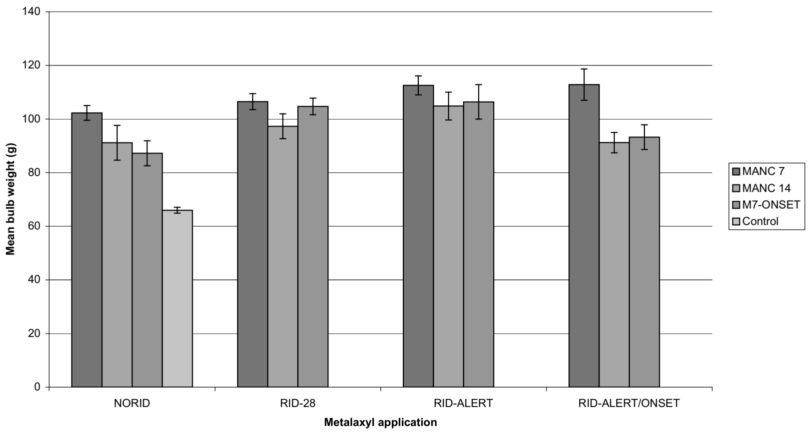

Figure 1. Effect of fungicide treatments on mean bulb weight (g) in 2001/02 season. Bars indicate the standard error for each treatment.

Weekly applications of mancozeb provided good control of downy mildew. All metalaxyl treatments enhanced disease control compared with mancozeb only. Factorial analysis of regimes showed a significant (p=0.005) increase in mean bulb weights between RID-ALERT regimes and those receiving no metalaxyl (NORID). Weekly mancozeb regime yields were significantly (p=0.001) higher than alternative mancozeb treatments (Wright *et al,.* 2002). Combining the three metalaxyl applications with fortnightly mancozeb sprays resulted in a saving of eight spray applications (40%) compared to the weekly mancozeb regime, without reducing mean bulb weight (Wright *et al.,* 2002).

In the 2002-03 season, trials were undertaken to investigate the efficacy of timing curative fungicides applied after different groupings of sporulation-infection events (two in 5-days, three in 5-days, five in 8-days, and seven in 11-days). Unfortunately this season was also a low risk one (Table 3) and little disease developed. Only two treatments (mancozeb applied weekly, and the two in 5 day curative regime) had significantly different results from the unsprayed control plots. The two in 5 day regime also received more spray applications (8) than recommended under current resistance management guidelines.

5. DISCUSSION

The use of a modified DOWNCAST model, combined with a threshold of three events in a 5-day period, provided suitable protection of crop loss due to onion downy mildew in a season of high disease risk, with a 40% reduction in the number of spray applications.

Epidemic severity can vary between years that have equal numbers of sporulation-infection events (Jesperson and Sutton 1987), depending on when in the season those events occur. Early season infections may result in a greater number of disease cycles than late season infections. By restricting curative spray applications to times when groups of sporulation-infection events occur, curative sprays are applied at times of greatest benefit. It is possible that in very high-risk years the allowed number of curative sprays may be exhausted early in the season, leaving the crop vulnerable towards the end of season. However, those late-season risk periods are more likely to be contained by the continuing protective fungicide programme, and are less likely to result in severe crop loss, because earlier sprays would have reduced inoculum.

The DOWNCAST model, as used by Wright *et al.* (2002), was not able to reduce the number of spray applications in the low risk 2000-01 season (Wright *et al.,* 2002). Improved models may achieve reductions in numbers of sprays through more accurate thresholds, either as groups of events, or as a disease severity index, as used for BOTCAST (Sutton *et al*., 1986). Another alternative is to use risk criteria with weather forecasts to predict periods of no sporulation and to delay application of protective fungicides until infection is likely (Beresford *et al.,* 1989). This is dependent on the interval between infection alerts being long relative to the standard spray programme, otherwise there is no scope to use a model to reduce numbers of fungicide applications.

6. ACKNOWLEDGEMENTS

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OVERWINTERING OF ROSE DOWNY MILDEW (*PERONOSPORA SPARSA*)

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1. INTRODUCTION

Whilst of sporadic occurrence in the UK, downy mildew caused by *Peronospora sparsa* Berk. is often an economically destructive disease of cultivated roses. Downy mildew epidemics can often be difficult to detect in their early stages as spore production, generally on the under surfaces of leaves, is sparse under most environmental conditions (Wheeler, 1981), and can occur on green tissues without any other obvious indication of infection (Xu and Pettitt, 2003). The first clear signs of disease are usually widespread defoliation of affected plants and the occurrence of typical angular-edged purplish-red to dark brown lesions on badly affected leaves.

The distribution of epidemics, the recurrence of disease in 'hot spots' in successive years and the tendency for severe symptoms to appear suddenly under favourable conditions all support the hypothesis that the pathogen overwinters, possibly as dormant mycelium, in cuttings and plants. However, the role of oospores in overwintering and initiating infections is less certain. The ability of downy mildews to form systemic infections in a wide range of herbaceous plants is well known (Spencer, 1981). This is especially important in woody perennials, which are vegetatively propagated from clonal stock. On raspberry, the downy mildew fungus (*Peronospora rubi*) readily infects leaves, flowers, developing fruits and stems. However, the spread of this pathogen into the vascular tissues of veins in raspberry was very limited. The fungus was confined to the cortex, extending only a few centimetres beneath infected nodes, whilst oospores were not seen in this tissue (Williamson *et al.*, 1995). Mycelium of *P. sparsa* may survive the winter in the cortex of rose stems (Wheeler, 1981; Misko and Postnikova, 1989), and systemic infections have been demonstrated in rose rootstocks cv. Manetti (*Rosa* x *noisettiana* Thory) in

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California, USA. Ellis *et al.* (1991) reported that *P. sparsa* can overwinter as root infections from severely affected shoots in blackberry, however, Harwood (1995) states that such symptomatic shoots have not been seen on UK roses. In recent years rose downy mildew has sometimes been observed to start on freshly planted 'bare-rooted' and apparently healthy plants. This is apparently a relatively new phenomenon in the UK and it is not known whether such manifestations of disease are due to systemic infections (either mycelium or oospores) or infections derived from infected plant debris containing oospores. In this chapter we present the results of investigations carried out under UK conditions on the possible modes of over-wintering of *P. sparsa* on floribunda and hybrid tea, bush roses.

2. MATERIALS AND METHODS

2.1 Mycelial growth *in planta*

Systemic infection was investigated by determining whether mycelia on leaves could invade vascular tissues and thus invade stems via leaf petioles. Detached leaves were inoculated *in vitro*. These infected leaves were then cleared, softened and stained at 1, 2, 3, 5, 7 and 14 days after inoculation to observe the mycelial development within the leaf. These prepared specimens were viewed using light and fluorescence microscopy to study the fungal structures with particular reference to the likelihood/frequencies of the fungus invading the vascular tissues after artificial inoculation of leaves, thus establishing possible systemic infections. Similar observations were also made on naturally-infected leaves sampled from both field and glasshouse production systems. For comparison, similar specimens were prepared from uninoculated plants of the same cultivars. Petiole segments were also collected from field-infected plants, cleared and assessed by light microscopy for the presence of *P. sparsa* structures.

2.2 Oospore production

Infected leaves either from artificial inoculation or field-infected plants (including leaf debris) were collected at various times to observe downy mildew structures. These infected leaves were cleared, softened and stained to observe the mycelial development within the leaf. Fungal structures inside the mesophyll were observed using scanning electron microscopy (SEM), light and fluorescence microscopy, with particular reference to the frequency and distribution of oospores.

Infected leaves from the field in various stages of decay and often with secondary infection (*Botrytis* spp.) were used to extract *P. sparsa* oospores. The most successful extraction procedure consisted of comminution of leaf tissues for 1-2 minutes in water in a blender followed by treatment with cellulase (40 mg cellulase in 10 ml 0.1M sodium phosphate buffer pH 5) at 30°C overnight. Following enzyme treatment, the suspension was comminuted a second time for 30 seconds and wet sieved. The fraction collected on the 20 μm sieve was re-suspended in distilled water. Oospore extracts were prepared from samples of infected leaves collected from five separate disease outbreaks (Table 1). Extracted spores were tested for viability using a version of the thiazolyl blue (MTT) procedure as used by Gunn (1999). A sample spore suspension was suspended in 1% aqueous thiazolyl blue and the colour of spore staining was observed microscopically after 24 h at room temperature. Germination tests were carried out on all extracts on 2% agar plates maintained at 18°C with a 10 h photoperiod. Assessments of infectivity of the five oospore extracts were performed by inoculating freshly-weaned *ex* tissue culture plants of the highly susceptible var. Remembrance. Plants were weaned in Levington compost in sealed propagators for 5 weeks before inoculation. Inoculation was by application of 2 ml of oospore extract per plant in an aerosol spray using a DeVilbis sprayer, giving approximately 50 oospores per plant on 10 replicate plants per extract. Plants were maintained in high humidity in sealed propagators in a 15°C constant temperature growth room for a further 2 month observation period during which the incidence of downy mildew symptoms was recorded.

2.3 Disease carry-over

Plants known to have been infected during the previous season's epidemics were collected from several nurseries and placed in controlled environment (CE) cabinets to monitor the subsequent occurrence of downy mildew. Newly diseased plants were also collected from the field, when still showing symptoms, and grown in polythene tunnels over winter and spring as well as being maintained in CE cabinets under misting conditions. Healthy plants were inoculated in CE cabinets and then maintained in the cabinets for several months. Disease development on these plants was monitored regularly. The position of diseased leaves was recorded in order to determine whether or not, following abscission or removal of old infected leaves, new leaves growing from the same nodes developed downy mildew symptoms.

3. RESULTS

3.1 Mycelial growth

Large quantities (>150) of cleared leaves and petiole sections, from various sources, were assessed and the invasion of vascular tissue by the fungus was not observed. Indeed, very little fungal mycelia were observed inside infected leaf tissues. These observations suggest that systemic infection was not as common for *P. sparsa* as previously thought under UK conditions.

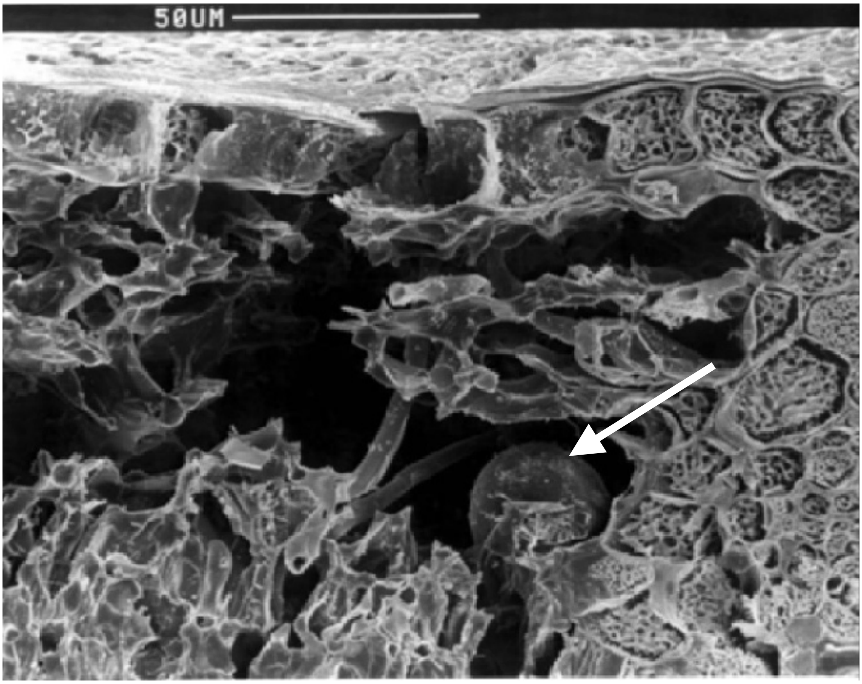

Figure 1. A scanning EM image of oogonium (arrow) of *P. sparsa* in infected rose leaf tissues.

3.2 Oospores

Oogonia were frequently observed in cleared diseased leaf tissues taken from field infected plants. In artificially inoculated plants, oogonia were often observed inside the mesophyll soon after infection, and their formation was recorded as early as one week after inoculation of detached leaves *in vitro*. Figure 1 is an example SEM image of an oogonium.

Oospores were frequently found in large numbers (>30 cm^{-2}) in infected leaves. The formation of oospores showed great variation from leaf to leaf, even on the same plant. Oogonia were normally observed within necrotic lesions and rarely seen in the 'green' parts of affected leaves. In contrast, conidiophores were largely present at the margins of lesions or in the green areas of leaves. Under optimum conditions, sporulation on the green parts of infected leaves tended to be sparser than on the necrotic areas, which was often a precursor to the development of necrosis.

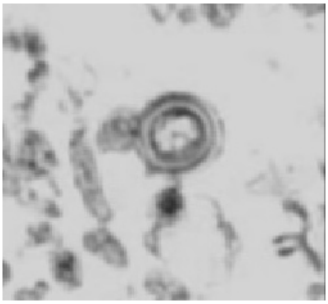

Figure 2. Image of an oospore of *P. sparsa* extracted from infected rose leaves and viewed by light microscopy.

Oospores were extracted successfully from infected leaves of varying ages and conditions, ranging from fresh lesions on still-attached leaves to totally dried, brown fallen leaves that often bore heavy secondary infections of *Botrytis* spp. as well as many other saprotrophs. Figure 2 shows an example image of an extracted oospore. The percentage viability of these spores as assessed by the MTT procedure varied between 32 and 80% (Table 1). Twenty samples of each extract were assessed microscopically and oospores were classed as not taking up stain (possibly dormant), staining dark brown or black (dead) and staining rose/pink (viable). Each sample was also checked for the presence of other structures, and a small number of conidia and conidiophores were observed, although none of these had taken up the MTT stain and they all appear to contain no cytoplasm. From the five oospore extracts assessed, viability appeared to decrease in the older, more decomposed and secondarily colonised leaf material, with fewer spores staining rose/pink and more staining black (samples 4 and 5, Table 1). Germination tests with the extracted oospores on water agar plates were unsuccessful. However, inoculations of freshly-weaned *ex* tissue culture rose plants resulted in infections in 2 out of 10 plants with one out of the five oospore extracts tested. This extract (sample

1, Table 1) was from the cleanest leaf tissue sample assessed, collected from artificially inoculated plants from growth cabinet experiments at East Malling and tested one year after collection.

Table 1. Summary results of the thiazolyl blue (MTT) germination test on oospores of *P. sparsa* extracted from infected rose leaves, collected from different sources.

Sample	Source	Secondary infection	Condition of leaves	Percentage of oospores stained		
				Rose/pink (viable)	Dark brown/black (dead)	No stain (dormant)
1	East Malling	No	Dried	80	8	12
2	Efford	No	Fresh	74	26	0
3	Kent	No	Frozen	62	22	16
4	Hertfordshire	Yes	Dried	32	45	23
5	Shropshire	Yes	Dried	45	31	24

3.3 Disease carry-over

When plants that showed severe mildew symptoms in the previous season were transferred, following the onset of dormancy and the removal of all leaf material, to either CE cabinets or polyethylene tunnels, downy mildew symptoms were consistently not observed in the new leaves developing in the new season. This was despite provision of frequent long wetting periods and ideal conditions for disease development. Similar plants where the leaves were not removed readily developed disease when mixed with plants with sporulating lesions under the same conditions. The majority of this disease appeared to arise from spread from leaf to leaf *via* conidia. Marking the positions of infected leaves in the first season showed that following the removal of all leaves, new leaves arising from the same nodes as previously infected leaves, did not develop any downy mildew symptoms. These results strongly indicate that mycelia did not spread from infected leaves to stems in the first season to act as an inoculum source for the following season.

Two important observations were made in these detailed studies with artificial inoculation in CE cabinets. Firstly, following inoculation the majority of the leaves that abscised and dropped showed no leaf spotting or necrosis, although often already bearing conidiophores. Secondly and in contrast, the majority of the leaves that did show symptoms remained attached to the affected plants.

4. DISCUSSION

It has been speculated widely that rose downy mildew overwinters as mycelium on wood or as systemic infections in stems, nodes and buds. This has been inferred from other downy mildew diseases and from the way downy mildew epidemics in rose appear to occur. Systemic infections do appear to be important in rose rootstocks in the downy mildew epidemics seen in California (Aegerter *et al.*, 2002). However, speculation on the importance of systemic infection in UK bush roses is not supported by the results of the present study. This is a very important consideration when developing local disease control strategies.

Frequent monitoring of disease on previously infected plants clearly indicated that if all the infected leaves were removed from diseased plants, new leaves arising from nodes previously bearing infected leaves did not develop new downy mildew symptoms. Histological observations also showed that the fungus failed to invade the vascular tissues in leaves and thus establish systemic infection. Similar observations were also recorded on downy mildew in *Rubus* (Williamson *et al.*, 1995). Indeed, limited amounts of fungal structures, other than oospores, were observed in infected leaves. Furthermore, experience of the rose micro-propagation industry with stock material indicates that if systemic infections do occur, they are rare. Oospores were frequently observed in infected leaves very soon after infection as also observed in *Rubus* (Williamson *et al.*, 1995). These results suggest that systemic infection is unlikely for rose downy mildew and that oospores are likely to be the primary overwintering inoculum. Of course, the present results do not preclude the possibility of the fungus infecting stems directly, which may act as a source of potential primary inoculum. However, such symptoms only occur in the most severe of disease outbreaks. Thus, it is important to dispose of infected leaf litter as one measure to reduce the level of primary inoculum. We have showed the potential of oospores as overwintering inoculum. However further research is needed to confirm this and to further understand several aspects related to oospore overwintering. These may include the effect of drying on oospore formation in dropped green leaves, the viability and pathogenicity of oospores in relation to environmental conditions, and potential biological control treatment of infected leaves to destroy oospores or prevent oospore formation.

5. ACKNOWLEDGEMENTS

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INFLUENCE OF ENVIRONMENTAL FACTORS ON THE DISEASE CYCLE OF WHITE RUST, CAUSED BY *ALBUGO CANDIDA*

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1. INTRODUCTION

White rust, caused by *Albugo candida* (Pers. ex Hook) O. Kuntze, is an important disease of a wide variety of *Brassica* crops. *A. candida* infects not only agronomically important crops but also can be found on various cruciferous weeds (Meekes *et al.,* this volume). *A. candida* is closely related to the downy mildews. *Albugo* is the only genus within the family Albuginaceae which, together with the Peronosporaceae, belongs to the order Peronosporales (Dick, 2002).

White rust has become a major disease of different Brassicaceae and in particular members of the *B. oleracea,* including Brussels' sprouts (*B. oleracea* var. *gemmifera*), cauliflower (*B. oleracea* var. *botrytis*) and broccoli (*B. oleracea* var. *italica*). In the Netherlands in 1997, these crops were grown on more than 7000 ha (Anonymous, 1998). Infection causes disfiguring white pustules on leaves, buttons and flowering stems leading to yield losses. Given the economic importance and the limited means to effectively control white rust, there is a need to understand better the abiotic and biotic factors that govern the development of this disease. Understanding the epidemiology of *A. candida* will contribute to the development of disease forecasting systems and provide necessary information to screen for resistance in host plants.

So far, most epidemiological studies have focused on white rust of *Brassica rapa* (syn. *B. campestris)* (Liu and Rimmer, 1990; Verma *et al.*, 1983) and mustard, *B. juncea* (Goyal *et al.*, 1996). Generally, disease

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development is favored by moist conditions and temperatures between 10-25°C (Howard *et al.*, 1994), and chilling appears to be required to initiate the release of zoospores from sporangia (Butler and Jones, 1961; Howard *et al.*, 1994). In particular, for white rust of Brussels' sprouts the temperature range for zoospore release was from 0-1°C to 25°C with the optimum around 10°C, but experimental data were not presented (Vanparijs, 1989). Similarly it was reported that at an optimum temperature range of 16-24°C only 3-4 h of leaf wetness was required for infection of cultivars of *B. oleracea* (Humpherson-Jones, 1991). Studies of white rust epidemiology on *B. rapa* indicated that disease development was most rapid at 21°C and that 14 days after infection more than 90% infection was achieved at all temperatures between 12 and 24°C (Verma *et al.*, 1983). At 3°C and at 29°C and above no infection occurred. The number of pustules on the adaxial surface of inoculated detached leaves was highest at 15°C decreasing at lower and higher temperatures (Verma *et al.*, 1983). These studies are largely descriptive and report the effect of temperature on a single component of the disease cycle for one host.

This study provides a more detailed analysis of the influence of temperature, leaf wetness period and inoculum density on zoospore release, latent period, production of sporangia and infection percentage. These epidemiological factors were studied using isolates of *A. candida* collected from different locations and different host plants.

2. MATERIALS AND METHODS

2.1 Host plants and fungal isolates

The cultivars listed in this study were Content (Brussels' sprouts), Fremont (cauliflowers) and Lord (broccoli). The *A. candida* isolates used in the experiments were collected from the field at locations listed in Table 1. Harvested sporangia were maintained at –20°C.

2.2 Influence of temperature on zoospore release

Seven-day-old seedlings of Brussels' sprouts, grown for 7 days at 20°C in steam-sterilized potting soil, were inoculated by placing a 10 µl droplet of a suspension of isolate FRA 000 on the upper surface of the cotyledons. The zoospore suspension was prepared by mixing frozen (-20°C) sporangia in demineralized water (Williams, 1985). The suspension was incubated for 3 hours at 15°C in the dark to release zoospores. After inoculation, the seedlings were placed at 100% R.H. for 24 hours (16 hours in the dark) at 20°C, and then at 20°C, 70% R.H. with a 16/8 hour light/dark cycle at low light intensity (± 1000 lux). Ten days after inoculation, newly formed sporangia were collected with a cyclone spore collector (Mehta and Zadoks, 1971) by rupturing the white blisters. The collected sporangia were suspended in demineralized water and adjusted to a final density of 5.0×10^4 sporangia/ml. Four ml of this suspension was added to 25 ml Erlenmeyer flasks and one flask was placed at each of 0, 5, 10, 15, 20 and 25°C in the dark. The number of empty and zoospore-containing sporangia were determined after 1, 2, 3, 4, 5, 6 and 24 hours using a haemocytometer to examine approximately 100 sporangia counted at each assessment. The experiment was conducted four times.

Table 1. Origin of isolates of *A. candida* used in the experiments (all isolates from the Netherlands).

Isolate	Source plant	Collection year	Location
FRA 000	Brussels' sprouts	1996	Friesland
ZHG 100	Brussels' sprouts	1996	Zuid-Holland
FLB 000	Brussels' sprouts	1996	Flevoland
NHB 200	Cauliflower	1997	Noord-Holland
NHB 300	Broccoli	1997	Noord-Holland

2.3 Influence of temperature on latent period and production of sporangia

Seedlings of Brussels' sprouts were grown for seven days at 20°C in steam-sterilized potting soil in 11x11x12 cm pots, each pot containing four seedlings. Eight pots were placed in a tray and six were inoculated with a suspension of *A. candida* isolate FRA000 as described above at a density of 4.0×10^4 sporangia/ml. The two remaining pots were inoculated with demineralized water only and served as control. The trays were placed at 5, 10, 15, 20 and 25°C, one tray for each temperature. The number of cotyledons showing white blisters were determined daily. Latent period was defined as the time period between inoculation and 50% of the cotyledons sporulating. The leaf area of the cotyledons was determined with an electronic leaf area meter (LI-3100, Li-cor, Inc. Lincoln, Nebraska, USA; calibrated for small surfaces). As soon as the first blisters started to erupt, the cotyledons were suspended in demineralized water and blisters were ruptured with a disecting needle. The numbers of sporangia released were counted with a haemocytometer. The experiment was repeated with isolates ZHG 100 and FLB 000 which were collected from different locations in the Netherlands (Table 1).

A similar experiment was conducted with true leaves instead of cotelydons. Leaves of three-week-old plants, with three unfolded leaves, were sprayed with a suspension of isolate FRA 000 using a DeVilbiss atomizer until run-off. The inoculum density was $1.0x10^4$ sporangia/ml; the suspension was incubated at 15°C for 3 hours to release zoospores. Plants were placed at five temperatures: 5, 10, 15, 20 and 25°C. One tray was placed at each temperature. From 5 days after inoculation, the number of infected plants and the number of pustules per plant were determined. Latent period was defined as the time between inoculation and 50% of the plants sporulating. The same experiment was conducted on cauliflower and broccoli with their respective isolates (Table 1).

2.4 Influence of leaf wetness period

Seeds of Brussels' sprouts were sown in 11x11x12 cm pots to give 9 seedlings (i.e. 18 cotyledons) and after seven days seedlings were inoculated with isolate FRA 000. In contrast to the method described above, inoculation was conducted immediately with the sporangial suspension, without allowing zoospores to be released during an incubation period. The inoculum density was $5.0x10^4$ sporangia/ml. Trays with seven treated pots and one untreated pot were placed in the dark at 5, 10, 15, 20 and 25°C and covered with a non-transparent lid to maintain 100% R.H.. Differences in leaf wetness period were obtained by removing the inoculum droplet with a Kleenex tissue after 0, 1, 2, 3, 4, 5 and 6 hours respectively. The non-transparent lid remained on the tray and was only removed 6 hours after inoculation when the last droplet was removed. One pot with nine seedlings was used for each leaf wetness period and one pot per tray served as a control inoculated with demineralized water. At each temperature two replicate trays were used and the experiment was conducted twice. The percentage of sporulating cotyledons was monitored daily at each temperature and used to determine the leaf wetness period that is necessary to obtain successful infection.

2.5 Influence of inoculum density

Different inoculum densities of isolate FRA000 were applied to cotyledons of seven-day-old seedlings of Brussels' sprouts. The densities were 1.0×10^2, 1.0×10^3, 1.0×10^4 and 1.0×10^5 sporangia/ml replicated twice in each of two trays. Each cotyledon received a droplet of 10 µl of the inoculum. The inoculum suspension was incubated for 3 hours at 15°C to release zoospores. The experiment was conducted at 5, 10, 15, 20 and 25°C. Observations started 5 days after inoculation and the percentage of sporulating cotyledons was determined per pot of nine seedlings (i.e. 18 cotyledons) for each temperature/density combination. The experiment was conducted twice.

2.6 Statistical analysis

Data were analysed with the Statistical Analysis System, version 6.12 (SAS Institute Inc., Cary, NC, USA). Data were tested for normality and homogeneity of variance and transformed when necessary. Regression analysis was performed with data on latent period, zoospore release and leaf wetness. When intercepts were not significantly different from zero (T-test, alpha=0.05), regression was forced through the origin. Data on production of pustules on true leaves and inoculum densities were analyzed by analysis of variance (ANOVA) and the means were compared by Tukey's test (alpha=0.05). Data that did not meet the conditions required for ANOVA were analysed with a non-parametric test (Kruskal-Wallis) followed by the Student-Newman-Keuls multiple range test.

3. RESULTS

3.1 Influence of temperature on release of zoospores

At 15 and 20°C, zoospores were released after 1 hour, whereas at 5°C it took 3 hours. Zoospore release in time followed a sigmoid curve with maximum release approached at 6 hours. No zoospore release occurred at 0 and 25°C. At 25°C there was physical deterioration and discolouration of sporangia after several hours of incubation. Despite this change in appearance, 25% and 7% of the sporangia that were incubated at respectively 0°C and 25°C for 24 hours still released zoospore when transferred to 15°C. At all temperatures tested, the percentage of sporangia releasing zoospores did not change between 6 and 24 hours of incubation.The relationship between temperature (T) and % zoospore release (Z) after 6 hours of incubation was given by the quadratic equation $Z = -0.32T^2 + 7.94T$ (r^2=0.94), with an optimum at 13.2°C (Figure 1).

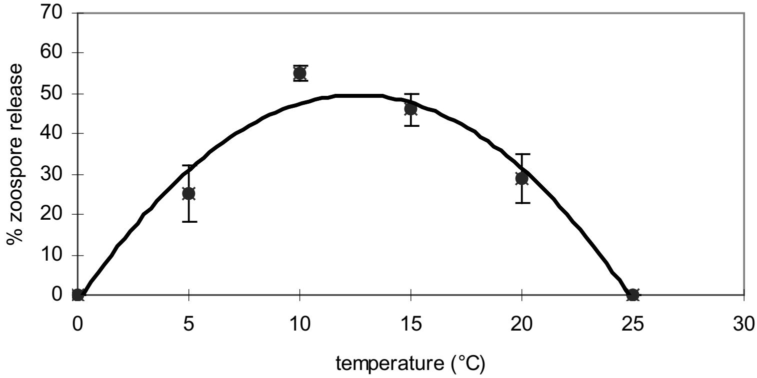

Figure 1. Percentage release of zoospores from sporangia of *A. candida* at different temperatures. The vertical bar is the standard error.

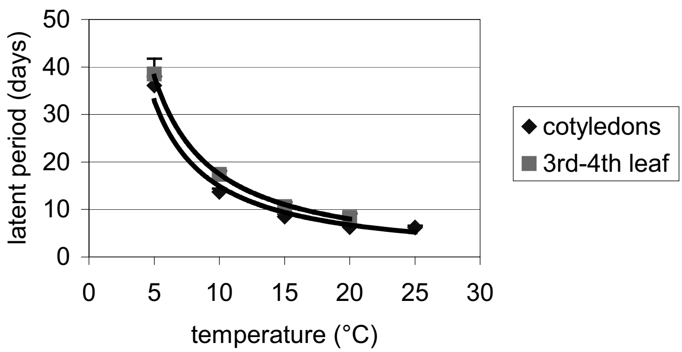

Figure 2. Latent periods (days) of white rust at different temperatures on cotyledons and true leaves of Brussels' sprouts. The vertical bar is the standard error.

3.2 Influence of temperature on latent period and production of sporangia

The relationship between latent period and temperature was identical for cotyledons and true leaves of Brussels' sprouts (Figure 2). For cotyledons, the latent period (LP) ranged between 37 days at 5°C and about

6 days at 25°C. The latent period on true leaves ranged from 39 days at 5°C to about 8 days at 20°C. No sporulation occurred on true leaves of plants grown at 25°C.

The relation between temperature and latent period was best described by the equation: $L = aT^b$ (Figure 2). Regression analysis of the linearized form $\ln(L) = \ln a + b \ln(T)$ was highly significant (r^2=0.94) with an estimated slope (b) of –0.82. Very similar relationships were obtained for other *A. candida* isolates of Brussels' sprouts and for other hosts on the different plant parts (Table 2).

Table 2. Estimated slope of the linearised relationship between latent period and temperature of different isolates* of *A. candida* on cotyledons and true leaves of different host plants.

Isolate	Surface	coefficient (b)
FRA000	cotyledons[1]	-0.82
FLB000	cotyledons[1]	-0.76
ZHG100	cotyledons[1]	-0.75
FRA000	true leaves[1]	-0.83
NHB200	true leaves[2]	-0.82
NHB300	true leaves[3]	-0.82

* 1) Brussels' sprouts, 2) cauliflower, 3) broccoli

Table 3. Production of sporangia and pustules of *A. candida* isolate FRA 000 on cotyledons and on true leaves of Brussels' sprouts at different temperatures (°C).

Temperature	Sporangia/cm^2 cotyledon x 10^5	Pustules/plant
5	1.26a*	15.5a**
10	1.45ab	9.2a
15	2.42b	14.8a
20	2.49b	20.3a
25	2.00b	-

Data in columns indicated with a different letter are significantly different according to

*) Student-Newman-Keuls multiple range test (α=0.05)

**) Tukey's test (α=0.05)

-) not determined

The production of sporangia on cotyledons ranged from 1.26×10^5 /cm² at 5°C to 2.49×10^5 /cm² at 20°C (Table 3). The effect of temperature over this range on the number of pustules/plant was not significant (α = 0.05). However, the number of newly formed sporangia on cotyledons of Brussels' sprouts grown at 5°C differed signicantly from the number produced on plants grown at 15, 20 and 25°C (α=0.01). The number of newly formed sporangia did not differ between FRA000, FLB000 and ZHG100, the three isolates collected from Brussels' sprouts plants grown at different locations in the Netherlands.

3.3 Influence of leaf wetness period

A leaf wetness period of only 1.8 hours was sufficient to obtain 10% of the cotyledons infected at 25°C (Figure 3). At temperatures ranging from 15-20°C, 2.3 to 2.8 hours of leaf wetness was sufficient to obtain 50% and 90% infection of cotyledons respectively. No cotyledons with white blisters were found at 5°C and with leaf wetness periods extending up to 6 hours.

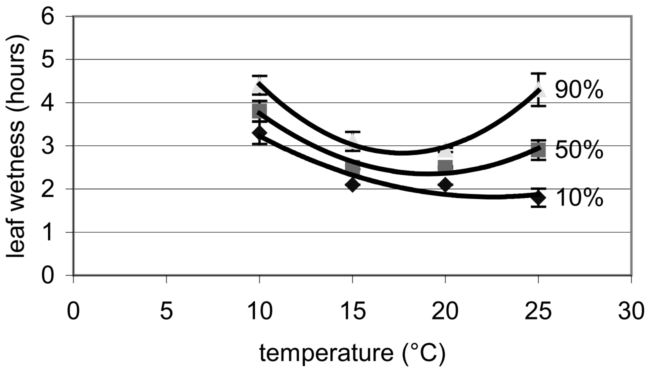

Figure 3. Leaf wetness periods (hours) resulting in 10, 50 and 90% infection of cotyledons of Brussels' sprouts at different temperatures. The vertical bar is the standard error.

The relation between temperature (T) and leaf wetness period (W) for the different infection percentages was well described with the quadratic equations (P<0.01):

10% infection: $W=0.009T^2-0.402T+6.32$ ($r^2=0.73$)
50% infection: $W=0.02T^2-0.64T+8.47$ ($r^2=0.74$)
90% infection: $W=0.03T^2-0.94T+11.16$ ($r^2=0.71$)

3.4 Influence of inoculum density

Percentage infection varied from 0.8% at low inoculum densities to approximately 90% at 10-20°C and high inoculum densities (Table 4).

Table 4. Influence of temperature (°C) and inoculum density on % infection of cotyledons of Brussels' sprouts inoculated with an isolate of *A. candida*.

	Inoculum density (sporangia/ml)			
Temperature	1 x 10^2	1 x 10^3	1 x 10^4	1 x 10^5
5	2.3a*	19.6a	54.8a	68.6a
10	17.3b	41.1b	93.5b	90.3b
15	19.1b	30.9b	88.6b	83.0b
20	18.6b	41.0b	93.9b	90.3b
25	0.8a	0a	55.4a	55.9a
Mean**	11.6a	26.6a	77.2b	77.6b

*) Data in columns indicated with a different letter are significantly different according to Tukey's test (α=0.05)
**) Data in row indicated with a different letter are significantly different according to Tukey's test (α=0.05)

Infection percentages significantly depended on both temperature and inoculum density (α=0.05). For all inoculum densities, % infection at 10, 15 and 20°C differed significantly from 5 and 25°C. Inoculation with densities of $1x10^4$ and $1x10^5$ sporangia/ml gave significantly higher % infection than the lower densities. The interaction between temperature and inoculum density was not statistically significant.

4. DISCUSSION

From these experiments it is clear that *Albugo candida* can develop as a pathogen within the temperature range of about 5-25°C. Only a short period of leaf wetness of 2-3 hours at 15-25°C is necessary to establish high levels of infection, similar to the 3-4 hours at 16-24°C mentioned by Humpherson-Jones (1991). White rust pustules containing sporangia develop within the above temperature range without the requirement for free water or a high relative humidity. All these characteristics make white rust a disease that can develop readily within European climates, where it has been more prevalent in the last decade and has caused considerable damage in several crops.

The results confirm to a large extent the results presented in earlier publications on specific components of the disease cycle (Howard *et al.*, 1994; Vanparijs, 1989). Vanparijs (1989) noted zoospore release between 0 and 25°C with an optimum around 10°C. In our experiments the optimum temperature for release of zoospores was about 13°C, but zoospore release did not occur at 0 and 25°C. At 25°C sporangia began to deteriorate physically. Development of the disease to sporulating pustules occurred within the range 5 and 25°C with a short latent period of 8 – 12 days between 10 and 20°C as described by Howard *et al.* (1994). Moist conditions are important during the first hours for infection, but are not necessary for subsequent sporulation. Our results are also similar to those obtained with *A. candida* isolates on *Brassica rapa* and *B. juncea* (Goyal *et al.*, 1996; Verma *et al.*, 1983); although from the second paper the temperature range for disease development seems to be shifted upwards to 9 to 27°C, and latency periods are 1 to 4 days shorter depending on temperature.

In many studies very few isolates of *A. candida* have been tested. In our study no significant differences were found between isolates from the same host plant from different geographical locations, or between isolates from different host plants, in the disease cycle on cotyledons and true leaves. Also, results with *A. candida* on shepherds' purse, *Capsella bursa-pastoris*, and radish, *Raphanus sativus*, confirm that white rust develops under similar conditions over a wide range of isolates and host plants (unpublished data).

Within and between experiments, the highest variation in sporulation was found when true leaves of Brussels' sprouts, cauliflower and broccoli were inoculated. To some extent, this variation might be explained by the presence or absence of a wax layer on the leaves which is not present on cotyledons. Under sub-optimal conditions a wax layer may develop and influence the capability of *A. candida* to penetrate the leaf. Also inoculum droplets may easily roll from the leaves before infection by zoospores can occur. Differences between cultivars in this physical characteristic could be one factor involved in quantitative resistance in the crop.

In contrast to studies reported with other Brassica crops (Verma and Bhownik, 1988), oospores do not seem to play a role in white rust epidemiology in *Brassica oleracea*. So far, no oospores have been found.

The year-round presence of crops belonging to *B. oleracea* in western Europe make asexual survival of *A. candida* possible and do not necessitate oospore formation as a means of surviving crop-free periods. The lack of genetic recombination in *A. candida* will also contribute to the limited differences found between isolates in our study.

This paper presents in detail conditions that influence the monocyclic processes of *A. candida* isolates from *B. oleracea*. The results can be used in the development of resistance assays and forecasting models to achieve better management of the disease in countries such as the Netherlands, where it is of increasing importance.

5. ACKNOWLEDGEMENTS

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HOST SPECIALISATION OF THE OOMYCETE *ALBUGO CANDIDA*

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1. INTRODUCTION

Albugo candida (Pers. ex Hook) Kuntze is a biotrophic oomycete pathogen of a wide variety of cruciferous crops. Recent phylogenetic analyses of ribosomal DNA sequences of a range of oomycetes have placed *A. candida* along with other *Albugo* species in the most basal taxon of the Peronosporomycetidae (Riethmuller *et al*., 2002). Riethmuller *et al*. (2002) suggested that the 'highly derived morphology' of the zoosporangiophores, the obligate parasitic behaviour on aerial parts of dicotyledonous plants and the adaptation to dry habitats indicate that the *Albugo* clade is highly evolved.

The disease caused by *A. candida* is often referred to as white rust or white blister due to the production of white sori formed on various parts of the host plant. The white sori, blisters or pustules, are the result of sub-epidermal growth, proliferation and expansion of zoosporangiophores leading to ruptures in the epidermis of the host tissue. Affected plant parts include roots, leaves, stems, inflorescence, fruits, and seeds. Typical symptoms include white blisters (Figure 1) and so-called stagheads (Goyal *et al*., 1996) that develop, in many cases, after systemic infection of flower heads. *A. candida* can cause substantial economic damage in several crops, including rapeseed (*Brassica napus* L.) (Petrie, 1988), turnip rape (*Brassica rapa*) (Petrie, 1994), and members of the *Brassica oleracea*, including Brussels sprouts (*Brassica oleracea* var. *gemmifera*), cauliflower (*Brassica oleracea* var. *botrytis*) and broccoli (*Brassica oleracea* var. *italica*). The latter three crops were grown in the Netherlands in 1997 on

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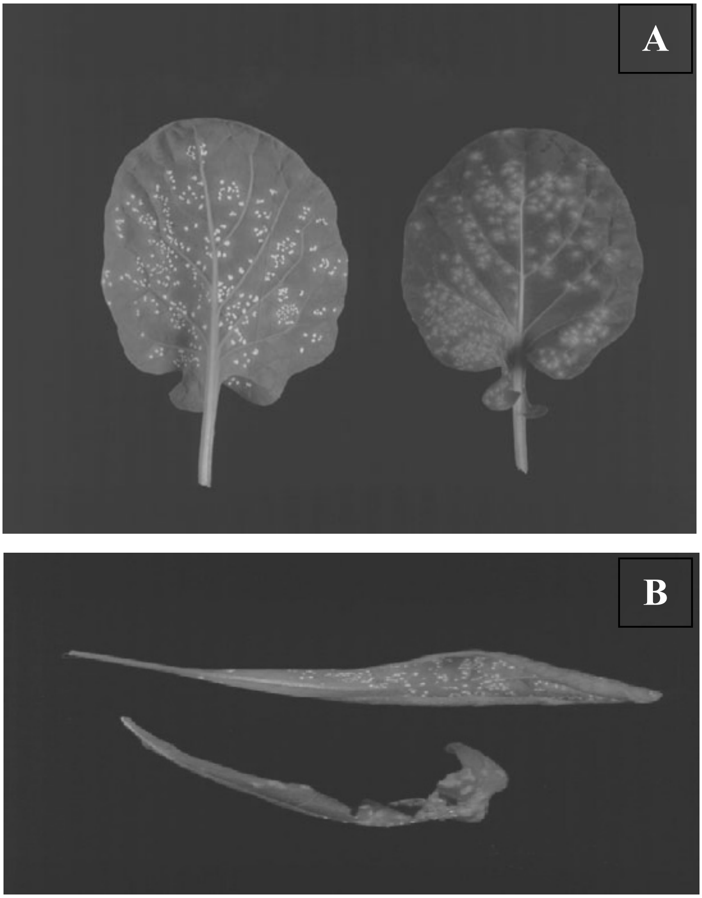

Figure 1. Typical white blisters or pustules caused by *Albugo candida* on (**A**) Brussels' sprouts (*B. oleracea* var. *gemmifera*) and (**B**) shepherd's-purse (*Capsella bursa-pastoris*).

more than 7000 ha (Anonymous, 1998) and infection with *A. candida* caused disfiguring white pustules on buttons and flowering stems leading to yield losses. *A. candida* infects not only agronomically important crops but also can be found on various wild cruciferous weeds (Jacobson *et al*., 1998).

Host specialisation has long been recognized in *A. candida*. Early studies have listed 241 species and 63 genera of crucifers that can be infected by *A. candida* (Biga, 1955 cited in Delwiche and Williams, 1977; Pound and Williams, 1963; Petrie, 1988; Rimmer *et al*., 2000). Host specialisation of *A. candida* has been inferred from studies in which isolates are collected from different hosts and subsequently cross-inoculated on a number of other host species and cultivars (Pound and Williams, 1963; Petrie, 1988). Pound and Williams (1963) already indicated that many authors have been hesitant to describe specialized races (pathotypes) of *A. candida* due to the fact that some of the results obtained in the cross-inoculation experiments did not reveal 'clear-cut positive or negative sporulation responses'. Given the economic importance and the limited means to effectively control white rust, there is a need to better understand host specialisation and the frequency and distribution of races of *A. candida*. Identification of specific races can lead to unravelling the molecular and biochemical basis of disease resistance in plants to *A. candida*, which is essential in plant breeding strategies (Borhan *et al.*, 2001; Adhikari *et al.*, 2003). Furthermore, knowledge of incompatible interactions between pathogen races and host plant species can be exploited to induce both local and systemic resistance in plants against compatible races of *A. candida* (Singh *et al.*, 1999; E.T.M. Meekes, C.F. Geerds and J.M.Raaijmakers, unpublished data).

Table 1. Races of *Albugo candida* and their corresponding host plant species (adapted from Pound and Williams, 1963).

A. candida 'race'	Primary host plant species
1	*Raphanus sativus*
2	*Brassica juncea*
3	*Armoracia rusticana*
4	*Capsella bursa-pastoris*
5	*Sisymbrium officinale*
6	*Rorippa islandica*

In their initial study, Pound and Williams (1963) described 6 races (Table 1). Since then, numerous other studies have described new host plants and races of *A. candida* (Verma *et al.*, 1975; Delwiche and Williams, 1977; Hill *et al.*, 1988; Verma *et al.*, 1999; Rimmer *et al.*, 2000). The purpose of this review is to provide a summary and detailed update of the host range and races of *A. candida* described to date.

2. RESULTS AND DISCUSSION

In addition to the *A. candida* races initially described by Pound and Williams (1963) at least 5 other races have been proposed, including race *Ac7* on *Brassica rapa* (syn. *Brassica campestris*) (Verma *et al.*, 1975), race *Ac8* on *Brassica nigra* (Delwiche and Williams, 1977), race *Ac9* on *Brassica oleracea* (Williams, 1985 cited in Hill *et al.*, 1988), race *Ac10* on *Sinapis arvensis* (Williams, 1985 cited in Hill *et al.*, 1988) and race *Ac11* on *Brassica carinata* (Williams, 1985 cited in Verma *et al*., 1999). Verma *et al*. (1999) proposed two additional races *Ac12* and *Ac13* on *Brassica juncea* and *Brassica rapa* subsp. *dichotoma*, respectively. *Ac12* and *Ac13* showed differential reactions on *B. carinata. Ac2* and *Ac12* showed differential reactions on *Brassica rapa* cv. Torch and *Brassica rapa* subsp. *trilocularis*. However, since for *Ac13* no comparison was made to *Ac7* and resistance among hosts and cultivars used in cross-inoculation experiments can occur, it is not fully clear yet whether races *Ac12* and *Ac13* are true new races or pathotypes instead. Although *A. candida* races isolated from Crucifereae are most virulent on those plant species from which they were originally isolated (referred to as 'homologous host'), they can infect certain genotypes of related *Brassica* species ('heterologous host') (Rimmer *et al*., 2000). This is illustrated in Table 2, in which the ability or inability of different *A. candida* races to cause symptoms on different host plants are listed. Whenever necessary, specific details on the outcome of cross-inoculation experiments will be given. The host plant species with their corresponding races of *A. candida* will be discussed in the alphabetical order given in Table 2.

• ***Arabidopsis***

Like many other Cruciferae, *Arabidopsis suecica* and *Arabidopsis thaliana* are susceptible to *A. candida*. However isolates collected from *Arabidopsis thaliana* did not cause disease in *Brassica* species nor in other wild crucifers, including *Capsella bursa-pastoris* and *Cardamine hirsuta*. Therefore, these *A. candida* isolates are most likely not race *Ac4*. The *A. candida* isolates obtained from *Arabidopsis thaliana* did, however, form tiny blisters on *Sinapis alba* (Holub *et al*., 1995). An *A. candida* isolate collected from *Cardamine pratensis* did not sporulate on wildtype *Arabidopsis thaliana* nor on the *eds1* mutant (Parker *et al*., 1996).

• ***Arabis***

A. candida isolates originating from *Arabis alpina* were only able to infect *Iberis amara*, *Lepidium sativum* and *Sinapis alba*. *A. candida* isolates from *Capsella bursa-pastoris* (*Ac4*), *Sisymbrium officinale* (Ac5), *Brassica rapa* (*Ac7*) or *Brassica oleracea* (*Ac9*) were not able to infect *Arabis alpina* (Napper, 1933).

• ***Brassica* species**

Brassica carinata is the homologous host of race *Ac 11* (Table 2, Williams, 1985 cited in Verma *et al*., 1999), but is also differentially susceptible to races *Ac1, Ac2, Ac7* and *Ac9*. Susceptibility varied considerably depending on the accession of *Brassica carinata* tested (Downey

Table 2. Latin and common names of host plant species of white rust (*Albugo candida* (Pers. ex Hook) Kuntze).

Host plant*	**Common name**	***A. candida* race****	**Reference**
Capparidaceae			
Capparis spp.	caper	-	Anonymous, 1960
Cruciferae			
Arabidopsis *suecica* (Fr.) Norrl.		not 2 or 4	Holub *et al*., 1995
thaliana L.	mouse-ear cress	not 2 or 4[1]	Holub *et al*., 1991; Holub *et al*., 1993; Holub *et al*., 1995[1]
Arabis spp.	rock cress	-	Anonymous, 1960
alpina L.	alpine rockcress	not 4, 5, 7 or 9[2]	Conners, 1967; Napper, 1933[2]
lyrata L.	lyrate rockcress	-	Jacobson *et al*., 1998
Armoracia rusticana Gaertn., Mey. & Scherb. = *A. lapathifolia* Gilib. Ex Usteri	horse radish	**3**[+] -	Anonymous, 1960; Conners, 1967; Pound and Williams, 1963[+]
Aurinia saxatilis (L.) Desv.	basket of gold, gold-dust, golden-tuft allysum	-	Burdiukova and Dudka, 1982
Barbarea			
stricta Andrz.	garden yellow rocket	-	Burdiukova and Dudka, 1982
vulgaris R.Br. = *B. arcuata* (Opiz ex Presl & Presl) Reichb.	yellow rocket	-	Anonymous, 1960; Burdiukova and Dudka, 1982
*Brassica****			
carinata A. Braun. (n = 17, genome bc) (*B. oleracea* x *B. nigra*)	Ethiopian mustard, Abyssinian cabbage/ mustard	**11** [+]; (1, 2, 7, 9)[3, 4, 5, 6]; not 2, 7, 8 or 10[4]	Downey and Rimmer, 1993[3]; Hill *et al*., 1988[4]; Petrie, 1988[5]; Williams, 1985[+] cited in Verma *et al*., 1999[6]

Table 2, continued.

Host plant*	**Common name**	***A. candida* race****	**Reference**
*Brassica****			
juncea (L.) Czern. (n = 18, genome ab) (*B. rapa* x *B. nigra*)	oriental (brown/leaf) mustard	**2** [+7], **2V** [+8], 8[9] (1, 7, 8, 9, 10).[8,9,10,11] not 1, 3, 4, 5, 6[7]	Anonymous, 1960; Conners, 1967; Downey and Rimmer, 1993[11]; Hill *et al*., 1988[9]; Lakra and Saharan, 1988; Nyvall, 1989; Petrie, 1988[10]; Petrie, 1994[+8]; Pound and Williams, 1963; Verma *et al*., 1975[+7]; Verma *et al*., 1999
napus L. (n = 19, genome ac) (*B. rapa* x *B. oleracae*)	(summer/winter) rape, oilseed rape	(2, 2V, 7, 7V, 9, 11)[12, 13, 14]	Anonymous, 1960; Conners, 1967; Dingley, 1969; Downey and Rimmer, 1993[12]; Liu and Rimmer, 1986; Napper, 1933[13]; Nyvall, 1989; Petrie, 1994[14]; Pidskalny and Rimmer, 1985[14]; Verma *et al*., 1975[12]
var. *napobrassica* (L.) Reichb. = *B. napo-brassica* DC. = *B. napobrassica* (L.) Mill.	swede, Swedish turnip, rutabaga	-	Dingley, 1969
nigra (L.) Koch (n = 8, genome b)	black mustard	(1, 2, 7)[15] **8** [+], (9)[15]	Anonymous, 1960; Burdiukova and Dudka, 1982; Delwiche and Williams, 1977[+]; Downey and Rimmer, 1993; Lakra and Saharan, 1988; Petrie, 1988[15]
oleracea L. (n = 9, genome c)		(7, 8)[16], **9** [+, 16], (11)[16]	Conners, 1967; Downey and Rimmer, 1993[16]; Parker *et al*., 1996; Pidskalny and Rimmer, 1985; Williams, 1985 cited in Hill *et al*., 1988[+].
var. *botrytis* L.	cauliflower	(2, 7)[17]	Anonymous, 1960; Conners, 1967; Dingley, 1969; Petrie, 1988[17]

Table 2, continued.

Host plant*	Common name	*A. candida* race**	Reference
*Brassica****			
oleracea var. *capitata* L.	(red, white) cabbage	(2, 7)[18]	Anonymous, 1960; Asselberg *et al*., 1996; Conners, 1967; Dingley, 1969; Petrie, 1988[18]
var. *gemmifera* Zenker	brussels sprouts	(2)[19]	Dingley, 1969; Petrie, 1988[19]
var. *gongylodes* L.	kohlrabi	(7)	Petrie, 1988
var. *ithalica* Plenck	broccoli	(1)	Petrie, 1988
var. *virides* L. = var. *acephala* DC. pro parte	kale, collard	-	Anonymous, 1960; Asselberg *et al*., 1996
rapa L. (n = 10, genome a)	wild turnip, turnip rape	(2)[20], **7**, **7V**[+], (8)[20]	Anonymous, 1960; Dingley, 1969; Downey and Rimmer, 1993; Napper, 1933[20]; Petrie, 1994[+]
subsp. *campestris* (L.) A.R. Clapham = *B. campestris* L.	wild turnip/bird rape	(2)[21], **7**[+]	Anonymous, 1960; Conners, 1967; Dingley, 1969; Nyvall, 1989; Petrie, 1988[21]; Pidskalny and Rimmer, 1985; Verma *et al*., 1975[+]
subsp. *chinensis* (L.) Hanelt = *B. chinensis* L.	chinese cabbage, pak-choi	not 2 [22]	Anonymous, 1960; Dar and Ghani, 1997; Lakra and Saharan, 1988[22]
subsp. *dichotoma* (Roxb.) Hanelt = *B. c.* var. *dichotoma* (Roxb.) G. Watt = *B. c.* var. *toria* Duthie & J.B. Fuller	brown sarson/toria	7	Lakra and Saharan, 1988; Petrie, 1988
subsp. *narinosa* (L.H. Bailey) Hanelt	tah tsai, Chinese savoy	not 2	Koike, 1996
subsp. *nipposinica* (L.H. Bailey) Hanelt = *B. c.* subsp. *nipposinica*	Japanese mustard	not 2	Koike, 1996

Table 2, continued.

Host plant*	**Common name**	***A. candida* race****	**Reference**
*Brassica****			
rapa subsp. *pekinensis* (Lour.) Hanelt = *B. pekinensis* (Lour.) Rupr. = *B. c.* var. *pekinensis*	pe-tsai, Chinese cabbage	not 1[23], not 2[23, 24]	Anonymous, 1960; Koike, 1996[24]; Lakra and Saharan, 1988[23]
subsp. *rapa* = subsp. *rapifera* Metzg. = *B. c.* subsp. *rapifera* (Metzg.) Sinskaya	turnip	(2), 7	Petrie, 1988
subsp. *trilocularis* (Roxb.) Hanelt = *B. c.* var. *sarson* Prain = *B. r.* subsp. *sarson* (Prain) Denford	yellow, brown sarson	(2)[25], 2[25], 7	Lakra and Saharan, 1988; Petrie, 1988[25]
tournifortii Gouan	African/Asian mustard, Mediterranean/wild turnip	not 1, 2, 7, 8 9, or 11	Lakra and Saharan, 1988
Camelina			
microcarpa Andrz. ex DC. = *C. sylvestris* Wallr.	small-seed/little pod, false flax	-	Anonymous, 1960; Burdiukova and Dudka, 1982; Conners, 1967; Petrie, 1994
sativa	(big-seed) false flax, gold-of-pleasure	-	Anonymous, 1960; Foller *et al*., 1998
Capsella bursa-pastoris (L.) Medik.	shepherd's-purse	**4** [+]	Anonymous, 1960; Conners, 1967; Dingley, 1969; Pound and Williams, 1963[+]
Cardamine spp.	bittercress, tooth wort	-	Anonymous, 1960
Cardamine			
diphylla (Michx.) Wood = *Dentaria diphylla* Michx.	crinkle/pepper root	-	Conners, 1967

Table 2, continued.

Host plant*	**Common name**	***A. candida* race****	**Reference**
Cardamine			
flexuosa With.	woodland bittercress	-	Burdiukova and Dudka, 1982
hirsuta L.	hairy bitter cress	-	Dingley, 1969
pratensis L.	cuckoo flower	-	Parker *et al*., 1996
Cochlearia officinalis L.	scurvy weed/grass	-	Anonymous, 1960
Coronopus			
didymus (L.) Smith	hog/lesser swine cress	-	Dingley, 1969
squamatus (Forssk.) Asch. = *Senebiera coronopus* (L.) Poir.	greater swine cress	-	Napper, 1933
Crambe abyssinica Hoch. (n = 45)	crambe	7	Petrie, 1988
Descurainia			
incana (Bern. ex Fish & Mey) Dorn = *D. richardsonii* Shulz = *Sisymbrium incanum* Bern. ex Fish & Mey	grey tansy mustard	-	Conners, 1967
incisa (Engelm.) Brit. = *Sisymbrium insicum* Engelm. ex Gray	cut-leaved tansy mustard	-	Anonymous, 1960
pinnata (Walt.) Brit.	tansy mustard	-	Anonymous, 1960
ssp. *brachycarpa* (Richardson) Detling = *Sisymbrium canescens* Nutt.	western / small pod / green tansy mustard	-	Petrie, 1994
sophia (L.) Webb = *Sisymbrium sophia* L.	flixweed	1, 2, 3, 4, 5 (6)[25]	Conners, 1967; Pound and Williams, 1963[25]
Draba spp. L.	whitlow-grass	-	Anonymous, 1960
nemorosa var. *leiocarpa* Lindblad	woodland draba	-	Petrie, 1994

Table 2, continued.

Host plant*	**Common name**	***A. candida* race****	**Reference**
Eruca sativa Mill.	arugula	not 1, 2, 3, 4, 5, 6 or 9[26, 27]; (2, 7)[28]	Pound and Williams, 1963[26]; Scheck and Koike, 1999[27]; Verma *et al*., 1999 [28]
Erysimum			
asperum (Nutt.) DC.	western wallflower	-	Anonymous, 1960
cheiranthoides L.	wormseed mustard	(1), 3, 4, not 2, 5 or 6	Pound and Williams, 1963
cheiri (L.) Crantz = *Cheiranthus cheiri* L.	(Aegean) wallflower	(4)[29]	Anonymous, 1960; Napper, 1933[29]
diffusum Ehrh.	grey wallflower	-	Burdiukova and Dudka, 1982
menziesii ssp. *eurekenze* (Hook.) Wettst.	menzies' wallflower	-[30]	Jacobson *et al*., 1998; Jacobson and Ojerio, 1996[30]
repandum L.	bushy wallflower, treacle mustard	-[31]	Burdiukova and Dudka, 1982; Jacobson and Ojerio, 1996[31]
Euclidium syriacum (L.) R.Br.	Syrian mustard	-	Burdiukova and Dudka, 1982
Hesperis matronalis L.	dame's rocket/violet	-	Anonymous, 1960
Iberis amara L.	rocket candytuft	4, 7[32]	Anonymous, 1960; Napper, 1933 [32]
Lepidium campestre (L.) R.Br.	basterd/field cress, field pepperweed, field peppergrass	-	Anonymous, 1960; Burdiukova and Dudka, 1982; Jacobson *et al*., 1998
densiflorum Schrad. = *L. apetalum* (auct. Non Willd.)	prairi peppergrass	-	Anonymous, 1960; Burdiukova and Dudka, 1982; Conners, 1967
Lepidium			
graminifolium L.	grassleaf pepperweed	-	Burdiukova and Dudka, 1982

Table 2, continued.

Host plant*	**Common name**	***A. candida* race****	**Reference**
Lepidium ? *oleraceum* Forst. f.	Cook's scurvy grass	-	Dingley, 1969
perfoliatum L	clasping pepperwort, yellow-flowered / clasping / round-leaved / peppergrass	-	Burdiukova and Dudka, 1982
ruderale L	peppergrass	-	Dingley, 1969
sativum L.	peppergrass, garden cress	4[33]	Anonymous, 1960; Conners, 1967; Dingley, 1969; Napper, 1933[33]
virginicum L.	poor man's pepper, peppergrass, Virginia pepperweed	-	Anonymous, 1960
Lobularia maritima (L.) Desv. = *Alyssum maritimum* (L.) Lam.	sweet alyssum	(4)	Napper, 1933
Malcolmia maritima (L.) R.Br.	virginia stock	-	Dingley, 1969
Myagrum perfoliatum L.	mite cress, muskweed	-	Burdiukova and Dudka, 1982
Nasturtium officinale R.Br. = *Rorippa nasturtium-aquaticum* (L.) Hayek	water cress	(3)[34], (4)[35]; not 1, 2, 4, 5 or 6[34]; not 7, 9 or 10[35]	Anonymous, 1960; Napper, 1933[35]; Pound and Williams, 1963[34]
Neslia paniculata (L.) Desv.	ball mustard	-	Conners, 1967
Raphanus raphanistrum L.	wild radish, jointed charlok	1[36]	Conners, 1967; Morris and Knox-Davies, 1980[36]

Table 2, continued.

Host plant*	**Common name**	***A. candida* race****	**Reference**
Raphanus			
sativus L. (n = 9, genome r)	radish	**1**[36, +37, 38, 39], not 2, 3, 4, 5, 6, 7, 8, 9 or 10[36, 37, 38, 39, 40]	Asselberg *et al*., 1996; Conners, 1967; Dingley, 1969; Downey and Rimmer, 1993; Lakra and Saharan, 1988[39]; Mukerji, 1975; Morris and Knox-Davies, 1980[36]; Napper, 1933[40]; Pidskalny and Rimmer, 1985[38]; Pound and Williams, 1963[+37]
var. *caudatus* (L.) L.H. Bailey = *R. caudatus* L.	rat-tail radish, mougri	1	Eberhardt, 1904 cited in Napper, 1933
Rapistrum rugosum (L.) All.		-	Burdiukova and Dudka, 1982
Rorippa *austriaca* (Crantz) Bess.	Australian yellow cress, Australian field cress	-	Burdiukova and Dudka, 1982
brachycarpa (Mey.) Hayek		-	Burdiukova and Dudka, 1982
islandica (Oeder) Borbas = *palustris* (L.) Bess.	water cress	(3, 4) **6**[+41]; not 1, 2 or 5[41]	Anonymous, 1960; Burdiukova and Dudka, 1982; Conners, 1967; Dingley, 1969; Pound and Williams, 1963[+41]
pyrenaica (Lam.) Reichb.		-	Burdiukova and Dudka, 1982
silvestris (L.) Bess.	creeping yellow cress	-	Burdiukova and Dudka, 1982
Sinapis			
alba L. (n =12, genome d)	white/yellow mustard	-	Burdiukova and Dudka, 1982
subsp. alba = *Brassica alba* (L.) Rabenh. = *B. hirta* Moench	white mustard	1, 2, 3, 4, (5, 6), 7, 9 and 10[42, 43]	Anonymous, 1960; Conners, 1967; Napper, 1933[42];Nyvall, 1989; Pound and Williams, 1963[43]; Verma *et al*., 1975
arvensis L.	charlock	**10**[+44]	Burdiukova and Dudka, 1982; Williams, 1985 cited in Hill *et al*., 1988[+44]

Table 2, continued.

Host plant*	Common name	*A. candida* race**	Reference
Sinapis			
arvensis L. subsp. *arvensis* = *B. kaber* (DC.) Wheeler			Anonymous, 1960; Conners, 1967
Sisymbrium altissimum L.	hedge/tumble mustard, tall rocket	(4)[45]	Anonymous, 1960; Conners, 1967; Pound and Williams, 1963[45]
irio L.	London rocket	-	Anonymous, 1960; Burdiukova and Dudka, 1982
officinale (L.) Scop.	(tall) hedgemustard	(1) 2, 3 (4) **5**$^{+}$ (6)[46]	Anonymous, 1960; Conners, 1967; Pound and Williams, 1963[+46];
volgense Bieb. ex Fourn.	Russian rocket	-	Burdiukova and Dudka, 1982
Syrenia montana (Pall.) Klokov		-	Burdiukova and Dudka, 1982
Thlaspi spp.		-	Anonymous, 1960
arvense L.	stinkweed, fanweed	(3, 4, 5)[47, 48]	Anonymous, 1960; Burdiukova and Dudka, 1982; Napper, 1933[47]; Pound and Williams, 1963[48]
Resedaceae:			
Reseda alba L.	white upright mignonette	-	Mukerji, 1975

* Species nomenclature according to: GRIN: USDA-ARS, National Genetic Resources Program, Germplasm Resources Information Network (http://www.ars-grin.gov/npgs/tax/index); database of the *Flora Europeae* of the Royal Botanic Garden of Edinburgh (http://rbg-web2.rbge.org.uk/FE/fe.html) and USDA Natural Resources & Conservation Service, plants database (http://plants.usda.gov).

** Bold numbers indicate primary host plants of *A. candida* races, numbers between brackets indicate disease incidence < 90%. Race 1 to 6 after Pound & Williams, 1963; race 7 after Verma *et al*., 1975; race 8 after Delwiche & Williams, 1977 - tentative classification; race 9 and 10 after William, 1985 cited in Hill *et al*., 1988 - tentative classifications; race 11 after Williams, 1985 cited in Verma *et al*., 1999 - tentative classification; - = no data.

*** Within cultivated species large differences exist between cultivars: reaction (resistant - susceptible) dependent on variety used.

and Rimmer, 1993). For example, race *Ac7* was able to infect 0 to 69% of plants from different accessions and the disease index (d.i.) ranged from 0 to 1.6 on a scale from 0 to 9. One *Brassica carinata* accession was also susceptible to races *Ac1* (37% incidence, d.i.=0.8) and *Ac2* (97% incidence, d.i.=5.8) and some other accessions were susceptible to either race *Ac1* (18% incidence, d.i.=0.7) or *Ac2* (10% incidence, d.i.=0.6) (Petrie, 1988). In experiments described by Hill *et al.* (1988), some *Brassica carinata* plants obtained from rapid cycling populations became infected with *A. candida* races *Ac1* or *Ac9,* but the disease index was relatively low (d.i.=1 on a scale from 0 to 9). None of the plants became infected with races *Ac2, Ac7, Ac8* or *Ac10* (Hill *et al*., 1988).

Brassica juncea is the homologous host of race *Ac2* as assigned by Pound and Williams (1963). They also reported that none of the plants of *Brassica juncea* showed symptoms when inoculated with *A. candida* isolates representing races *Ac1, Ac3, Ac4, Ac5* or *Ac6*. Other studies indicated that, dependent on the cultivar, *Brassica juncea* is to some extent susceptible to races *Ac1, Ac7, Ac9* and *Ac10* (d.i.=1, 5, 1, and 1, respectively, on a scale from 0 to 9). A rapid cycling population of *Brassica juncea* was very susceptible to *A. candida* race *Ac8* (d.i. 9) (Hill *et al*., 1988). Also Petrie (1988) found that 2 to 29% of the plants of five *Brassica juncea* cultivars (Success, Burgonde, Commercial Brown, Lethbridge 22A and Newton) showed symptoms when inoculated with race *Ac1*, but again the severity was very low (d.i.< 1). Some cultivars of *Brassica juncea* were susceptible to *A. candida* race *Ac1, Ac7* and *Ac8*, but most culitvars were resistant (Downey and Rimmer, 1993). *A. candida* race *Ac2* caused major problems in *Brassica juncea* until the release of resistant cultivars Domo, Cutlass (oriental mustard) and Scimitar (brown mustard). However a “new” race referred to as 2V infected 100% of the plants of these cultivars. Race *Ac2* is now subdivided in two pathotypes designated 2A and 2V (Petrie, 1994; Rimmer *et al*., 2000).

Only some genotypes of *Brassica napus* were susceptible to races *Ac7*, *Ac9* and *Ac11*, but most were resistant (Downey and Rimmer, 1993). *Brassica napus* was susceptible to several isolates collected from *Brassica oleracea* (race *Ac9*) (Napper, 1933). It was also susceptible to race *Ac7V*: for the six cultivars, 1 - 21% of the plants were infected. Four of these six cultivars were also susceptible to race *Ac2V* (disease incidence ranged from 2 to 19%). Nevertheless *Brassica napus* remains a good source of resistance against *A. candida* races *Ac2* and *Ac7* (Petrie, 1994).

Brassica nigra is the homologous host of race *Ac8* as assigned by (Delwiche and Williams, 1977). Races *Ac1*, *Ac2*, *Ac7* and *Ac9* were also able to infect some genotypes of *Brassica nigra* (Downey and Rimmer, 1993).

Brassica oleracea is the homologous host of race *Ac9* (Williams, 1985 cited in Hill *et al*., 1988). So far, all genotypes of *Brassica oleracea* tested were susceptible to *A. candida* race *Ac9* (Downey and Rimmer, 1993). Some genotypes of *Brassica oleracea* were, to a certain degree, susceptible to races *Ac1, Ac2, Ac7, Ac8* and *Ac11*. Dependent on variety and cultivar, susceptibility to races other than *Ac9* ranges from 0 to 85% of the plants infected. For example, 85% of the plants of *Brassica oleracea* var. *botrytis* cv. Super Snowball were susceptible to race *Ac7* (d.i.=2.2 on a scale from 0 to 9) and 81% of the plants of the same cultivar were susceptible to race *Ac2* (d.i.=1.7). Only 2% of the plants of *Brassica oleracea* var. *capitata* cv. Houston Evergreen were infected by race *Ac7*, and 5-11% of the plants of cv. Mammoth Red Rock were infected by races *Ac2* and *Ac7* (severity trace to 0.6); seven other cultivars were resistant. For *Brassica oleracea* var. *gemmifera* cv. Long Island Improved, 2% of the plants were susceptible to race *Ac2* (severity: trace) and 13% of the plants of *Brassica oleracea* var. *gongylodes* cv. Early Purple Vienna were susceptible to *Ac7* (severity: 0.28) (Petrie, 1988). For *Brassica oleracea* var. *ithalica* cv. Italian Green Sprouting, 6% of the plants were susceptible to race *Ac1* (severity: trace) (Petrie, 1988).

Brassica rapa (syn. *Brassica campestris*) is the homologous host of *A. candida* race *Ac7* (Verma *et al*., 1975). Most genotypes of *Brassica rapa* are resistant to race *Ac2*, but some are susceptible. Similarly, most genotypes of *Brassica rapa* are resistant to race *Ac8*, but some are susceptible. In *Brassica rapa,* resistance is governed by both major and minor resistance genes (Downey and Rimmer, 1993). *Brassica rapa* is subdivided into several subspecies and examples of infection levels change accordingly. For instance, 6 to 59% of plants of different cultivars of *Brassica rapa* subsp. c*ampestris* were susceptible to race *Ac2* (Petrie, 1988). However, *Brassica rapa* subsp. *chinensis* was not susceptible to several isolates of races Ac2 and Ac1, nor to *A. candida* originating from *Brassica tournifortii* (Lakra and Saharan, 1988). *A. candida* isolates from Japanese mustard (*Brassica rapa* subsp. *nipposinica*) and tah tsai (*Brassica rapa* subsp. *narisona*) were not able to cause symptoms on red mustard (*Brassica juncea* var *rugosa*). Similarly, *A. candida* from Chinese cabbage (*Brassica rapa* subsp. *pekinensis*) was not able to infect red mustard, but it did infect Japanese mustard and tah tsai (Koike, 1996). *Brassica rapa* subsp. *pekinensis* was not infected by *A. candida* race *Ac1* or several isolates of race *Ac2* (Lakra and Saharan, 1988). *Brassica rapa* subsp. *rapa* (cv. Snowball) was susceptible to *Ac2*, but only 6% of plants were infected (Petrie, 1988). Dependent on the accession of *Brassica rapa* subsp. *trilocularis*, 0 to 100% of the plants was infected by race *Ac2*, and especially accessions from Nepal were very susceptible to *Ac2* (Petrie, 1988). Similar to *Ac2*, resistance in *B. rapa* cvs Tobin, AC Parkland and

Reward, was overcome by a new pathotype designated as 7V. Race *Ac7* is now subdivided in two pathotypes designated 7A and 7V (Petrie, 1994; Rimmer *et al*., 2000).

A. candida from *Brassica tournifortii* was able to infect *Brassica rapa* subsp. *rapa*, but not other subspecies of *Brassica rapa*, nor *Brassica carinata*, *Brassica juncea*, *Brassca napus*, *Brassica nigra*, *Brassica oleracea*, *Eruca sativa* or *Raphanus sativus*. Nor were isolates of *Raphanus sativus*, *Brassica juncea* and *Brassica rapa* subsp. *chinensis* able to infect *Brassica tournifortii* (Lakra and Saharan, 1988). These results indicate that the isolate obtained from *Brassica tournifortii* is not likely to belong to races *Ac1* and *Ac2*, and it might also not belong to *Ac7, Ac8, Ac9* or *Ac11*.

• ***Coronopus***

A. candida originating from *Coronopus squamatus* was able to infect *Iberis amara*, *Lepidium sativum* and *Sinapis alba* (Napper, 1933).

• ***Crambe***

Crambe abyssinica was susceptible to *A. candida* race *Ac7*. Although 97% of the plants were infected, conspicuous purple rings were formed around some smaller blisters, which could indicate a resistance response (Petrie, 1988).

• ***Descurainia***

Descurainia sophia was highly susceptible to several races: disease incidence ranged from 90 to 100% for races *Ac1* (90%), *Ac2* (98%), *Ac3* (100%), *Ac4* (100%) and *Ac5* (96%). Only 3% of the plants became infected with race *Ac6* (Pound and Williams, 1963).

• ***Eruca***

Plants of *Eruca sativa* seem highly resistant, since none of the plants became infected with races *Ac1* to *Ac6* (Pound and Williams, 1963), although one isolate per race and one accession were tested. Verma *et al.* (1999) found low levels of infection by races *Ac2* and *Ac7*: 3% of the plants was infected by *Ac2* (d.i. = 1 on a scale from 0 to 4), and 13% with race *Ac7* (d.i.= 1). *A. candida* originating from *E. sativa* did not infect cauliflower (*Brassica oleracea* var. *botrytis*), Chinese cabage (*Brassica rapa* subsp. *pekinensis*), Japanese mustard (*Brassica rapa* subsp. *nipposinica*), red mustard (*Brassica juncea* var. *rugosa*) and tah tsai (*Brassica rapa* subsp. *narinosa*) (Scheck and Koike, 1999).

• ***Erysimum***

Erysimum cheiranthoides is differentially susceptible to several races of *A. candida*: 100% of the plants showed symptoms when inoculated with race *Ac3* or *Ac4*, but only 1% was susceptible to *Ac1*. No infection was found when inoculated with races *Ac2*, *Ac5* or *Ac6* (Pound and Williams, 1963), although only one accession was tested. *E. cheiri* showed symptoms when inoculated with *A. candida* originating from *Arabis alpina* and with some, but not all, isolates originating from *Capsella bursa-pastoris* (Napper, 1933). *Erysimum menziesii* ssp. *eurekense* showed susceptibility to *A. canida* isolates obtained from different hosts. Isolates originating from *Erysimum menziesii* subsp. *eurekense* were not pathogenic to *Erysimum repandum* (Jacobson and Ojerio, 1996).

• ***Iberis, Lepidium, Lobularia***

Iberis amara was infected by *A. candida* originating from *Arabis alpina*, *Brassica rapa*, *Capsella bursa-pastoris*, *Coronopus squamatus*, *Sinapis alba* subsp. *alba*, and *Sisymbrium officinale*. This indicates that *Iberis amara* is susceptible to races *Ac4, Ac5* and *Ac7*. Similar results were obtained for *Lepidium sativum*. Only one isolate of *A. candida* originating from *Capsella bursa-pastoris* (*Ac4*) was able to infect *Lobularia maritima* (Napper, 1933).

• ***Nasturtium***

Only 6% of the *Nasturtium officinale* plants were infected with race *Ac3* and they were not susceptible to races *Ac1, Ac2, Ac4, Ac5* or *Ac6* (Pound and Williams, 1963), although only one isolate per race and one accession were tested. Napper (1933) found that only one out of eleven isolates of *A. candida* originating from *Capsella bursa-pastoris* (*Ac4*) was able to infect *Nasturtium officinale*. *A. candida* isolates originating from *Arabis alpina* (3 isolates), *Brassica oleracea* (4 isolates, Ac9), *Brassica rapa* (Ac7), *Coronopus squamatus*, *D. tenuifolia* (3 isolates), *Sinapis arvensis* subsp. *arvensis* (*Ac10*), or *Sisymbrium officinale* (*Ac5*) were not able to cause symptoms on *Nasturtium officinale* (Napper, 1933).

• ***Raphanus***

Raphanus sativus is the homologous host of race *Ac1* (Pound and Williams 1963). *A. candida* from *Raphanus raphanistrum* infected *Raphanus sativus* cv. Saxa (47%) and cv. Long Scarlet (82%). Conversely, *A. candida* from *Raphanus sativus* (*Ac1*) infected 89% of the plants of *Raphanus raphanistrum*. *A. candida* from *Raphanus raphanistrum* or *Raphanus sativus* were not able to infect *Brassica oleracea* var. *botryris* cv. Snowball, *Brassica oleracea* var. *capitata* cv. Cape Spitz or *Brassica oleracea* var. *gemmifera* cv. Long Island (Morris and Knox-Davies, 1980). *A. candida* from *Raphanus sativus* was able to infect *Raphanus sativus* var. *caudatus* and *Sinapis alba* (Eberhardt, 1904 cited by Napper, 1933). *Raphanus sativus* accessions seem to be resistant to other *A. candida* races, since none of the plants were infected with races *Ac2,* to *Ac6* (Pound and Williams, 1963), although only one isolate per race and one accession was tested. All genotypes tested (Downey and Rimmer, 1993) were resistant to *Ac2*, *Ac7*, *Ac8*, or *Ac9*, but for race *Ac8* and *Ac9* testing was limited. Lakra and Saharan (1988) and Napper (1933) found similar results: *Raphanus sativus* was only susceptible to its homologous race, not to *Ac2* or to *A. candida* originating from *Brassica rapa* subsp. *chinensis* or *Brassica tournifortii* or to *A. candida* from *Arabis alpina* (3 isolates), *Brassica oleracea* (4 isolates, *Ac9*), *Brassica rapa* (*Ac7*), *Capsella bursa-pastoris* (11 isolates, *Ac4*), *C. squamatus*, *D. tenuifolia*, *Sinapis arvensis* susp. *arvensis* (*Ac10*) or *Sisymbrium officinale* (*Ac5*).

- ***Rorippa***

Rorippa islandica is the homologous host of race *Ac6*. For *Rorippa islandica*, 98% of the plants tested was susceptible to *Ac6*, whereas only 23% were susceptible to *Ac3* and 1% to race *Ac4*. None of the plants were susceptible to races *Ac1*, *Ac2* or *Ac5* (Pound and Williams, 1963), although only one isolate per race and one accession was tested.

- ***Sinapis***

Sinapis arvensis is the homologous host of race *Ac10* (Williams, 1985 cited in Hill *et al*., 1988). *A. candida* originating from *Sinapis arvensis* was able to infect some plants of *Brassica rapa*, *Brassica nigra*, *Brassica juncae* and *Brassica napus* but disease ratings were low: 1 to 3 on a scale of 1 - 9 (Hill *et al*., 1988). *Sinapis alba* subsp. *alba* was highly susceptible to several races of *A. candida*, including isolates originating from *Arabis alpina* (3 isolates), *Brassica oleracea* (4 isolates, *Ac9*), *B. rapa* (*Ac7*), *Capsella bursa-pastoris* (11 isolates, *Ac4*), *Coronopus squamatus*, *D. tenuifolia*, *Sinapis arvensis* subsp. *arvensis* (*Ac10*), and *Sisymbrium officinale* (*Ac5*) (Napper, 1933). Pound and Williams (1963) found similar results: 78% of the plants of *Sinapis alba* subsp. *alba* were susceptible to *Ac5* and 19% to *Ac6*, whereas 100% of the plants became infected with races *Ac1* to *Ac4*. In conclusion, *Sinapis* appears to be a general host of multiple races of *A. candida*.

- ***Sisymbrium***

Sisymbrium officinale is the homologous host of race *Ac5*. It is only susceptible to several other races of *A. candida* when plants are young (cotyledons not yet fully enlarged). For young plants, 36% were susceptible to *Ac1*, 98% to *Ac2*, 96% to *Ac3*, 20% to *Ac4*, 98% to *Ac5* and 4% to *Ac6*. Another species within the genus, *Sisymbrium altissimum*, was slightly susceptible to *A. candida Ac4*: 3% of the plants showed symptoms (Pound and Williams, 1963).

• ***Thlaspi***

Thlaspi arvense was slightly susceptible to *Ac3* (5% of the plants infected), *Ac4* (16%) and to *Ac5* (2%) (Pound and Williams, 1963). It was also susceptible to *A. candida* originating from *Capsella bursa-pastoris* (susceptible to 5 of 11 isolates tested, *Ac4*), *Coronopus squamatus* and *Sisymbrium officinale* (Ac5) (Napper, 1933).

3. CONCLUDING REMARKS

In this study, a summary and update on the host range and different races of the oomycete pathogen *Albugo candida* was given. Identification of specific races is essential in plant breeding strategies (Borhan *et al.*, 2001; Adhikari *et al.*, 2003) and may also be exploited to induce both local and systemic resistance in plants against *A. candida*. In their initial study, Pound and Williams (1963) described 6 races (Table 1). Since then at least 5 other *A. candida* races have been proposed, including race *Ac7* on *Brassica rapa*, *Ac8* on *Brassica nigra*, *Ac9* on *Brassica oleracea*, *Ac10* on *Sinapis arvensis* and *Ac11* on *Brassica carinata* (Table 2). *Sinapis* appears to be a general host of multiple races of *A. candida*. At the time Pound and Williams (1963) described the first 6 races of *A. candida*, they already indicated that the description of specialized races (pathotypes) of *A. candida* is rather difficult given that results obtained in cross-inoculation experiments did not reveal 'clear-cut positive or negative sporulation responses'. The results of our survey confirm and extend this lack of clear-cut responses, making it difficult to describe new races in addition to the current 11 races. Progress in molecular analysis and phylogeny of oomycete pathogens may further help to develop specific molecular markers to identify and discriminate between different races of *A. candida.*

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TEMPORAL AND SPATIAL VARIATION IN VIRULENCE OF NATURAL POPULATIONS OF *BREMIA LACTUCAE* OCCURRING ON *LACTUCA SERRIOLA*

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1. INTRODUCTION

Plant pathogenic fungi and their populations include a very large and heterogeneous group of organisms that occupy positions of great importance in both agriculture (McDermott and McDonald, 1993) and natural plant communities (Burdon, 1993; Burdon and Thrall, 1999). Understanding the role pathogens play in shaping the genetic structure of plant populations and communities requires an understanding of the pathogen´s diversity and origins, and the evolutionary interplay that occurs between hosts and their pathogens (Burdon and Silk, 1997). The structure and changes in genetic variation and co-evolution of host and pathogen populations must be considered as a complex phenomenon from the spatial and temporal viewpoint (Burdon, 1997; Burdon and Thrall, 1999). So far only a limited number of host-pathogen interactions in natural plant communities have been characterized in detail (Burdon and Jarosz, 1988; Burdon, 1997). The same is true for oomycete pathogens (Lebeda and Schwinn, 1994; Drenth and Goodwin, 1999; Lebeda *et al*., 2002). Such variation was described for *Peronospora parasitica* and *Albugo candida* on naturally growing populations of *Arabidopsis thaliana* (Holub and Beynon, 1997). Currently the most intensively studied wild pathosystem for oomycetes is probably *Lactuca* spp.–*Bremia lactucae* (Lebeda, 2002; Lebeda *et al*., 2002; Lebeda and Petrželová, 2003).

Bremia lactucae Regel (lettuce downy mildew) is an important pathogen of cultivated lettuce (*Lactuca sativa* L.) and occurs on at least 200 wild Asteraceae species (Crute and Dixon, 1981; Lebeda and Syrovátko,

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1988; Lebeda *et al.*, 2002). Distribution of *B. lactucae* is worldwide, occurring on all continents, mostly where the lettuce crop is grown (Marlat, 1974). The economic impact of lettuce downy mildew on the lettuce crop can be very high (Crute, 1992a). However, there is incomplete information about the distribution, interaction and impact of the fungus on wild *Lactuca* species (Lebeda *et al.*, 2002; Lebeda and Zinkernagel, 2003b). Recently, *Lactuca serriola* L. is regarded as the most common naturally growing host species of *B. lactucae* (Lebeda, 1998a, 2002; Lebeda and Syrovátko, 1988; Lebeda *et al.*, 2001, 2002; Petrželová and Lebeda, 2000, 2003). For example, a high frequency (ca 80%) of naturally infected *L. serriola* populations were found in the Czech Republic (Lebeda, 2002; Lebeda and Petrželová, 2003; Petrželová and Lebeda, 2003), whilst in some other countries (eg south of France, Italy, some parts of USA) with different environmental conditions the occurrence is very low or the pathogen is absent (Lebeda *et al.*, 2001; Lebeda, unpubl. results).

Most of the host-pathogen interactions between *Lactuca* spp. and *B. lactucae* are race-specific (Lebeda *et al.*, 2002; Lebeda and Zinkernagel, 2003b). The first genetic studies showed that resistance in *L. sativa* to downy mildew was inherited simply. Further detailed classical and molecular genetic studies have shown that race-specificity is based on a complementary gene-for-gene relationship. The specificity is determined by dominant resistance *Dm* genes (or R-factors) in the host which are matched by dominant factors for avirulence in the fungus (Crute, 1992b; Lebeda and Schwinn, 1994). More than 40 host resistance genes (R-factors) have been identified so far (Reinink, 1999; Sicard *et al.*, 1999; Lebeda *et al.*, 2002). At least 13 of these resistance genes originated or were located in *L. serriola* (Lebeda *et al.*, 2002). Numerous additional resistance genes might be present in *L. serriola* accessions (Lebeda, 1986; Farrara and Michelmore, 1987; Lebeda and Boukema, 1991; Bonnier *et al.*, 1992), but there is limited information about their distribution in natural populations of *L. serriola.*

B. lactucae is highly variable oomycete pathogen. The variation was first categorized in terms of physiological races (Crute and Dixon, 1981), and later as specific virulence determinants (virulence factors, virulence phenotypes) based on the interpretation of the host-pathogen interaction in a gene-for-gene relationship (Crute, 1987). This approach provided a basis for the application of population studies in pathogen populations (Lebeda, 1982a). A population genetics approach was broadly used in studies of virulence variation of pathogen populations occurring on lettuce crops (Lebeda, 1981, 1982b; Crute, 1987; Lebeda and Zinkernagel, 2003a). However, until now this approach was not used for detailed studies of *B. lactucae* occurring in populations of naturally growing *L. serriola*. Only individual isolates originating from natural populations of *L. serriola* have been investigated for specific virulence variation (Lebeda, 1984, 1986, 2002; Lebeda and Boukema, 1991). Recently, Lebeda and Petrželová (2003) undertook the first study focused on distribution of *B. lactucae* virulence phenotypes (v-phenotypes) in a wild pathosystem.

The present paper presents a more detailed and comprehensive continuation of previous studies (Lebeda, 2002) and is focused on detailed analysis of spatial and temporal aspects of virulence structure of *B. lactucae* populations occurring on naturally growing *L. serriola* populations in the Czech Republic. In addition, the virulence structure of pathogen populations occurring in wild (native) and crop pathosystems was compared.

2. MATERIALS AND METHODS

2.1 Pathogen isolates and v-factor determination

A total of 132 isolates of *B. lactucae* collected from infected and naturally growing plants of *L. serriola* during the period 1998-2000 (Lebeda and Petrželová, 2003) were used for v-factor analysis (Table 1). Isolates were maintained and multiplied on seedlings of *L. serriola* accessions (PI 273617, LSE/57/15 or LS-102). In 1999, *B. lactucae* isolates were collected also from lettuce crops (*L. sativa*) grown commercially in the Czech Republic. Altogether 37 isolates were used for determination of v-factors on the basic differential set of *Lactuca* spp. genotypes (Guénard *et al*., 1999; van Ettekoven and van der Arend, 1999; Lebeda and Zinkernagel, 2003b), and 19 v-factors (v1-v18, v36-v38) were established (Figure 4). Isolates originating from *L. sativa* were maintained and multiplied on cvs Hilde or Cobham Green.

Virulence of isolates was examined by screening on the differential set of 49 *L. sativa* and *L. serriola* genotypes. Thirty-five cultivars of *L. sativa*, twelve accessions of *L. serriola* and two interspecific hybrids were included into the differential set (Lebeda, 1997; Lebeda and Zinkernagel, 2003b). *L. serriola* (LSE/57/15) served as a universally susceptible control. Seed samples of the differentials originated from the *Lactuca* spp. collection, which is kept by Department of Botany, Faculty of Science, Palacký University in Olomouc in cooperation with the Gene Bank

Department - Workplace Olomouc, Research Institute of Crop Production in Prague, Czech Republic. A summary of the differential genotypes arranged according to the race-specific resistance *Dm* genes and R-factors was given by Lebeda and Zinkernagel (2003b). The presence of resistance genes and factors in the differentials was established according to recent knowledge of the genetics of the *Lactuca* spp.–*B. lactucae* interaction (Bonnier *et al.*, 1994; Illot *et al.*, 1989; Lebeda, 1997; Guénard *et al.*, 1999; Lebeda and Zinkernagel, 2003b). In total, 26 *L. sativa* and *L. serriola* genotypes carrying one or two race-specific *Dm* genes and/or R-factors were used for basic differentiation of isolates and determination of v-factors. For more detailed characterization of variation amongst the studied isolates, the set was enlarged with several additional cultivars and accessions either carrying a combination of more race-specific *Dm* genes and R-factors or with their resistance being not clearly specified yet.

Table 1. Origin of *Bremia lactucae* isolates from *Lactuca serriola* in the Czech Republic used for v-factor analysis.

Region	District (abbreviation)	Year of collecting/ number of isolates		
		1998	1999	2000
Jihomoravský	Brno-country (BC)	16	0	0
	Břeclav (BV)	0	0	6
Moravskoslezský	Nový Jičín (NJ)	0	0	6
Olomoucký	Olomouc (OL)	4	30	40
	Prostějov (PV)	3	0	1
	Přerov (PR)	0	0	25
Středočeský	Rakovník (RA)	0	1	0
Total number of isolates		23	31	78

2.2 Growing of differential set, inoculation and sporulation assessment

Seedlings of *L. sativa* and *L. serriola* differentials were grown in plastic boxes (320 x 265 x 60 mm) lined with moistened filter paper and closed with a glass cover. Tests were carried out at the stage of fully expanded cotyledon leaves (7-9 days after sowing). Inoculum was prepared by washing 2 to 3-day-old spores from seedlings of *L. serriola* (LSE/57/15, PI 273617 or LS-102) with distilled water. The density of inoculum was approximately 10^5 spores ml^{-1}, and was sprayed onto seedlings using a glass chromatography sprayer. Inoculated seedlings were incubated at a temperature of 10 to 15°C covered with black foil, for the first 12-24 h after inoculation, than transferred to a 12 h photoperiod. Each test comprised 20 to 25 seedlings per *Lactuca* spp. genotype (accession). Any tests giving ambiguous results were repeated.

The degree of infection was assessed quantitatively in two-day intervals from day 6 to 14 days after inoculation using the 0-3 scale described by Dickinson and Crute (1974):

0 = symptomless, no sporulation;
1 = limited sporulation, isolated sporophores present;
2 = < 50 % cotyledon area covered with sporophores;
3 = > 50 % cotyledon area covered with sporophores.

The final value of sporulation intensity was expressed as a percentage of the maximum scores according to Towsend and Heuberger (1943). Isolates were examined for the presence of 27 v-factors and their reaction to three additional *Lactuca* spp. genotypes and a *L. serriola* x *L. sativa* hybrid (Table 2). In individual isolates, a v-factor was regarded as present if the final sporulation intensity on a genotype carrying the corresponding *Dm* gene or R-factor was more than 30%.

2.3 Pathogen virulence variability at the population level

Basic theoretical and methodological aspects relating to the genetic analysis of *B. lactucae* populations were described elsewhere (Lebeda, 1981, 1982a; Lebeda and Zinkernagel, 2003a). Variation in virulence and its temporal changes were quantified by the relative frequencies of v-factors (virulences), which were expressed as the ratios between the numbers of isolates with a given v-factor and the total numbers of isolates analyzed for their presence. Frequencies of occurrence of v-factors at surveyed localities were counted for the interpretation of data concerning distribution of virulence. They were expressed as the ratios between the number of localities at which isolates with given v-factors occurred, and the total number of localities investigated for the presence of a v-factor. The results concerning the virulence of individual isolates and their interpretation were analysed separately (Lebeda and Petrželová, 2003).

A survey of virulence of *B. lactucae* isolates obtained from *L. sativa* in 1999 was undertaken according the basic methodological principles described above. The frequencies of v-factors in the crop

Table 2. Geographic distribution of v-factors in the Czech population of *Bremia lactucae* on *Lactuca serriola* (1998-2000).

Region (District)* Site of origin	Virulence factors 1	2	3	4	5/8	6	7	10	11	12	13	14	15	16	17	18 36 37 38	23	24	25	26	27	28	29	30	Reaction to genotypes SP	TI	CR	PI	No. of isolates	Year
Jihomoravský (BC)																														
Brno - Hády	.	.	.	.	5/8	.	7	.	.	.	.	.	15	.	.	.	23	24	25	26	27	.	29	30	-	+	-	+	1	1998
Hajany I	1	2	.	4	5/8	6	7	10	.	12	13	14	.	16	.	.	23	.	.	.	.	.	.	.	+	-	-	+	2	1998
	1	2	.	4	5/8	6	7	10	11	12	13	14	.	16	.	.	23	.	.	.	.	.	.	.	+	-	-	+	1	1998
Hajany II	1	2	3	4	5/8	6	7	10	11	12	13	.	.	16	.	.	23	.	.	.	.	.	.	.	+	-	-	+	1	1998
	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	-	-	-	+	1	1998
Ořechov	1	2	.	4	5/8	.	7	10	.	12	13	14	15	16	.	.	23	24	25	26	27	28	29	30	+	-	-	+	1	1998
	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	1998
Pravlov	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	1998
	.	.	.	.	.	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	4	1998
Rd to Mor. Bránice	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	2	1998
(from Silůvky)	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	1998
Jihomoravský (BV)																														
Lednice	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	-	-	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	2	2000
Sedlec u Mikulova	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	3	2000
Moravskoslezský (NJ)																														
Odry	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	2	2000
	.	.	.	.	.	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	2	2000
Rd to Palačov (from Starý Jičín)	.	.	.	.	.	.	7	.	.	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000

KEY: * see Table 1; Rd = road; **.** = v-factor absent; - = avirulent; + = virulent;
SP = cv. Spartan Lakes; TI = cv. Titan; CR = *L. serriola x L. sativa* (CS-RL); PI = *L. serriola* (PI 273617).

Table 2, continued. Geographic distribution of v-factors in the Czech population of *Bremia lactucae* on *Lactuca serriola* (1998-2000). See start for key.

Region (District)* Site of origin	Virulence factors 1	2	3	4	5/8	6	7	10	11	12	13	14	15	16	17	18 36 37 38	23	24	25	26	27	28	29	30	Reaction to genotypes SP	TI	CR	PI	No. of isolates	Year
Olomoucký (OL)																														
Blatec & Tážaly	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	1999
	.	.	.	.	.	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
	.	.	.	.	.	.	7	.	11	.	.	14	15	16	17	.	23	24	25	26	27	28	29	30	-	-	+	-	1	2000
Bolelouc	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	+	+	1	1999
Bystrovany	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	2	1999 & 2000
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Dolany I	.	.	.	.	.	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Dolany II	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
Grygov I	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	1998
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	-	+	-	1	1998
Grygov II	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	3	1999
	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	+	+	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	6	2000
Hlásnice	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	1999
Hlušovice I	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Hlušovice II	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
Horní Sukolom	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	+	+	2	1999
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	.	.	23	24	25	26	27	.	29	30	-	+	-	+	1	1999
Krčmaň	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	27	28	29	30	-	+	+	-	1	1999
Mrsklesy	.	.	.	.	5/8	.	7	.	.	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	-	+	-	1	1999

Table 2, continued. Geographic distribution of v-factors in the Czech population of *Bremia lactucae* on *Lactuca serriola* (1998-2000). See start for key.

Region (District)* Site of origin	Virulence factors 1	2	3	4	5/8	6	7	10	11	12	13	14	15	16	17	18 36 37 38	23	24	25	26	27	28	29	30	Reaction to genotypes SP	TI	CR	PI	No. of isolates	Year
Olomoucký (OL)																														
Nový Dvůr	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	1999
	.	.	.	.	5/8	.	7	.	.	.	.	.	15	.	.	.	23	24	25	26	27	.	29	30	-	+	-	+	1	1998
	.	.	.	.	5/8	.	7	.	.	.	.	.	15	.	.	.	.	24	25	26	27	.	29	30	-	+	-	-	1	1998
	.	.	.	.	.	.	7	.	.	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	-	+	-	1	2000
Olomouc-Droždín I	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	+	+	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Olomouc-Droždín II	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	2000
Olomouc-Holice I	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	+	+	1	1999
Olomouc-Holice II	.	.	.	.	5/8	.	7	.	.	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	2	1999 & 2000
	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
Olomouc-Holice III	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	1999
Olomouc-Holice IV	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
Olomouc-Hodolany	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Přáslavice	.	.	.	.	.	.	7	.	11	.	.	.	15	.	.	.	23	24	25	26	27	.	29	30	-	+	-	+	1	1999
	.	.	.	.	.	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Rd to Blatec/Tážaly	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	1999
(from Grygov)	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	-	+	-	1	2000
Rd to Velký Týnec -Čechovice (from Svésedlice)	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	+	-	+	-	1	2000

Table 2, continued. Geographic distribution of v-factors in the Czech population of *Bremia lactucae* on *Lactuca serriola* (1998-2000). See start for key.

Region (District)* Site of origin	Virulence factors 1	2	3	4	5/8	6	7	10	11	12	13	14	15	16	17	18 36 37 38	23	24	25	26	27	28	29	30	Reaction to genotypes SP	TI	CR	PI	No. of isolates	Year
Olomoucký (OL)																														
Rd to Blatec (from Vrbátky)	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	.	.	23	24	25	26	27	.	29	30	-	+	-	+	1	1999
Samotišky I	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Samotišky II	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Savín	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	1999
Svésedlice	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	+	+	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	+	+	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	27	28	29	30	-	+	+	-	1	1999
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Tovéř	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	+	+	1	2000
Tršice	.	.	.	.	.	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	2	2000
Velká Bystřice I	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Velká Bystřice II	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	.	24	25	26	.	28	29	30	-	+	+	-	1	2000
	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
Velký Týnec I	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	1999
Velký Týnec II	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	-	+	-	1	2000
Velký Týnec-Čechovice	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Olomoucký (PV)																														
Smržice	.	.	.	.	.	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	2000
Soběsuky	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	3	1998

Table 2, continued. Geographic distribution of v-factors in the Czech population of *Bremia lactucae* on *Lactuca serriola* (1998-2000). See start for key.

Region (District)* Site of origin	Virulence factors																								Reaction to genotypes				No. of isolates	Year
	1	2	3	4	5/8	6	7	10	11	12	13	14	15	16	17	18 36 37 38	23	24	25	26	27	28	29	30	SP	TI	CR	PI		
Olomoucký (PR)																														
Hrabůvka	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Hustopeče nad Bečvou	.	.	.	.	.	.	7	.	.	.	.	.	15	.	.	.	23	24	.	26	27	.	29	30	-	+	-	-	1	2000
Klokočí	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
Lipník nad Bečvou	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	2	2000
Pavlovice	.	.	.	.	.	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Předmostí u Přerova	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Přerov - Čekyně I	.	.	.	.	.	.	7	.	.	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	-	+	-	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	3	2000
Přerov - Čekyně II	.	.	.	.	5/8	.	7	.	11	.	.	.	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Přerov - Penčice	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	-	+	-	1	2000
Přerov - Vinary	.	.	.	.	.	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	1	2000
	.	.	.	.	.	.	7	.	11	.	.	14	15	16	17	.	.	24	25	.	27	28	29	30	-	-	+	-	1	2000
Radslavice	.	.	.	.	.	.	7	.	.	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
Rd to Velká (from Hrabůvka)	.	.	.	.	5/8	.	7	.	.	.	.	.	15	16	17	.	23	24	25	26	27	28	29	30	+	-	-	+	2	2000
Šišma	.	.	.	.	5/8	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
	.	.	.	.	.	.	7	.	11	.	.	14	15	16	17	.	.	24	25	26	27	28	29	30	-	-	+	-	1	2000
Teplice nad Bečvou	.	.	.	.	5/8	.	7	.	.	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	-	+	-	1	2000
	.	.	.	.	.	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	2000
Středočeský (RA)																														
Krupá	.	.	.	.	5/8	.	7	.	11	.	.	.	15	.	17	.	23	24	25	26	.	28	29	30	-	+	+	-	1	1999

pathosystem (*L. sativa*) were compared with the frequencies of v-factors in the wild pathosystem (*L. serriola*).

3. RESULTS

3.1 Virulence structure of pathogen population

Isolates of *B. lactucae* originating from *L. serriola* exhibited great variation in their virulence to *Lactuca* spp. differentials. For quantification of virulence variation in populations of *B. lactucae*, relative frequencies of v-factors were determined. To describe the variation in more detail, relative frequencies of several other virulences (cvs Spartan Lakes (SP) and Titan (TI), interspecific hybrid *L. serriola* x *L. sativa* (CS-RL) (CR), *L. serriola* (PI 273617) (PI)), which appeared to be of a greater importance for the differentiation of the isolates studied, have been included also. Results are summarized in Figure 1. With the exception of v18, v36, v37 and v38 matching newly introduced R-factors (Lebeda, 1997; Maisonneuve *et al*., 1999; Lebeda and Zinkernagel, 2003a), all other known v-factors have been detected during this investigation, but there were considerable differences in the frequencies of their occurrence. Generally, the highest frequencies (0.58 to 1.00) were recorded for v-factors or virulences matching resistance genes localized in *L. serriola* or in *L. sativa*, how ever resistance was derived from *L. serriola*. Only v7 was present in the maximum frequency (1.00) during all observation periods, i.e. it could be considered as genetically fixed in the pathogen population. Very close to the maximum frequency were v-factors v15, v17, v24, v25, v26, v28, v29 and v30. Relatively high values were observed for several other v-factors and/or virulences (v5/8, v11, v16, v23, v27, vCR). On the contrary, very low frequencies (0.008 to 0.038) were recorded for v1, v2, v3, v4, v6, v10, v12 and v13. These matched resistance genes localized in *L. sativa* cultivars, and were detected only in several isolates collected in 1998. The most variable frequency was for the presence of v5/8, v11, v14, v16, v17, v23, v27, and also the virulences designated as vSP, vCR, vTI and vPI.

3.2. Temporal and spatial virulence structure and their changes

In order to record changes in virulence structure of the pathogen population, relative frequencies of v-factor occurrence and other virulences were counted for each year during the period 1998-2000 (Figure 2). These data showed a considerable increase in the relative frequencies of some v-factors (v11 and v14) and virulences (vCR). On the other hand, the previously more widely distributed v-factors v5/8 and v23, and virulences vSP and vPI, showed a considerable decrease within the population of *B. lactucae* (Figure 2).

The geographic distribution of individual v-factors and virulences in the Czech Republic are given in the Table 2. The presence of v-factors was verified at a total of 63 localities. The greatest differences among localities were found in the presence or absence of v5/8, v11, v14, v16, v23 and v27 and virulences vSP, vCR, vTI and vPI.

Variation among individual and spatially isolated populations of *B. lactucae* is best characterized by relative frequencies of v-factors occurrence at surveyed localities (Figure 3). From Figure 3 it is evident that the frequencies at localities recorded for some v-factors (v5/8, v11, v14, v23, vTI (v6+36), vSP, vCR, vPI) were slightly higher than the frequencies calculated for their occurrence in the studied set of isolates (Figure 1). These differences show wider distribution of these v-factors among individual populations of the pathogen, but on the other hand it is also evidence for existence of variation among isolates within individual populations.

3.3. Comparison of virulence structure in wild and crop pathosystems

A comparative study of the virulence structure of *B. lactucae* populations in wild and crop pathosystems was undertaken in 1999. Only frequencies of the most important and common v-factors occurring in crop pathosystems were considered. The results (Figure 4) showed that there are substantial differences in the virulence structure of both pathosystems. Some v-factors (v1-v4, v6, v10, v12, v13, v18, v36, v37, v38) are completely missing in pathogen populations from *L. serriola*, whilst another (v14) is very reduced. Only a few v-factors (v5/8, v7, v16) occurred in both pathosystems in more or less equal frequencies. The remaining v-factors detected (v15, v17) were predominant in the wild pathosystem (Figure 4). These results demonstrate that both pathosystems differ substantially in relation to the structure of virulence of the pathogen populations.

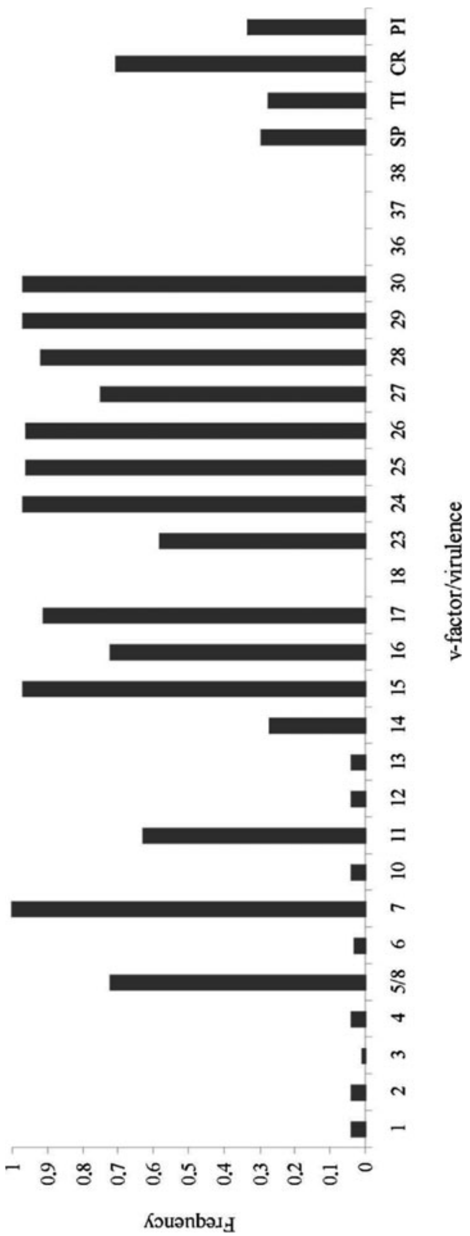

Figure 1. Frequencies of v-factors in Czech population of *Bremia lactucae* on *Lactuca serriola* (1998-2000).

v2, v3, v4, v10, v12, v13, v14: v-factors matching *Dm* genes or R-factors in cvs of *L. sativa*.
v7, v15, v16, v17, v23, v24, v25, v26, v27, v28, v29, v30: v-factors matching *Dm* genes or R-factors in *L. serriola*.
v5/8, v6, v11, v18, v38: v-factors matching *Dm* genes or R-factors in cvs of *L. sativa*, how ever derived from *L. serriola*.
v36, v37: v-factors matching resistance in cvs of *L. sativa*, however derived from *L. saligna*.
vSP: virulence to cv. Spartan Lakes, corresponding resistance gene for the reaction not known (*Dm*1+?).
vTI: virulence to cv. Titan, corresponding resistance gene for the reaction not known (*Dm*6+R36+?).
vCR: virulence to *L. serriola* x *L. sativa* (CS-RL), corresponding resistance gene for the reaction not known (R18+?).
vPI: virulence to *L. serriola* (PI 273617), corresponding resistance gene for the reaction not known (R?).

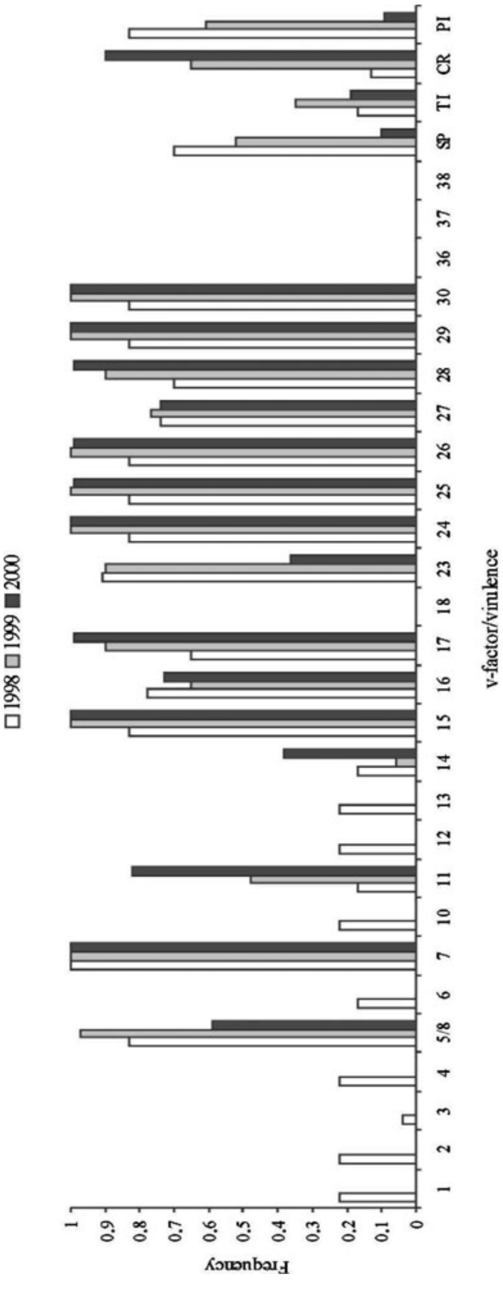

Figure 2. Temporal changes in the frequencies of v-factors (see Figure 1) in Czech population of *Bremia lactucae* during the period 1998–2000.

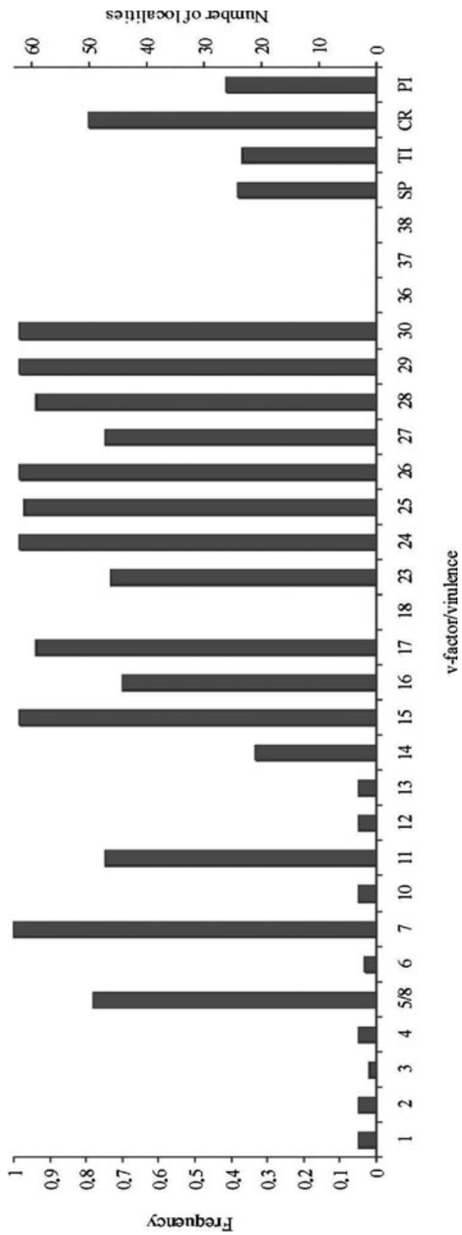

Figure 3. Frequencies of v-factors (see Figure 1) of *Bremia lactucae* at localities (1998-2000).

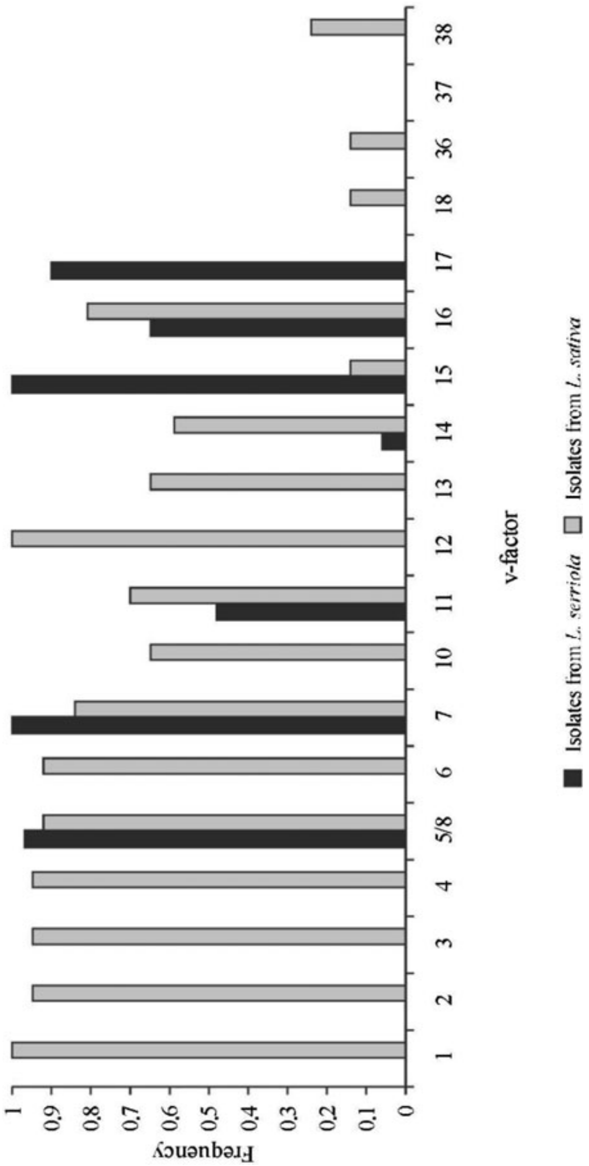

Figure 4. Comparison of frequencies of most important v-factors (see Figure 1) of *Bremia lactucae* in wild and crop pathosystems (1999).

4. DISCUSSION

Many native plants suffer frequent infections by many fungi (Ellis and Ellis, 1997), including downy mildews, however despite that they remain common. This is because both the host and pathogen have evolved very effective survival strategies. Little is known of the strategies required to ensure survival of the variety of selection pressures that may have been involved in their evolution. The association between a host and a parasite is considered to be the result of a co-evolutionary process (Simms, 1996). During this process, those host adaptations that tend to limit parasite attack are postulated to lead to the selection of parasite adaptations that enable the parasite to cope with those adaptations and *vice versa* (Clarke and Akhkha, 2002). However, forces responsible for interactions and differences between natural and crop plant pathosystems could be different (Robinson, 1976). In oomycetes there is very limited knowledge about these interactions. The *Lactuca* spp.–*B. lactucae* pathosystem, however, could be considered as an excellent model for such studies (Lebeda, 2002; Lebeda *et al*., 2002). This is because of the relatively good knowledge of the genetics of the host-pathogen interactions (Crute, 1992b).

There is no detailed information about the resistance structure of natural populations of *L. serriola* to *B. lactucae.* However, the virulence structure of natural populations of *B. lactucae* is very specific and different from populations in the crop (*L. sativa*) pathosystem (Crute, 1987; Lebeda, 1984, 1989, 2002; Lebeda and Jendrulek, 1987; Lebeda and Zinkernagel, 2003a). The majority of the tested isolates of *B. lactucae* originating from *L. serriola* expressed clear genetic affinity to *L. serriola* accessions, because v-factors matching resistance located in or derived from *L. serriola* prevailed considerably in the isolates studied (Lebeda, 2002). However, variation was found in the presence of individual v-factors or virulences among pathogen individuals (isolates) and populations (Lebeda and Petrželová, 2003). Virulence factors matching resistance located in *L. sativa* were recorded only very rarely. A surprisingly high frequency was recorded for virulence to the interspecific hybrid *L. serriola* x *L. sativa* (CS-RL) (*Dm*18+?), which has been so far resistant to most isolates of *B. lactucae* from *L. sativa* and *L. serriola* (Lebeda, 1997, 1998b; Lebeda and Blok, 1991; Lebeda and Zinkernagel, 1999). Nevertheless, this resistance was recently broken down by some isolates originating from *L. sativa* and first identified in Germany (Lebeda and Zinkernagel, 2003a,b). Another unexpectedly high frequency was recorded for factor v36 matching resistance in cv. Titan (*Dm*6+R36) derived from *L. saligna* (Lebeda, 1997;

Lebeda and Zinkernagel, 2003a,b). Virulence to these two accessions has been reported only for some German isolates of *B. lactucae* originating from *L. sativa* (Lebeda, 1997, 1998b; Lebeda and Zinkernagel, 1999, 2003a).

So far there is limited information available on geographic differences in virulence among populations of *B. lactucae*. Research of this topic has been realized only in the crop *L. sativa*-*B. lactucae* pathosystem (Crute, 1987, 1992b; Lebeda, 1982b; Lebeda and Zinkernagel, 2003a). In this pathosystem it was proved that geographic differences in virulence are a consequence of utilization of different resistance genes in the various areas of lettuce cultivation. The differences are evident not only between areas, but also between countries and continents (Crute, 1987; Lebeda and Jendrulek, 1987). It is expected that differences in virulence population structure are the result of long lasting co-evolution between host and pathogen populations (Lebeda and Schwinn, 1994; Lebeda and Zinkernagel, 2003a). In the wild pathosystem (*L. serriola*-*B. lactucae*) no detailed research aimed at spatial distribution and temporal changes of virulence has been published previously and the present paper is the first comprehensive contribution to this topic. These data have revealed existence of clear geographic differentiation in virulence in natural populations of *B. lactucae*. Within the area of southern Moravia (district Brno-county) a relatively high frequency of v-factors (e.g. v1, v2, v3, v4, v6, v10, v12, v13 and v14) was recorded. These v-factors are more likely to be typical for pathogen isolates originating from *L. sativa* and were not recorded at any other area investigated for the natural distribution of *B. lactucae*.

From recent data it is evident that virulence structure is a dynamic phenomenon, because qualitative and quantitative shifts in v-factor composition were recorded in populations of *B. lactucae* on natural populations of *L. serriola* during the period of our observations (Figure 2). Considerable changes in the frequencies of some v-factors were recorded. Also the spectrum of v-factors changed substantially and become broader in comparison with some previous results (Lebeda, 1984, 2002). The same phenomenon was reported for populations of *B. lactucae* on lettuce (Lebeda and Schwinn, 1994; Lebeda, 1998b; Lebeda and Zinkernagel, 1999, 2003a). In crop pathosystems, the genetic structure of resistance of host populations and thereby also the distribution of virulence in pathogen populations is largely determined by plant breeders and growers. The main cause of the virulence shift was due to utilization of new resistance genes in lettuce breeding programmes (Lebeda, 1998b; Lebeda and Zinkernagel, 2003a).

In the wild pathosystem (*L. serriola - B. lactucae*) the genetic background of such changes and potential selection pressures have not been studied yet. However, it is difficult to expect that the main factor of selection pressure will be the composition of the host population because it is not directly influenced by human activity (plant breeding). Nevertheless, some changes in genetic background of host populations may be expected as an effect of migration. It is very well known that achenes of *L. serriola* are transported for very long distances and the host plant is able to grow in very different habitats (Lebeda *et al*., 2001).

B. lactucae is very adaptable to changes in the genetic structure of the host population (Lebeda, 1998b; Lebeda *et al*., 2002; Lebeda and Zinkernagel, 2003a), and could be considered as a pathogen with a high evolutionary potential (McDonald and Linde, 2002). It therefore poses a very great risk in relation to breakdown of resistance genes. The frequencies of particular v-factors of *B. lactucae* mostly correlate strongly with the frequencies of corresponding resistance genes and with the time of their introduction and occurrence in a specific locality (Lebeda, 1979, 1981; Lebeda and Zinkernagel, 2003a). However, more detailed information about the structure and distribution of race-specific resistance in populations of weedy growing *Lactuca* spp. (Lebeda *et al*., 2001, 2002) is needed to compare them with the distribution of corresponding virulence in populations of *B. lactucae*. Nevertheless, in some other wild pathosystems (e.g. *Senecio vulgaris*-*Erysiphe fischeri*) very broad variations were found in pathogen virulence, host resistance and symptom expression within one host-parasite population (Bevan *et al*., 1993a,b,c). Crute (1990) identified at least three different resistance phenotypes in British *L. serriola* populations, some with the presence of race-specific resistance and different levels of field resistance. In the Czech host populations studied here, some partial information about spatial and temporal variation of resistance of *L. serriola* has been obtained. There is evidence for the existence of intra- and inter-population variability of *L. serriola* resistance to *B. lactucae* (Lebeda, unpubl. results). An increase or decrease of virulence frequencies recorded in natural populations of *B. lactucae* over a period of time may signal that the changes in resistance structure had occurred in populations of the host *L. serriola*. More detailed research on this topic is underway (Lebeda and Petrželová, 2003). Data obtained in other pathosystems displaying gene-for-gene interactions (e.g. *Linum marginale*-*Melampsora lini*) have provided some evidence of coordinated changes in the host and pathogen populations (Burdon and Thrall, 2000; Thrall and Burdon, 2003).

The knowledge of ways in which natural pathosystems may interact with crop pathosystems is extremely limited. Recently these types of interactions were considered for powdery mildew pathosystems (Clarke and Akhkha, 2002). The data presented here for *B. lactucae* and *L. serriola/L. sativa* have demonstrated clearly that these pathosystems have a completely different virulence structure. Preliminary evidence for this was also obtained from v-phenotype analysis (Lebeda and Petrželová, 2003), where only a few isolates of *B. lactucae* resembled the known races (v-phenotypes) of the pathogen, whilst the majority of isolates was completely different. Recent data also demonstrate that the pathogen populations on naturally growing *L. serriola* plants are more or less independent from the epidemiological and genetical points of view. It was hypothesized previously (Lebeda, 1989; Lebeda and Blok, 1990; Lebeda, 2002; Petrželová and Lebeda, 2003) that there may be some connection (genotype and gene flow) between both pathosystems. It seems that the pathogen virulence structure in natural *L. serriola* populations is primarily influenced by the structure of host population resistance and their general genetical background. This is also supported by our previous and recent data on the response of *L. serriola* accessions to *B. lactucae* isolates originating from *L. serriola* (Lebeda *et al*., 2002). Recent data suggest that there is probably no direct epidemiological linkage between both pathosystems.

5. ACKNOWLEDGEMENTS

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DOWNY MILDEW OF PEARL MILLET: PRESENT SCENARIO IN INDIA

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1. INTRODUCTION

Pearl millet [*Pennisetum glaucum* (L.) R. Br.] is an important crop grown in varying ecological environments of low to high productivity levels in the semi-arid tropics. It is not only a staple food of millions of rural people inhabiting these regions but is also valued source of fodder to cattle. The world cropped area under pearl millet is about 26 million hectares, out of which India has 9.4 m ha with a total production of about 6.5 m tonnes, with a productivity level of 750 kg/ha (Anon., 2002). Although the crop is quite hardy, it suffers from various biotic stresses. In particular, downy mildew caused by *Sclerospora graminicola* (Sacc.) Schroet. is widespread and destructive.

Historically, the disease was of minor consequence in India. However, the introduction of hybrid varieties during the late 60s has triggered the progression of the disease. The magnitude of grain yield reduction largely depends upon disease severity and stage of crop growth during infection. Thakur *et al.* (1978) reported 13-65% loss in grain yield. Recently estimated yield losses of up to 30 per cent have been recorded (Singh *et al.*, 1993; Singh, 1995). A significant correlation (r=0.99) between downy mildew incidence and yield was reported by Mayee and Siraskar (1982). Though the area under pearl millet cultivation has remained more or less static (Figure 1), production and productivity has shown a declining trend. This decline was more pronounced during the years of downy mildew epidemics i.e 1968-69, 1979-80, 1985-86 and 1991-92. The popular hybrids HB-3, NHB-3, BJ-104, BK-560, MBH-110 and MLBH-104 succumbed to downy mildew and were subsequently withdrawn (Table 1). Recently ICMH-451 (MH-179), PG-5822, Kanchan 321 and Badshah have also exhibited susceptibility to this disease.

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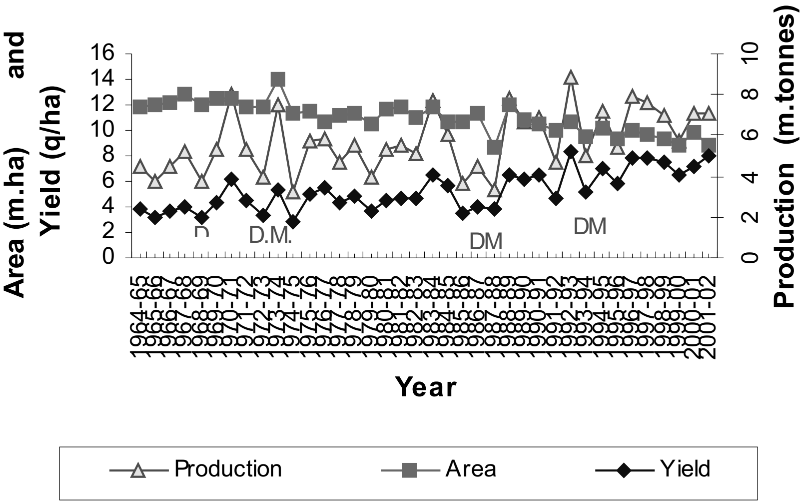

Figure 1. Area, production and yield of pearl millet in India (1965-2002).
Source: Agricultural Statistics at a Glance, 2002.

Table 1. Development of hybrids and their subsequent breakdown in India.
Source: All India Coordinated Pearl Millet Improvement Project Reports.

Year	**MS lines used**	**Hybrid developed**	**Status**
1965-70	Tift-23A	HB-1, HB-2, HB-3, HB-4	Succumbed to downy mildew (DM) during 70's
1970-75	5071-A	NHB-3, NHB-4, NHB-5	Downy mildew epidemic after two years
1975-80	5141-A, 5054-A	BJ-104, BK-560, GHB-27, CJ-104, GHB-30	Succumbed to DM in 1985
1980-85	111A, MS-2, 126 D2A	PHB-10, PHB-14, MBH-110, GHB-1399	Occurrence of pollen shedders; succumbed to DM
1985-90	81-A, 841-A, 843-A	MH-179, HHB-50, HHB-60, PUSA-23, HHB-67, HHB-68	Fairly resistant to DM
1990-2000	89111-A, 81-A	HHB-94, RHB-58	as above

2. SYMPTOMS

Typical symptoms of downy mildew infection begin with yellowing or chlorosis of leaves starting from the base to the tip of leaves which gradually covers a greater proportion of the later formed leaves. A fine whitish downy growth of the fungus is characteristically observed on leaves become brown and necrosis occurs leading to shredding along the veins of the leaves (Thakur, 1992). In the case of early infection, the nodal buds are stimulated leading to development of lateral shoots and thereby giving a bunchy and malformed appearance to the whole plant. In systemically infected plants, leafy growths are produced in place of floral parts. This is the characteristic green ear phase of the disease and results in total grain loss. Severely infected plants generally remain stunted and do not produce panicles and often die at the seedling stage.

3. BIOLOGY AND EPIDEMIOLOGY

3.1 Pathogen

Sclerospora graminicola reproduces by both asexual (sporangia, zoospores) and sexual (oospores) processes. Sporangia are produced on infected leaves and germinate to release motile zoospores (Singh *et al.*, 1993). Sporangia take 35-120 minutes to germinate releasing 2-6 zoospores through a sporangial pore at 20ºC. Germ tube formation starts between 2-20 minutes of zoospore release and growth of the germ tube lasts for 120-180 minutes (Ahuja *et al.*, 1986). A minimum of 3 h at 25ºC and 95% RH is essential to initiate sporulation (Singh *et al.*, 1993). In nature sporangia are produced during the early hours. Maximum numbers of sporangia were trapped between 02.00 to 04.00, and negligible numbers were trapped during the midday hours (Singh and Dang, 2004).

Oospores are thick walled resting spores produced in a scattered fashion in the interveinal leaf mesophyll. Oospores vary in diameter with an average of 33 μm and do not germinate immediately but have a prolonged resting period. The oospores enable the pathogen to survive long, hot dry crop free periods (Williams, 1984). The pathogen is predominantly heterothallic and sexual cross compatibility between isolates of Indian and African continents have been reported (Michelmore *et al.*, 1982; Idris and Ball, 1984). Different methods for oospore germination have been described (Hiura 1930; Pande 1972; Bhat 1973; Sundaram and Guraha 1977) but most of these are difficult to reproduce (Dang, 1981). However, Panchbhai *et al.* (1991) reported up to 76 per cent germination of oospores when treated with 2-5 per cent sodium hypochlorite (NaOCl).

3.2 Pathogenic variability

Considerable variability occurs in *S. graminicola*. The host cultivar HB-3 was highly resistant to downy mildew at Mysore, but was highly susceptible in other places in the country (Bhat, 1973). Host cultivar reactions to different isolates of *S. graminicola* were demonstrated by Ball (1983). Four pathotypes were reported (Table 2) on the basis of differential reactions. (Thakur *et al.*, 1999). Pathogenic variability is found within a population, between seasons, among single oospore and single zoospore isolates of *S. graminicola* (Thakur and Shetty, 1993), and among isolates from several cultivars of pearl millet (Thakur *et al.*, 1999). Furthermore DNA finger printing of specific isolates revealed distinct restriction fragment length polymorphism (RFLP) (Sastry *et al.*, 1995; Thakur *et al.,* this volume). Pathogenic variability was assessed on 15 isolates of *S. graminicola* collected from various cultivars in India. Five major groups were identified using amplified fragment length polymorphism (AFLP) (Sivaramakrishanan *et al.*, 2003). The pathotypes have evolved by host cultivar directed selection under field conditions (Thakur *et al.*, 1992).

Table 2. Downy mildew incidence (%) caused by zoospores of different pathotypes of *S. graminicola.* Source: All India Coordinated Pearl Millet Improvement Project Report, 1994.

Pathotypes	**Host Differential cvs**				
	NHB-3	**Kalucomba**	**7042**	**MHB-110**	**852-B**
Pathotype-1 (HB-3 Mysore)	89.2	42.0	75.0	15.0	43.0
Pathotype-II (Gulbarga Karnatka)	0.0	46.0	0.0	0.0	0.0
Pathotype-III (MHB-110 A. bad)	26.0	7.0	20.0	76.0	0.0
Pathotype-IV (7042, ICRISAT)	82.4	16.5	85.0	7.0	0.0

3.3 Role of sexual and asexual spores

Asexual spores are ephemeral and die within a matter of hours after maturity. Therefore, rapid dispersal and infection is essential. Wind speed, relative humidity, temperature, sunshine and leaf wetness can influence viability and dispersal of spores. Sporangia remained viable for 2.5-6 h depending up on temperature, relative humidity and wind speed (Shetty, 1987). A negative correlation between wind speed and the number of sporangia trapped over a bajra field was observed (Singh and Dang, 2004). The shoot tip of young seedlings is the most susceptible site for infection. Sporangia play a significant role in the secondary spread of the disease (Dang, 1981).

Oospores are the primary inoculum source. They remain viable in the soil for 3-4 years, however in infected host tissues viability has been reported for up to 10 years (Williams, 1984). Oospore production in infected tissues is inversely proportional to the number of sporangia produced and production is more if plants are raised under stress conditions (Safeeulla, 1975). Survival depends upon various soil factors (Nene and Singh, 1975). Oospores remain pathogenic and require an overwintering period for germination (Thakur, 1981). One year old oospores stored under field conditions cause maximum infection to the pearl millet crop (Thakur and Gangopadhyay, 1986). The factors affecting oospore viability and germination and subsequent infection and their impact in epidemiology are yet to be ascertained.

3.4 Role of seed transmission

Seed plays an important role in pathogen transmission and dispersal. Oospores carried on the external surface of the seed and inside the leaf debris provide a major source of infection (Shetty *et al.*, 1978). A positive correlation was observed between oospore indices of individual seed samples and the corresponding downy mildew incidence under natural field conditions (Thakur and Gangopadhyay, 1986). Mycelium survives partly in the degenerated layers of the pericarp cells of the funicular regions of the seed and infects the emerging seedlings (Ahuja *et al.*, 1989). Surface sterilized seeds from infected heads produced 0.6 per cent infected seedlings (Dang, 1981), indicating the internal seed-borne nature of the disease. Later studies indicated that drying the seeds below 20 per cent moisture and sowing healthy certified seeds could help avoid seed-borne infection (Williams, 1984).

3.5 Role of environmental factors

Various meteorological factors influence disease progress and epidemic development. Work conducted at the International Crop Research Institute for the Semi Arid Tropics (ICRISAT, India) shows that with suitable environmental conditions (temperature 24-28°C, relative humidity more than 90 per cent during late night and early morning), a severe downy mildew epidemic in a susceptible cultivar can develop within 2-3 weeks (Bandopadhyay, 1992). Five years of studies in the northern part of the country revealed that progress of the disease was fastest when maximum temperature was around 30-32ºC and a minimum of 20-23ºC, with relative humidity more than 85 per cent in the morning hours combined with frequent showers (Dang, 2002). Regression analysis revealed a non-linear relationship between disease index and days after sowing (Table 3) (Dang, 2002), explaining up to 99 per cent of the variability. Further validation of these relationships under field conditions is in progress (Dang *et al.*, 2003). Further studies on microclimatic conditions affecting disease epidemic in the field will help to develop an improved disease prediction system.

Table 3. Non-linear equations for progress of downy mildew for different dates of sowing during crop season of 1988 to 1993. Source: Dang, 2002.

Date of sowing	**Equation**	**Coefficient of determination (R^2)***
30th June	$Y=109.81-141.17e^{-0.026d}$	0.99
10th July	$Y=97.74-128.65e^{-0.032d}$	0.98
20th July	$Y=95.77-114.68e^{-0.040d}$	0.99
30th July	$Y=98.25-80.3e^{-0.034d}$	0.98

*Significant at 1% level where: Y = % disease index; d = days after sowing.

4. DISEASE MANAGEMENT

Effective control of downy mildew of pearl millet is possible through the use of resistant varieties and to a lesser extent by cultural and chemical control. Thus greater emphasis in this chapter is given to breeding for disease resistance.

4.1 Host plant resistance

4.1.1 Source of resistance

Host plant resistance provides a practical economic means of disease control under farmers' field conditions. Resistance breeding has received major attention under the All India Co-ordinated Pearl Millet Improvement Programme (AICPMIP) and at ICRISAT and consequently disease resistant high yielding cultivars are available. Well planned and intensive research over the last three decades has resulted in development of highly effective field and laboratory screening techniques (Thakur and Kanwar, 1977; Williams *et al.*, 1981; Singh *et al.*, 1997), identification of several sources of resistance and development of several downy mildew resistant cultivars in India (Singh *et al.*, 1993; Rathi and Panwar, 1997). The evaluation of a large number of germplasm accessions has resulted in the identification of a sufficient number of sources of resistance (Singh, 1990). The stability of resistance is evaluated at downy mildew hot spots at different centers of AICPIMP and international pearl millet downy mildew nurseries. Pearl millet accessions that have shown high degree of resistance include IP8749-1, IP6147-2, IP6147-4, P181-2, IP8877-3, P462-4, 700481-5-3, P2895-3, P2910-2, P1449-2, IP8695-1, IP8695-4, P3281, RCO11-2, IP8630-1, 75-3 (Singh, 1990) and ICML-12, ICML-13, ICML-14, ICML-15 and ICML-16 (Singh *et al.*, 1990).

4.1.2 Inheritance of resistance

A number of studies aimed at understanding the mechanism of resistance has been conducted, however the results are still inconclusive. This may be due to high level of out-crossing exhibited by both pearl millet and *S. graminicola*. However, several reports indicate that resistance is dominant over susceptibility and is controlled by few or more genes and is quantitative (Das, 1981; Dang *et al.*, 1990; Kataria *et al.*, 1994; Singh, 1995). Further, in a number of recent studies it has been reported that A_1 cytoplasm is not responsible for downy mildew susceptibility (Yadav *et al.*, 1993; Yadav, 1996; Kumar, 2002). Therefore, there is a need to investigate the inheritance pattern due to nuclear genes, with the availability of precise inoculum techniques and highly homozygous resistance and susceptible sources (Singh, 1995; Thakur and Mathur, 2002). Furthermore, the use of molecular marker genetic linkage maps has yielded useful results (Jones *et al.*, 1995; Hash and Witcombe, 1994).

4.1.3 Breeding for resistance

Conventional breeding procedures such as pure line selection, pedigree selection, backcross breeding, induced mutation and recurrent selection have been used with varying degrees of success (Thakur and Mathur, 2002). Recently, marker assisted selection (MAS) procedures have been used, in which selection is for the presence of molecular tags tightly linked to gene(s) controlling resistance. Marker assisted back-cross improvement of downy mildew resistance has been conducted using ICMP-451 as the donor parent and H-77/833-2 as the recurrent parent and a resistant version of HHB-67 has been developed (Sharma, 2001). Hence, MAS can be used to pyramid major genes including resistant genes and it is possible for the breeder to conduct many cycles of selection in a year (Hash and Witcombe, 1996; Witcombe and Hash, 2000).

A number of resistant hybrids and varieties have been developed at different coordinated centers and ICRISAT, including some promising hybrids HHB-67, HHB-68, HHB-94. ICMH-423, Pusa-23 and Pusa 322, MLBH-104, ICMH-356, RHB-30 RHB-90, RHB-121 and a few open pollinated varieties HC-4, HC-10, HC-20, WCC-75, ICMV-221, ICMP-451, ICTP-8203, ICMB-7703 (Table 4). In addition, several downy mildew resistant hybrids like Nandi-5, JKBH-26, Pro-agro-9930, Pro-agro-9444 and Pro-agro-7701 developed by the private sector are extensively cultivated. A new concept in the production of hybrids is the development of top cross hybrids. In these hybrids, the pollen parent is a genetically diverse population instead of an inbred line. It is expected that these hybrids will have more durable resistance than single cross hybrids. Recently top cross hybrids namely JBH-1 (Madhya Pradesh) and HHB-146 (Haryana) have been released.

4.2 Other control measures

4.2.1 Cultural

Cultural methods include crop sanitation, overplanting and roguing of diseased plants, manipulation of planting dates, crop rotation, host nutrition, inter and mixed cropping, deep ploughing and soil solarisation, use of trap crop, and avoidance of monoculture (Jeger *et al.*, 1998). Seed infested with oospores is a primary source of infection and is important in dissemination of the pathogen from one place to another. Use of clean seed for sowing, removal and burning of plant debris soon after harvesting, and deep ploughing of the field in summer to expose the oospores to high temperature have been found effective in the reduction of primary infection (Thakur, 1992).

The late sown crop has been reported to have higher downy mildew incidence than the early-sown crop (Chahal *et al.*, 1978). Early planting has resulted in reduced disease incidence and increased yield, probably because of relatively high temperature and low humidity, and absence of sporangial inoculum in the northern parts of India.

Roguing and destruction of infected plants between 15-30 days after sowing has been recommended as a means of reducing secondary infection (Thakur, 1980). However, this is a cumbersome process requiring labour and is not being practiced by farmers. Transplanting 20 days old seedlings resulted in less downy mildew. The direct seeded hybrid HB-3

Table 4. Downy mildew reaction (% incidence) of released hybrids/varieties/lines in India (1992-2003).
Source : All India Coordinated Pearl Millet Improvement Project Reports, 1992-2003

No.	Genotype	Year											
		1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003
1	HHB-67	6.7	10.6	11.0	6.9	14.1	2.8	2.6	4.8	7.4	8.3	6.7	20.3
2	GHB-30	2.9	2.9	2.8	3.0	-	-	-	-	-	-	-	-
3	HHB-60	2.5	4.0	4.0	4.5	6.0	-	-	-	3.2	-	-	-
4	Pusa-23	2.3	2.9	4.1	5.1	6.1	1.4	1.8	6.3	1.4	0.57	-	-
5	HHB-50	5.8	2.7	10.3	7.8	-	-	-	-	-	-	-	-
6	843-A	9.7	10.0	19.0	18.0	-	-	-	-	-	-	-	-
7	81-A	4.8	6.0	6.8	8.0	-	-	5.0	8.0	6.0	3.0	-	-
8	H-77-833-2	12.2	18.9	18.2	20.0	-	70.3	88.1	87.1	79.7	94.3	-	-
9	H90/4-5	2.5	3.3	7.2	5.4	-	-	-	-	-	-	-	-
10	Pusa 605	-	-	-	-	-	-	3.5	2.3	1.8	1.4	-	-
12	GHB-306	-	-	-	-	-	5.7	5.5	12.3	15.4	7.2	-	-
13	HHB-94	-	-	-	-	-	-	-	5.6	8.4	3.4	-	-
14	Sabauri	-	-	-	-	-	0.2	0.2	3.1	1.4	3.5	-	-
15	Sharadha	-	-	-	-	-	-	0.3	4.3	2.0	2.2	-	-
16	ICTP-8203	+	+	+	+	+	4.7	4.5	8.3	6.4	1.9	-	-
17	PB-106	+	+	+	+	+	+	+	+	+	+	0.9	-
18	RHB-90	+	+	+	+	+	+	+	+	+	+	0.9	-
19	HHB 117	+	+	+	+	+	+	+	+	0.0	1.3	3.2	4.6
20	RHB 121	+	+	+	+	+	+	+	+	+	+	1.8	2.3
21	HHB 146	+	+	+	+	+	+	+	+	0.0	0.0	0.5	0.0
22	HB-3	87.7	79.2	79.4	80.6	88.4	93.4	91.9	91.1	82.1	95.0	81.1	100.0

\- = The disease reaction is not available
\+ = Hybrids-not released

had a 65% downy mildew incidence compared with only 22% incidence in the transplanted crop (Thakur, 1986).

4.2.2 Chemical control

The systemic fungicide metalaxyl has been found highly effective in controlling pearl millet downy mildew. It is absorbed through leaves, stem and roots and can be applied in various formulations (Williams, 1984). Fungicidal seed treatment followed by foliar spray is very effective and economical. Seed treatment with Apron SD-35 @ 2g a.i./kg completely protected susceptible pearl millet plants (HB-3) up to 30 days. A metalaxyl foliar spray at 20 days or applied twice at 20 and 30 days of plant growth gave less disease at harvest time. Seed treatment followed by a foliar spray of Ridomil-25W.P. @ 1g a.i./litre of water is very effective in controlling the disease even in downy mildew 'sick' plots. These treatments improved plant growth and yield significantly (Dang *et al.*, 1983). The disease control efficacy of metalaxyl used as seed treatment was altered when fungicide treated seeds were sown in fertilizer amended soil. Disease control was 48% when fungicide-treated seeds were sown in unamended soil, whereas in P and K amended soil, only 17-25% disease control was recorded (Gangopadhyay and Dang, 1988). Ridomil-25 W.P. has now been replaced by a mixed formulation of metalaxyl with the protective fungicide mancozeb (Ridomil MZ). Seed treatment with Apron SD-35 @ 2g a.i./kg seed followed by foliar application with Ridomil-MZ (2g a.i./litre) has been recommended for complete control of the disease (Singh and Shetty, 1990). Recently a liquid formulation of metalaxyl @ 3ml/kg of seed (Apron XL-35ES) has been found effective in controlling the disease for 25 days (M.S. Panwar, *unpublished*)

5. CONCLUSIONS

Downy mildew of pearl millet remains an important constraint in sustainable pearl millet production in India and other countries. Resistant cultivars remain the most economical and practical means of disease control. The resistance of hybrids/varieties must be regularly monitored for the incidence of downy mildew. Need-based application of metalaxyl seed treatment should gain momentum and curb the progression of the disease. The recent development of top cross hybrids is likely to be more durable than single cross hybrids. To augment these efforts, cultural practices with a focus on crop rotation and sanitation should be adopted over a wider scale.

6. LOOKING AHEAD

Though impressive research over the years has been carried out on various aspects of pearl millet downy mildew, there are several questions which have remained unanswered. In the coming years, therefore, we need to focus on the following aspects for developing and implementing viable and meaningful disease management strategies:

- determination of whether cultivar responses across locations are due to the target pathogen or environmental variation;
- identification and exploitation of specific resistant genes through gene pyramiding and their deployment through harnessing the more precise techniques now available;
- elucidation of the relative role of sexual and asexual spores, micro-climate and soil conditions for disease development; thereby facilitating effective development of disease prediction models.

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GENETIC AND PATHOGENIC VARIABILITY AMONG ISOLATES OF *SCLEROSPORA GRAMINICOLA*, THE DOWNY MILDEW PATHOGEN OF PEARL MILLET

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1. INTRODUCTION

Downy mildew, caused by the oomycete pathogen *Sclerospora graminicola* (Sacc.) Schröet, is economically the most important disease of pearl millet (*Pennisetum glaucum* (L.) R. Br.) in Asia and Africa. The pathogen induces systemic infection in pearl millet plants that manifests itself through foliar chlorosis and panicle malformation (also called "green-ear" or "crazy top"). The fungus is heterothallic (Michelmore *et al*., 1982) and reproduces both by sexual and asexual means, and is therefore highly variable (Ball and Pike, 1984; Thakur *et al*., 1999). The commercial cultivation of a number of genetically homogeneous single-cross F_1 hybrids of pearl millet in India has led to increased virulence in *S. graminicola* populations, thus shortening the useful life of the hybrid cultivars (Thakur *et al*., 1999). Monitoring virulence changes in the pathogen population, identifying resistance to specific and multiple pathotypes, and directing pearl millet breeding programmes towards strategic utilization and deployment of resistance genes form the basis of a long-term downy mildew management research at ICRISAT (Thakur, 1999).

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Pathogenic and genetic variability in *S. graminicola* has been studied through the collaborative International Pearl Millet Downy Mildew Virulence Nursery (IPMDMVN) project, disease surveys in farmers' fields, the use of host differential lines in greenhouse experiments, and the use of different DNA marker techniques (Thakur, 1999). Because of the lack of well-defined R genes and host differential lines, application of molecular marker technologies would be useful for studying the genetic changes and the associated virulence changes in the pathogen populations, if any. Further, virulence types are subject to selection by host genotype, which limits their use in diversity studies (Kolmer *et al*., 1995). Therefore, the molecular markers will be able to detect genetic changes that may occur at various loci in the fungal genome, including those closely associated with virulence. Several molecular markers, such as restriction fragment length polymorphism (RFLP), simple sequence repeats (SSR), random amplified polymorphic DNA (RAPDs), and internally transcribed spacers (ITS) of rDNA have been used to study genetic variability in pathogen populations. Recently, amplified fragment length polymorphism (AFLP) has been found to be a better method for detecting genetic variability among fungal pathogens (O'Neill *et al*., 1997; Restropo *et al*., 1999). In this study, we assessed the genetic variability among the isolates of *S. graminicola* using AFLP, as well as virulence reactions on host differential lines and the influence of the host genotypes on genetic changes in the isolates, and how these three studies correlate.

2. MATERIALS AND METHODS

2.1 *Sclerospora graminicola* isolates

Fifteen isolates of *S. graminicola* collected from different pearl millet genotypes in several regions in India were maintained either on the same host cultivar from which spores were collected, or on an alternative susceptible genotype in greenhouse isolation chambers for a number of asexual generations (Table 1). We assume that through successive asexual generations on a single host, the population becomes genetically homogeneous. Three of the 15 isolates (*Sg* 139, *Sg* 150 and *Sg* 151) were maintained simultaneously on two different host genotypes, thus making the effective number of isolates only 12. Each isolate was maintained on pot-grown seedlings of a specific host genotype in a polyacrylic isolation chamber (0.15 m^3) in the greenhouse to eliminate any chance of contamination from the other isolates or other pathogenic spores. For collection of spores, Mira cloth (Calbiochem, USA) pieces (2 × 5 cm) were

Table 1. Isolates of *Sclerospora graminicola* (*Sg*) collected from different pearl millet genotypes and locations in India and maintained at ICRISAT-Patancheru.

Pathogen isolate*	Collection host	Location	Maintenance host
Sg 008 (1992)	NHB 3	Patancheru, AP	NHB 3
Sg 013 (1992)	7042S	Patancheru, AP	7042S
Sg 021 (1993)	MLBH 104	Ghari, Maharashtra	7042S
Sg 048 (1994)	7042/HB3	Mysore, Karnataka	852 B
Sg 139 (1997)	Nokha local	Jodhpur, Rajasthan	N*okha* local
Sg 139 (1997)	Nokha local	Jodhpur, Rajasthan	7042S
Sg 140 (1997)	7042/HB3	Jamnagar, Gujarat	7042S
Sg 149 (1997)	Kushwadara	Gwalior, MP	7042S
Sg 150 (1997)	MBH 110	Jalna, Maharashtra	834B
Sg 150 (1997)	MBH 110	Jalna, Maharashtra	MBH 110
Sg 151 (1997)	81A	Durgapura, Rajasthan	*Nokha* local
Sg 151 (1997)	81A	Durgapura, Rajasthan	7042S
Sg 153 (1997)	7042S/NHB 3	Patancheru, AP	7042S
Sg 200 (1998)	ICMH 451	Ramgarh, Gujarat	ICMP 451
Sg 212 (1998)	PG 5822	Pura ki Dhani, Rajasthan	ICMP 451

*Year of collection in parentheses.

wetted in sterile distilled water and wiped gently over the sporulating pearl millet leaves. These Mira cloth pieces were transferred to sterilized small glass vials and the samples were stored at 4°C overnight.

2.2 DNA extraction

For each isolate, 4 pieces of Mira cloth containing spores from the infected plants in the greenhouse were soaked in 5 ml sterile distilled water. The water-suspended spores were collected in a 2 ml tube by centrifugation at 12,000 x g for 10 min. Genomic DNA was extracted as described by Sastry *et al*. (1995).

2.2.1 AFLP analysis

AFLP analysis was carried out using a commercial kit (Life Technologies, USA) following the manufacturer's protocols, with slight modifications. Four *Eco*RI (+2) primers and six *Mse*I (+3) primers were used in 12 combinations for amplification. Selective primers provided in the kit were used and the amplification was carried out according to the manufacturer's protocol. The *Eco*RI primer was labeled with [γ-^{32}P]-ATP (3000 Ci/mmol) and the PCR products in 5.0 µl sub-samples were separated by electrophoresis on 6% denaturing polyacrylamide DNA sequencing gel containing 7.5 M urea. Autoradiograms were obtained using Kodak X-Omat film.

2.3 Virulence assay

In a greenhouse experiment, pot-grown seedlings of 12 pearl millet genotypes were spray inoculated with sporangial suspensions (1×10^5 sporangia ml^{-1}) of S. *graminicola* isolates. The experiment was conducted in a completely randomized block design in three replications with 30-40 seedlings in each replicate. Sporangial inoculum for each isolate was obtained from pot-grown seedlings of respective host genotypes and maintained in isolation chambers in a greenhouse. Scoring for downy mildew infection in seedlings was done two weeks after inoculation and incidence assessed as percentage of infected seedlings with chlorosis and sporulation on leaves. The experiment was repeated once.

2.4 Statistical analysis

Analysis of variance of downy mildew incidence data was done using GENSTAT statistical package (Rothamsted Experiment Station, Herpenden, Herts AL52JQ, UK) on both original and transformed scales for both the tests. Since error mean squares on both scales were similar, the mean incidence data were presented on the original scale.

The presence or absence of each band in the AFLP gel or autoradiogram was scored as 1 or 0, respectively. In the case of virulence studies, disease incidence of < 20 % was taken as 0 and >20% as 1 for cluster analysis. Cluster analysis of the AFLP and virulence data was based on similarity indices between pathogen isolates using the Clustan Graphics software package (Clustan Limited, UK).

3. RESULTS

3.1 Genetic characterization of isolates

In AFLP analysis, five of the 12 primer combinations showed high levels of polymorphism for three of the 15 isolates of *S. graminicola,* each maintained on two host genotypes (Table 2). A total of 50 to 70 AFLP markers were scored for each of the primer combinations and the percentage polymorphism (number of polymorphic bands/total number of bands scored) varied from 25 to 60 among the isolates. A representative autoradiogram of AFLP using the primer combination E-AT and M-CTA is shown in Figure 1. The dendrogram based on the similarity index data of AFLP markers identified five major groups among the 15 isolates (Figure 2): Group I with one isolate, Group II with three isolates, Group III with five isolates, Group IV with three isolates, and Group V with three isolates.

The AFLP patterns of the same isolates maintained on different host genotypes for a varying number of asexual generations showed significant differences (Table 2). The isolate *Sg* 139, although multiplied on pearl millet genotype 7042S, could be distinguished from the same isolate maintained for several generations on *Nokha* local by 1-7 polymorphic AFLP markers in different primer combinations (Table 2). Although the grouping of these isolates would indicate the closeness of *Sg* 139 isolates maintained on either *Nokha* local or 7042S (Figure 2), the molecular changes in the pathogen isolates as revealed by the AFLP patterns are indicative of the potential of the isolates for genetic adaptation to different hosts. In contrast, *Sg* 151 collected from pearl millet genotype 81A, but

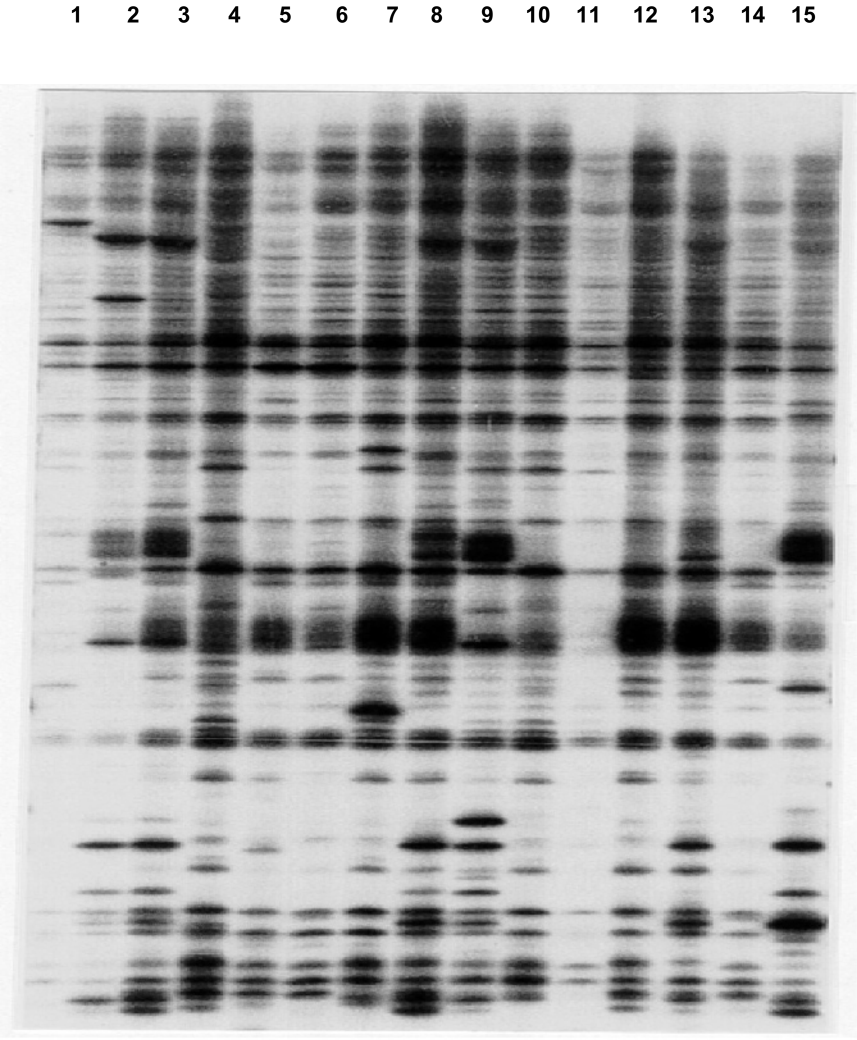

Figure 1. A representative autoradiogram showing the AFLP analysis of 15 isolates of *S. graminicola*, using the primer combination E-AT and M-CTA (the order of the isolates is the same as in Table 1).

maintained on *Nokha* local and 7042S, was placed into two different groups (Groups II and III, Figure 2) suggesting a significant genetic change as a

Table 2. Level of polymorphism for the same pathogen isolates of *S. graminicola* maintained on different host genotypes of pearl millet.

Primer combination used		Number of polymorphic bands observed					
		Sg 139 maintained on		Sg 151 maintained on		Sg 150 maintained on	
EcoRI primer	MseI primer	7042S	Nokha local	7042S	Nokha local	834B	MBH 11(
E- AT	M- CTG	3	1	9	2	2	10
E- AT	M- CAG	6	1	6	5	1	8
E- AT	M- CTA	1	2	7	3	0	10
E- TA	M- CTG	1	2	6	4	0	12
E- TA	M- CAG	1	0	2	1	1	1

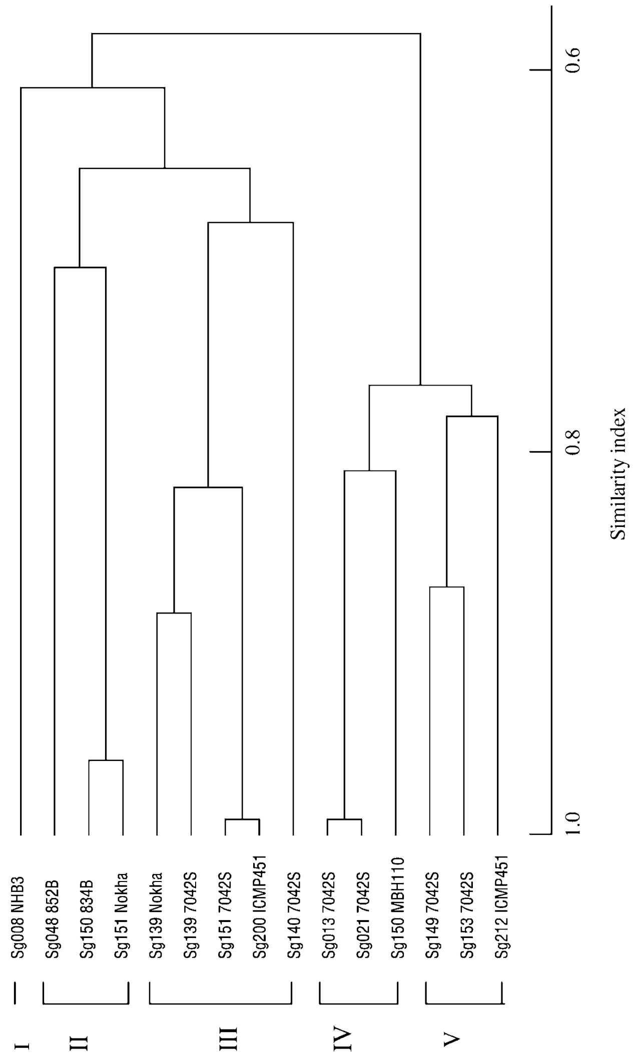

Figure 2. Dendrogram showing the grouping of the 15 isolates of *S. graminicola* based on AFLP markers.

result of growing this isolate on different host genotypes. In these groupings, the number of polymorphic AFLP markers varied between 3 and 11 with various primer combinations (Table 2). Similarly, *Sg* 150 maintained on 834B and MBH 110 was in two separate groups (Groups II and IV, Figure 2).

3.2 Pathogenic characterization of isolates

Downy mildew incidence on 12 pearl millet genotypes by 15 *S. graminicola* isolates showed a high variability, ranging from 0 to 99%. Using a 20% incidence level for the cut-point for resistance (R) and susceptibility (S), *Sg* 048 was most virulent (S reaction in 11 of the 12 pearl millet genotypes), followed by *Sg* 151, *Sg* 150 and *Sg* 139 (S reaction in 6 genotypes), and the least virulent were *Sg* 008 and *Sg* 013 (S reaction only in 2 genotypes) (Table 3). Isolates of *Sg* 151 maintained on two host genotypes showed a differential virulence reaction; the one maintained on *Nokha* local was more virulent (S reaction in 10 genotypes) than that maintained on 7042S (S reaction in 5 genotypes). Cluster analysis of the virulence data classified the 15 isolates into five major groups (Figure 3).

4. DISCUSSION

The genetic changes revealed by the polymorphic AFLP patterns among the different pathogen isolates of *S. graminicola* suggest that an isolate collected from a specific cultivar may adapt to a new cultivar after several cycles of growth on that host. The genetic changes in the isolate may not, however, be reflected in virulence changes. The two isolates, *Sg* 008 and *Sg* 013, collected and maintained on two different host genotypes NHB 3 and 7042S, respectively, for several asexual generations showed different AFLP profiles but similar virulence reactions, suggesting genetic changes in the genomic regions other than those involving virulence genes. In contrast, *Sg* 151, maintained on *Nokha* local and 7042S showed different genetic and virulence groupings, suggesting a significant influence of the specific host genotype on the pathogen virulence. It is possible that these cumulative genetic changes occurring over several asexual generations may be reflected in the avirulence gene of the pathogen resulting finally in altered virulence reaction. In those cases where the isolate was maintained on the original collection host for several asexual generations, there were fewer polymorphic AFLP markers than in those maintained on different hosts, suggesting that some of these genetic changes may have taken place in the pathogen to assist in the adaptation to the new host genotype.

Table 3. Phenotypic reactions (R and S)[a] and downy mildew incidence (%) on 12 pearl millet host differentials to 15 isolates of *S. graminicola* in two greenhouse tests in 2000-2001.

Isolate	Host differentials											
	IP 18292	IP 18293	700651	P310-17	P 7-4	MBH 110	852B	NHB3	834B	ICMP 451	843B	7042S
Sg 008-NHB3	R	R	R	R	R	R	R	S	R	R	R	S
Sg 013-7042S	R	R	R	R	R	R	R	S	R	R	R	S
Sg 021-7042S	R	R	R	R	R	R	R	S	R	S	S	S
Sg 048-852B	S	S	S	S	S	S	S	S	R	S	S	S
Sg 139-Nokha	S	S	R	R	R	R	R	S	R	S	S	S
Sg 139-7042S	S	S	R	R	R	R	R	S	R	S	S	S
Sg 140-7042S	R	R	R	R	R	R	R	S	R	R	S	S
Sg 149-7042S	R	R	R	R	R	R	R	S	R	S	S	S
Sg 150-MBH110	R	R	S	R	R	S	S	R	S	R	R	S
Sg 150-834B	R	S	R	R	R	S	S	R	S	R	R	S
Sg 151-Nokha	S	S	S	S	S	R	S	S	R	S	S	S
Sg 151-7042S	S	R	R	R	R	R	R	S	R	S	S	S
Sg 153-7042S	R	R	R	R	R	R	R	S	R	S	S	S
Sg 200-ICMP451	R	R	R	R	R	R	R	S	R	S	S	S
Sg 212-ICMP451	R	R	R	R	R	R	R	S	R	S	S	S

[a]R indicates resistance (<20% downy mildew incidence) and S indicates susceptible (>20% downy mildew incidence) based on mean of 2 tests and 3 replications in each test, with 30-40 seedlings per replication.

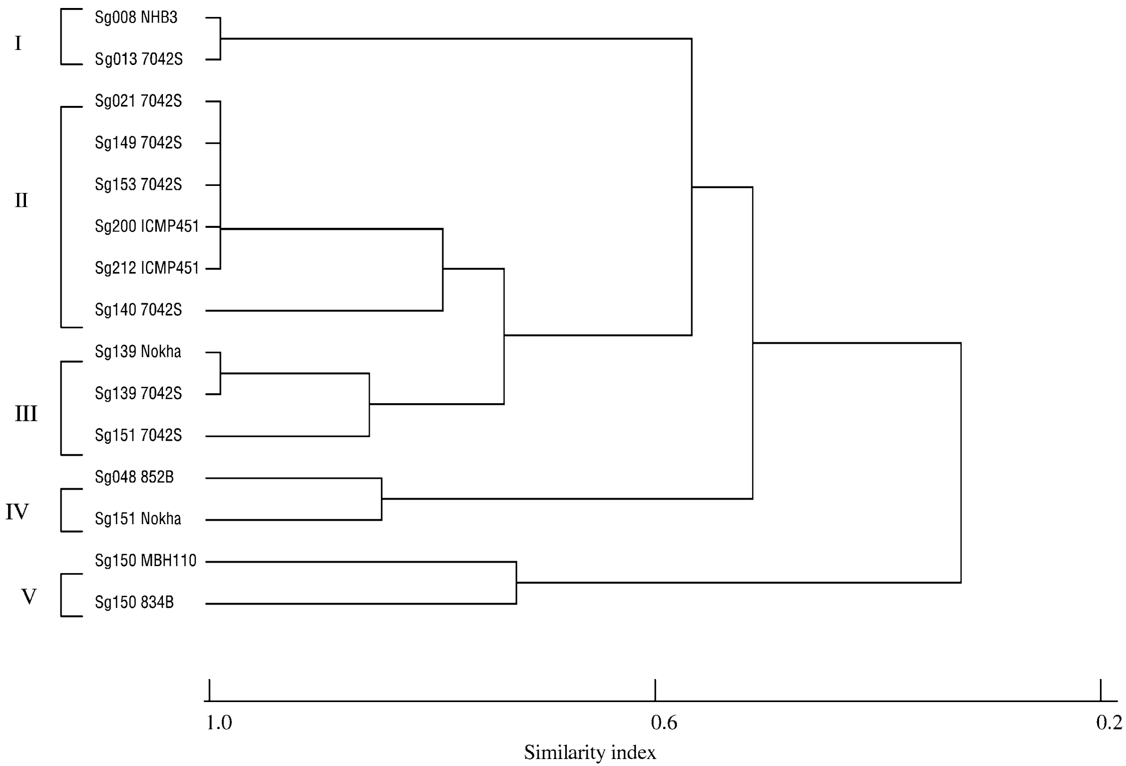

Figure 3. Dendrogram showing the grouping of the 15 isolates of *S. graminicola* based on virulence reaction against 12 host differentials.

However, the number of cycles required for the genetic change in the pathogen isolate depends on the nature of the pathogen isolate and the host genotype. This is illustrated in the case of isolate *Sg* 151, which was grouped differently in virulence reaction- and AFLP-based clustering.

One of the reasons for differential groupings of isolates by virulence reaction and AFLP markers could be that the AFLP is based on a larger coverage of the pathogen genome, including the avirulence genes, whereas the virulence reaction is mostly confined only to the region of avirulence gene. Similar studies by other workers have also shown poor correlation between AFLP grouping and virulence analysis (Zhong and Steffenson, 2001). In other cases (Levy *et al.*, 1993; Kolmer, 1993) genetic variability of the fungal isolates using the molecular markers and virulence were highly correlated. Thus, the initial AFLP patterns may represent alleles (restriction sites) that are prevalent when the spores are grown on one host, but switching to another host will allow a previously rare genotype to be selected and become predominant. It has been shown that rapid selection for virulent pathotypes occurs on different hosts from asexual spore population (Thakur *et al.*, 1992).

Genetic differences between any two pathogen isolates were little affected by maintaining on the same host genotype. The two isolates *Sg* 200 and *Sg* 212, maintained on the same host genotype ICMP 451, showed 12 and 52 polymorphic bands differences respectively. This would suggest that *Sg* 212, the isolate from Rajasthan, was genetically different from the isolate *Sg* 200 from Gujarat as they had different collection hosts from different locations. In contrast, *Sg 200*, collected and maintained on ICMP 451, and *Sg* 151, collected from 81A and maintained on 7042S, grouped together, suggesting genetic similarity between the isolates. Another significant observation is that the field isolate from Patancheru, Sg153, and the one purified from it, *Sg* 013, were in different subgroups under a major group, suggesting the occurrence of genetic heterogeneity in the pathogen population from the same field.

The emergence of a new virulent pathotype in an asexual population need not be due to genetic recombination alone, with other factors such as host cultivar-directed selection for specific virulence in a variable pathogen population also contributing (Leonard, 1987; Thakur *et al.*, 1992). Specificity may be the outcome of a number of genetic interactions between the plant and pathogen. When a host-selected pathogen is used to infect a new host cultivar, genes that are involved in the early recognition events and virulence may undergo changes that may condition the infection process. Some of these genetic changes are reflected in the virulence characteristics of the different pathogen isolates.

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GENETIC AND MOLECULAR CHARACTERISATION OF *PLASMOPARA HALSTEDII* ISOLATES FROM HUNGARY

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1. INTRODUCTION

Plasmopara halstedii (Farl.) Berl. et de Toni, the downy mildew pathogen of sunflower (*Helianthus annuus* L.) causes devastating disease of this crop worldwide. Mildewed sunflower plants do not usually produce viable seed and, once they are systemically infected, they cannot recover from the disease. The pathogen colonises the underground tissues of young seedlings resulting in a typical systemic infection of the whole plant. Under favourable conditions, the fungus sporulates on affected leaves from which sporangia may spread wind-blown causing secondary local infections on adjacent plants. Recently, an increasing number of such local infections give rise to secondary systemic infections whereby the fungus may enter the newly developing seed in a latent form permitting seed transmission unnoticed.

Over the last 14 years, a dramatic change has become evident in the *P. halstedii* populations in many countries, as far as virulence and sensitivity to phenylamide fungicides are concerned. In Europe, an increasing number of pathotypes, each with distinct virulence structure, have been identified, and at least 5 such variants have already been distinguished from Hungary (Virányi and Gulya, 1995). Furthermore, two alternative hosts of this fungus, the common cocklebur (*Xanthium strumarium*) and the common ragweed (*Ambrosia artemisiifolia*) have recently been found in Hungary, with both having the potential of serving as an additional source of inoculum and/or gene pool for new pathogenic forms (Virányi, 1984; Walcz *et al.,* 2000).

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Since fungal diversity of this kind has consequences in both disease epidemiology and breeding for resistance, there is a need to identify the virulence structure of local fungal populations and to monitor the changes over time. The traditional methodology for virulence testing (Gulya *et al.,* 1998; Tourvieille *et al.,* 2000b) requires a standard set of sunflower differential lines, considerable bench space, and the results may sometimes be unclear. Consequently, alternative methods, such as the use of DNA polymorphism, would help in advancing studies of population diversity. Molecular methods have been used for studying variability of oomycete fungi (Cooke *et al.,* 2002). However, little is known so far on the molecular nature of *P. halstedii* and the published data show very little variation between pathotypes and isolates (Roeckel-Drevet *et al*., 1997; Tourvieille *et al.,* 2000a; Intelmann and Spring, 2002).

In our research programme we have been studying the diversity of the Hungarian populations of *P. halstedii* collected from cultivated sunflower and from other hosts. We aimed to differentiate the virulence phenotypes of this fungus identified to date in our country by means of DNA polymorphism. For this, we used the RAPD method (Williams *et al.,* 1992), which is suitable for detecting polymorphism despite the absence of sequence information and with only nanograms of DNA required for the tests. Prior to use of this technique, however, we needed to work out a reliable methodology of producing contaminant-free fungal biomass, extracting and purifying total DNA, selecting among primers suitable for amplification, and optimising PCR conditions. Preliminary results gained so far are reported in this paper.

2. MATERIALS AND METHODS

2.1 Production of fungal material

A selected number of field isolates of *P. halstedii*, collected in Hungary (Figure 1) representative of the five pathotypes (Table 1) was the subject of investigation. Genetically uniform material was obtained by single sporing using a modified methodology of Spring *et al*. (1998), which is described below. Single sporangial isolates (SSI) were then propagated on their respective compatible sunflower line and the freshly produced sporangia were re-tested for virulence by inoculating each on a set of standard differentials (Gulya *et al.,* 1998).

Table 1. Ten core field isolates used in the study, representing each pathotype found in Hungary.

Isolate	Year	Pathotype	Location
Bi02	2000	100	Bicsérd
911	1995	100	Yugoslavia
130	1993	700	Sajóhídvég
98	1991	700	Mezőtúr
29-13	1998	700	Bácsalmás-Gödöllő
145	1994	710	Tenk
129	1993	730	Sajóhídvég
101	1991	730	Kisunyom
114	1992	330	Kiszombor
68	1989	330	Kunszentmárton

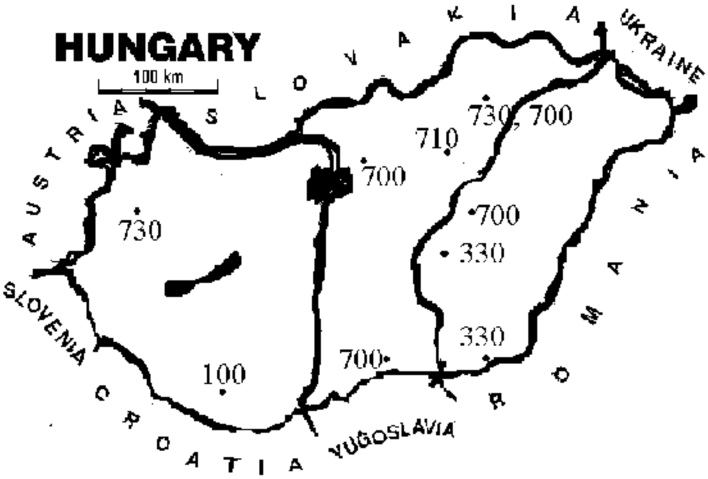

Figure 1. The isolates used in the studies were collected from the main sunflower growing regions in Hungary (numbers indicate pathotype).

2.2 Single sporing

We developed a methodology for the application of genetic analysis to the Hungarian *P. halstedii* populations, based on the method of Spring *et al.* (1998). By means of a dissecting microscope, droplets of sporangial suspensions each containing one single sporangium were isolated and transferred to sunflower leaf segments. After an incubation period of 7-8 days, a proportion of the leaf segments supported fungal sporangia, and these were then further propagated before use. This technique allowed us to get genetically pure material (single sporangial isolates, SSI) without using special equipment such as a micromanipulator.

2.3 DNA extraction

Freshly produced sporangia were collected from sunflower cotyledon leaves with a home-made vacuum collector and stored deep-frozen prior to DNA extraction. Thirty mg of sporangia were mechanically ruptured by using an Ultra Turrax (IKA-T8) in extraction buffer (50 mM Tris-HCl pH 8.0, 100 mM EDTA, 0,15 M NaCl, 3% w/v Sarcosym), followed by equal volume of phenol and chlorophorm/isoamyl alcohol (24:1) treatment. DNA was precipitated with isopropanol and re-suspended in MQ water. Samples were treated with Rnase and the quantity measured by spectrophotometer. Chemicals were purchased from Sigma-Aldrich Co.

2.4 RAPD-PCR procedure

A number of UBC (University of British Columbia, Canada) primers were tested for their suitability for getting amplification products with the DNA samples examined. PCR was carried out in a Biometra Thermocycler 3. PCR reaction was carried out in 25 µl as 1.5 mM $MgCl_2$, 1 x buffer, 50 ng template DNA, 200 µM of each dNTP's, 0.5 µM primer, 1 unit Taq Polymerase (Frementas) running at 94°C for 5 min, followed by 39 cycles of 94°C 15 s, 37°C 1 min, 72°C 1 min and a final elongation at 72°C 5 min. PCR products were separated in 1.3% agarose gel in 0.5% TBE buffer and bands were stained with ethidium bromide and visualised using an UV transilluminator. Data were analysed by MDS (multidimensional scaling analysis) using the SPSS 8.0 software programme.

3. RESULTS AND DISCUSSION

3.1 Single sporing

The single sporing technique allowed us to obtain genetically pure material (single sporangial isolates, SSI) with an acceptable output of 6-8%.

3.2 Diversity of virulence

Some changes in virulence pattern were evident when comparing the virulence pattern of a series of field isolates (sub-populations) with their single sporing progeny. Segregation for virulence had occurred among the majority of field isolates and their SSI (Table 2; compare with Table 1), indicating the parental isolates were mixtures of virulence phenotypes. Similar findings have been reported by Molinero-Ruiz, *et al.* (2002) from Spain.

Table 2. Number of single sporangium isolates (SSI) obtained for each field isolate, and their pathotype.

Field isolate	Number of single sporangium isolates	Pathotype
Bi02	3	100
	1	300
911	1	100
	1	300
	1	330
	1	730
130	5	700
	1	710
98	1	700
29-13	7	700
	2	710
145	3	710
	2	700
129	1	710
101	1	730
	2	700
114	2	330
61	6	330
	1	730

3.3 DNA extraction

Disruption of sporangia by means of an Ultra Turrax homogenisator proved to be the most efficient method for extracting total DNA from *P. halstedii,* with respect to reducing the loss of material and permitting quick release of the cell contents. The different extraction buffers and methods (Roeckel-Drevet *et al.,* 1997; Tourvieille *et al.,* 1998; Intellman and Spring, 2002) we compared did not alter the outcome of the procedure.

3.4 RAPD-PCR procedure

Among a total of 66 primers tested on ten single sporangial isolates (SSI), each representing one of the 10 pathotypes examined (Table 1), 43 appeared to be suitable for producing amplification products (Figure 2). Of these, 15 primers gave polymorphic bands, and there were 55 such bands. However, RAPD analysis carried out with 9 SSIs representing pathotype 700 and 4 SSIs belonging to pathotype 710, using 12 of the above primers gave negative results, i.e. practically no polymorphism within or between pathotypes could be detected**.** The extremely low level of molecular variation between SSIs found corroborates the results described by Roeckel-Drevet *et al.* (1997) using RAPD. Furthermore, Tourvieille *et al.* (2000a) obtained comparable results (80 to 100 % similarity) even when including a wide range of *P. halstedii* isolates of various geographic origins.

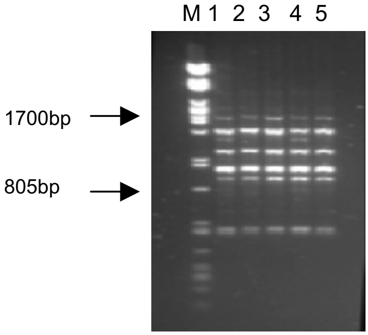

Figure 2. RAPD profiles of the five different pathotypes. Lanes: M, molecular weight marker lambda-Pst; 1, Bi02; 2, 130; 3, 145; 4, 101; 5, 61. Primer: UBC 706.

3.5 MDS analysis

A two-dimensional analysis of RAPD profiles obtained with the ten core (field) isolates distinguished 4 groups (Figure 3). One group contained isolates 114, 29-13, 145, and 129 representing four different pathotypes (330, 700, 710, and 730, respectively). A second group consisted of isolates 911 (pathotype 100), and 98 and 130 (pathotype 700). In the third group there were two isolates, 61 and 101, belonging to pathotypes 330 and 730, respectively. Finally, isolate Bi02 (pathotype 100) was the only member of group 4. This pathotype was once the only pathogenic form throughout Europe, but has seemed to disappear from Hungary during the last decade.

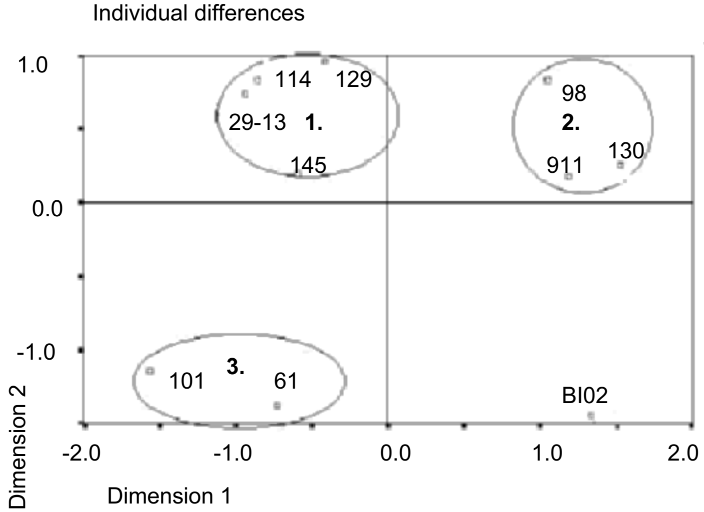

Figure 3. Grouping of isolates by MDS analysis (groups 1, 2, and 3 as indicated; group 4 contains isolate B102 alone).

In conclusion, the polymorphisms we obtained for a number of RAPD primers did not correlate with any virulence phenotype in *P. halstedii*. On the basis of the work presented here, we conclude that finding molecular markers linked with virulence is likely to be difficult, particularly if changes in virulence are assumed to arise from by single point mutations within an avirulence gene, confirming the results of others and as demonstrated for a number of fungi (Knogge, 1996). Because of the spatial and temporal changes of *P. halstedii* populations, however, there is a need to follow the appearance of new variants in nature. Although we couldn't find any specific marker to differentiate virulence phenotypes, we are planning to continue this work using more sensitive and accurate methods. This will include ITS analysis (Says-Lesage *et al*., 2002), IGS analysis (Appel and Gordon, 1996), DNA analysis by minisatellite and simple-sequence repeat primers (Intelmann and Spring, 2002), and isozyme analysis of isolates partly following the methodology of Goodwin *et al.* (1994). Once this work is concluded, we hope to be able to detect minute molecular differences suitable for characterising the diversity of populations of this pathogen in Hungary.

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RESPONSE OF WILD AND WEEDY *CUCURBITA* L. TO PATHOTYPES OF *PSEUDOPERONOSPORA CUBENSIS* (BERK. & CURT.) ROSTOV. (CUCURBIT DOWNY MILDEW)

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1. INTRODUCTION

The genus *Cucurbita* includes ca 14 species native to the New World from the United States south to Argentina. It includes at least five different species domesticated before European contact (Sanjur *et al.*, 2002). In many parts of the world, these domesticated species are widely cultivated as vegetables and, to a lesser extent, as oilseeds, animal forages, and ornamentals. Wild populations of *Cucurbita* often possess disease-resistance genes that are unknown or extremely rare in domesticated populations. This has been demonstrated for resistance to many viruses and fungal pathogens that infect *Cucurbita* (Rhodes, 1964; Provvidenti, 1990; McCreight and Kishaba, 1991; Munger, 1993; Provvidenti, 1993). Extensive research has also been conducted to transfer disease-resistance genes from wild species into modern *Cucurbita* cultivars (Contin and Munger, 1977; de Vaulx and Pitrat, 1979; Washek and Munger, 1983; Whitaker and Robinson, 1986; Herrington *et al.*, 1988a, 1988b, 1989; Robinson *et al.*, 1988; Tasaki and Dusi, 1990).

Pseudoperonospora cubensis (cucurbit downy mildew) is one of the most important foliar pathogens infecting cucurbits (Palti and Cohen, 1980; Thomas, 1996; Lebeda and Widrlechner, 2003). It is widely distributed throughout the world and can inflict major production losses in both open field and protected culture. *P. cubensis* is characterized by large variation in

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pathogenicity, which was recently reviewed by Lebeda and Widrlechner (2003). There is limited information on interactions between various *Cucurbita* species and *P. cubensis;* only three species (*C. maxima, C. moschata, C. pepo*) have been reported as natural hosts of the pathogen (Palti and Cohen, 1980). Limited research and breeding of *Cucurbita* species for resistance to *P. cubensis* have been conducted (Lebeda, 1992). It was suggested that host-parasite specificity between *Cucurbita pepo* and *P. cubensis* is controlled by race-specific factors (Lebeda and Křístková, 1993). Host-parasite specificity among *Cucurbita* species and various *P. cubensis* isolates is a complex phenomenon deserving closer analysis (Lebeda and Křístková, 1993; Lebeda and Widrlechner, 2003).

Considering the past success of efforts to identify disease-resistance genes in wild *Cucurbita* species and our general desire to clarify these host-parasite relationships, we designed the present study to evaluate a set of wild and weedy *Cucurbita* populations, representing a wide phylogenetic cross-section of the genus with relationships to all five domesticated species, for their responses to diverse *P. cubensis* pathotypes.

2. MATERIALS AND METHODS

The first step of this work was to survey the holdings of the National Plant Germplasm System (NPGS) of the United States to develop a representative set of *Cucurbita* species for evaluation. This set focused on wild and weedy taxa, given that many domesticated *Cucurbita* populations have already been evaluated (Chauhan, 1984; Bains and Sharma, 1986; Lebeda and Křístková, 1993; Wessel-Beaver, 1993; Ríos-Labrada *et al*., 1997; Lebeda and Křístková, 2000; Keinath and Du Bose, 2000). Seed samples of this initial set were provided by five NPGS units and included 97 accessions representing 14 taxa and 10 *Cucurbita* species (Table 1).

These 97 accessions (Table 1) were grown in a growth chamber and/or a glasshouse in plastic pots filled with garden soil under day/night temperatures of 25°C/15°C. Five plants were grown from each accession. The plants were regularly irrigated and once per week the nutritional solution Kristalon, version “Fruit and Flower” (Hydro Agri Rotterdam, The Netherlands; N 15%, P_2O_5 5%, K_2O 30%, MgO 3%) was added. No other chemical treatments were applied to the plants during cultivation. Well developed leaves (3 or 4 true leaves) of 6 to 8-week old plants were used for the screening. Tests were carried out on five leaf discs (20 mm in diameter) per plant (Lebeda, 1991).

Altogether, 11 isolates of *P. cubensis* originating from the Czech Republic (8 isolates) and from France, Spain and the Netherlands (one isolate from each country) were used (Table 2). These isolates represent 9

Table 1. Set of *Cucurbita* spp. accessions used for screening.

Taxon	No. of accessions	Supplier[1]
C. argyrosperma C. Huber var. *palmeri* (L.H. Bailey) L. Merrick & D.M. Bate	18	PGRCU
C. argyrosperma C. Huber subsp. *sororia* (L.H. Bailey) L. Merrick & D.M. Bates	25	PGRCU
C. cylindrata L.H. Bailey	1	PGRCU
C. digitata A. Gray	1	PGRCU
C. ecuadorensis H.C. Cutler & Whitaker	3	PGRCU
C. ficifolia Bouche	4	PGRCU
C. foetidissima Kunth	5	WRPIS
C. maxima Duchesne subsp. *andreana* (Naudin) Filov	3	PGRU
C. okeechobeensis (Small) L.H. Bailey subsp. *martinezii* (L.H. Bailey) T.W. Walters & D.S. Decke	7	PGRCU
C. pedatifolia L.H. Bailey	1	PGRCU
C. pepo L.	6	NCRPIS
C. pepo L. var. *fraterna* (L.H. Bailey) Filov	4	NCRPIS
C. pepo L. var. *ovifera* (L.) Harz	1	NCGRP
C. pepo L. var. *texana* (Scheele) Filov	18	NCRPIS

[1] NCGRP = National Center for Genetic Resources Preservation, Fort Collins, Colorado;
NCRPIS = North Central Regional Plant Introduction Station, Ames, Iowa;
PGRCU = Plant Genetic Resources Conservation Unit, Griffin, Georgia;
PGRU = Plant Genetic Resources Unit, Geneva, New York;
WRPIS = Western Regional Plant Introduction Station, Pullman, Washington.

P. cubensis pathotypes (Lebeda and Gadasova, 2002; Lebeda and Widrlechner, 2003) differing substantially in their pathogenicity. For the purpose of this study, no pathogen isolates were available from the New World centers of origin or diversity for *Cucurbita*. The isolates were maintained and multiplied on leaf segments or leaf discs of *Cucumis sativus* cv. Marketer 430, which also served as a susceptible control. Long-term maintenance of pathogen isolates was achieved through storage of cultures (leaves bearing conidiophores) in a low-temperature freezer (-80 C).

Table 2. Set of *Pseudoperonospora cubensis* isolates used for screening of *Cucurbita* spp. accessions (modified from Lebeda and Gadasová, 2002; Lebeda unpublished results).

Differential genotype	*P. cubensis* isolate (PC)/reaction pattern										
	3/00[1]	6/97	1/88	11/00	2/95	6/96	1/00[2]	14/00	12/00	2/00	1/97
Cucumis sativus	+	+	+	+	+	+	+	+	+	+	+
C. melo subsp. *melo*	+	-	+	-	-	-	+	+	+	+	+
C. melo var. *conomon*	-	-	-	-	-	-	-	-	-	-	+
C. melo var. *acidulous*	-	-	-	-	-	-	+	-	+	+	+
Cucurbita pepo var. *pepo*	-	-	-	-	-	-	-	-	-	-	-
C. pepo var. *texana*	-	+	+	+	+	+	-	+	+	+	+
C. pepo var. *fraterna*	-	-	-	-	-	-	-	-	-	-	-
C. maxima	-	-	-	+	+	+	-	+	+	+	+
Citrullus lanatus	-	-	-	-	-	-	-	-	-	+	+
Benincasa hispida	-	+	+	+	+	+	+	+	+	+	+
Luffa cylindrica	-	-	-	-	-	-	-	-	+	-	-
Lagenaria siceraria	-	+	+	+	+	+	+	+	+	+	+

PC = general designation of *P. cubensis* isolates, coding = isolate number/year of collecting, with origin of isolate, [1] = France, [2] = The Netherlands, others = Czech Republic;
- = resistant reaction, no visible symptoms of sporulation or very sparse sporulation;
+ = susceptible reaction (moderate or abundant sporulation).

Inoculation, incubation and disease assessment was carried out following methods described elsewhere (Lebeda, 1986; Lebeda, 1991; Lebeda and Křístková, 1993; Lebeda and Widrlechner, 2003). Final evaluation of phenotypic expression of host-parasite interaction was made 14 days after inoculation according to a 0-4 scale (Lebeda, 1991).

3. RESULTS AND DISCUSSION

From the perspective of phenotypic expression, we observed extensive variation in the response of the studied set of *Cucurbita* spp. and accessions to 11 isolates of *P. cubensis*. Here we will summarize and interpret only a portion of those results. In total, 57 different reaction patterns were recorded with 13 accessions completely resistant, 12 accessions completely susceptible, and 32 accessions expressing pathotype (race) specific patterns.

Table 3. List of taxa and accessions of *Cucurbita* spp. with completely resistant reaction.

Taxon	Accession
C. argyrosperma var. *palmeri*	PI 512201
C. argyrosperma subsp. *sororia*	PI 438832, PI 442345, PI 442348, PI 442358, PI 489696, PI 512209, PI 512218, PI 512219, PI 512221, PI 512222, PI 512223,
C. foetidissima	PI 442200

Table 4. List of taxa and accessions of *Cucurbita* spp. with completely susceptible reaction.

Taxon	Accession
C. maxima subsp. *andreana*	G 5285, G 29253
C. pepo	PI 173681
C. pepo var. *fraterna*	PI 614683
C. pepo var. *ovifera*	NSL 91999
C. pepo var. *texana*	PI 614694, PI 614696, PI 614697, PI 614698, PI 614699, PI 614700, PI 614701

Few *Cucurbita* species (*C. argyrosperma, C. foetidissima*) and accessions exhibited a completely resistant reaction (Table 3) to the 11 isolates. Similarly, relatively few species (*C. maxima, C. pepo*) and accessions expressed a completely susceptible reaction (Table 4). These results showed that in most host-parasite interactions there are various levels of resistance based on clear expression of pathotype and/or race-specificity. However, this was complicated by frequent observation of incomplete resistance characterized by limited sporulation of the pathogen (data not presented here).

Occurrence of pathotype-specific patterns in wild and weedy *Cucurbita* species is summarized in Table 5, which shows that most of the studied taxa displayed pathotype and/or race-specificity. The most variable taxa include *C. argyrosperma, C. foetidissima, C. okeechobensis* and *C. pepo* (Table 5). Specific reaction patterns are not presented in this paper.

Table 5. Occurrence of pathotype (race)-specific patterns in wild and weedy *Cucurbita* spp.

Taxon	No. of tested accessions	No. of reaction patterns	Ratio patterns/ accession
C. argyrosperma var. *palmeri*	18	16	0.89
C. argyrosperma subsp. *sororia*	25	12	0.48
C. cylindrata	1	1	1.00
C. digitata	1	1	1.00
C. ecuadorensis	3	2	0.67
C. ficifolia	4	4	1.00
C. foetidissima	5	4	0.80
C. maxima subsp. *andreana*	3	2	0.67
C. okeechobensis subsp. *martinezii*	7	6	0.86
C. pedatifolia	1	1	1.00
C. pepo	6	5	0.83
C. pepo var. *fraterna*	4	4	1.00
C. pepo var. *texana*	18	9	0.50
C. pepo var. *ovifera*	1	1	1.00

For the first time, pathotype specificity has been described for *C. maxima* and *C. moschata* (Bains and Sharma, 1986; Thomas *et al.*, 1987), which expands our knowledge of pathotype specificity based only on *C. pepo* cultivars (Lebeda and Křístková, 1993). Notably, *C. pepo* cultivars with the fruit types acorn, straightneck and ornamental gourd are more susceptible to *P. cubensis* when compared with zucchini, cocozelle and vegetable marrow (Lebeda and Křístková, 2000). Our recent results support those findings, and demonstrate broad genetic variation in resistance to *P. cubensis* among in wild and weedy *Cucurbita* spp., including *C. pepo*. This at least partially contradicts the conclusions of Paris (2001), who considered *C. pepo* deficient in genes for disease resistance. Data reported in this paper should be valuable for future research and exploitation in breeding cucurbits for disease resistance.

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POTENTIAL AND LIMITS FOR THE USE OF NEW CHARACTERS IN THE SYSTEMATICS OF BIOTROPHIC OOMYCETES

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1. INTRODUCTION

Within the past 10 years, few groups of organisms have been regrouped taxonomically to a similar extent as the oomycetes. Biochemical properties like the cellulosic cell wall and the lysine synthesis pathway raised doubts much earlier on the relatedness with eumycotic fungi, with which the oomycetes share little more than a superficial similarity in their hyphoidal phenotypic organization and in their osmotrophic nutrition. In addition, their diploid life cycle which ends in the fusion of gametangia shortly after meiosis has taken place in the oogonia and antheridia (reviewed by Tommerup, 1981) is unparalleled in any other fungus-like entity. The unique cytological feature of the anteriorly directed flagellum with tripartite hairs finally unravelled the oomycetes as part of the Stramenopiles (Patterson, 1989) or Straminipila, according to the formal diagnosis of the kingdom by Dick (2001). Molecular data supported this view (Leipe *et al.*, 1994). Except for the equivocal discussion about which of the diverse groups of the Straminipila the oomycetes show a closer phylogenetic affinity to, and whether the osmotrophic or phototrophic nutrition is ancestral in that group (for refs. see Dick, 2002a), this rooting is widely accepted nowadays. Most current efforts focus more on the internal phylogeny of the oomycetes, or Peronosporomycetes, following the nomenclature of Dick (2001). With a delay of about 10 years, in comparison to the systematics of angiosperms, molecular tools have become employed more to unravel paraphyletic taxa within the Peronosporomycetes and to form new monophyletic clades, thus inevitably disturbing the traditional system and nomenclatural hierarchy to an almost confusing extent (de Queiroz and Gauthier, 1994; Dick, 2002a).

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It is the scope of this review to shed light on current developments in the systematics of the Peronosporomycetes from a methodological point of view rather than commenting on the taxonomic rearrangements themselves. This will encompass particularly the problems of sample assessment for systematic studies of obligate biotrophic Peronosporomycetes, and therefore will put special emphasis on the downy mildews and similar parasitic associations between plants and the white rusts (Albuginaceae) and parts of the Pythiales. The strengths and weaknesses of the tools currently used in taxonomic analysis of oomycetes are also compared. Finally, it is argued that efforts to improve on the present species classification are still worthwhile, since this provides the basis of any reliable phylogeny.

2. GENERAL PROBLEMS IN TAXONOMICAL STUDIES OF BIOTROPHIC OOMYCETES

The low degree of morphological diversity in oomycetes has been a recurrent problem for taxonomists since the beginning of research in that field. Hall (1996) has reviewed broadly the deficiencies of the morphologic and particularly of the morphometric species concept. Nevertheless, it is still today the first and most convenient step on the way to species determination. Host preference of the parasitic oomycetes, as propagated by Gäumann (1923), was widely accepted as an additionally important and often even more reliable criterion for the classification, since it is based on a broad array of physiological and biochemical requirements. Unfortunately, the latter remain mostly unclear with respect to the mechanisms involved, and the limits of host specificity were seldom tested through infection experiments. This argument was frequently stressed to explain the equivocal classification found in many taxa of the Peronosporomycetes. Thus inadequate species definition led to a wealth of binomials, many of which are accepted by few other than the authorities who were responsible for their creation (for reviews see Constantinescu, 1991; Dick, 2002b). On the other hand, a reliable differentiation of taxa is the inevitable prerequisite for each systematic study (Brasier, 1989).

Ways of escaping from this problem are usually sought by employing alternative characters on the physiological, biochemical or chemical levels (Hall, 1996). However, in biotrophic organisms this aim is fundamentally hampered by the inability to cultivate them on artificial media in order to produce sufficient biomass and fresh samples necessary for the investigation of such properties. Moreover, large parts of the target organisms are embedded in the tissue of their hosts from which they are very difficult to separate (eg see El-Gariani and Spencer-Phillips, this volume). This leaves asexual spores and sporophores (referred to here as sporangia and sporangiophores) as the only accessible structures of the target organism free from contamination by host tissue. Sample amounts of such reproductive structures from field collections of downy mildews are often so minute that not much more than a few sporangiophores and sporangia can be gained (Figure 1). It is one of the major challenges for taxonomists in this field of research to develop techniques which are appropriate to deal with these shortcomings.

Besides traditional reasons, the continuing prevalence of morphometric data and typical host-pathogen association in species determination of pathogenic oomycetes is clearly based on the fact that sample amount and condition is not a significant factor affecting applicability of these convenient techniques. This argument may appear trivial, but both parameters, the amount and the condition of material under investigation, have so far been the factors most limiting the use of other potential taxonomic characters such as karyotypes, specific chemical compounds and isoenzyme patterns. A fourth category, the use of DNA-based characters, has now overcome the obstacle of quantitative sample limitation through the use of PCR-based analyses. The employment of this technique, however, is still restricted mostly to either relatively fresh material or samples especially conserved for it. As a consequence, studies on the molecular phylogeny of biotrophic oomycetes often rely on accessions with only a single or few species. This circumstance can fundamentally undermine the validity of the work if the species were misidentified. This has been documented recently for more than 10% of the samples in the Ascomycetes genus *Phoma* and up to 18% of the Helotiales species used for ITS and SSU-sequencing, respectively, of the nuclear ribosomal DNA, and documented in public data-bases (Table 1; Bridge *et al.*, 2003).

In contrast to the taxonomic work in many other groups of organisms, sampling of biotrophic oomycetes is often done more accidentally than intentionally. Except for some agronomically important "celebrities" (eg the downy mildews *Bremia lactuca* on lettuce, *Plasmopara viticola* on vine, *Pseudoperonospra cubensis* on cucumber and *Sclerospora graminicola* on pearl millet), which are studied intensively for epidemiological reasons, most sister taxa pathogenic to uncultivated host plants are difficult to collect from the field at a certain time, unless they are extremely frequent and occur on very common hosts like a few white rusts (eg *Albugo candida* on *Capsella bursa-pastoris*) and some downy mildews (eg *Peronospora chenopodii* on *Chenopodium album*). A possible resolution for this problem is the use of herbarium specimens. Such

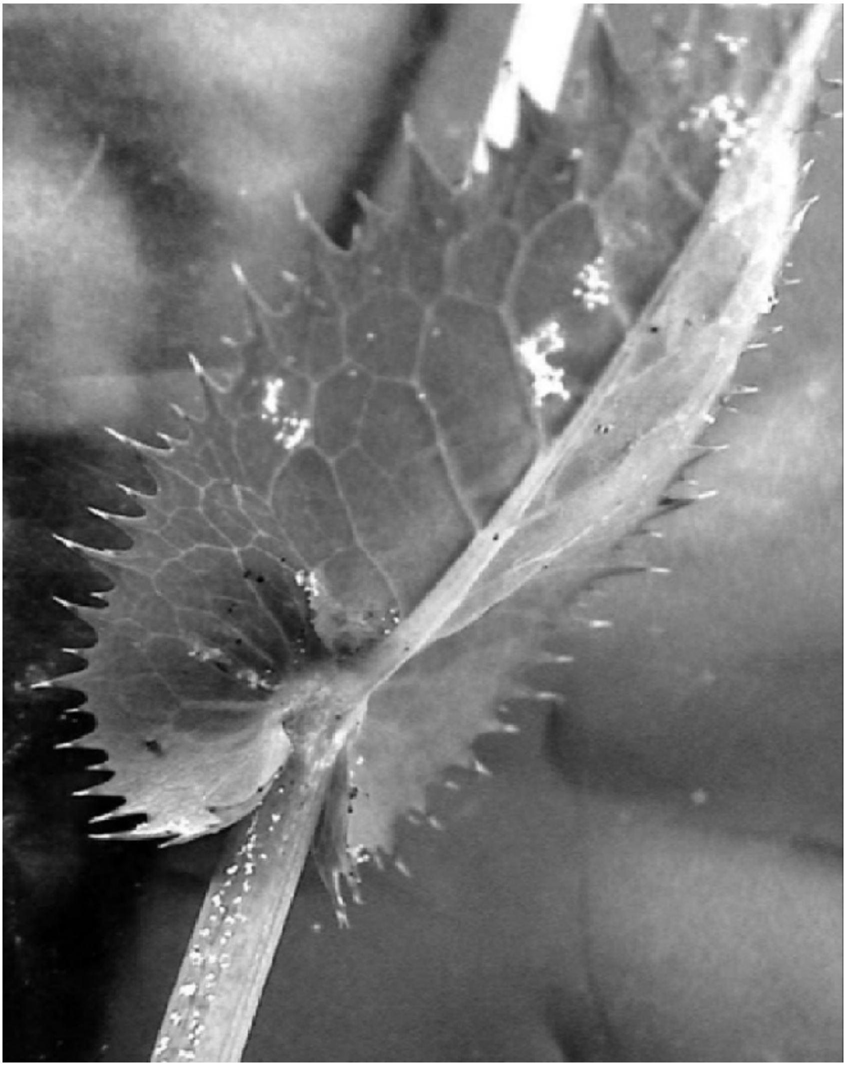

Figure 1. Scattered sporulation of *Bremia lactucae* on the stem and leaf surface of *Sonchus arvensis*.

collections, traditionally used as an indispensable source for taxonomic work in angiosperms, are often the unique basis for comparative infra-generic studies of pathogenic oomycetes. In general, voucher specimens contain preserved material of the pathogen attached to its host. It is usual that the host tissue is mostly in a much better condition than the sparse and fragile hyphoidal structures of the pathogen presented on the plant surface. With careful handling, however, sufficient sporangiophores and sporangia normally remain intact between the relief-forming structures of the plant surface. Unfortunately, in recent times the deposition of voucher specimens has been neglected in some studies, perhaps due to the conviction that they would be destroyed anyway and worthless for later investigations. The opposite is true if efforts are enhanced to develop appropriate techniques for the use of such material for discovering markers other than morphological characters as well.

Table 1. A critical comparison by Bridge *et al.* (2003) of sequence data of the 18S rDNA (SSU) from species of the Helotiales and of the ITS region from *Amanita* and *Phoma* species as deposited in the EMBL database. Data sets were selected to contain depositions from different laboratories world-wide and elaborated at different times.

	Helotiales	*Amanita*	*Phoma*
Number of sequences	100	51	55
Misidentified organisms *	18	8	6
Dubious/chimeric samples	4	5	3
Insufficient length of sequence to align	5	5	14
Study unpublished	21	11	33
Sample un-vouchered in public collections	30	19	30

* Detailed information on mis-identifications is provided in the original paper.

3. MICROMORPHOLOGY – A TAXONOMIC SUPPLEMENT THAT MAKES SENSE?

Hall (1996) has emphasised that ultrastructural characters were chronically under-studied in downy mildews. This is certainly true and has not much changed since 1996. In contrast to morphometric characters which were frequently shown to be little more than rough guides for species classification due to their high variability even in the same individual (eg Delanoe, 1972; Skidmore and Ingram, 1985; for other examples see refs. in Hall, 1996), ultrastructural features have unequivocally demonstrated their taxonomic and phylogenetic value in many fields of plant systematics (for refs see Spring and Buschmann, 1998). The most prominent examples are the use of pollen structure (Faegri and Iversen, 1975), trichome morphology (Hummel and Staesche, 1962) and epicuticular wax ultrastructure (Barthlott, 1993).

Very few studies have been published on ultrastructural characters of biotrophic oomycetes. Shaw (1981) presented an interesting tree for the phylogeny of Peronosporales in which the mode of operculate germination was integrated with other characters.

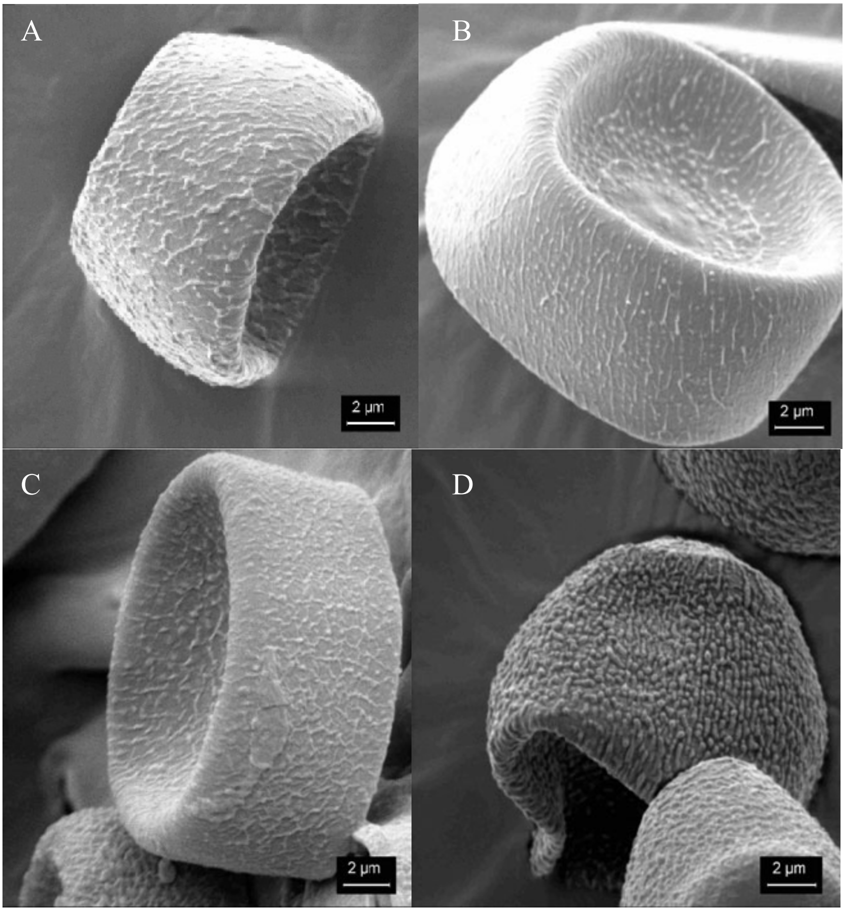

Figure 2. Typical surface ornamentations on sporangia of *Albugo* spp.: A, *A. amaranthi* from *Amaranthus retroflexus*, irregular ornamentation; B, *A. hydrokotyles* from *Hydrokotyles laxiflora*, striate ornamentation;, C, *A. tragopogonis* from *Helianthus annuus*, reticulate ornamentation; D, *A. candida* from *Capsella bursa-pastoris*, nodular ornamentation in lines. SEM micrographs made by M. Thines (2003).

The structural distinctivness of the haustoria has been reviewed by Bhandari and Mukerji (1993) and was a major argument for the separation of *Hyaloperonospora* from *Peronospora* in a recent revison made by

Constantinescu and Fatehi (2002). The latter authors additionally found other micromorphological characters, like the presence or absence of callose plugs, the branching type of sporangiophores and the surface appearance of sporangia which helped to distinguish *Perofascia* from *Peronospora* s. str. and *Hyaloperonospora*. All of these characters were visible in light microscopy, except for the surface structure for which scanning electron microscopy (SEM) (Shiraishi *et al.*, 1975) was necessary. Hall (1996) referred to a few other examples in the genera *Peronospora* and *Basidiophora* where surface ornamentation of sporangia proved to be useful for species delimitation. However, these characters have not yet been widely investigated.

We have recently tested the potential of SEM on a broader range of downy mildew taxa with the intention of checking the variability of characters, and also to explore additional features for use in routine classification. The overall experience showed that easily accessible structures of pathogenic oomycetes on the host surface provide a far broader array of characteristics than expected before this investigation. Thus, ornamentation is not only divided into smooth or verrucose (c.f. Constantinescu and Fatehi, 2002), but there are several additional types like irregular, striate, reticulate and nodular structures (Figure 2) or sharp and fading transitions from smooth to ornamented surface parts (Figure 3, A and B). Multiple sampling of different isolates, wherever possible, has revealed a low degree of intra-specific variability in surface ornamentation among populations in the white rust group, and a survey of *Albugo* is in preparation for publication. However, this cannot be generalized for other taxa. In different genera, other characters may be significant for taxonomic studies. Besides the already mentioned operculate structure for germination, the connecting site of sporangia and sterigmata could be of interest as well (Figure 3). The latter, for instance, can be acuminate (eg some *Peronospora* ssp.) or papillate (eg some *Plasmopara* ssp.), and solid at the tip or with a hole.

In any case, even if the degree of possible convergent developments in independent groups has not yet been sufficiently explored (with the exception of opening structures in zoosporangia) and synapomorphies may be difficult to be traced, the value of ultrastructural characters for classification can hardly be disputed. One of the unchallenged advantages of these phenotypic characters lies in the fact that they can be documented on dried herbarium specimens even decades after their collection, and this requires a minimum amount of sample. In combination with molecular tools, ultracytological characters could become an important instrument for the pre-selection of samples in phylogenetic studies.

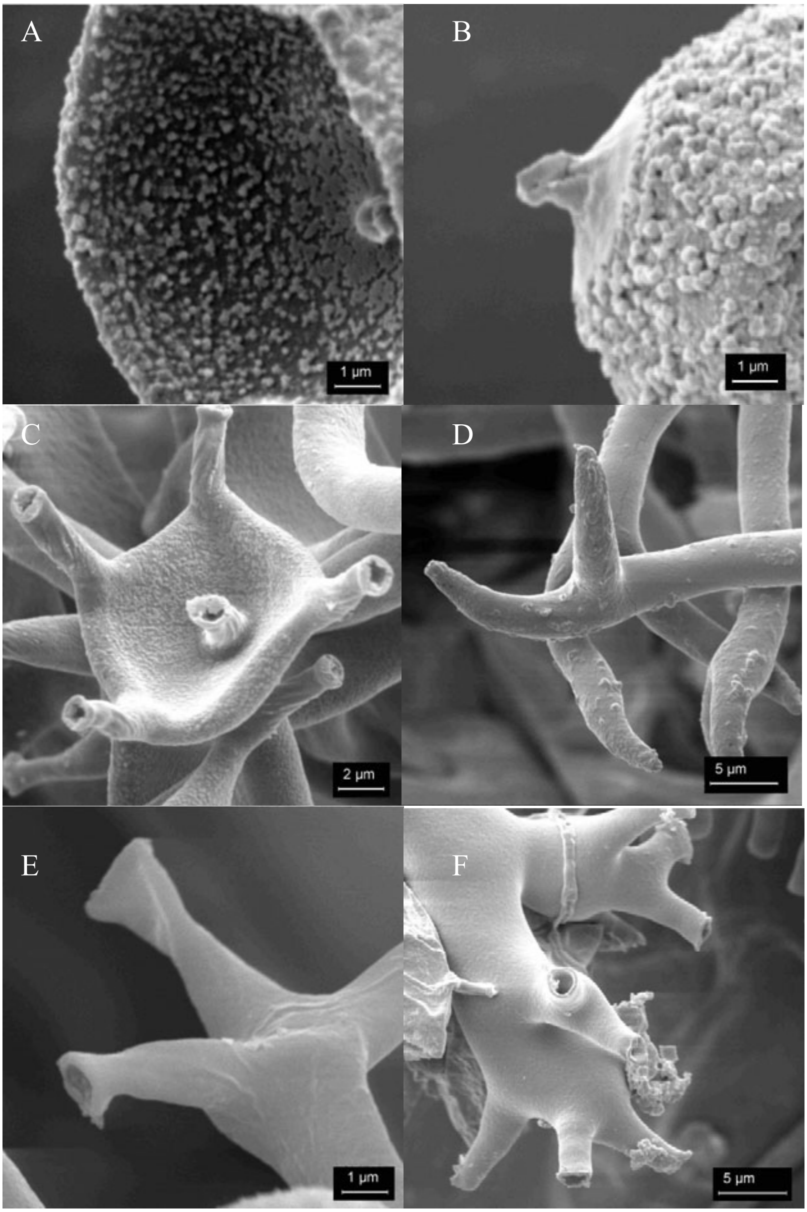

Figure 3. Ultrastructure of sporangia and sterigmata from oomycetes giving examples for the variablity in the connecting site of sporangia and sterigmata. A and C, *Bremia lactucae* from *Lactuca sativa*; B, *Pseudoperonospora cubensis* from *Cucumis sativus*; D, *Peronospora chenopodii* from *Chenopodium album*; E, *Plasmopara epilobii* from *Epilobium parviflorum;* F, *Plasmopara pygmaea* from *Anemone nemorosa*. SEM micrographs, M. Thines (2003).

4. CHEMICAL CHARACTERS IN THE TAXONOMY OF OOMYCETES

Chemical characters contributed to modern plant systematics to no lesser degree than the ultrastructural features, particularly between 1970 and 1990, before protein-based and DNA-based techniques became available (see Spring and Buschmann, 1998). Except for the general discovery of the cell wall polymers and their phylogenetic implications, there appear to have been no similar investigations of oomycetes with respect to lower molecular weight compounds. This is surprising in viewing the fact that separation techniques and spectroscopic methods for structure identification have improved enormously in recent years. Indeed they are now able to deal with microgram amounts of sample as has been shown with chemotaxonomic studies on plant trichomes (Spring, 2000). In contrast to eumycotic fungi, where metabolites have been screened primarily for pharmaceutical reasons or for biotechnological benefits, the biosynthetic capacity of oomycetes has not been prospected to date. In biotrophic groups, besides the lack of extractable material, this may also partly stem from the superficial impression that such organisms prioritise obtaining nutrients from their hosts, so are less able to produce significant amounts of own specialised metabolites. Albeit, even the very early descriptions mention that oomycetes accumulate large amounts of lipophilic compounds in vacuoles that sometimes occupy 80-90% of their oospore volumes (see Figure 4 B). Fatty acids were shown not only to have a high content of energy, but also valuable taxonomic characteristics. This was particularly useful in microorgansims with similar deficits as oomycetes in differentiating morphological features, as for instance in the cyaonobacteria (Cohen *et al.*, 1995, Romano *et al.*, 2000), bacteria (Wells *et al.,* 1993), microalgae (Volkman *et al.*, 1991) and yeasts (Cottrell *et al.*, 1986; Viljoen *et al.*, 1989).

A first attempt to employ gas chromatography for the analyses of fatty acids in extracts from sporangia of *Plasmopara halstedii* recently revealed an unexpected structural diversity and a high quantitative constancy in the patterns of independent field isolates (Spring and Haas, 2002). Neither pathotypes nor isolates from distant regions in Europe and in North America showed significant differences. On the other hand, the fatty acid profile was clearly distinctive in comparison to other oomycetes and other microorganisms associated with sunflower. The identity of the fatty acid profile in an isolate from perennial sunflower *H.* x *laetiflorus* supported the con-specifity with isolates from annual sunflower as had been concluded

Figure 4. Lipid vesicles (arrowed) in gametangia (A), oospores (B) and sporangia (C) of *Plasmopara halstedii* (C; reproduced from Spring and Zipper, 2000).

from cross-infection experiments and sequencing of the large subunit (LSU) of nuclear ribosomal DNA (Spring *et al*., 2003).

These results encouraged an expansion of such chemotaxonomic studies on a broader range of downy mildews and white rusts. Lipid vesicles, that are most prominent in the oospores of *P. halstedii*, occur in hyphae and sporangia as well (Figure 4), thus providing the opportunity to access sample material easily from sporulation at the plant surface. Since microgram to low milligram amounts of material are sufficient for GC-based fatty acid profiling, the technique is nearly as sensitive as most currently employed DNA analyses. The striking constancy of species specific patterns is exemplified with two GC traces obtained from 50 µg of sporangia of the sunflower white rust, *Albugo tragopogonis* (Figure 5). Structure identification can be performed either by GC-MS or indirectly by co-chromatography of fatty acid standards. Depending on the quality of resolution, patterns may even be used for fingerprint comparison without further structural information (similar to DNA-based analyses with RFLP, RAPD etc).

Besides the already mentioned advantages of the fatty acid chemotaxonomy on biotrophic oomycetes in terms of sensitivity, character stability and economical aspects, it should be pointed out that some factors, possibly restricting its applicability, have yet to be explored. Thus the influence of storage conditions for sample material on the chemical stability of fatty acids in general, and on the highly unsaturated compounds in particular, has to be investigated. At present, samples from air dried material and storage at room temperature revealed no significant alterations over a period of ca. one year (Spring and Haas, unpublished results). For the use of most herbarium specimens, however, this time frame would be insufficient. The question of whether fatty acid patterns could serve as classification characters only, or if they could serve for phylogenetic approaches as well, has still to be assessed. This will depend on elucidation of the biosynthetic pathway, in which the participation and the subcellular compartmentation of different desaturases and elongases is still unclear for oomycetes. The fast progress in this field, at least for various more easily accessible groups of organisms, has been reviewed recently by Sayanova and Napier (2004) who reported on the biosynthetic routes of polyunsaturated fatty acids in bacteria, plants and animals. Tracing the phylogeny of these enzymes in oomycetes could become an exciting project leading directly from metabolite studies into the use of metabolomics for systematics.

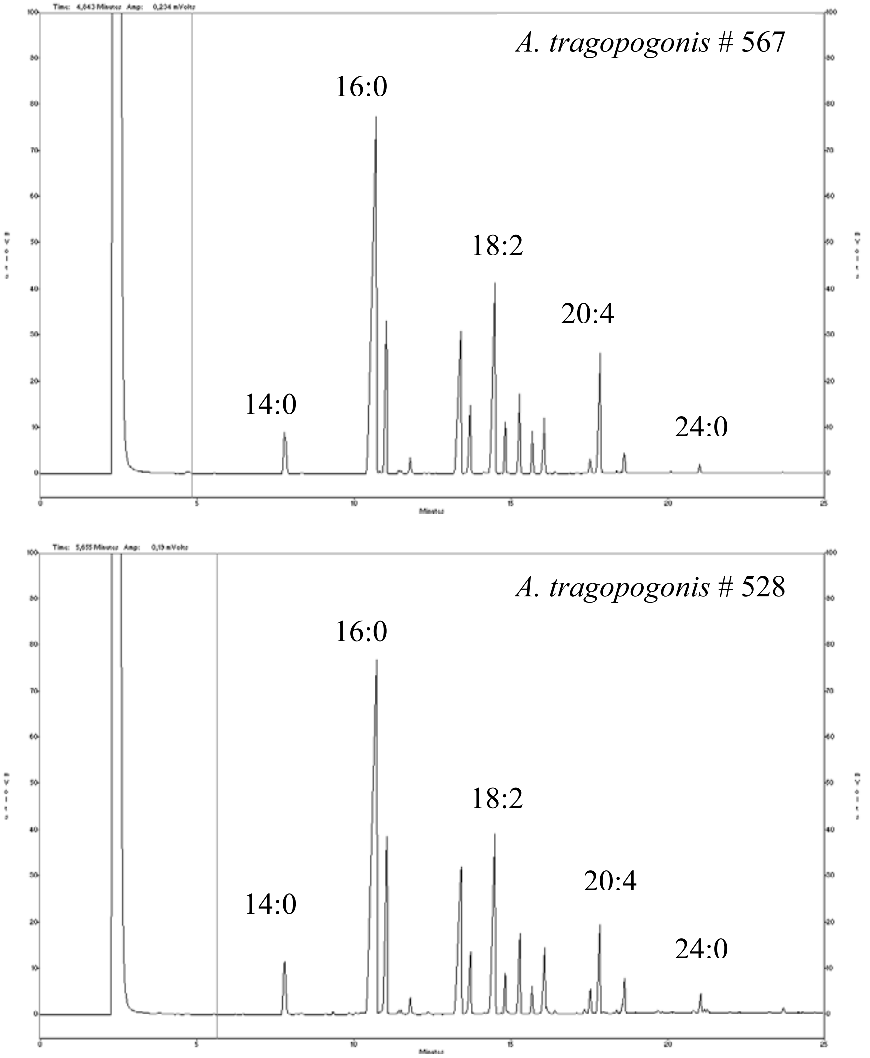

Figure 5. The fatty acid profiles of *Albugo tragopogonis* as analysed by GC. Samples were collected in Bethlehem, South Africa (# 528) and in Plieningen, Germany (# 567) from sunflower, *Helianthus annuus*. Fatty acids are designated as number of carbon atoms:number of double bonds. The injected amount represents the content of fatty acids in the total lipid fraction extracted from ca. 50 µg of sporangia.

5. MOLECULAR TOOLS AND THEIR USE ON DIFFERENT TAXONOMIC LEVELS

The regular use of molecular tools in the systematics of oomycetes started about ten years later than in angiosperms and other prominent groups of organisms. Unlike *Arabidopsis* for plants or *Drosophila* for insects, no such model organism has intentionally been selected by the scientific community for a modulated comprehensive genomic analyses in oomycetes (although *Phytophthora* is slowly gaining this key position). For that reason, sequence information in general as well as in specific genes is still very fragmentary. Nevertheless, enormous progress has been made very recently and "genetic research on oomycetes has entered an exciting phase" (Kamoun, 2003). From the earlier mentioned limitations on the amount and condition of sample material, it is clear that these data mostly derive from taxa readily cultivated *in vitro* like *Pythium* spp. and *Phytophthora* spp. (see review of Birch and Whisson, 2001). Thus, material-intensive techniques such as the use of protein banding patterns (successfully employed for species differentiation in *Phytophthora*; Kaosiri and Zentmyer, 1980; Erselius and de Vallavieille, 1984), isozyme studies (intensively used for the comparison of *Phytophthora* species and epidemiological studies on potato late blight; Oudemans and Coffey, 1991; Fry *et al.*, 1993) and the use of RFLP-based DNA analyses (for *Phytophthora* see reviews by: Brasier, 1989; Förster and Coffey, 1989) have not yet been employed on a significant range of biotrophic oomycetes.

PCR-based techniques offered a possibility to get over the obstacle of minute sample amounts and to avoid time-consuming propagation of field isolates on susceptible host tissue. However, though PCR has become a powerful tool in systematic studies of pathogenic oomycetes, there is still a lack of specific primers suitable for phylogenetic surveys in this group of organisms. In practice, the work has almost entirely focused on different parts of the nuclear ribosomal gene (Figure 6), with a few exceptions like the mitochondrial cytochrome C oxidase gene (COX2) employed by Hudspeth and coworkers (Hudspeth *et al.*, 2000, 2003; Cook *et al.*, 2001). While Dick *et al.* (1999) used the 18S rDNA (small subunit, SSU) to justify the separation of the Peronosporomycetes into the sub-class taxa Saprolegniomycetidae and Peronosporomycetidae, Riethmüller *et al.* (1999) used sequence analysis of the large subunit (LSU) of the 28S rDNA to compare internal relationships of the Saprolegniomycetidae and related groups. Similar studies for other groups of the Peronosporomycetes followed soon after (Leclerc *et al.*, 2000; Petersen and Rosendahl, 2000). An important advantage of the technique applied for the amplification of the LSU rDNA was that primers specific for the pathogen were developed which did not attach to the host DNA. In this way, DNA extractions from infected plant tissue could be used, making the separate harvest of sporangia and sporangiophores unnecessary. In a recent taxonomic study, this advantage allowed us to compare field isolates of *Plasmopara halstedii*, the downy mildew of the annual sunflower, with other species of the genus and to check the con-specific nature of pathogenic populations found on related host species (Spring *et al.*, 2003). It quickly became clear, however, that the highly conserved regions of the ribosomal DNA lacked sufficient variability to resolve the phylogeny of taxa on lower ranking levels (Göker *et al.*, 2003a).

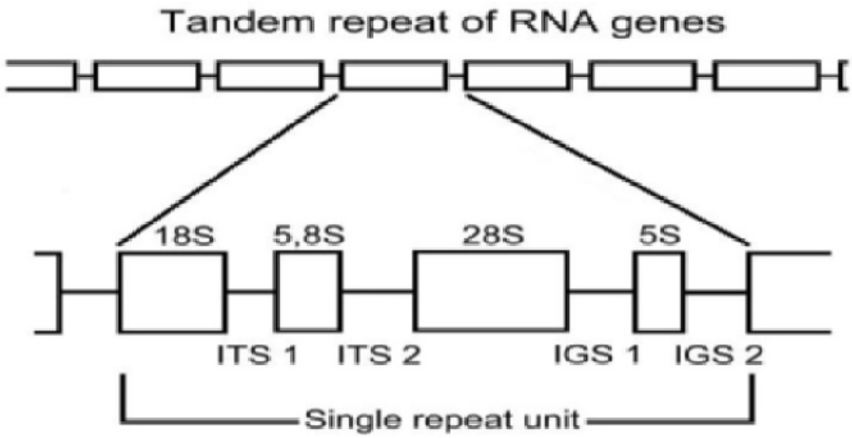

Figure 6. Scheme for the arrangement of coding and non-coding parts of the nuclear ribosomal DNA (modified from Drenth and Irwin, 2001).

Hence, non-coding regions of the genome were employed for studies of the infra-generic relationships of oomycetes and the internal transcribed spacer (ITS) region of the nuclear ribosomal DNA became the candidate of first choice (Leclerc *et al.*, 2000; Cooke *et al.*, 2000). This tool has clearly supported the erection of the new genera *Hyaloperonospora* Constant. and *Perofascia* Constant. (Constantinescu and Fatehi, 2002; Göker *et al.*, 2003b; Voglmayr, 2003) which Constantinescu had separated from *Peronospora*-like pathogens of the Brassicaceae and related hosts due to conspicuous morphological characters. Phylogenetic trees based on ITS data now shed light on the paraphyletic status of several genera including

Phytophthora, *Pythium* and *Plasmopara* (Voglmayr, 2003), and additional surprises might be expected in the near future.

Nevertheless, even such a variable region like the ITS proved to be insufficient to resolve the relationship in some groups at the generic level like *Peronospora s.str.* and *Pseudoperonospora* (Voglmayr, 2003), or within parts of *Peronospora* (Cooke *et al.*, 2002) and *Hyaloperonospora* (Göker *et al.*, 2003b). In addition, there is an increasing demand for the molecular-based intra-specific differentiation of oomycetes, particularly those parasitising crop plants. Recent attempts to distinguish physiological races of the sunflower downy mildew pathogen, *Plasmopara halstedii*, by ITS sequencing failed to show useful race-correlation (Bachofer, 2003). Interestingly, no other non-coding part of the nuclear genome has so far been used successfully for phylogenetic studies in pathogenic oomycetes. The IGS region (inter-genomic spacer of the nuclear ribosomal DNA) was used recently to characterize sub-specific populations of the soybean brown stem rot pathogen, *Phialophora gregata* (Deuteromycetes) (Chen *et al.*, 2000). However, attempts to employ universal primers derived from eumycotic IGS on *Plasmopara* isolates failed to amplify DNA, except for a few samples which had been contaminated slightly by spores of *Alternaria alternata* (Deuteromycetes) (Bachofer and Spring, unpublished data). On the other hand, progress has been made recently in the amplification of mitochondrial IGS from *Phytophthora infestans* (Wattier *et al.*, 2003).

In the case of population studies it seems more promising to screen for polymorphisms on the entire genome than just on a single region. Besides RAPD (eg Roeckel-Drevet *et al.*, 1997) and AFLP studies, the inter simple sequence repeat technique (iSSR) has been employed for classifications on the subspecific level (eg Intelmann and Spring, 2002), as it is exemplified here for *Plasmopara halstedii*. Moreover, molecular tools are developing rapidly and new techniques, like the analysis of single nucleotide polymorphism (SNP) or retroelements (eg Ty3/*Gypsy*) and ISTR (inverse sequence tagged repeats) analyses, have now been introduced as for instance in studies of *Phytophthora* (eg Judelson, 2002).

With the rapidly growing knowledge in molecular genetics of pathogenic oomycetes (Kamoun, 2003) we certainly may expect soon a much broader data base of sequence information. The usefulness of coding and non-coding genomic regions on the different taxonomic levels (Table 2) is unpredictable and will have to be explored individually. With respect to the increased sensitivity of PCR-based approaches, it should be taken into account that the care taken for appropriate pathogen sampling should be increased simultaneously. Besides the already mentioned problems with sample contamination through other organisms, the genetic heterogeneity of the pathogen sample itself (eg see Komjáti *et al.,* this volume) could also produce flawed results, at least at the intra-specific level. Equivocal results could, for instance, derive from simultaneous multiple race infections on the same host plant, and may only be avoided by cloning the material prior to extracting the DNA (cf Intelmann and Spring, 2002). Such time consuming and difficult measures may appear over-cautious for the aim of many studies, but they will certainly be necessary when it comes to the questions of speciation, inheritance, ploidy levels and the role of hybridization in biotrophic oomycetes. Molecular techniques will help to address these exciting fields of systematics in the near future even in this difficult to handle group of organisms.

Table 2. Molecular-based techniques and their potential use for systematic analysis in biotrophic oomycetes as demonstrated for selected groups.

	Useful at taxonomic level of:	Example of application and ref.
Isozyme analysis	genus to sub-species; inheritance studies	*Phytophthora*; Oudemans & Coffey 1991
RFLPs	genus to sub-species	*Phytophthora*; Förster & Coffey, 1989
SSU rDNA sequencing	kingdom to genus	Peronosporomycetes; Dick *et al.*, 1999
LSU rDNA sequencing	kingdom to genus	Peronosporomycetes; Riethmüller *et al.*, 1999, 2002; Petersen & Rosendahl, 2000
COX2 sequencing	kingdom to genus	Peronosporomycetes; Hudspeth *et al.*, 2000, 2003
ITS sequencing	family to species	Peronosporoaceae; Göker *et al.*, 2003b, Voglmayr, 2003
RAPDs	species to sub-species	*Plasmopara;* Roeckel-Drevet *et al.*, 1997
AFLPs	species to sub-species	*Plasmopara;* Roeckel-Drevet *et al.*, 1997
iSSRs	species to sub-species	*Plasmopara;* Intelmann & Spring, 2002
SNPs	species to sub-species; crossing anaylsis	*Phytophthora;* Niepold, 2003
Retrotransposons	species to sub-species	*Phytophthora;* Judelson, 2002

6. CONCLUSIONS

Within the past decade molecular tools in general, and PCR-based techniques in particular, have drastically broadened the possibilities to characterize difficult to analyse groups of organisms and to unravel their phylogeny. However, despite this fast and enormous progress the use of molecular characters in the systematics of pathogenic oomycetes is still in its infancy. This stems mainly from the difficulties inherent in accessing material from the mostly biotrophic taxa in an appropriate condition and in sufficient amounts. It can also not be ignored that, despite the economic relevance of several species pathogenic to some important crop plants, there are not many experts worldwide who are able to provide or investigate taxonomically well defined samples of pathogenic oomycetes from uncultivated host plants. Moreover, Dick (2002b) has recently reviewed the problems that may arise from attempts to link the molecular phylogeny to the classical parts of taxonomy and systematics. To overcome these problems, it will be necessary to focus future efforts on several aspects which are summarized here briefly.

The provision of samples investigated for molecular phylogenetic studies has to be improved considerably wherever possible, both in terms of quantity as well as quality. Spot-checking of single accidentally accessed populations should be avoided, even if the correctly calculated phylogenetic tree obtained from such data does not betray this weakness of its branches. Similarly, efforts should be made to explore further genomic regions for comparative investigations and to test the coherence of their calculated tree topologies. In terms of quality, traditional principles of taxonomy (particularly in relation to species classification) should be recognized as an equally important task as the seemingly more valuable goal of studying phylogeny. Additional characters are required to enable a quick and reliable sample classification as the basis of subsequent phylogenetic studies. Ultrastructural features and fatty acid profiles are candidates which could possibly fill this gap. As a consequence, the deposition of voucher specimens remains inevitable and should be adapted to the requirements of other than just morphological re-investigation. *Vice versa*, further technical improvements in the use of molecular and chemical characters should enable the accessibility of herbarium material. This would allow vouchers of type specimens to be re-investigated for the new characters, in order to avoid further confusion with respect to the use of correct binomials.

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A SEEDLING BIOASSAY TO DETECT THE PRESENCE OF *PLASMOPARA HALSTEDII* IN SOIL

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1. INTRODUCTION

Plasmopara halstedii (Farl.) Berl. & de Toni, the causal agent of sunflower downy mildew (SDM), is a soil-borne fungus found on all continents except Australia (Gulya *et al*., 1997). Unlike many other downy mildew diseases, which are foliar infections, the primary mode of infection in SDM is a root infection of young seedlings, which results in a systemic infection, culminating in either seedling death or an extremely stunted plant with minimal seed yield. Although foliar infections due to airborne sporangia are observed, these local lesions do not develop into systemic infections, and thus are of minimal significance.

The geographic distribution and incidence of *Plasmopara halstedii* is most often documented by observation of diseased sunflower plants. Growing season surveys commonly estimate that SDM occurs in 5% to 30% of fields in the north central United States (Gulya, 1996; 2003). As infection, and thus disease occurrence, is highly influenced by waterlogged soils after planting (thus facilitating oospore germination and zoospore motility), the incidence of this pathogen is often underestimated when environmental conditions are not optimal for infection. Similarly, when late season surveys are made to enumerate plant diseases, the incidence of seedling diseases such as SDM is again underestimated since the infected plants have died and are no longer present.

The ability to determine whether a field contains a pathogen is of value both to growers and to researchers. While many laborious methods are available to recover oospores of various pathogens from soil samples (Burr and Stanghellini, 1973; Pratt and Janke, 1978; Magarey, 1989; Gaag and Frinkling, 1997), there is still the need for a simple, diagnostic method

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to ascertain the presence or absence of an oomycete such as *P. halstedii*. Soil bioassays to detect the presence of other oomycetes have been developed (Malvick *et al*., 1984; Purwantara *et al*., 1996; Windels and Nabben-Schindler, 1996), but none have been developed specifically for *P. halstedii*. Armed with knowledge of the presence of *P. halstedii* in a particular field, a grower could decide whether it was necessary to select a downy mildew-resistant hybrid. In the case of commercial seed production, managers could determine the potential risk of SDM for phytosanitary certification of seed crops.

The objective of this study was thus to develop a relatively simple method, requiring only greenhouse or growth chamber facilities, to qualitatively detect the presence of *P. halstedii* in soil. If the procedure could be modified to make the test semi-quantitative, this would be useful but was not our primary objective.

2. MATERIALS AND METHODS

2.1 Initial trials

Preliminary trials with soil samples collected from fields with high incidence of SDM demonstrated the practicality of a seedling bioassay (Gulya and Radi, 2003). The initial method consisted of collecting a bulked soil sample from 6 to 10 sites in a field, to a depth of 10 cm. Small (10 x 10 x 5 cm) plastic trays (T & O Plastics, Minneapolis, MN, USA) with drainage holes were filled with candidate soil samples, and 72 seeds of an untreated oilseed sunflower hybrid were planted to a depth of 20 mm with the aid of a dibble board, ensuring even placement of seeds. The amount of soil required to fill one flat was approximately 3 liters, or 1.5 to 1.8 kg, depending upon soil type and moisture. The trays were watered with deionized water after planting, and daily thereafter. Trays were held in a greenhouse maintained at 18 - 27^{0}C with a 16 hr photoperiod maintained with supplemental lighting. Seed germination was assessed daily by careful exhumation of seedlings, and when the radicles reached 1 cm in length, a flooding regime was started. Fiberglass pans measuring 25 x 12 x 1.5 cm were placed beneath groups of two trays, and filled to the brim with deionized water. After 8 h, the pans were tilted and the trays allowed to drain overnight. The flooding process was repeated for four consecutive days to ensure that all seedlings had the opportunity for infection when their radicles were at the optimal stage (i.e. 1 - 2 cm). The seedlings were grown for 14 days following the final day of flooding (18 days after planting), at which time the trays were transferred to chambers maintained at 15-17^{0}C with 100% relative humidity. While symptoms of downy mildew infection (chlorosis on cotyledons and true leaves, and stunting) are visible starting at 7 to 9 days after infection, the presence of fungal sporulation (induced by incubation at 100% RH) is much easier to observe. Disease incidence in the bioassay was calculated as the number of systemically infected seedlings divided by the number of emerged seedlings, and expressed as a percentage, hereafter referred to as SDM infectivity index (SDMII) to avoid confusion with disease incidence in field surveys.

2.2 Effect of diluting soil samples with sand

In an attempt to minimize the volume of soil necessary for the seedling bioassay, an experiment was conducted to study the effect of diluting soil samples with sand. Soils from six sunflower fields observed to have sunflower downy mildew were collected in the fall of 2001. The soils were pulverized and uniformly mixed with washed, ungraded river sand in the following proportions (soil/sand, vol/vol): (1) undiluted soil, (2) 50:50, (3) 25:75, and (4) 12:88. Seedling bioassays were done on three replications of each of the 24 samples (6 fields x 4 dilutions), and the experiment was repeated once.

2.3 Use of the seedling bioassay to assess *Plasmopara halstedii* incidence

Three large-scale studies were done to assess the incidence of SDM in commercial fields in north central United States using the soil bioassay.

(1). In September 2001, soils were collected from 64 sunflower fields during a survey conducted by personnel of the annual yield survey coordinated by the National Sunflower Association (NSA). Surveyors were instructed to collect a 4-liter composite soil sample from three to five locations in a field. The soils were stored in sealed plastic bags at room temperature until processed in mid-winter. The soil samples were handled as described above, with the following details. Two replications of each soil sample were assayed in two experiments, using four trays. To ensure sufficient soil for both experiments, all soil samples were diluted with sand at a 1:1 (v/v) ratio.

(2). In June 2002, crop scouts were asked to survey sunflower fields within the first month after planting, and to record the percent SDM incidence as well as to collect soil samples from the same fields. After running the soil bioassay on these samples, disease incidence, as recorded by the scouts, was compared with SDMII as determined by the soil bioassay.

(3). In September 2002, participants in the NSA sunflower survey were asked again to collect soil samples from fields in the six states being surveyed (Gulya, 2003). The samples were held at room temperature for four months and processed as described above in mid-winter. The incidence of SDM recorded by field observations was compared with SDMII from soil bioassays.

2.4 Effect of soil storage conditions prior to the bioassay

This study was initiated to determine whether a maturation period was necessary with fall-collected soil samples to enhance the infectivity of *P. halstedii* oospores in the soil. The test examined the effects of storage duration and temperature and was started in the fall of 2002. Soil samples from nine sunflower fields known to have SDM were collected in October 2002, placed in sealed plastic buckets, and held at room temperature (20ºC), at 4ºC, and at -20ºC. At 2 weeks and 1, 2, 3 and 4 months, an aliquot of soil was removed, mixed with sand for a 50% dilution, and three replications of the bioassay run on each of the nine samples stored at three different temperatures.

3. RESULTS

3.1 Effect of soil dilution with sand on *P. halstedii* infectivity

The SDMII or infectivity of field soils and the emergence of sunflower seedlings was significantly affected by dilution (Table 1). Seedling establishment or stand was greatest in soil diluted to 50% with sand, possibly because many of the soils were fine textured and the addition of sand aided aeration. However, increasing amounts of sand above 50% did not increase the emergence rate. The recovery of *P. halstedii* (SDMII) was highest in the undiluted soil, and significantly less in all dilutions. There were no significant differences, however, in SDMII between the 12%, 25% or 50% dilution rates. Soil samples in all subsequent experiments were diluted to a 50% sand level, primarily to enable us to collect smaller soil samples. Diluting all soil samples to the same level would still allow SDMII comparisons between soil samples, despite the fact that the actual infectivity would be less than that observed with undiluted soil samples.

Table 1. Effect of diluting field soil samples with sand on the emergence of sunflower seedlings (stand) and recovery of *Plasmopara halstedii* (DM) in the seedling bioassay. Means followed by the same letter are not significantly different at the P=0.05 level according to the Duncan's multiple range test.

	Soil percentage			
	100	50	25	12
Stand %	63b	75a	63b	67ab
% DM	43a	35b	35b	31b

3.2 Assessment of *Plasmopara halstedii* incidence using the soil bioassay

In the summer of 2001, crop scouts surveyed 79 fields in eastern and central North Dakota (ND) and found SDM in 33% of the fields. After testing soil samples from the same general area with the soil bioassay, we recovered *P. halstedii* from 97% of the samples. SDMII/sample ranged from 0 to 93%, and averaged 28% in the positive samples. Thus in 2001, a mid-summer survey found SDM in 33% of the fields while the soil bioassay detected the pathogen in 97% of fields from the same general area. In the summer of 2002, the crop scouts surveyed 197 sunflower fields in eastern and central ND and observed SDM in 16% of the fields. When bioassays were done on 123 soil samples within one week of collection, *P. halstedii* was recovered from only 21% of the samples. SDMII/sample ranged from 0.5% to 52%, and averaged 5% in positive samples, which was much lower than observed in the samples collected in the fall of 2001. This prompted us to investigate whether summer collected samples might not be as infective as overwintered samples due to incomplete oospore maturation.

In the fall 2002 survey, 477 fields were inspected in eight states for all sunflower diseases, and SDM was observed only in the states of ND and South Dakota (SD). SDM incidence was 8% in ND and 1.5% in SD (or 4.6% incidence considered all 477 fields) based on the presence of infected plants at the end of the growing season. A total of 194 soil samples were returned from the survey, and of these 63% were positive for *P. halstedii.* SDMII in the positive samples ranged from 0.7% to 100% and averaged 37%. *P. halstedii* was recovered from samples in all states, with SDMII ranging from 16 to 100% (Table 2). Thus, in 2002 the soil bioassay confirmed the presence of *P. halstedii* in 63% of assayed soils in comparison to field disease observations in only 4.6% of the same fields.

Table 2. Percent of fall-collected soil samples from which *Plasmopara halstedii* was recovered using the soil bioassay during two years of sampling in seven Midwestern US states (--, not sampled).

	No. of samples	US mean	Percent positive samples, by State						
			ND	SD	MN	KS	CO	NE	TX
2001	64	92	97	82	100	--	--	--	--
2002	194	63	71	69	67	46	16	92	100

3.3 Effect of storage temperature and duration on infectivity of soil

While this study was initiated with soil from nine fields observed to have some incidence of downy mildew, the soil bioassays revealed the presence of *P. halstedii* in only two fields. Thus, the results summarized here pertain to only two of the nine fields. There was very low SDMII (i.e. recovery of *P. halstedii)* from soils processed two to eight weeks after collection (Table 3). There was a significant increase in SDMII in samples stored for 12 weeks, compared to shorter durations, and another substantial increase when soils were stored for 16 weeks. Temperature during the storage period also had a profound effect upon SDMII. Soils stored frozen at –4ºC had very low infectivity throughout the 16 week study, and at its conclusion, were 15-fold lower than samples stored at either 4ºC or 20ºC. There was no significant difference in SDMII between soils stored at 4ºC or 20ºC.

Table 3. Effect of storage temperature and duration on the recovery of *Plasmopara halstedii* (SDMII) from soil samples in the seedling bioassay. Means followed by the same letter are not significantly different at the P=0.05 level according to the Duncan's multiple range test.

Storage temp., ºC	Time in storage (weeks)					Mean
	2	4	8	12	16	
20	0.5	0.8	5.5	18.8	67	18.5a
4	0.2	0.5	1.0	12.8	67	16.4a
-10	0.2	0.0	0.8	2.4	4.4	1.6b
Mean	0.3a	0.4a	2.5a	11.3b	46.2c	

4. CONCLUSIONS AND DISCUSSION

A seedling bioassay to detect the presence of *Plasmopara halstedii* in soil samples has been developed which can be used as a diagnostic tool or in research studies to monitor the incidence of this pathogen. The bioassay has limitations as currently employed, as summer or fall collected samples must be stored for at least 16 weeks to achieve high levels of infectivity. We have not tested spring-collected samples to compare the effect of natural overwintering.

Using the soil bioassay, we have demonstrated that both summer and fall disease surveys observing infected plants greatly underestimate the actual presence and distribution of this soil-borne pathogen. While more time consuming, the soil bioassay will give a more accurate appraisal of the occurrence of *P. halstedii*. In recent experiments, we have used metalaxyl-treated sunflower seed and non-treated seeds in paired flats with the same soil samples to demonstrate the occurrence of metalaxyl-resistant isolates in one step.

Various modifications may help improve the efficiency of this soil bioassay. Soil moisture and freeze/thaw cycles have been observed to enhance oospore germination with other oomycetes (Duncan and Cowan, 1980; Burruano *et al*., 1992; McQuilken *et al*., 1992) and these factors will be investigated in our continuing studies. The effect of temperature on oospore maturation varies with between different genera and species. *Phytophthora* oospores exhibit greater germination when stored at low or sub-freezing temperatures (Duncan, 1985; Medina and Platt, 1999), while Spring and Zipper (2000) concluded the *P. halstedii* oospore germination was not enhanced by storage at sub-freezing temperatures. The stimulatory effect of host plant root exudates (Pratt, 1978; Gowda and Bhat, 1986) and soil extracts (Hord and Ristaino, 1992; Stromberg *et al*., 2001) on oospore germination has been noted with several oomycetes, and these stimulatory effects may well play a role in the efficacy of the *P. halstedii* soil bioassay.

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EICOSAPENTAENOIC ACID, A POSSIBLE MARKER FOR DOWNY MILDEW CONTAMINATION IN SUNFLOWER SEEDS

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1. INTRODUCTION

Plasmopara halstedii (Farl.) Berl. & Toni is a serious and economically important pathogen of cultivated sunflower in almost all parts of the world (Gulya *et al.*, 1997). Assumed to be endemic to North America, the pathogen was reported to have reached sunflower fields in Yugoslavia and other eastern European countries from 1941 (Sackston, 1981). Within a few decades sunflower downy mildew then spread into countries of nearly all continents, except for Australia. It is most likely that seeds are the natural "vehicle" for long distance and intercontinental dispersal of *P. halstedii*. In addition, the intensified trade and seed exchange within the past 20 years appears to be the driving force for the contemporary spread of new pathotypes and fungicide tolerant genotypes of the pathogen throughout most areas of sunflower cultivation. Measures to prevent this, by keeping sunflower plants grown from imported seeds under constant pathological surveillance for the first two growing seasons, were proposed many years ago (Leppik, 1962). These methods are impractical in current times, where breeders, for instance, try to raise two generations of sunflower per year by shipping seeds from the northern to the southern hemisphere and *vice versa*. Moreover, downy mildew contaminated seeds seldom give rise to plants with typical symptoms known from systemic infections (Cohen and Sackston, 1974; Döken, 1989; Spring, 2001). It is for such reasons that diagnostic markers are needed urgently to identify the contamination of sunflower seeds with *P. halstedii* and to prevent further seed-borne propagation of the pathogen.

Various attempts have been made recently to develop diagnostic tools for the identification of *P. halstedii* in plant tissue and seed samples by means of molecular markers. Thus, Roeckel-Drevet *et al.* (1999) generated

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an oligonucleotide primer for the selective amplification of *P. halstedii* DNA and Bouterige *et al.* (2000) tried to revive a former idea of Liese *et al.* (1982) employing an enzyme-linked immunosorbent assay (ELISA) test for tracing the pathogen. However, experiments that prove a satisfactory sensitivity with seed samples of defined contamination have not been published yet.

The results of our recent study on the fatty acid composition of sunflower downy mildew, its host plant and other co-inhabiting microorganisms (Spring and Haas, 2002) has encouraged us to address this problem with an alternative phytochemical approach. Lipid vesicles in the cytoplasm of *P. halstedii* occur in hyphae and sporangia. In oospores, the vesicles occupy most of the volume and seem to represent an important endogenous reserve to fuel oospore germination as an energy supply during the period between zoospore formation and infection of a new host. The chemical composition of these lipids has now been elucidated by gas chromatography and mass spectroscopy. This has revealed the presence of large amounts of polyunsaturated fatty acids, in particular of eicosapentaenoic acid (C20:5) which comprised up to one third of the total lipid content. In contrast, eicosapentaenoic acid was not found in the uninfected host plant and in eumycota species that are frequently associated with sunflower (eg *Alternaria alternata*, *Botrytis cinerea*, *Phomopsis helianthi*, *Sclerotinia sclerotiorum*) (Spring and Haas, 2002). Screening experiments with *P. halstedii* isolates of different pathotypes (5 types tested) and geographic origin (8 isolates from Germany, France and USA) showed high qualitative and quantitative consistency in the fatty acid profiles. This implies that the specific fatty acid composition of *P. halstedii* could serve as a diagnostic marker for the presence of the pathogen in sunflower tissue and achenes.

2. MATERIALS AND METHODS

Seeds used for the investigation were collected from 19 sunflower fields in southern Germany during 5 seasons. When possible, samples from healthy, systemically and non-systemically infected plants (late, symptomless flower head infection as described by Spring, 2001) were collected separately and stored in darkness at room temperature, at 5°C and at –18°C.

Fatty acid analyses were performed using pericarps of uninfected and infected achenes of sunflower. The status of infection was checked by means of microscopic investigation, either by splitting the achenes into halves (using one for microscopy, the other for lipid extraction) or by inspecting randomly selected achenes from each sample. The tissue of the embryo, which usually does not carry the pathogen (Döken, 1989), was removed and infection was recorded when structures (hyphae, sporangia, oospores) of the pathogen were observed in the inner parenchyma of the pericarp. Staining with resorcin blue (specific for callose) eased the recognition of downy mildew structures from collapsed host cells and eumycotic hyphae. A minimum of 10 achenes was used for each sample. In some experiments, healthy and infected seeds were mixed. Aliquots of infected and uninfected material (w:w) were used at ratios above 1:50 in the proportions given in the results.

Lipid extraction was performed exclusively from pericarp material in order to avoid the large amounts of fatty acids in the host's cotyledons. The extraction procedure with $CHCl_3$/MeOH (2:1, v:v) followed the method previously described by Spring and Haas (2002). The formation of methyl esters was achieved by means of BF_3/MeOH (14%) according to standard procedures (Morrison and Smith, 1964; Christie, 1984). The fatty acid methyl esters were analysed by GC on a Shimadzu GC-17A gas chromatograph, equipped with a Varian-Chrompack CP-Sil 8 CB capillary column (25 m x 0.32 mm), on-column injector and flame ionisation detector (FID). The operating conditions were as follows: detector temperature, 350°C; helium carrier gas, 25 ml/min; column temperature initially set at 160°C for 2 min and subsequently increased by 8°C/min to 340°C and maintained at this temperature until separation was completed. Peak identification was performed by comparison of retention times with known standards. Alternatively, samples were analysed by HPLC on a Hypersil ODS (5 µm) column (4 x 250 mm) with 90 % acetonitrile (1.3 ml/min) and peak detection at 213 and 226 nm.

3. RESULTS AND DISCUSSION

Fatty acid patterns obtained with GC from the pericarps of different sunflower lines and cultivars (e.g. HA 89, HA 821, HAR 5, HA 335, Albena, cv. Giganteus) were almost uniform. They resembled those known from sunflower oil in consisting of high amounts of palmitic acid (C16:0), stearic acid (C18:0), oleic acid (C18:1) and linoleic acid (18:2), while linolenic acid (C18:3), arachidic acid (C20:0) and behenic acid (C22:0) represented minor components (Table 1). In contrast, samples of infected seeds additionally contained characteristic fatty acids of *P. halstedii,* among which eicosapentaenoic acid (C20:5) was the most prominent (Figure 1). The latter, which makes up to one third of the total fatty acid content in pure samples of sunflower downy mildew sporangia, was not detected in samples from other microorganisms frequently associated with sunflower (Spring and Haas, 2002). The relative content of C20:5 fatty acid in samples of infected achenes varied considerably from a minimum of 1% to a maximum of 15% and averaged 7.2% (SD 4.95; SEM 1.08; n = 21) in 21 independent samples. Infected achenes from non-systemically infected plants revealed insignificantly lower contents of eicosapentaenoic acid when compared with samples from systemically infected plants. The high degree of variation in the C20:5 content between samples most likely reflects different extents of pathogen invasion of the achenes. For example, Döken (1989) has shown that even systemically infected plants produced up to 28% of seeds bearing no signs of infection.

Table 1. The fatty acid composition (ratio of total peak area in GC analyses) of pathogenic fungi associated with sunflower (Spring and Haas, 2002).

	Fatty Acid (carbon atoms:number of double bonds)									
	14:0	16:0	18:0	18:1	18:2	20:0	20:1	20:5	22:0	22:1
H. annuus pericarp*	tr	9.6	8.7	22.6	44.1	0.5	-	-	1.0	-
Plasmopara halstedii [#]	1.9	17.4	0.9	11.0	22.0	0.6	1.9	30.9	0.8	3.9
Alternaria alternata	-	16.5	3.5	26.3	50.5	-	-	-	-	1.0
Botrytis cinerea	-	13.5	2.5	46.6	36.5	-	-	-	-	-
Fusarium oxysporum	tr	22.3	7.1	22.5	37.5	-	-	-	2.7	-
Phomopsis helianthi	0.7	28.6	7.8	25.2	33.1	0.4	tr	-	0.5	-
Sclerotinia sclerotiorum	0.5	25.6	4.3	18.0	44.2	-	-	-	-	-

tr = trace amounts (< 0.1 %); - = not detected;
* sunflower cultivar HA89;
[#] mean value of 8 accessions.

Experiments testing the degradation of fatty acids upon the storage of seed samples have only been started recently. However, preliminary results did not indicate significant decomposition of specific fatty acids, especially polyunsaturated ones. The relative content of C20:5 fatty acid in an infected seed sample stored over 6 months at room temperature, 5°C and –18°C ranged from 13.9% to 12.4% and 15.3%, respectively. Several other samples were stored for 2-3 years at room temperature before extraction of their lipids, and still clearly indicated the presence of *P. halstedii* through the occurrence of eicosapentaenoic acid in GC analysis. From these results it appears that the use of fatty acid characters for diagnostic tests is less dependent on the freshness of plant material than is reported for DNA-based methods (Says-Lesage *et al.*, 2000).

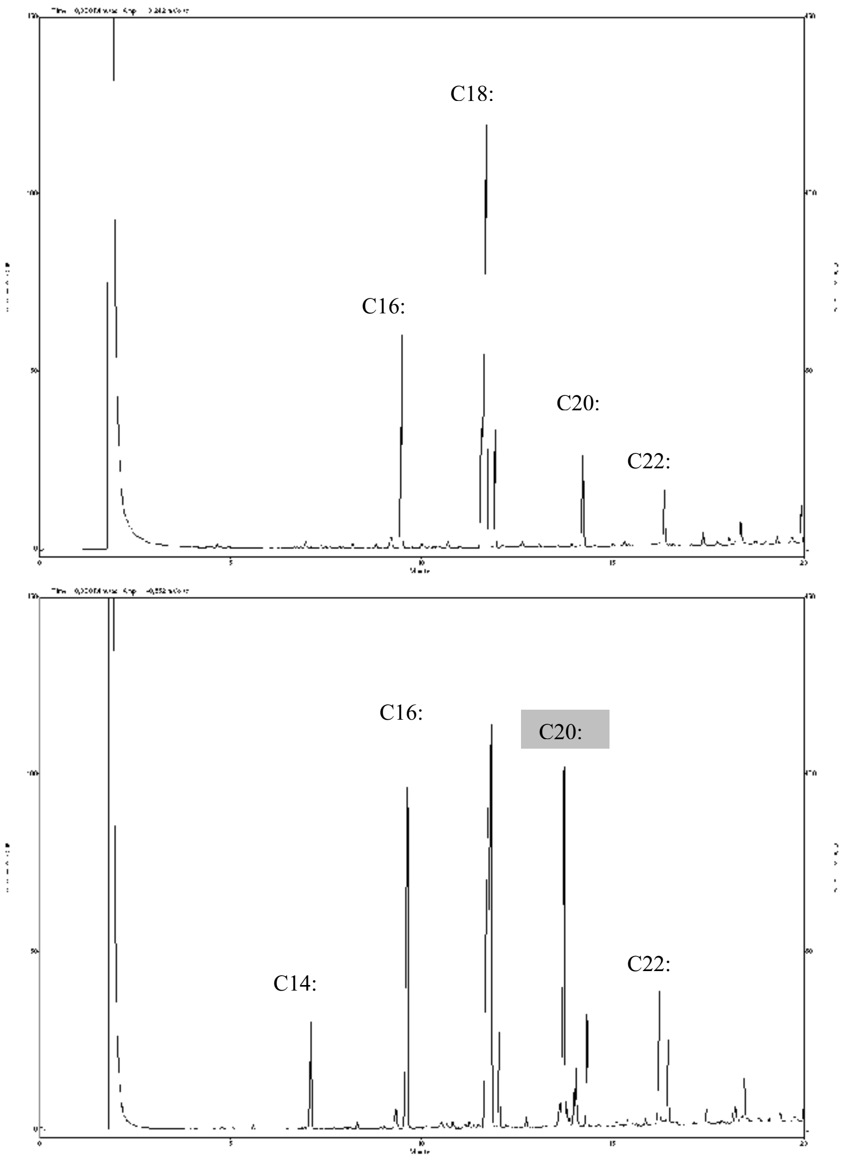

Figure1. The fatty acid profiles of healthy (top) and *Plasmopara halstedii* infected (bottom) sunflower achenes in GC analysis**.**

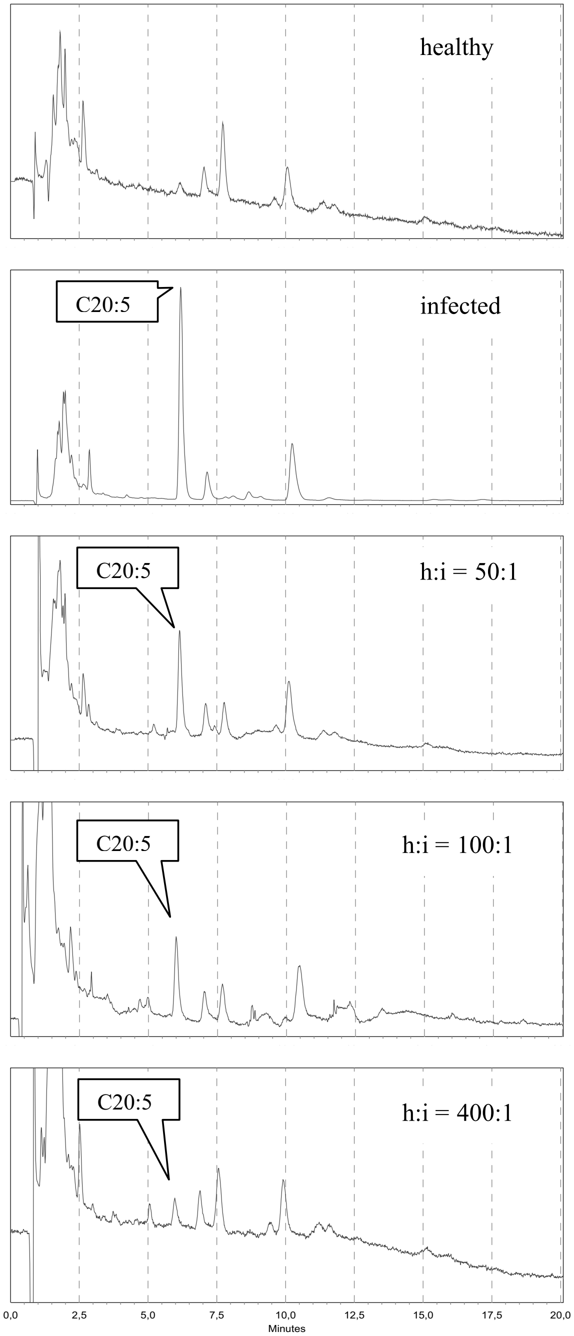

Figure 2.
HPLC-based detection of eicosapentaenoic acid (C20:5) in pericarp samples of healthy (h) and infected (i) seed samples as well as in mixed samples at h:i ratios between 50:1 and 400:1.

Mixing experiments with infected and uninfected achenes revealed that the ability to trace the contamination of seed samples with downy mildew infected achenes by means of the C20:5 fatty acid in GC analysis, was limited to a ratio of approximately 1:50. Beyond this point the signal of eicosapentaenoic acid either failed to reach the threshold of detection or was masked by increasing background signals. This problem could not be overcome by concentrating the extract or by pre-fractionation of saturated and unsaturated fatty acids with silica/$AgNO_3$ (data not shown). As an alternative, UV detection appeared suitable for a more selective identification of polyunsaturated fatty acids like C20:5 in lipid extracts that predominantly consisted of saturated or low unsaturated compounds. With HPLC analysis it was possible to separate eicosapentaenoic acid from other unsaturated fatty acids in pericarp extracts (eg C18:1, C18:2), and to detect *P. halstedii* contamination of seed samples down to a ratio of 1 infected achene in 400 healthy ones (Figure 2). For each seed sample, the limits of detection are dependent on the severity of infection in the seeds. Thus, in slightly infected samples the diagnostic threshold will be higher.

The severity of infection limits the value of any diagnostic technique, whether it is based on the sporulation of the pathogen on germinated seedlings (Cohen and Sackston, 1974), on the expression of proteins for immuno-assays (Liese *et al.*, 1982), or on pathogen-specific nucleic acids (Roeckel-Drevet *et al.*, 1999; Says-Lesage *et al.*, 2000). Except for the seedling test, which is extremely time consuming and expensive, all other techniques so far employed suffer from the risk that the diagnostic marker for the pathogen tends to be absorbed by the overwhelming amount of background host tissue. Molecular approaches, and particularly those based on PCR, may be of greatest sensitivity. However, the fatty acid analysis appears to be a promising alternative, until the efficacy of molecular techniques for identifying sunflower downy mildew in seed samples has been proven.

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ISOLATION OF VIABLE *PERONOSPORA VICIAE* HYPHAE FROM INFECTED *PISUM SATIVUM* LEAVES AND ACCUMULATION OF NUTRIENTS *IN VITRO*

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1. INTRODUCTION

A variety of methods have been used to isolate the inter- and intracellular infection structures of rust, powdery mildew and downy mildew fungi from infected plant tissues. These include enzymic maceration (Crucefix *et al*., 1987; Beale *et al*., 1990; Clark and Spencer-Phillips, 1990) or mechanical disruption followed by either density gradient centrifugation (Gil and Gay, 1977; Tiburzy *et al*., 1992; Cantrill and Deverall, 1993) or lectin affinity chromatography (Hahn and Mendgen, 1992). Enzymic maceration and mechanical disruption have also been used to extract viable oospores of *Peronospora viciae* and *Plasmopara halstedii* from plant tissues (Van Der Gaag and Frinking, 1996; Spring and Zipper, 2000).

Mechanical disruption is suitable for hyphae of septate fungi but not aseptate oomycetes. Workers with downy mildews have used cellulase and Macerozyme R10 or R200 to isolate hyphae of *Bremia lactucae* (Crucefix *et al*., 1987) and *P. viciae* (Beale *et al*., 1990; Clark and Spencer-Phillips, 1990), but they were not in viable condition. Subsequently, Ashton and Spencer-Phillips (1993) isolated *P. viciae* hyphae using Macerozyme R200, and some were shown to be viable when stained with hydroethidine. Their protocol has been enhanced in the present work so that viable hyphae are isolated routinely and free from other microorganisms, enabling investigation of physiological processes such as nutrient uptake.

Nutrient transfer has been studied in a number of haustorial biotrophic pathogens (Spencer-Phillips, 1997; Hall and Williams, 2000;

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Mendgen and Hahn, 2002). Evidence suggests that either glucose or sucrose is the main host carbon compound transferred in powdery mildew infections (Manners, 1989; Aked and Hall, 1993; Clark and Hall, 1998). Furthermore, glucose was taken up by a suspension of mycelium *Erysiphe graminis* and *Erysiphe pisi* (Hall *et al*., 1992; Clark and Hall, 1998), and subsequent evidence confirms that glucose and not sucrose is transferred from the host to the powdery mildew mycelium in *E. graminis* infections (Sutton *et al*., 1999). Elegant and extensive work with the rust fungus *Uromyces fabae* has shown that haustoria are the major site of sugar and amino acid uptake from infected plant cells for this pathogen (Mendgen *et al*., 2000; Hahn and Mendgen, 2001; Mendgen and Hahn, 2002). These data also indicate that glucose is the main sugar accumulated by rust haustoria.

The position of intercellular hyphae in non-haustorial biotrophic infections is often suggestive of nutrient exchange, with hyphae commonly in close contact with host cell walls. In *Claviceps purpurea*, for example, hyphae lie adjacent to cells of the bundle sheath and phloem (Shaw and Mantle, 1980), and these hyphae are in a highly advantageous position to obtain nutrients. Similar observations have been made with hyphae of haustorial pathogens such as rust fungi (reviewed in Spencer-Phillips, 1997), and indeed evidence now suggests that amino acids may be accumulated by intercellular hyphae of *U. fabae* (Voegele and Mendgen, 2003). Experiments with the downy mildew pathogen *B. lactucae* infecting lettuce cotyledons showed that it could utilize glucose prior to penetration, while leucine uptake was delayed until haustoria were established (Andrews, 1975). More recent work indicated that intercellular hyphae of *P. viciae* are able to accumulate carbon from sucrose supplied to the apoplast of infected pea leaves (Clark and Spencer-Phillips, 1993). These experiments were conducted by feeding ^{14}C-labelled sugars to infected leaves and isolating hyphae, which were no longer viable, for analysis by autoradiography.

The present paper describes protocols both for isolating viable hyphae of the pea downy mildew pathogen, *P.viciae*, and for their use in experiments to investigate uptake of ^{14}C-labelled sugars. Preliminary data presented provide the first demonstration that downy mildew hyphae can accumulate sugars *in vitro*, and support earlier evidence (reviewed by Spencer-Phillips, 1997) that haustoria may not be essential for carbon accumulation by these pathogens.

2. MATERIALS AND METHODS

2.1 Infected plants

Seeds of *Pisum sativum* cvs Krupp Pelushka and Livioletta were soaked overnight in distilled water before being sown in 10 cm diameter pots containing Levington F2S compost, and then incubated in a regime of 16 h light at 20 °C and 8 h dark at 14 °C. Leaves of 10 days old plants were inoculated with conidia of a *P. viciae* isolate originally obtained from Dr David Kenyon at the National Institute of Agriculture Botany (Cambridge, UK). The inoculum was prepared by washing spores from sporulating infections and adjusted with sterile distilled water (SDW) to give 5×10^4 spores ml^{-1}, and then sprayed onto whole plants using a chromatography sprayer (Fisher Scientific Ltd, UK). Plants were incubated for 4 to 6 d in propagators with lids sealed with tape to ensure maximum relative humidity, in a regime of 14 h light and 10 h dark at 12 °C.

2.2 Isolation of endophytic hyphae

Isolation of hyphae was achieved using a medium containing Macerozyme R200 (Yakult, Japan) in phosphate buffered saline, based on a method developed by Ashton (1994). The maceration medium consisted of 25 ml 0.02 M phosphate buffer ($NaH_2PO_42H_2O$ and $Na_2HPO_42H_2O$) at pH 5.8, 0.15 M sodium chloride, 1% w/v dextran sulfite, 0.1 % w/v Macerozyme R200, and either 16 μg ml^{-1} Chloramphenicol, 1 mg ml^{-1} Streptomycin or a combination of 130 μg ml^{-1} Ampicillin and 52 μg ml^{-1} Rifampicin. The different solutions were sterilised by passing through a 0.2 μm Millipore syringe filter (Whatman Inc., UK), and all experimental procedures were performed under aseptic conditions.

Whole leaflets from plants 4, 5 and 6 d post-inoculation were surface sterilised by immersion in 40% v/v ethanol, then 1% v/v sodium hypochlorite, both for 30 s, and finally washed three times with sterile distilled water before infiltration with the maceration medium. This was injected into the mesophyll by using a 1 ml syringe without needle applied to the underside of the leaflet close to the petiole, until the whole lamina appeared water soaked (Figure 1a). Thirty injected leaflets were put into a 100 ml flask containing 25 ml macerating medium, and then incubated for 3 h at 25 °C, whilst mixing with air bubbled through 1 M sodium hydroxide solution to remove CO_2. The flasks were then shaken vigorously by hand for 3 minutes to complete the maceration process. Leaf tissue not macerated was removed with sterile forceps, and the remaining suspension transferred to a sterilised 50 ml tube, and centrifuged at 150 g (r = 12 cm) for 5 min. The supernatant was removed with a sterile pipette, and the pellet was washed twice in phosphate buffer (0.02 M, pH 5.8) before application of vital stains and incubation in ^{14}C-labelled sugars. Aliquots of cell and washing suspensions were plated onto potato dextrose agar and nutrient agar in Petri dishes, and incubated at 17 °C for 3–14 days to assess microbial contamination.

Hyphae were also isolated from infected leaves that had been inoculated with *P. viciae* conidia suspended in the fungicide Benlate (Synchemicals, UK) at 0.2% w/v, in order to inhibit growth of contaminating fungi.

2.3 Viability of isolated hyphae

Cell and mycelial suspensions were treated with the vital stains hydroethidine, fluorescein diacetate, Neutral Red, tetrazolium chloride and tetrazolium bromide to determine hypha and plant cell viability. The total number of isolated hyphae and lengths of viable portions were determined using a Nikon Optiphot microscope for 10 replicate 60 µl volumes (3 x 20 µl samples for each replicate) of cell suspensions from each of 10 separate isolation experiments.

A stock solution of hydroethidine (Polysciences Inc., USA) was made by dissolving 7 mg in 10 ml N, N-dimethyl acetamide. This was diluted for staining by adding 20 µl to 10 ml phosphate buffered saline (0.02 M phosphate buffer, 0.15 M NaCl, pH 7.4), then mixed immediately and filtered through a 0.2 µm Millipore filter. The cell suspension was centrifuged as before, the pellet gently resuspended in the stain solution and incubated at room temperature for 15 min in the dark. The stain solution was removed by centrifugation, the pellet washed once with phosphate buffer (0.02 M, pH 5.8), and finally resuspended in a small volume of buffer. A drop of the suspended cells was placed on a slide, covered with a cover slip, excess liquid removed and then examined by fluorescence microscopy using the Optiphot microscope and filter block UV10 (excitation, 330-380 nm; barrier transmission > 420 nm).

A stock solution of fluorescein diacetate (Sigma Chemical Co., UK) was prepared using 0.05 g in 10 ml acetone, from which fresh working solution was made for each experiment using 40 µl of stock solution diluted in 2 ml phosphate buffer (0.02 M, pH 5.8). Slides were prepared by mixing equal volumes of cell suspension and working solution, and viewed using the Optiphot microscope and filter block B (excitation, 450–490 nm; barrier transmission > 520 nm).

Neutral Red (Sigma Chemical Co., UK) stain was prepared as a 0.1% w/v solution in distilled water. A drop was added to a few drops of cell suspension on a slide and examined after 5 min with bright field optics.

Tetrazolium chloride (2,3,5-triphenyl tetrazolium chloride) was dissolved in phosphate buffer (0.05 M, pH 7.5) to give a 0.8 % w/v solution. This was added to a small test tube containing the cell suspension to give a 2:1 v/v mixture, and incubated for 1 h in the dark at either room temperature or 4°C. The suspension was then washed once with phosphate buffer (0.05 M, pH 7.5) and examined with bright field optics. A stock solution of tetrazolium bromide (3-(4,5-dimethylthiazol-2)-2,5-diphenyl-2H-tetrazolium bromide) was prepared containing 0.1% w/v dissolved in phosphate buffer (0.01 M, pH 6.2). Equal volumes of this solution and cell suspension were mixed and incubated at room temperature for 1 h, before examination with bright field optics.

2.4 ^{14}C-sugar uptake by isolated hyphae

For nutrient uptake experiments, hyphae of *P. viciae* isolated from 30 leaves were resuspended in 8 ml of phosphate buffer (pH 5.8). A 0.5 ml aliquot of hyphal suspension was added to 0.5 ml phosphate buffer (pH 5.8) plus 3.7 µl of 0.63 mM D-[U-^{14}C]glucose or -fructose at specific activity of 12 G Bq / m mol, or 6.6 µl of 0.35 mM [U-^{14}C]sucrose specific activity of 20.9 G Bq / m mol (product codes CFB. 96, CFB. 47 and CFB. 146, respectively; Amersham Pharmacia Biotech, UK). This gave a final sugar concentration of 2.3 µM. After incubating isolated hyphae with labelled sugar at 12 °C for 40 min, the suspension from each tube was filtered through a 0.45 µm HAWP 02500 membrane filter (Millipore, Ireland). Filters were washed once with sterile ice-cold phosphate buffer containing 10 times the concentration of the corresponding unlabelled sugar, followed by two washes with buffer alone, placed in scintillation vials and 300 µl Solusol (National Diagnostics, UK) were added. The suspension was incubated for 3 h before adding scintillation solution (Eco Safe Economy-Biodegradable Liquid Scintillation Cocktail, Meridian, UK). All the vials were incubated for 48 h at room temperature and radioactivity was measured by liquid scintillation counting using a scintillation counter (1215 Racbeta II, LKB Wallac, Finland), and corrected for quenching by chlorophyll and other constituents of the cell suspension (El-Gariani, 2003). Control incubations included sodium azide (NaN_3) or heat-treated hyphae, in order to quantify passive adsorption of radiolabelled sugars. For both treatments, hyphal suspension was prepared as described above, then either incubated with 10 mM NaN_3 at room temperature for 30 min or heated in boiling water for 5 min followed by incubation in phosphate buffer alone for 30 min prior to the start of the experiment. The relative proportions of label incorporated into soluble and insoluble metabolites were investigated by treating hyphae with ethanol following incubation with ^{14}C-glucose. Hyphae in the incubation suspension were pelleted and soluble components removed by suspending in 1 ml of 80 % v/v ethanol for 5 min, before they were resuspended in 1 ml phosphate buffer for scintillation counting.

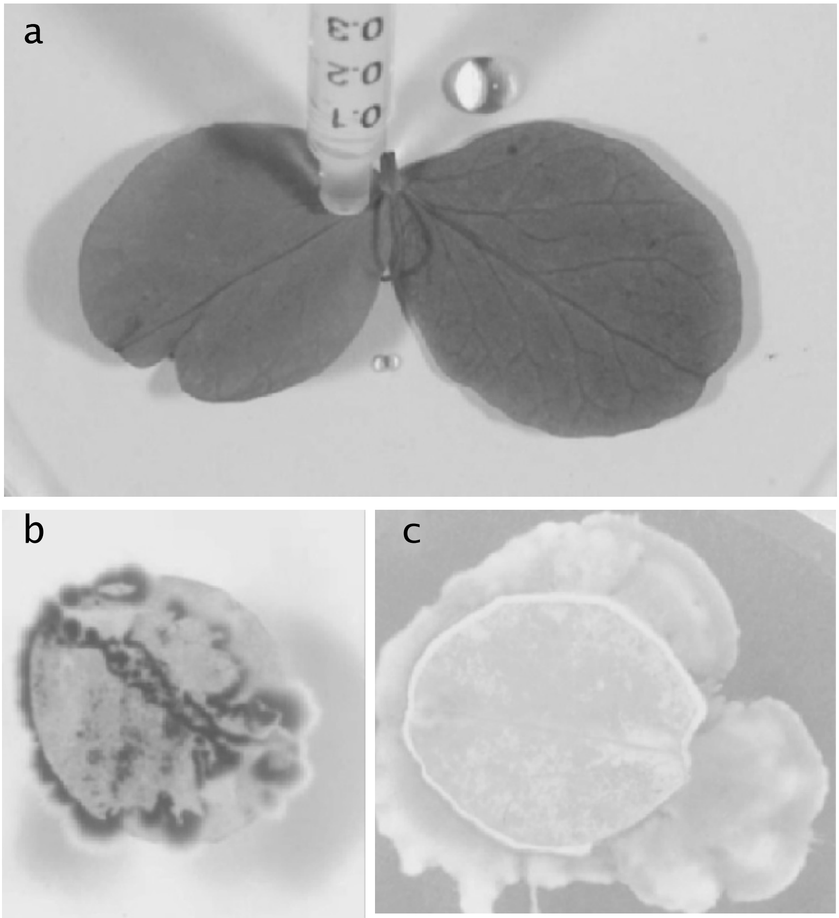

Figure 1. Leaflets of *Pisum sativum* cv. Leviolleta 4 d post-inoculation with *Peronospora viciae* conidia. (a) Infiltration of leaflets with maceration medium: leaflet on right has been infiltrated, giving a water-soaked appearance; leaflet on left about to be infiltrated, showing position of syringe. (b & c) Non-sterilised leaflets showing microbial growth after incubation on potato dextrose agar (b) and nutrient agar (c): the fungus in (b) was identified as *Cladosporium cladosporioides*, whilst the bacterium in (c) was not identified.

The statistical significance of differences between treatments was determined at the 5 % probability level by analysis of variance.

3. RESULTS

A large number of preliminary experiments using leaves of *P. sativum* cvs Liviolleta and Krupp Pelushka were undertaken (El-Gariani, 2003) to establish optimal conditions for isolating viable hyphae of *P. viciae*. These showed that leaves of both cultivars were macerated to the same extent. Subsequent experiments used cv. Liviolleta, as cv. Krupp Pelushka was no longer available commercially. Whilst leaflets 4 days post-inoculation were macerated completely in 3 h, the maceration medium had little effect on leaves with 5 and 6 day old infections over the same maceration period. Therefore 4 day old infections were used in the protocol for routine isolation of hyphae.

Various epiphytic microorganisms usually colonise plant tissues. These potential contaminants should be avoided in any assay to determine nutrient uptake by isolated *P. viciae* hyphae. *Cladosporium cladosporioides* was a fungus commonly associated with *P. viciae* on *P. sativum*, and typically grew from non-sterilised leaves plated on potato dextrose agar (Figure 1b). Likewise, bacteria typically grew from leaves plated on nutrient agar (Figure 1c). To suppress these microorganisms, experiments were undertaken to select an appropriate method of surface sterlisation. Antibiotics were also used. The outcome was the surface sterilisation protocol described above. This eliminated epiphytic microorganisms so that none grew when surface sterilised leaflets were placed on the agar media. The viability of *P. viciae* hyphae was not affected by this protocol, but endophytic bacteria were not eliminated. Hyphal viability was decreased, however, when the concentrations of the sterilisation agents and the time of application to pea leaves were increased.

The predominant bacterial contaminants produced white to yellowish colonies when cultured on nutrient agar (Figure 1c), and presumably included endophytes as they sometimes grew from surface sterilised leaves. Four antibiotics were tested at different concentrations in the maceration medium to prevent this source of contamination. Preliminary results showed that Streptomycin at concentrations up to 1 mg ml^{-1} did not reduce bacterial growth, whilst Chloramphenicol at 16 μg ml^{-1} reduced bacterial contamination, but also killed the *P. viciae* hyphae. Bacterial growth was completely suppressed using a combination of 130 μg ml^{-1} Ampicillin and 52 μg ml^{-1} Rifampicin, which also resulted in a similar proportion of viable hyphae to control isolations without antibiotics. The bacterial growth was not suppressed when either Rifampicin or Ampicillin alone was applied at these concentrations.

Occasionally, hyphae of *C. cladosporioides* were observed when the cell suspension from macerated leaves was viewed by microscopy, but these hyphae were non-viable as no colonies grew when plated out on the agar media. Microscopy also showed large hyphal fragments of *P. viciae*. These had variable width, contained granular cytoplasm and had haustoria that were either completely or incompletely detached from plant cells (Figure 2a).

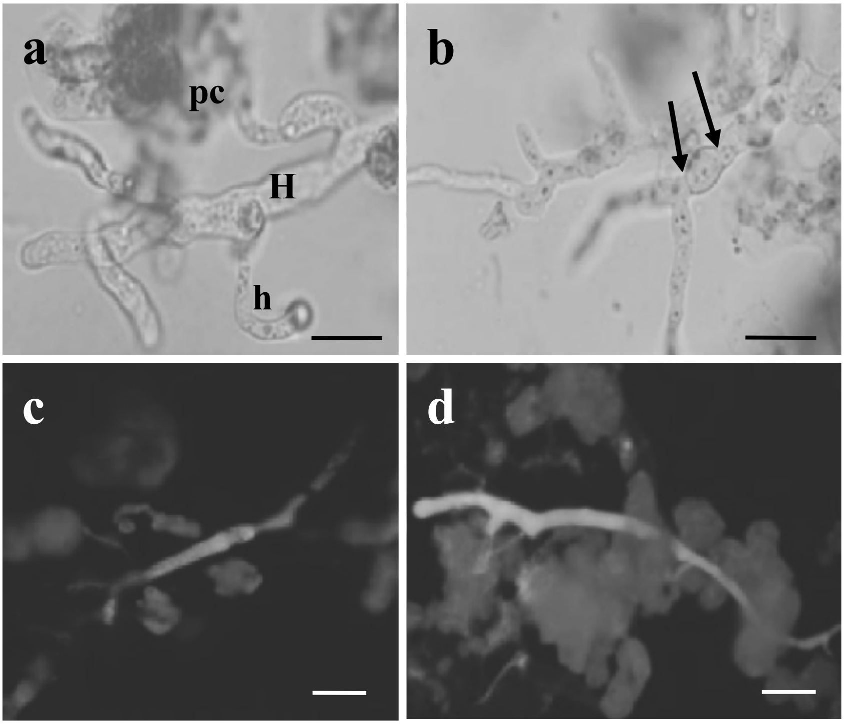

Figure 2. Hyphae of *P. viciae* and cells of *P. sativum* cv. Liviolleta prepared from leaf tissue enzymically macerated 4 d after inoculation. (a) Hypha (H) with granular cytoplasm and haustoria (h) that were sometimes completely detached from plant cells. (b) Hyphae isolated from leaves inoculated with conidia treated with Benlate; note septa indicated by arrows. (c) Viable hypha showing blue fluorescence after staining with hydroethidine; non-viable plant cells show a red auto-fluorescence. (d) Pink auto-fluorescence of hypha stained with hydroethidine shows it is non-viable. Scale bars = 25 µm.

Hyphae isolated from leaves inoculated with conidia treated with Benlate had an altered morphology (Figure 2b). They were stunted with an increased frequency of branching. Hyphae were septate, and haustoria were not observed and appeared to be absent. Furthermore, sporulation of *P. viciae* was reduced compared to plants inoculated with conidia in SDW alone.

Some hyphae incubated in hydroethidine showed a bright blue fluorescence indicating viability (Figure 2c), whilst others showed an orange-red or pink auto-fluorescence (Figure 2d) indicating that they were not viable. No plant cells showed the blue fluorescence but instead their chlorophyll produced a red auto-fluorescence. With fluorescein diacetate (FDA), a yellow-green fluorescence indicative of viable cytoplasm was observed in hyphae but not plant cells, which again showed red auto-fluorescence. A similar yellow-green fluorescence, however, was also seen in control hyphae not stained with FDA. Initial experiments using Neutral Red showed that it was a useful viability stain for *P. viciae* conidia that contain large vacuoles. Neutral Red stained the cytoplasms of all plant cells, but did not readily differentiate between viable and non-viable hyphae. Tetrazolium chloride and tetrazolium bromide stained neither plant cells nor hyphae with the red colour reported to be characteristic of viable cytoplasm.

These staining experiments confirmed that plant cells were not viable, and that hydroethidine was the most appropriate stain for determining the viability of isolated hyphae. The hyphae remained viable when incubated with either glucose or sucrose at 12 °C for 3 h and 4 °C for 5 h, but no viable hyphae were observed after 24 h at 4 °C.

The mean length of viable *P. viciae* hyphae ranged between 218.8 ± 39.8 and 343.9 ± 96.5 μm in 10 replicate experiments (10 replicate 60 μl volumes of cell suspension for each experiment), with an overall mean of 290.9 ± 12.9 μm (Figure 3). One way analysis of variance showed that there was a significant difference (at 95 % confidence level) in length of viable hyphae between the experiments. The total number of hyphae isolated in 0.5 ml of cell suspension (the volume used in sugar uptake experiments) was calculated from the numbers counted in the 60 μl samples. This ranged from 133 to 350, and the number that were viable ranged from 25 to 83, in 10 separate experiments, with a mean of approximately 190 hyphae. Of these, approximately 43 would be viable, giving a total length of viable hyphae of 12509 μm. The average diameter of hyphae determined from 150 measurements was 5.3 ± 0.15 μm. The length of viable hyphae was determined for each sugar uptake experiment and used to calculate the total surface area in the incubation solution, so that uptake of label could be expressed as p moles min^{-1} cm^{-2} of plasma membrane.

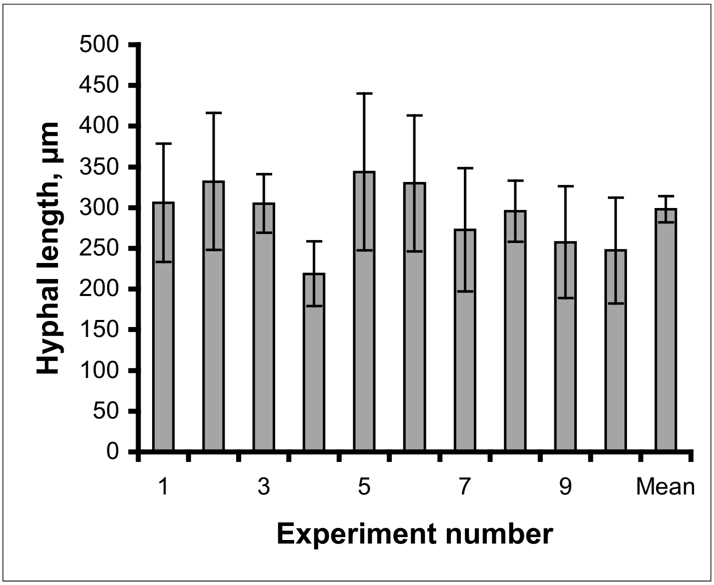

Figure 3. Length of *P. viciae* hyphae isolated by the routine protocol 4 d post-inoculation in each of 10 experiments, presented as means (± standard error) of 10 replicate 60 µl samples. Mean = mean of total of 35 hyphae.

The rate of uptake of label from glucose ranged between 27.81 ± 3.86 and 443.61 ± 63.24 p mole cm^{-2} min^{-1} in a total of 8 experiments. In the experiments shown in Figure 4, uptake from glucose was 118.47 ± 15.71 p mole cm^{-2} min^{-1}, and was eight times greater than the uptake of label from fructose and sucrose. Uptake of these two sugars was also not significantly different than in the presence of azide. In contrast, azide inhibited uptake of label from glucose by approximately 81 % (Figure 4), and heat treatment inhibited uptake by 97 % (data not shown).

Treatment of hyphae with 80 % ethanol after glucose uptake reduced label to a mean of 46 % from 4 experiments (38 % in Figure 4). Glucose plus ethanol treatment was significantly different ($p \leq 5$ %) from glucose plus azide, and if the azide control data are subtracted from the glucose alone and glucose plus ethanol wash data, this indicates that 23 % of label accumulated was insoluble in the experiments shown in Figure 4.

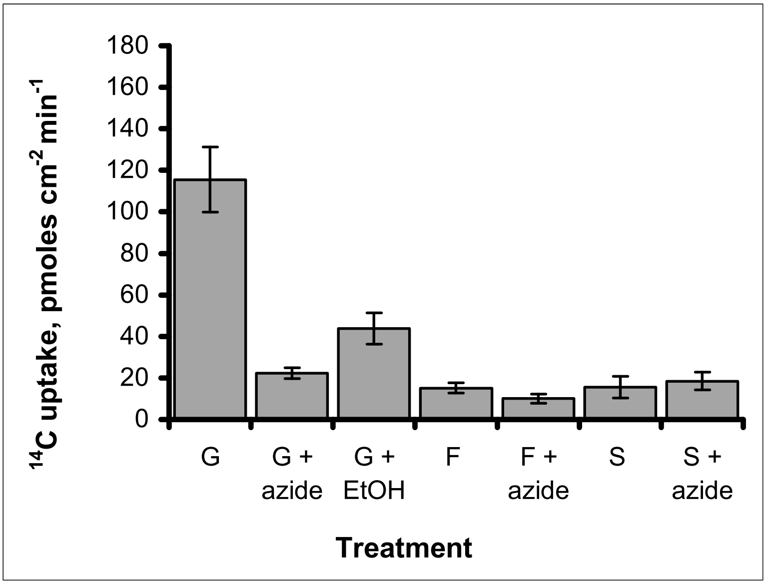

Figure 4. Accumulation of ^{14}C from 2.3 µM ^{14}C -glucose (G), -fructose (F) and -sucrose (S) by *P. viciae* hyphae *in vitro*, in the absence and presence of 10 mM NaN_3 (azide). For glucose, hyphae were washed in 80 % ethanol after incubation in labelled sugar (G + EtoH). Data presented as mean ± SE of three experiments each with five replicates, expressed as uptake cm^{-2} surface area of viable hyphae min^{-1} during a 40 min uptake period.

4. DISCUSSION

This study has demonstrated that a macerating medium containing Macerozyme R200 in phosphate buffered saline, amended with a combination of the antibiotics Ampicillin and Rifampicin, resulted in the isolation of viable hyphae but non-viable plant cells. These results agree with Ashton (1994), who found that maceration media containing phosphate buffered saline resulted in no plant cells remaining viable. In a previous study with *B. lactucae*, Crucefix *et al*. (1987) found that hyphae were non-viable after isolation from lettuce leaves when cellulases and hemicellulases were used in the maceration medium. These enzymes may have affected the fungal cell wall, causing loss of viability. Thus the present results are significant because this is the first time that downy mildew hyphae have been isolated routinely from infected tissues in a viable condition.

A range of vital stains was examined to find which was most appropriate for *P. viciae* hyphae. Hydroethidine accumulated in the cytoplasm of living cells and was retained to show a blue fluorescence in viable hyphae following exposure to UV light (Bucana *et al*., 1986; Ashton, 1994), while non-viable hyphae appeared an orange-red or pink colour. The hydoethidine viability test was used routinely to examine isolated hyphae prior to incubation with ^{14}C labelled sugars. Use of hydroethidine showed that *P. viciae* hyphae can remain viable when extracted from infected leaves by maceration for up to 3 h at 12 °C. This period of time is more than needed for sugar uptake experiments. One disadvantage is that visualization of hydroethidine staining requires UV light, so any hyphae become non-viable during microscopic examination (Ashton, 1994). For this reason, other vital stains were assessed, because they could be used to test viability of hyphae that could then be used in experiments.

Fluorescein diacetate (FDA) has been applied to determine viability of conidia, mycelium and oospores of several species of fungi which include *Beauvaria bassiana, Paecilomces fumosoroseus* and *Phytophthora megasperma*, where it was found to be a reliable viability test (Cohen, 1984; Schading *et al*., 1995). Results here showed that isolated *P. viciae* hyphae gave a yellow-green fluorescence, but as this was similar to unstained hyphae it was not a reliable viability test. Neutral Red is accumulated into vacuoles of viable cells, which are then the only structures to appear red. If the cells are dead or damaged then the whole cell will stain a red colour (Gahan, 1984; Weber, 2002). Initial experiments using Neutral Red showed that it was a useful viability stain for *P. viciae* conidia that contain large vacuoles. In contrast, it was less useful for hyphae, which had reduced vacuole size after maceration (Ashton, 1994). In the present work Neutral Red frequently did not accumulate in hyphal vacuoles, but stained the whole cytoplasm instead. This may be due to the hyphae being killed by the high light intensity of the microscope and the vacuolar membranes lysing to release their dye content into the cytoplasm (Ashton, 1994).

Tetrazolium bromide has been used to determine the viability of *Phytophthora megasperma* and *P. viciae* oospores (Cohen, 1984; Van Der Gaag, 1994). These studies found that viable oospores stained red, with black or non-coloured oospores indicating non-viability. In the present results, both tetrazolium bromide and chloride produced non-coloured hyphae, which may have lost viability as a result of the duration of incubation in the dyes. These problems with FDA, Neutral Red and the tetrazolium stains meant that hydroethidine remained the most useful vital stain.

Isolated hyphal fragments were 219–344 μm in length and 5.3 μm in diameter. This contrasts with the data of Clark and Spencer-Phillips (1990) who found that hyphae isolated from the different cv. Krupp

Pelushka were between 90-450 µm long and 2-25 µm wide. However, the isolation method used by these authors was different, and resulted in non-viable hyphae. The variation in number, total length and viability of hyphae isolated in each sample is a factor that would affect nutrient uptake experiments by introducing large variability between samples. For this reason, the length of viable hyphae used for each sugar uptake experiment was measured.

Prior to isolating viable *P. viciae* hyphae, it was necessary to sterilize the leaves to eliminate other microbial growth. Sodium hypochlorite and ethanol were used in a surface sterilisation regime to suppress at least epiphytic microorganisms. Contaminants have been avoided previously by surface sterilisation of infected leaves, followed by antibiotic treatment during the enzymatic maceration stage (Street *et al*., 1986; Clark and Spencer-Phillips, 1990; Van Der Gaag and Frinking, 1996). This worked well in the present work, although hyphal viability decreased when the concentration and time of application were increased. This suggested that under these conditions the sterilising agents penetrated the plant tissue.

Inoculation of plants with conidia treated with Benlate resulted in hyphae of *P. viciae* appearing septate. Other workers found that fungicides applied to *Phytophthora* spp. induced similar phenotypic and morphological changes but did not suppress fungal growth (Kuhn *et al.,* 1991; Groves and Ristaino, 2000; Staples, 2001). They also reported that the form of development induced by Benlate included enhanced or altered patterns of branching, which may account for the apparent increased branching observed here for *P. viciae*. The lack of haustoria following Benlate treatment further supports the view that downy mildew fungi do not need to produce haustoria for successful colonisation of plant tissues (Spencer-Phillips, 1997).

Antibiotics were also used to control endophytic bacteria. Whilst Chloramphenicol suppressed contamination, it reduced viability of *P. viciae* hyphae at the concentrations used in this study. This was not surprising, as Chloramphenicol inhibits protein synthesis in oomycetes and generally inhibits growth of the Peronosporales (Tsao, 1983 cited in Van Der Gaag and Frinking, 1996). A combination of 130 µg ml^{-1} Ampicillin and 52 µg ml^{-1} Rifampicin, which did not reduce viability of *P. viciae* hyphae significantly, completely controlled bacteria. Van Der Gaag and Frinking (1996) also found that these antibiotics suppressed bacterial growth and did not affect oospore germination of *P. viciae*. Streptomycin was found to inhibit bacterial growth only partly, and so was not useful. The results of this study indicated that it was possible to achieve satisfactory disinfestation of *P. sativum* leaves using the surface sterilisation protocol and combination of antibiotics reported above. This approach was adopted in the routine protocol for isolating viable hyphae used for sugar uptake experiments. Again, this is a significant advance on the work of Ashton (1994), whose method did not control microbial contamination.

As mentioned earlier, the length of viable hyphae was measured for each sugar uptake experiment, due to the variability in number of hyphae isolated between replicates. Despite this complication, the maceration method for isolating hyphae provides a new approach to study nutrient uptake by these pathogens. The experiments with fructose and sucrose showed that they were not accumulated to a significant amount, in contrast to label from glucose. This indicates that downy mildew hyphae can accumulate carbon from sugars in the absence of viable plant cells, and that glucose appears to be the preferred sugar. Thus *P. viciae* is predicted to have a hexose transporter that transports glucose but not fructose. This is different to dikaryotic hyphae of the rust fungus *U. fabae*, where a hexose transporter expressed specifically at the haustorial plasma membrane transports both glucose and fructose (Voegele *et al*., 2001).

Clark and Spencer-Phillips (1993) found that ^{14}C from sucrose within the leaf apoplast was accumulated by *P. viciae*. It is possible that downy mildews have evolved a sucrose transport system at the host-pathogen interface, given that sucrose is the most abundant soluble carbohydrate in the transport stream of many vascular plants. Indeed there is evidence that sucrose is taken up intact by some fungi (Lam *et al*., 1994; Reinders and Ward, 2001). However, the present results suggest that in the experiments by Clark and Spencer-Phillips (1993), ^{14}C-sucrose supplied to the apoplast may first have been converted to glucose and fructose by extracellular invertase activity, with glucose being the major sugar accumulated by intercellular hyphae. The ability to accumulate sugars *in vitro* supports the notion that haustoria may not be essential for carbon accumulation by downy mildews (Spencer-Phillips, 1997).

In conclusion, we have reported a novel method for isolating viable hyphae of a downy mildew fungus, which can be used in physiological experiments. The results presented demonstrate how this approach can be applied for investigating nutrient uptake processes. The method can also be used to validate molecular data, for example after key proteins and their genes have been identified and deployed to generate functional transformants.

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BENZOTHIADIAZOLE-INDUCED RESISTANCE TO *PLASMOPARA HALSTEDII* (FARL.) BERL. *ET* DE TONI IN SUNFLOWER

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1. INTRODUCTION

Downy mildew of sunflower caused by *Plasmopara halstedii* (Farl.) Berl. *et* de Toni is one of the most destructive diseases of this host worldwide. The pathogen can be effectively controlled by using resistant plant cultivars and seed dressing. Recently, however, protection against the sunflower downy mildew has been affected by several factors. One of these is the high variability of *P. halstedii*. Nowadays, fourteen different pathotypes (races) are known all over the world (Gulya *et al.,* 1996) and at least six of these exist in Hungary (Kormány and Virányi, 1997; Komjáti *et al.,* this volume). Tolerance (resistance) of the pathogen to the fungicide metalaxyl is another problem, with which the plant growers and breeders have to cope (Albourie *et al.*, 1998).

Beside the traditional control strategies, therefore, alternative or supplementary methods are needed to provide effective protection against the disease. One possible solution is the use of systemic acquired resistance (SAR), i.e. the activation of the defence system of the plants.

In the present work, we examined a commercially available immunactivator, Bion 50 WG (benzo(1,2,3)thiadiazole-7-carbothioic acid S-methyl ester) to test its effectiveness in this pathosystem. Although Bion has already been found to reduce infection of sunflower by *P. halstedii* (Tosi *et al.*, 1999), we expanded our study to include different compatible and incompatible host-parasite combinations. In addition, the histology of genetic and systemic acquired resistance was also compared.

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2. MATERIALS AND METHODS

The sunflower cultivar GK70, and inbred lines DM2 and RHA274 as well as pathotypes (races) of *Plasmopara halstedii* 100, 700 and 710 were used to get four compatible (GK70+100, GK70+700, DM2+710, RHA274+700) and one incompatible (RHA274+100) combinations for examination.

In the first experiment, pre-germinated seeds (25 for each treatment) were inoculated with sporangia of the downy mildew fungus (50,000 sporangia/ml) using the whole seedling inoculation technique (Cohen and Sackston, 1973), one day prior to or after treatment with Bion 50 WG (a commercial product) at 320 mg/l. Seedlings were soaked in Bion solution for 30 minutes. In the second experiment, however, only pre-treatment was undertaken and Bion concentrations at 160 mg/l and 320 mg/l were used. The germlings were then planted into pots and grown in a climatic chamber (experiment 1) or in a glasshouse (experiment 2) for a month. Eight days after planting, plants were covered with plastic bags and sprayed with distilled water to provide a saturated environment for fungal sporulation.

Disease assessement was made by recording the occurrence and the intensity of sporulation, by using a 0-4 scale where the proportion of leaf area covered by sporangia was graded as follows: 0, no sporulation; 1-4, sporulation appearing on <25 %, 25-75 %, >75 % and 100 % of the total leaf area, respectively (Oros and Virányi, 1987). After three weeks from planting, typical symptoms such as leaf chlorosis and/or damping-off were also recorded. The height of plants was measured at 11, 13, 17 and 22 days after planting, because downy mildew also causes severe stunting of affected plants.

Histological examination of Bion treated and untreated host-pathogen combinations was undertaken by fluorescence microscopy (Olympus, Japan; filter block BX 50, excitation 460-490 nm, transmission >515 nm). A series of cross-sections of sunflower stems were examined to detect fungal elements (hyphae, haustoria) and host cell response (necrosis, and secondary cell division). Sections were examined unstained. Infection and necrosis were assessed on a 0-4 scale based on their appearance in one, two, three and four quarters of the cross-sections both in the cortical and pith parenchyma.

Experiments were repeated twice. Data were subjected to analysis of variance (α=0.05) using Fisher's multiple comparison with the MINITAB 10.2 statistical package.

3. RESULTS

Bion treatment at 320 mg/l significantly reduced fungal sporulation on cotyledons (Figure 1) and leaf chlorosis on true leaves (data not shown) in all compatible host-pathogen combinations examined. Pre-treatment, however, was more effective as compared to treatment after inoculation. There was no infection found in the incompatible interaction (RHA274+100) as far as the appearance of symptoms either on cotyledons or on true leaves is concerned.

No significant difference in disease severity could be found between Bion treatments at 160 or 320 mg/l (Figure 2).

The incidence of damping-off as one of the most visible symptoms of the downy mildew of sunflower was evaluated on 22-day old plants, and the results are presented in Table 1. Both pre- and post-application of Bion effectively reduced the number of plants damped-off. Pre-treatment, however, was more effective compared to treatment after inoculation in the RHA274+700 host-pathogen combination (Table 1). Furthermore, Bion 50 WG applied at 160 and 320 ppm reduced the number of plants damped-off to the same extent.

Figure 3 represents the average plant height in two compatible interactions (results were very similar in all other compatible combinations). Statistical differences were found in plant height between 0-control (non-treated and uninoculated) and any other treatment. Moreover, plants treated after inoculation with *P. halstedii* (post-treated) were significantly shorter than those in other treatments. No statistical difference could be detected between the height of plants treated either with 160 or 320 mg/l Bion prior to inoculation (data not shown).

Fluorescence microscopy of cross-sections of sunflower stems revealed a relatively higher rate of necrosis (cell death) in incompatible than in compatible interactions (Figures 4 and 5). In addition, genetic resistance coupled with Bion pre-treatment resulted in strong restriction of the fungus (Figure 4). In compatible host-pathogen combinations, however, depending on host genotype, the fungus could readily colonise parenchyma tissues at various rates (Figure 6) and Bion treatment significantly decreased the development of fungal structures, i.e. the appearance of hyphae and haustoria, as well (Figure 5). Furthermore, cell necrosis, intensive fluorescence and secondary cell division found to be associated with infected and Bion-treated plants resembled that usually appearing in genetically resistant inoculated sunflowers (Figure 7).

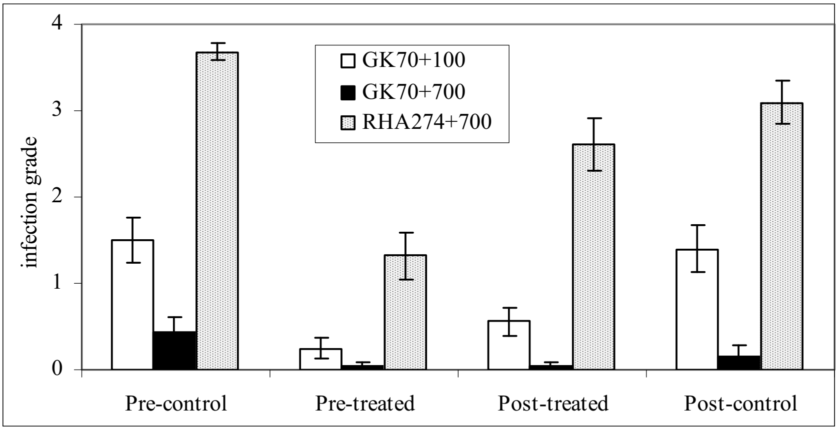

Figure 1. Infection grade (sporulation on cotyledons; 4=100%, see text) of 9-day old plants in three compatible combinations. Host varieties: GK70 and RHA274. *Plasmopara halstedii* pathotypes: 100 and 700. Pre-control: soaked in distilled water 1 day before inoculation. Pre-treated: soaked in Bion solution (320 mg/l) 1 day before inoculation. Post-treated: soaked in Bion solution 1 day after inoculation. Post-control: soaked in distilled water 1 day after inoculation. Bars represent 95% confidence intervals.

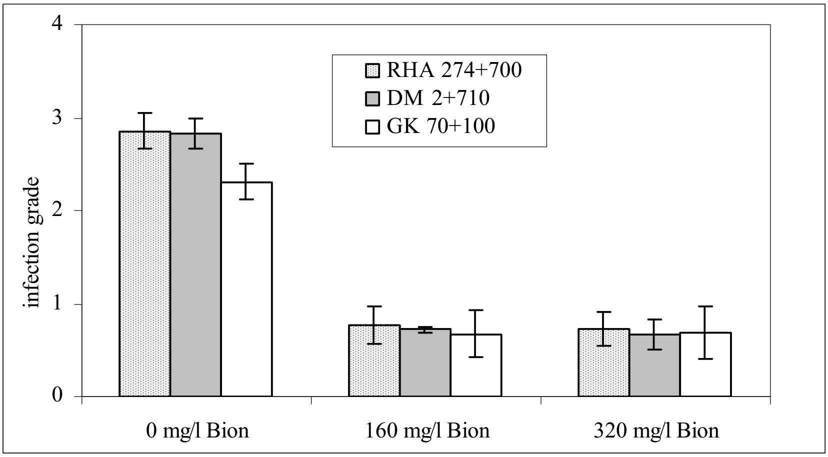

Figure 2. Effect of Bion concentrations on disease development of pre-treated plants (see Figure 1) in three compatible interactions (infection grade calculated as the average of sporulation intensity on cotyledons and true leaves). Bars represent 95% confidence intervals.

Table 1. Effect of Bion treatment on the occurrence of damping-off symptoms in different host-pathogen combinations 22 days from planting. For explanation of treatments see Fig. 1.

Host-parasite combinations	Ratio of damped-off plants (%)			
	Pre-control	Pre-treatment	Post-treatment	Post-control
GK70+100	47	0	0	15
GK70+700	40	7	0	21
RHA274+100	0	0	0	0
RHA274+700	100	40	80	100

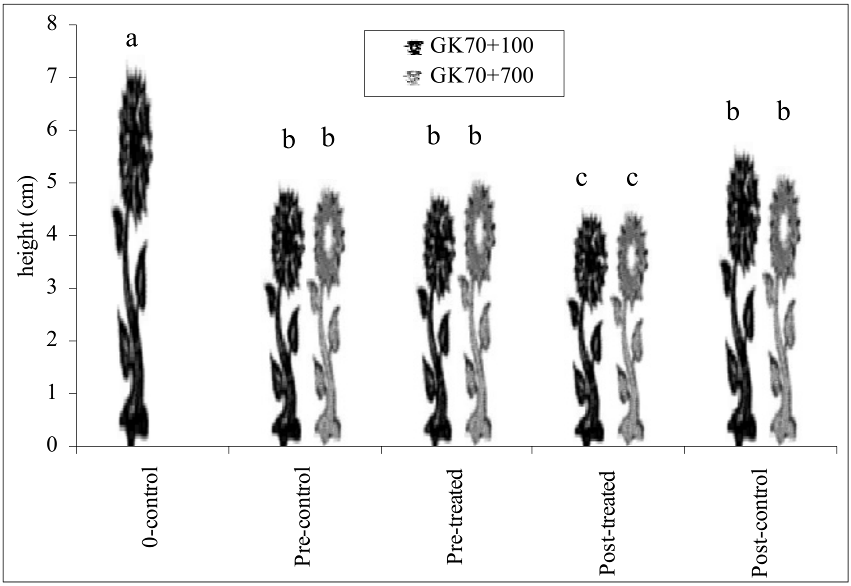

Figure 3. Average plant height during the one-month examination period in two compatible interactions. Letters a, b and c indicate significant differences between treatments (α=0.05). For explanation of treatments see Fig. 1; 0-control=non-treated and uninoculated.

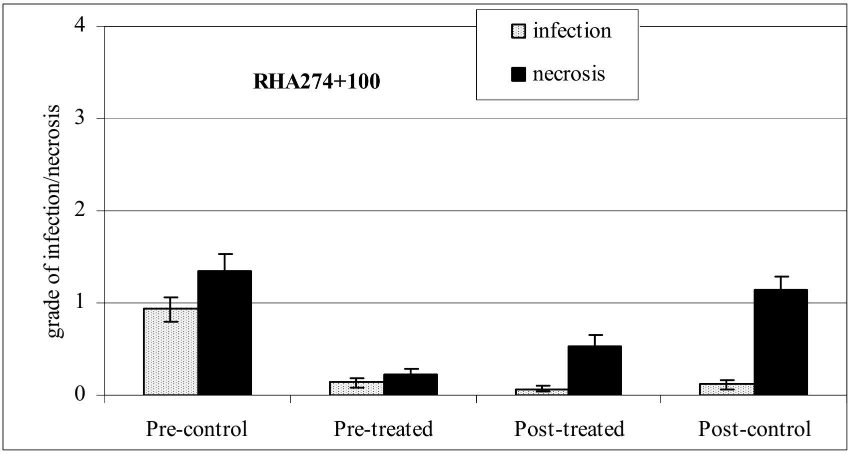

Figure 4. Effect of Bion treatment on host response (necrosis; see text) and pathogen development (hyphae and haustoria; see text) in an incompatible interaction. For explanation of treatments see Fig. 1.

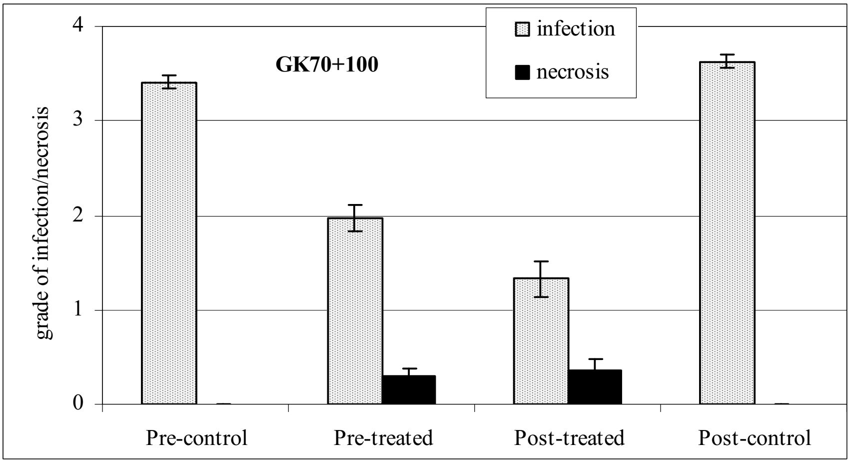

Figure 5. Effect of Bion treatment on host response and pathogen development in a compatible interaction. For explanation of grades and treatments see Figs 1 and 4.

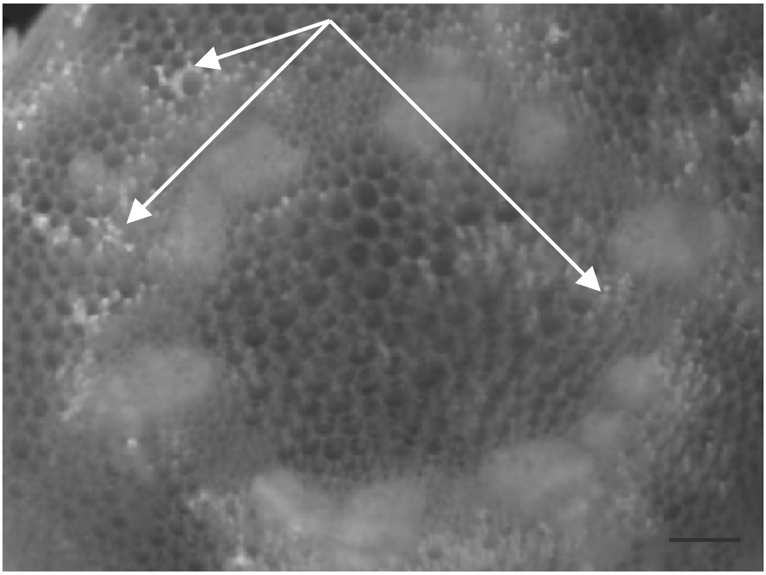

Figure 6. Plasmopara halstedii (pathotype 100)-sunflower (cv. GK 70) compatible interaction. Arrows show autofluorescence of cell wall around intercellular hyphae (scale bar = 400 μm).

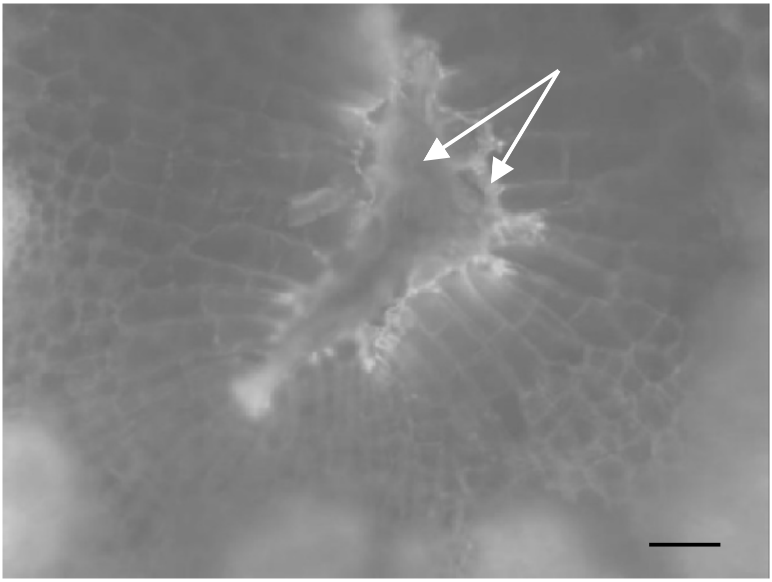

Figure 7. Plasmopara halstedii (pathotype 100)-sunflower (RHA 274) incompatible interaction. Arrows show cell necrosis and intensive fluorescent response (scale bar = 100 μm).

4. CONCLUSIONS

In the present study, the effect of Bion 50 WG, an immunactivator, on protecting sunflower against *P. halstedii* has been investigated using compatible and incompatible host-pathogen interactions. Systemic acquired resistance (SAR) induced by Bion appeared to be able to restrict downy mildew symptoms, seen as fungal sporulation, leaf chlorosis and damping-off, significantly. Pre-treatment of the seedlings with Bion was more effective than treatment after inoculation, corroborating earlier findings by Sticher *et al.* (1997) and Tosi *et al.* (1999) who assumed the immunactivator needed a lag period prior to infection for sufficient effectivity.

A comparison of the effect of two different Bion concentrations showed that the lower application rate (160 mg/l) was as effective as the higher rate (320 mg/l). Nevertheless, 160 mg/l is the approximate concentration generally used in the field against powdery mildew of wheat and barley.

In our experimental conditions, the effect of Bion lasted at least for a month, but it was unable to counteract dwarfing of inoculated plants. Interestingly, reduced growth of plants after Bion application had already been described by Heil *et al.* (2000), a phenomenon possibly due to allocation of energy for induced resistance.

Several workers have investigated the histopathology of sunflower downy mildew including compatible and incompatible interactions (Virányi and Dobrovolszky, 1980; Virányi and Mohamed, 1985; Mouzeyar *et al.*, 1993, 1994, 1995; Heller *et al.*, 1997; Mazeyrat *et al.*, 1999), and described the resistant response of the host following inoculation. However, this is the first paper dealing with the appearance of SAR at host tissue level. Fluorescence microscopy revealed characteristic alterations of affected parenchymatous cells of induced plants that resembled those found in inoculated genetically resistant sunflower genotypes.

In conclusion, the immunactivator Bion 50 WG effectively restricted downy mildew development in sunflower in our experiments. Similarly, this compound has proved to be very active against a root-parasitic weed, *Orobanche cumana* (Sauerborn *et al.*, 2002), so it seems that Bion 50 WG has some potential as one component of an integrated programme for disease management in sunflower crops.

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